

Reinforcement Learning: Theory and Algorithms

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WORKING DRAFT

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Notation

This section collects notation and conventions that will be used throughout the book.

General Mathematical Notation

- For an integer $K \geq 1$, we slightly abuse notation and use 0-based indexing:

$$[K] := \{0, 1, 2, \dots, K-1\}.$$

- We use $\mathbb{1}\{\mathcal{E}\}$ to denote the indicator function for an event $\mathcal{E}$, which takes value 1 if $\mathcal{E}$ is true and 0 otherwise.
- We use $\tilde{O}(\cdot)$ notation to hide polylogarithmic factors; i.e., $f(x) = \tilde{O}(g(x))$ if $f(x) = O(g(x) \cdot \text{poly}(\log x))$. Unless specified otherwise, $\log$ denotes the natural logarithm.
- For a set $\mathcal{X}$, let $\Delta(\mathcal{X})$ denote the set of probability distributions over $\mathcal{X}$.

Vectors and Norms

- For a vector v , the expressions $(v)^2$, $\sqrt{v}$, and $|v|$ denote the component-wise square, square root, and absolute value operations.
- Inequalities between vectors are understood elementwise. For vectors v, v' , we write $v \leq v'$ if $v(j) \leq v'(j)$ for all coordinates j .
- For a vector v , we refer to its j -th component as either $v(j)$ or $[v]_j$.
- For real-valued matrix A , we denote $\|A\|_2 = \sup_{x: \|x\|_2=1} \|Ax\|_2$ which denotes the maximum singular value of A . We denote $\|A\|_F$ as the Frobenius norm $\|A\|_F^2 = \sum_{i,j} A_{i,j}^2$ where $A_{i,j}$ denotes the i, j 'th entry of A .
- For a vector x and a matrix M , unless stated otherwise, $\|x\|$ and $\|M\|$ denote the Euclidean and spectral norms, respectively. For a positive semi-definite matrix M , we use $\|x\|_M := \sqrt{x^\top M x}$ (with dimensions understood from context).
- We denote the ℓ_∞ -norm (max norm) by $\|x\|_\infty := \max_i |x_i|$. (Note: In the context of value functions V and Q , contraction arguments typically rely on the ℓ_∞ -norm).

Probability

- For a real-valued function f and a distribution $\mathcal{D}$, define the variance as

$$\text{Var}_{\mathcal{D}}(f) := \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2] - (\mathbb{E}_{x \sim \mathcal{D}}[f(x)])^2.$$

Reinforcement Learning Context

- We write $s \in \mathcal{S}$ for states and $a \in \mathcal{A}$ for actions. Rewards are real-valued; when convenient we treat the reward function as a vector $r \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ indexed by (s, a) .
- The discount factor is $\gamma \in [0, 1)$; for a policy π , the (discounted) value function is $V^\pi : \mathcal{S} \rightarrow \mathbb{R}$ and the action-value function is $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$.
- We denote a *history* (or *trajectory prefix*) at time t as $\tau_t := (s_0, a_0, r_0, \dots, a_{t-1}, r_{t-1}, s_t)$, which represents the information available to the agent at decision time t . We let $\mathcal{H}$ denote the set of all such finite histories.
- We overload notation so that for a distribution μ over $\mathcal{S}$,

$$V^\pi(\mu) := \mathbb{E}_{s \sim \mu} [V^\pi(s)].$$

- We overload notation and let P also denote a matrix of size $|\mathcal{S}||\mathcal{A}| \times |\mathcal{S}|$ with entries $P_{(s,a),s'} := P(s' \mid s, a)$. We also define P^π to be the transition matrix on state-action pairs induced by a deterministic policy π , namely

$$P_{(s,a),(s',a')}^\pi := \begin{cases} P(s' \mid s, a) & \text{if } a' = \pi(s'), \\ 0 & \text{if } a' \neq \pi(s'). \end{cases}$$

With this notation,

$$\begin{aligned} Q^\pi &= r + \gamma P V^\pi, \\ Q^\pi &= r + \gamma P^\pi Q^\pi, \\ Q^\pi &= (I - \gamma P^\pi)^{-1} r. \end{aligned}$$

- For a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, define the greedy policy and its induced value as

$$\begin{aligned} \pi_Q(s) &:= \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a), \\ V_Q(s) &:= \max_{a \in \mathcal{A}} Q(s, a). \end{aligned}$$

- For a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, the *Bellman optimality operator* $\mathcal{T} : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined by

$$\mathcal{T}Q := r + \gamma P V_Q. \tag{0.1}$$

- For a stationary policy π , the *Bellman evaluation operator* $\mathcal{T}^\pi : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined by

$$\mathcal{T}^\pi Q := r + \gamma P^\pi Q.$$

- In the finite-horizon setting (with time-dependent transitions P_h and rewards r_h), we define the time-dependent operators $\mathcal{T}_h, \mathcal{T}_h^\pi : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ (which map the value at step $h+1$ to step h) as:

$$\begin{aligned} \mathcal{T}_h Q &:= r_h + P_h V_Q, \\ \mathcal{T}_h^\pi Q &:= r_h + P_h^\pi Q. \end{aligned}$$

- For a policy π and initial distribution μ , the *discounted state-action occupancy measure* $d_\mu^\pi \in \Delta(\mathcal{S} \times \mathcal{A})$ is defined as:

$$d_\mu^\pi(s, a) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s, a_t = a \mid s_0 \sim \mu).$$

We similarly define the state occupancy measure $d_\mu^\pi(s) := \sum_{a \in \mathcal{A}} d_\mu^\pi(s, a)$.

Part I

Fundamentals

Chapter 1

Markov Decision Processes

1.1 Discounted (Infinite-Horizon) Markov Decision Processes

In reinforcement learning, the interactions between the agent and the environment are often described by an infinite-horizon, discounted Markov decision process (MDP)

$$M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu),$$

specified as follows:

- A state space $\mathcal{S}$, which may be finite or infinite. For mathematical convenience, we will assume that $\mathcal{S}$ is finite or countably infinite.
- An action space $\mathcal{A}$, which also may be discrete or infinite. For mathematical convenience, we will assume that $\mathcal{A}$ is finite.
- A transition function $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, where $\Delta(\mathcal{S})$ is the space of probability distributions over $\mathcal{S}$ (i.e., the probability simplex). $P(s' \mid s, a)$ is the probability of transitioning into state s' upon taking action a in state s . We use $P_{s,a}$ to denote the vector $P(\cdot \mid s, a)$.
- A reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$. $r(s, a)$ is the immediate reward associated with taking action a in state s . More generally, $r(s, a)$ could be a random variable (with a distribution that depends on s, a). While we largely focus on the deterministic case, extensions to stochastic rewards are often straightforward.
- A discount factor $\gamma \in [0, 1)$, which defines an effective horizon for the problem.
- An initial state distribution $\mu \in \Delta(\mathcal{S})$, which specifies how the initial state s_0 is generated.

In many cases, we will assume that the initial state is fixed at s_0 , i.e. μ is a distribution supported only on s_0 .

1.1.1 The objective, policies, and values

Policies. In a given MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$, the agent interacts with the environment according to the following protocol: the agent starts at some state $s_0 \sim \mu$; at each time step $t = 0, 1, 2, \dots$, the agent first observes the current state s_t , then selects an action $a_t \in \mathcal{A}$, receives the immediate reward $r_t = r(s_t, a_t)$, and finally observes the next state $s_{t+1} \sim P(\cdot \mid s_t, a_t)$.

It will be convenient to work with the *history* (or trajectory prefix) available at decision time t , which ends with the currently observed state:

$$\tau_t := (s_0, a_0, r_0, s_1, \dots, a_{t-1}, r_{t-1}, s_t).$$

We will refer to τ_t interchangeably as a *history* or a *trajectory* (prefix). Note that τ_t contains exactly the information available to the agent when choosing a_t .

In the most general setting, a policy specifies a decision-making strategy in which the agent chooses actions adaptively based on the history of observations; precisely, a policy is a (possibly randomized) mapping $\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$, where $\mathcal{H}$ is the set of all finite trajectories of the form above (i.e. histories of all lengths that end in a state). At time t , the action is sampled as $a_t \sim \pi(\cdot \mid \tau_t)$.

A *stationary* policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ specifies a decision-making strategy in which the agent chooses actions based only on the current state, i.e. $a_t \sim \pi(\cdot \mid s_t)$. A deterministic, stationary policy is of the form $\pi : \mathcal{S} \rightarrow \mathcal{A}$.

Values. We now define values for (general) policies. For a fixed policy and a starting state $s_0 = s$, we define the value function $V_M^\pi : \mathcal{S} \rightarrow \mathbb{R}$ as

$$V_M^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid \pi, s_0 = s \right].$$

The expectation is with respect to the randomness of the trajectory (state transitions and any stochasticity in π). Since $r(s, a) \in [0, 1]$, we have $0 \leq V_M^\pi(s) \leq 1/(1 - \gamma)$.

Similarly, the action-value (or Q-value) function $Q_M^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is defined as

$$Q_M^\pi(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid \pi, s_0 = s, a_0 = a \right],$$

and $Q_M^\pi(s, a)$ is also bounded by $1/(1 - \gamma)$.

Goal. Given a state s , the goal of the agent is to find a policy π that maximizes the value:

$$\sup_{\pi} V_M^\pi(s), \tag{1.1}$$

where the supremum is over all (possibly non-stationary and randomized) policies. As we shall see, there exists a stationary deterministic policy which is simultaneously optimal for all starting states s .

We drop the dependence on M and write V^π when it is clear from context.

Example 1.1 (Navigation). Navigation is perhaps the simplest example of reinforcement learning. The state is the agent's current location. The four actions might be moving one step east, west, north, or south. In the simplest setting the transitions are deterministic. Taking the north action moves the agent one step north of its location. The agent might have a goal state g , and the reward is 0 until the agent reaches the goal, and 1 upon reaching g . Since $\gamma < 1$, there is an incentive to reach the goal earlier. As a result, the optimal behavior corresponds to finding a shortest path from the initial to the goal state. If a policy reaches the goal in d steps (and then stays there), the value from the initial state is γ^d .

Example 1.2 (Conversational agent). This is another natural reinforcement learning problem. The state can be the transcript of the conversation so far, along with additional context (e.g., task context, user profile information, or other relevant world state). Actions depend on the domain; in the most basic form, an action is the next utterance.

Sometimes, conversational agents are designed for task completion, such as a travel assistant, tech support agent, or virtual receptionist. In these cases, there may be a predefined set of *slots* that the agent must fill before it can propose

a solution (e.g., travel dates, origin, destination, and mode of travel). Actions can be natural language queries aimed at filling these slots.

In task-completion settings, reward is naturally defined as a binary indicator of whether the task was completed (e.g., whether travel was successfully booked). Depending on the domain, the reward can be refined to reflect quality or cost. In more open-ended conversational settings, the reward may reflect user satisfaction.

Example 1.3 (Strategic games). This is a popular category of applications, where reinforcement learning has achieved human-level performance in Backgammon, Go, Chess, and various forms of Poker. Typically, the state is the current game position, actions are legal moves, and reward is the eventual win/loss outcome (or a more detailed score, when defined). Technically, these are multi-agent settings, yet many successful approaches use methods that do not explicitly model multiple agents.

1.1.2 Bellman Consistency Equations for Stationary Policies

Stationary policies satisfy the following consistency conditions:

Lemma 1.4. *Suppose that π is a stationary policy (possibly randomized). Then V^π and Q^π satisfy the following Bellman consistency equations: for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,*

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)], \\ Q^\pi(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^\pi(s')]. \end{aligned}$$

In particular, if π is deterministic, then $V^\pi(s) = Q^\pi(s, \pi(s))$.

We leave the proof as an exercise to the reader.

It is helpful to view V^π as a vector of length $|\mathcal{S}|$, and Q^π and r as vectors of length $|\mathcal{S}| \cdot |\mathcal{A}|$. We overload notation and let P also refer to a matrix of size $(|\mathcal{S}| \cdot |\mathcal{A}|) \times |\mathcal{S}|$ where the entry $P_{(s, a), s'}$ is equal to $P(s' | s, a)$.

We also define P^π to be the transition matrix on state-action pairs induced by a stationary policy π , specifically:

$$P_{(s, a), (s', a')}^\pi := P(s' | s, a) \pi(a' | s').$$

In particular, for deterministic policies we have:

$$P_{(s, a), (s', a')}^\pi := \begin{cases} P(s' | s, a) & \text{if } a' = \pi(s'), \\ 0 & \text{if } a' \neq \pi(s'). \end{cases}$$

With this notation, it is straightforward to verify:

$$\begin{aligned} Q^\pi &= r + \gamma P V^\pi, \\ Q^\pi &= r + \gamma P^\pi Q^\pi. \end{aligned}$$

Corollary 1.5. *Suppose that π is a stationary policy. Then*

$$Q^\pi = (I - \gamma P^\pi)^{-1} r, \tag{1.2}$$

where I is the identity matrix.

Proof. To see that $I - \gamma P^\pi$ is invertible, observe that for any non-zero vector $x \in \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|}$,

$$\begin{aligned} \|(I - \gamma P^\pi)x\|_\infty &= \|x - \gamma P^\pi x\|_\infty \\ &\geq \|x\|_\infty - \gamma \|P^\pi x\|_\infty && \text{(triangle inequality for norms)} \\ &\geq \|x\|_\infty - \gamma \|x\|_\infty && (P^\pi \text{ is row-stochastic}) \\ &= (1 - \gamma)\|x\|_\infty > 0, && (\gamma < 1, x \neq 0) \end{aligned}$$

which implies $I - \gamma P^\pi$ is full rank. □

The following lemma will also be useful:

Lemma 1.6. *We have that*

$$[(1 - \gamma)(I - \gamma P^\pi)^{-1}]_{(s,a),(s',a')} = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}^\pi(s_t = s', a_t = a' \mid s_0 = s, a_0 = a),$$

so we can view the (s, a) -th row of this matrix as an induced distribution over state-action pairs when following π after starting with $s_0 = s$ and $a_0 = a$.

We leave the proof as an exercise to the reader.

1.1.3 Bellman Optimality Equations

A remarkable and convenient property of discounted MDPs is that there exists a stationary and deterministic policy that simultaneously maximizes $V^\pi(s)$ for all $s \in \mathcal{S}$. This is formalized in the following theorem.

Theorem 1.7. *Let Π be the set of all (possibly non-stationary and randomized) policies. Define*

$$\begin{aligned} V^*(s) &:= \sup_{\pi \in \Pi} V^\pi(s), \\ Q^*(s, a) &:= \sup_{\pi \in \Pi} Q^\pi(s, a). \end{aligned}$$

These quantities are finite since $V^\pi(s)$ and $Q^\pi(s, a)$ are bounded between 0 and $1/(1 - \gamma)$.

There exists a stationary and deterministic policy π such that for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$\begin{aligned} V^\pi(s) &= V^*(s), \\ Q^\pi(s, a) &= Q^*(s, a). \end{aligned}$$

We refer to such a π as an optimal policy.

Proof. For any $\pi \in \Pi$ and time t , π specifies a distribution over actions conditioned on the history of observations. For the purposes of this proof, it is helpful to formally let S_t , A_t , and R_t denote random variables, distinguishing them from realizations denoted by lower-case symbols.

We first show that conditioned on the one-step history $(S_0, A_0, R_0, S_1) = (s, a, r, s')$, the maximum future discounted return from time 1 onwards does not depend on s, a, r . More precisely, we claim that

$$\sup_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma V^*(s'). \quad (1.3)$$

Fix any policy π . Define the “offset” policy $\pi_{(s,a,r)}$ as follows: for all $t \geq 0$ and all histories $(s_0, a_0, r_0, \dots, s_t)$,

$$\begin{aligned} \pi_{(s,a,r)}(A_t = b \mid S_0 = s_0, A_0 = a_0, R_0 = r_0, \dots, S_t = s_t) \\ := \pi(A_{t+1} = b \mid S_0 = s, A_0 = a, R_0 = r, S_1 = s_0, A_1 = a_0, R_1 = r_0, \dots, S_{t+1} = s_t). \end{aligned}$$

By a change of variables on the time index and the definition of $\pi_{(s,a,r)}$, we have

$$\mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \mid \pi_{(s,a,r)}, S_0 = s' \right] = \gamma V^{\pi_{(s,a,r)}}(s').$$

Moreover, for each fixed (s, a, r) , the set $\{\pi_{(s,a,r)} : \pi \in \Pi\}$ is equal to Π itself (shifting a policy by one step is a bijection on the class of all history-dependent randomized policies). Therefore,

$$\sup_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma \cdot \sup_{\pi \in \Pi} V^{\pi_{(s,a,r)}}(s') = \gamma \cdot \sup_{\pi \in \Pi} V^{\pi}(s') = \gamma V^*(s'),$$

proving (1.3).

We now define a stationary deterministic policy $\tilde{\pi}$ by choosing, for each state s , any maximizer of the one-step lookahead objective:

$$\tilde{\pi}(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \mathbb{E} \left[r(s, a) + \gamma V^*(S_1) \mid S_0 = s, A_0 = a \right].$$

We claim that $V^{\tilde{\pi}}(s) = V^*(s)$ for all s .

Fix an initial state s_0 . Using the law of iterated expectations,

$$\begin{aligned} V^*(s_0) &= \sup_{\pi \in \Pi} \mathbb{E} \left[r(S_0, A_0) + \sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid S_0 = s_0 \right] \\ &= \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s_0, A_0, R_0, S_1) \right] \mid S_0 = s_0 \right] \\ &\leq \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \sup_{\pi' \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi', (S_0, A_0, R_0, S_1) = (s_0, A_0, R_0, S_1) \right] \mid S_0 = s_0 \right] \\ &\stackrel{(a)}{=} \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \gamma V^*(S_1) \mid S_0 = s_0 \right] \\ &= \sup_{a_0 \in \mathcal{A}} \mathbb{E} \left[r(s_0, a_0) + \gamma V^*(S_1) \mid S_0 = s_0, A_0 = a_0 \right] \\ &\stackrel{(b)}{=} \mathbb{E} \left[r(s_0, A_0) + \gamma V^*(S_1) \mid S_0 = s_0, \tilde{\pi} \right]. \end{aligned}$$

Here, step (a) uses (1.3), and step (b) follows from the definition of $\tilde{\pi}$.

Applying the same argument recursively (conditioning on S_1 , then S_2 , etc.) yields, for all $t \geq 0$,

$$V^*(s_0) \leq \mathbb{E} \left[r(S_0, A_0) + \gamma r(S_1, A_1) + \cdots + \gamma^t V^*(S_{t+1}) \mid S_0 = s_0, \tilde{\pi} \right].$$

Taking $t \rightarrow \infty$ and using dominated convergence (since V^* is bounded), we obtain $V^*(s_0) \leq V^{\tilde{\pi}}(s_0)$. Since by definition $V^{\tilde{\pi}}(s_0) \leq \sup_{\pi \in \Pi} V^{\pi}(s_0) = V^*(s_0)$, we conclude $V^{\tilde{\pi}}(s_0) = V^*(s_0)$. As s_0 was arbitrary, this proves the first claim.

For the same policy $\tilde{\pi}$, an analogous argument starting from $Q^*(s, a)$ shows $Q^{\tilde{\pi}}(s, a) = Q^*(s, a)$ for all (s, a) , completing the proof. $\square$

This shows that we may restrict ourselves to stationary and deterministic policies without any loss in performance. The following theorem, also due to Bellman [1956], gives a precise characterization of the optimal action-value function.

Theorem 1.8 (Bellman optimality equations). *We say that a vector $Q \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ satisfies the Bellman optimality equations if, for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,*

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} Q(s', a') \right].$$

For any $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, we have $Q = Q^*$ if and only if Q satisfies the Bellman optimality equations. Furthermore, any deterministic policy defined by

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q^*(s, a)$$

is an optimal policy (with ties broken in an arbitrary but fixed manner).

Before we prove this claim, we introduce a few definitions. Let π_Q denote a greedy policy with respect to a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$:

$$\pi_Q(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a),$$

where ties are broken in some arbitrary but fixed manner. With this notation, the optimal policy can be written as

$$\pi^* = \pi_{Q^*}.$$

We also use the following notation to map a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ to a vector of length $|\mathcal{S}|$:

$$V_Q(s) := \max_{a \in \mathcal{A}} Q(s, a).$$

The Bellman optimality operator $\mathcal{T}_M : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined as

$$\mathcal{T}Q := r + \gamma P V_Q. \quad (1.4)$$

With this notation, the Bellman optimality equations can be written concisely as

$$Q = \mathcal{T}Q.$$

Thus, the previous theorem states that $Q = Q^*$ if and only if Q is a fixed point of the operator $\mathcal{T}$.

Proof. We begin by showing that

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a). \quad (1.5)$$

Let π^* be an optimal stationary deterministic policy, which exists by Theorem 1.7. Consider the (non-stationary) policy that takes action a in state s at time 0 and then follows π^* thereafter. By definition of V^* as the supremum over all (possibly non-stationary) policies,

$$V^*(s) \geq Q^{\pi^*}(s, a) = Q^*(s, a),$$

where the last equality uses that π^* is optimal for all starting states. Since a is arbitrary, this gives $V^*(s) \geq \max_a Q^*(s, a)$. Conversely, since π^* is deterministic, Lemma 1.4 yields

$$V^*(s) = V^{\pi^*}(s) = Q^{\pi^*}(s, \pi^*(s)) \leq \max_a Q^{\pi^*}(s, a) = \max_a Q^*(s, a),$$

proving (1.5).

First, we show the forward direction, i.e. that Q^* satisfies $Q^* = \mathcal{T}Q^*$. For any (s, a) ,

$$\begin{aligned} Q^*(s, a) &= \sup_{\pi \in \Pi} Q^\pi(s, a) \\ &= r(s, a) + \gamma \sup_{\pi \in \Pi} \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^\pi(s')] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\sup_{\pi \in \Pi} V^\pi(s') \right] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^*(s')] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a'} Q^*(s', a') \right], \end{aligned}$$

where the third line uses that the transition distribution does not depend on π , and the last line uses (1.5). This proves sufficiency.

For the converse, suppose $Q = \mathcal{T}Q$ for some Q . We now show that $Q = Q^*$. Let $\pi = \pi_Q$ be a greedy policy with respect to Q . The identity $Q = \mathcal{T}Q$ means

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[V_Q(s')] = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[Q(s', \pi(s'))],$$

which in matrix form is $Q = r + \gamma P^\pi Q$. Thus,

$$Q = (I - \gamma P^\pi)^{-1} r = Q^\pi,$$

using Corollary 1.5.

It remains to show that π is optimal. Let π' be any other deterministic stationary policy. First note that

$$[(P^\pi - P^{\pi'})Q^\pi]_{s,a} = \mathbb{E}_{s' \sim P(\cdot | s, a)}[Q^\pi(s', \pi(s')) - Q^\pi(s', \pi'(s'))] \geq 0,$$

since π is greedy with respect to $Q = Q^\pi$. Using $Q^\pi = r + \gamma P^\pi Q^\pi$ and $Q^{\pi'} = (I - \gamma P^{\pi'})^{-1} r$, we have

$$\begin{aligned} Q^\pi - Q^{\pi'} &= (I - \gamma P^{\pi'})^{-1} \left((I - \gamma P^{\pi'}) - (I - \gamma P^\pi) \right) Q^\pi \\ &= \gamma (I - \gamma P^{\pi'})^{-1} (P^\pi - P^{\pi'}) Q^\pi \\ &\geq 0, \end{aligned}$$

where the last step uses that $(I - \gamma P^{\pi'})^{-1}$ has nonnegative entries (indeed, $(I - \gamma P^{\pi'})^{-1} = \sum_{t=0}^{\infty} \gamma^t (P^{\pi'})^t$). Thus $Q^\pi \geq Q^{\pi'}$ for all deterministic stationary π' . Since Theorem 1.7 establishes that an optimal policy exists within the set of stationary deterministic policies, and π dominates all such policies, π must be globally optimal. Therefore $Q = Q^\pi = Q^*$, completing the proof. $\square$

1.2 Finite-Horizon Markov Decision Processes

In some settings, it is more natural to work with finite-horizon (and hence time-dependent) Markov Decision Processes; see the discussion below. A finite-horizon, time-dependent MDP $M = (\mathcal{S}, \mathcal{A}, \{P_h\}_{h=0}^{H-1}, \{r_h\}_{h=0}^{H-1}, H, \mu)$ is specified by:

- A state space $\mathcal{S}$, which may be finite or infinite.
- An action space $\mathcal{A}$, which may be finite or infinite.
- A time-dependent transition kernel $P_h : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, where $\Delta(\mathcal{S})$ is the set of probability distributions over $\mathcal{S}$. Here $P_h(s' | s, a)$ is the probability of transitioning to state s' after taking action a in state s at time step h . (The stationary setting is recovered when P_h does not depend on h .)
- A time-dependent reward function $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, where $r_h(s, a)$ is the immediate reward obtained by taking action a in state s at time step h . More generally, one may allow $r_h(s, a)$ to be stochastic, but we focus on deterministic rewards for simplicity.
- A positive integer horizon H .
- An initial state distribution $\mu \in \Delta(\mathcal{S})$, which specifies how s_0 is generated.

A (possibly non-stationary and randomized) policy π specifies, at each time h , a distribution over actions conditioned on the observed history up to time h . (Equivalently, one may restrict attention to Markov policies $\pi(s, h)$ without loss of optimality in finite-horizon MDPs; see the theorem below.)

For a policy π , state s , and time $h \in \{0, \dots, H\}$, define the value function $V_h^\pi : \mathcal{S} \rightarrow \mathbb{R}$ as the expected cumulative reward from time h onward:

$$V_h^\pi(s) = \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(S_t, A_t) \mid \pi, S_h = s \right],$$

where r_t denotes the stage- t reward function and the expectation is over the trajectory induced by π and the transition kernels. Similarly, define the state-action value (Q-value) function $Q_h^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ by

$$Q_h^\pi(s, a) = \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(S_t, A_t) \mid \pi, S_h = s, A_h = a \right].$$

We also write $V^\pi(s) = V_0^\pi(s)$, and note that $V_H^\pi(s) \equiv 0$ for all s .

In terms of finite horizon *Bellman operators*, we can write this recursion as follows: for a function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, define the stage- h *optimal* Bellman operator $\mathcal{T}_h$ by

$$(\mathcal{T}_h f)(s, a) := r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} f(s', a') \right].$$

Under this definition, the optimal Q-functions satisfy the recursion:

$$\begin{aligned} Q_H^* &\equiv 0 \\ Q_h^* &= \mathcal{T}_h Q_{h+1}^* \quad \text{for } h = H-1, \dots, 0 \end{aligned}$$

Given a starting state s , the goal is to find a policy maximizing the value, i.e.,

$$\max_{\pi} V^\pi(s), \tag{1.6}$$

where the maximum is over all (possibly non-stationary and randomized) policies.

Theorem 1.9 (Bellman optimality equations: finite horizon). *For $h \in \{0, \dots, H\}$, define*

$$Q_h^*(s, a) = \sup_{\pi} Q_h^\pi(s, a),$$

where the sup is over all (possibly non-stationary and randomized) policies, and define $Q_H^(s, a) \equiv 0$ (terminal condition) for all (s, a) .*

Then $\{Q_h^\}_{h=0}^H$ is the unique collection of functions satisfying, for all $h \in \{0, \dots, H-1\}$ and $(s, a) \in \mathcal{S} \times \mathcal{A}$,*

$$Q_h(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} Q_{h+1}(s', a') \right]. \tag{1.7}$$

Moreover, any policy π^ defined by*

$$\pi^*(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_h^*(s, a), \quad h = 0, \dots, H-1,$$

is an optimal policy.

We leave the proof as an exercise to the reader. (It follows by backward induction.)

	Poly?	Strongly Poly?
Value Iteration	$ \mathcal{S} ^2 \mathcal{A} \frac{L(P, r, \gamma) \log \frac{1}{1-\gamma}}{1-\gamma}$	X
Policy Iteration	$(\mathcal{S} ^3 + \mathcal{S} ^2 \mathcal{A})\frac{L(P, r, \gamma) \log \frac{1}{1-\gamma}}{1-\gamma}$	$(\mathcal{S} ^3 + \mathcal{S} ^2 \mathcal{A}) \cdot \min \left\{ \frac{ \mathcal{A} ^{ \mathcal{S} }}{ \mathcal{S} }, \frac{ \mathcal{S} ^2 \mathcal{A} \log \frac{ \mathcal{S} ^2}{1-\gamma}}{1-\gamma} \right\}$
LP Algorithms	$ \mathcal{S} ^3 \mathcal{A} L(P, r, \gamma)$	$ \mathcal{S} ^4 \mathcal{A} ^4 \log \frac{ \mathcal{S} }{1-\gamma}$

Table 1.1: Computational complexities of various approaches (universal constants are omitted) to find an *exact* solution. Polynomial-time algorithms may depend on the bit complexity $L(P, r, \gamma)$, while strongly polynomial algorithms do not. Note that value iteration and policy iteration have iteration complexity scaling as $1/(1-\gamma)$; thus, if γ is treated as part of the input (encoded in $L(P, r, \gamma)$), these methods are not polynomial-time in the bit-length model. Similarly, policy iteration is strongly polynomial only when γ is treated as a fixed constant. In contrast, the linear programming approach yields both polynomial-time (in the bit-length model) and strongly polynomial algorithms; the latter can be obtained via interior point methods. See the text and Section 1.6 for references. Here, $|\mathcal{S}|^2|\mathcal{A}|$ is the assumed runtime per iteration of value iteration, and $|\mathcal{S}|^3 + |\mathcal{S}|^2|\mathcal{A}|$ is the assumed runtime per iteration of policy iteration; these are consistent with cubic-time linear system solves.

Discussion: stationary vs. time-dependent models. For the purposes of this book, it is natural to study both models: we typically assume stationary dynamics in the infinite-horizon discounted setting and time-dependent dynamics in the finite-horizon setting.

From a theoretical perspective, the finite-horizon *time-dependent* formulation is often more convenient: backward induction avoids contraction arguments, and many sharp statistical rates admit simpler proofs in this setting.

From a practical perspective, explicitly time-dependent models are less common because they lead to policies and value functions whose tabular representations scale as $O(H|\mathcal{S}||\mathcal{A}|)$ and $O(H|\mathcal{S}|)$, respectively, i.e., a factor $O(H)$ larger than in the stationary case. In practice, one often folds temporal information into the state (e.g., augmenting the state with the time index), which recovers a stationary MDP on an expanded state space and can be combined naturally with function approximation to represent policies and value functions compactly.

1.3 Computational Complexity

This section concerns the problem of computing an optimal policy when the MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ is known; this is the classical *planning* problem in reinforcement learning. While much of this book is concerned with statistical limits, understanding computational limits is also informative. We will consider algorithms that return both exact and approximately optimal policies. In particular, we will distinguish polynomial-time (and strongly polynomial-time) algorithms.

Assume that (P, r, γ) are specified with rational entries. Let $L(P, r, \gamma)$ denote the total bit-length of the description of M . We measure running time in the standard bit-complexity model in which basic arithmetic operations $+$, $-$, $\times$, $\div$ on numbers with $O(L(P, r, \gamma))$ bits take unit time.¹ In this setting, one may hope for an algorithm that returns an *exact* optimal policy with running time polynomial in $|\mathcal{S}|$, $|\mathcal{A}|$, and $L(P, r, \gamma)$.

It is also useful to understand which algorithms are *strongly* polynomial. Here, one does not explicitly restrict (P, r, γ) to be rational. An algorithm is strongly polynomial if it returns an optimal policy with running time polynomial in $|\mathcal{S}|$ and $|\mathcal{A}|$ only (with no dependence on $L(P, r, \gamma)$).

¹Equivalently, one can state bounds that scale polynomially with the bit-length of the numbers manipulated by the algorithm. We use the simplified unit-cost model to keep the dependence on $L(P, r, \gamma)$ explicit.

The next two subsections cover classical *iterative* algorithms that compute Q^* . We then discuss the linear programming approach.

1.3.1 Value Iteration

Perhaps the simplest algorithm for discounted MDPs is to iteratively apply the Bellman optimality operator $\mathcal{T}$. Starting from some initialization Q , we repeat

$$Q \leftarrow \mathcal{T}Q.$$

This algorithm is referred to as *Q-value iteration*.

Lemma 1.10 (Contraction). *For any $Q, Q' \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,*

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty.$$

Proof. First, for any $s \in \mathcal{S}$, we claim

$$|V_Q(s) - V_{Q'}(s)| \leq \max_{a \in \mathcal{A}} |Q(s, a) - Q'(s, a)|.$$

Assume $V_Q(s) \geq V_{Q'}(s)$ (the other case is symmetric), and let $a \in \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a)$ be a greedy action for Q at s . Then

$$\begin{aligned} V_Q(s) - V_{Q'}(s) &= Q(s, a) - \max_{a' \in \mathcal{A}} Q'(s, a') \\ &\leq Q(s, a) - Q'(s, a) \leq \max_{a \in \mathcal{A}} |Q(s, a) - Q'(s, a)|. \end{aligned}$$

Using this bound,

$$\begin{aligned} \|\mathcal{T}Q - \mathcal{T}Q'\|_\infty &= \gamma \|PV_Q - PV_{Q'}\|_\infty = \gamma \|P(V_Q - V_{Q'})\|_\infty \\ &\leq \gamma \|V_Q - V_{Q'}\|_\infty = \gamma \max_s |V_Q(s) - V_{Q'}(s)| \\ &\leq \gamma \max_s \max_a |Q(s, a) - Q'(s, a)| = \gamma \|Q - Q'\|_\infty, \end{aligned}$$

where the inequality $\|Px\|_\infty \leq \|x\|_\infty$ uses that each component of Px is a convex combination of the entries of x . $\square$

The following result bounds the suboptimality of the greedy policy induced by an approximate Q -function.

Lemma 1.11. (*Q-error amplification*) *For any $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,*

$$V^{\pi_Q} \geq V^* - \frac{2\|Q - Q^*\|_\infty}{1 - \gamma} \mathbf{1},$$

where $\mathbf{1}$ denotes the vector of all ones.

Proof. Fix a state s and let $a = \pi_Q(s)$. Then

$$\begin{aligned} V^*(s) - V^{\pi_Q}(s) &= Q^*(s, \pi^*(s)) - Q^{\pi_Q}(s, a) \\ &= (Q^*(s, \pi^*(s)) - Q^*(s, a)) + (Q^*(s, a) - Q^{\pi_Q}(s, a)). \end{aligned}$$

For the first term, using that a is greedy for Q ,

$$Q(s, \pi^*(s)) \leq Q(s, a) \implies Q^*(s, \pi^*(s)) - Q^*(s, a) \leq (Q^*(s, \pi^*(s)) - Q(s, \pi^*(s))) + (Q(s, a) - Q^*(s, a)) \leq 2\|Q - Q^*\|_\infty.$$

For the second term, Bellman consistency for π_Q gives

$$Q^*(s, a) - Q^{\pi_Q}(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^*(s') - V^{\pi_Q}(s')] \leq \gamma \|V^* - V^{\pi_Q}\|_\infty.$$

Combining,

$$V^*(s) - V^{\pi_Q}(s) \leq 2\|Q - Q^*\|_\infty + \gamma \|V^* - V^{\pi_Q}\|_\infty.$$

Taking the supremum over s on the left-hand side yields

$$\|V^* - V^{\pi_Q}\|_\infty \leq 2\|Q - Q^*\|_\infty + \gamma \|V^* - V^{\pi_Q}\|_\infty,$$

and rearranging gives

$$\|V^* - V^{\pi_Q}\|_\infty \leq \frac{2}{1-\gamma} \|Q - Q^*\|_\infty.$$

Equivalently, $V^{\pi_Q} \geq V^* - \frac{2}{1-\gamma} \|Q - Q^*\|_\infty \mathbf{1}$. □

Theorem 1.12. (*Q-value iteration convergence*) Set $Q^{(0)} = 0$, and for $k = 0, 1, \dots$ define

$$Q^{(k+1)} = \mathcal{T}Q^{(k)}.$$

Let $\pi^{(k)} = \pi_{Q^{(k)}}$. For

$$k \geq \frac{1}{1-\gamma} \log\left(\frac{2}{(1-\gamma)^2 \epsilon}\right),$$

we have

$$V^{\pi^{(k)}} \geq V^* - \epsilon \mathbf{1}.$$

Proof. Since $\|Q^*\|_\infty \leq 1/(1-\gamma)$, and $Q^{(k)} = \mathcal{T}^k Q^{(0)}$ with $Q^* = \mathcal{T}Q^*$, the contraction property (Lemma 1.10) gives

$$\begin{aligned} \|Q^{(k)} - Q^*\|_\infty &= \|\mathcal{T}^k Q^{(0)} - \mathcal{T}^k Q^*\|_\infty \leq \gamma^k \|Q^{(0)} - Q^*\|_\infty \\ &= \gamma^k \|Q^*\|_\infty \leq \frac{\gamma^k}{1-\gamma} \leq \frac{\exp(-(1-\gamma)k)}{1-\gamma}, \end{aligned}$$

where we used $\gamma^k \leq \exp(-(1-\gamma)k)$. Choosing k as stated ensures $\|Q^{(k)} - Q^*\|_\infty \leq \frac{(1-\gamma)\epsilon}{2}$, and applying Lemma 1.11 yields $V^{\pi^{(k)}} \geq V^* - \epsilon \mathbf{1}$. □

Iteration complexity for an exact solution. The previous results guarantee that Q -value iteration converges geometrically to Q^* , and hence yields an ϵ -optimal policy after $O\left(\frac{1}{1-\gamma} \log \frac{1}{\epsilon}\right)$ iterations. To obtain an *exact* optimal policy in the setting where (P, r, γ) are rational, one can combine this convergence with a standard bit-complexity separation argument. In particular, when the data (P, r, γ) have total description length $L(P, r, \gamma)$, the value of any *fixed* deterministic stationary policy can be written as a rational function of these inputs whose numerator and denominator have bit-length polynomial in $L(P, r, \gamma)$. Consequently, among the finite set of deterministic stationary policies, either two policies have exactly the same value (a true tie) or their values differ by at least $2^{-\text{poly}(L(P, r, \gamma))}$. Thus, once $\|Q^{(k)} - Q^*\|_\infty$ is smaller than a corresponding threshold, any greedy policy with respect to $Q^{(k)}$ must be optimal (up to arbitrary tie-breaking). This yields the polynomial-time bound in Table 1.1 when γ is treated as a fixed constant. Value iteration is not strongly polynomial: as $\gamma \uparrow 1$, the number of iterations required to reach the separation scale can grow arbitrarily large, and moreover the algorithm may never certify exact optimality in finite time without explicitly appealing to such bit-length arguments.

1.3.2 Policy Iteration

Policy iteration for discounted MDPs starts from an arbitrary deterministic stationary policy π_0 and repeats, for $k = 0, 1, 2, \dots$:

1. *Policy evaluation.* Compute the value of the current policy, e.g., compute V^{π_k} by solving the linear system $V^{\pi_k} = r(\cdot, \pi_k(\cdot)) + \gamma P^{\pi_k} V^{\pi_k}$, and then obtain Q^{π_k} via $Q^{\pi_k}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_k}(s')]$. (Equivalently, one may compute Q^{π_k} directly using Equation 1.2.)
2. *Policy improvement.* Update the policy to be greedy w.r.t. Q^{π_k} :

$$\pi_{k+1} = \pi_{Q^{\pi_k}}.$$

The first step is called *policy evaluation* and the second *policy improvement*.

Lemma 1.13 (Monotonicity and contraction of policy iteration). *For all $k \geq 0$,*

1. (**Sandwich bound**) $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k} \geq Q^{\pi_k}$ (elementwise).
2. (**Geometric contraction to optimality**)

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty.$$

Proof. We repeatedly use that $\mathcal{T}$ is monotone and a γ -contraction in $\|\cdot\|_\infty$ (Lemma 1.10), and that $Q^* \geq Q^\pi$ elementwise for any policy π .

Step 1: $\mathcal{T}Q^{\pi_k} \geq Q^{\pi_k}$. For deterministic π_k , we have $V^{\pi_k}(s) = Q^{\pi_k}(s, \pi_k(s))$. Moreover, by definition, $V_{Q^{\pi_k}}(s) = \max_a Q^{\pi_k}(s, a) \geq Q^{\pi_k}(s, \pi_k(s)) = V^{\pi_k}(s)$. Therefore, for every (s, a) ,

$$\mathcal{T}Q^{\pi_k}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_{Q^{\pi_k}}(s')] \geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_k}(s')] = Q^{\pi_k}(s, a).$$

Step 2: $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$. Let $\pi' = \pi_{k+1} = \pi_{Q^{\pi_k}}$. Define the (policy-specific) Bellman evaluation operator on state values,

$$(\mathcal{T}^{\pi'} V)(s) := r(s, \pi'(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi'(s))}[V(s')].$$

This operator is monotone and a γ -contraction in $\|\cdot\|_\infty$, with unique fixed point $V^{\pi'}$.

By greediness of π' w.r.t. Q^{π_k} ,

$$(\mathcal{T}^{\pi'} V^{\pi_k})(s) = r(s, \pi'(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi'(s))}[V^{\pi_k}(s')] = Q^{\pi_k}(s, \pi'(s)) = \max_a Q^{\pi_k}(s, a) = V_{Q^{\pi_k}}(s) \geq V^{\pi_k}(s).$$

Applying $\mathcal{T}^{\pi'}$ repeatedly and using monotonicity yields an increasing sequence $V^{\pi_k} \leq (\mathcal{T}^{\pi'})V^{\pi_k} \leq (\mathcal{T}^{\pi'})^2 V^{\pi_k} \leq \dots$, which converges to the fixed point $V^{\pi'}$. Hence,

$$V^{\pi_{k+1}} = V^{\pi'} \geq V_{Q^{\pi_k}} \quad (\text{elementwise}).$$

Now, for any (s, a) ,

$$Q^{\pi_{k+1}}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_{k+1}}(s')] \geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_{Q^{\pi_k}}(s')] = \mathcal{T}Q^{\pi_k}(s, a),$$

which proves $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$ and completes the sandwich bound.

Step 3: contraction. Since $Q^* = \mathcal{T}Q^*$ and $Q^* \geq Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$ elementwise,

$$0 \leq Q^* - Q^{\pi_{k+1}} \leq Q^* - \mathcal{T}Q^{\pi_k} = \mathcal{T}Q^* - \mathcal{T}Q^{\pi_k}.$$

Taking $\|\cdot\|_\infty$ and applying Lemma 1.10 gives

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \|\mathcal{T}Q^* - \mathcal{T}Q^{\pi_k}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty,$$

as claimed. $\square$

With Lemma 1.13, a geometric convergence bound follows immediately.

Theorem 1.14 (Policy iteration convergence). *Let π_0 be any initial deterministic stationary policy, and let $\{\pi_k\}$ be generated by policy iteration. Then for all $k \geq 0$,*

$$\|Q^* - Q^{\pi_k}\|_\infty \leq \gamma^k \|Q^* - Q^{\pi_0}\|_\infty \leq \frac{\gamma^k}{1 - \gamma}.$$

In particular, for

$$k \geq \frac{\log\left(\frac{1}{(1-\gamma)\epsilon}\right)}{1 - \gamma},$$

we have the elementwise performance guarantee

$$Q^{\pi_k} \geq Q^* - \epsilon \mathbf{1}.$$

Iteration complexity for an exact solution. Policy iteration improves the value monotonically and converges geometrically, but unlike value iteration it can terminate with an *exact* optimal policy in finitely many iterations because the policy changes only when the greedy action changes. A naive worst-case bound on the number of iterations is the number of deterministic stationary policies, $|\mathcal{A}|^{|\mathcal{S}|}$, since policy iteration never revisits a policy. A modest refinement improves this to $\frac{|\mathcal{A}|^{|\mathcal{S}|}}{|\mathcal{S}|}$. More strongly, when one measures complexity in the *strongly polynomial* model (unit-cost real arithmetic) and treats γ as fixed, substantially better bounds are known: policy iteration finds an optimal policy in a number of iterations polynomial in $|\mathcal{S}|$ and $|\mathcal{A}|$ (see Table 1.1 and Section 1.6 for references and precise statements).

1.3.3 Value Iteration for Finite-Horizon MDPs

We now specify the value iteration algorithm for finite-horizon MDPs. In the finite-horizon (time-dependent) setting, dynamic programming via backward induction computes the optimal value functions exactly, and the natural analogues of value iteration and policy iteration coincide.

Let $Q_H(s, a) \equiv 0$ for all (s, a) . For $h = H - 1, H - 2, \dots, 0$, define

$$Q_h(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)} \left[\max_{a' \in \mathcal{A}} Q_{h+1}(s', a') \right]. \quad (1.8)$$

Equivalently, one may initialize $Q_{H-1}(s, a) = r_{H-1}(s, a)$ and then run the same recursion for $h = H - 2, \dots, 0$.

By Theorem 1.9, the resulting functions satisfy $Q_h = Q_h^*$ for all $h \in \{0, \dots, H\}$, and the policy

$$\pi(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_h^*(s, a)$$

(with arbitrary tie-breaking) is optimal.

1.3.4 The Linear Programming Approach

It is helpful to understand an alternative approach to computing an optimal policy when the MDP is known. In particular, consider a discounted MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$ where P , r , and γ are specified by rational numbers. From a bit-complexity perspective, the iterative algorithms above are not, in general, polynomial-time in the description length of the input because their iteration complexity typically depends polynomially on $1/(1 - \gamma)$, while a rational value of $1 - \gamma$ may be represented with only $O(\log \frac{1}{1-\gamma})$ bits. In contrast, the LP approach yields polynomial-time algorithms in the standard bit model.

The Primal LP

Consider the following linear program in variables $V \in \mathbb{R}^{|\mathcal{S}|}$:

$$\begin{aligned} \min \quad & \sum_{s \in \mathcal{S}} \mu(s) V(s) \\ \text{subject to} \quad & V(s) \geq r(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) V(s') \quad \forall s \in \mathcal{S}, a \in \mathcal{A}. \end{aligned}$$

The feasible region is exactly the set of value functions that dominate their Bellman optimality backup. The optimal value function V^* is the (unique) minimal feasible vector (componentwise). If μ has full support, then V^* is also the unique optimizer of the above objective (since increasing any coordinate increases the objective).

Computational complexity for an exact solution. Table 1.1 summarizes standard runtime bounds for solving this LP when the coefficients are rational and we measure complexity in the bit model. There are also specialized interior-point methods for discounted MDP LPs with strong guarantees; see Section 1.6 for references and further discussion.

Policy iteration and simplex. Policy iteration can be interpreted as a pivoting method for closely related LP formulations (and, in particular, has a well-known connection to simplex-type methods under appropriate pivot rules). While the simplex method is not strongly polynomial for general LPs, policy-iteration/simplex variants admit strong polynomial-time guarantees for discounted MDPs when the discount factor is treated as fixed; see Ye [2011].

The Dual LP and the State–Action Polytope

We next introduce the discounted occupancy measure, which yields a convenient dual view. For a (possibly randomized) stationary policy π and a fixed starting state s_0 , define the distribution over state–action pairs

$$d_{s_0}^\pi(s, a) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s, a_t = a | s_0). \quad (1.9)$$

It is straightforward to verify that $d_{s_0}^\pi$ is a probability distribution over $\mathcal{S} \times \mathcal{A}$ (since $(1 - \gamma) \sum_{t \geq 0} \gamma^t = 1$). For a starting distribution μ , we also define the averaged occupancy measure

$$d_\mu^\pi(s, a) := \mathbb{E}_{s_0 \sim \mu} [d_{s_0}^\pi(s, a)].$$

Flow (consistency) constraints. The discounted occupancy satisfies a linear recursion. For a fixed start state s_0 , one has for every $s \in \mathcal{S}$,

$$\sum_{a \in \mathcal{A}} d_{s_0}^\pi(s, a) = (1 - \gamma) \mathbb{1}\{s = s_0\} + \gamma \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} P(s | s', a') d_{s_0}^\pi(s', a'). \quad (1.10)$$

Averaging (1.10) over $s_0 \sim \mu$ yields the corresponding relation for d_μ^π :

$$\sum_{a \in \mathcal{A}} d_\mu^\pi(s, a) = (1 - \gamma)\mu(s) + \gamma \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} P(s | s', a') d_\mu^\pi(s', a') \quad \forall s \in \mathcal{S}. \quad (1.11)$$

Motivated by (1.11), define the *state–action polytope*

$$\mathcal{K}_\mu := \left\{ d \in \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|} \left| d(s, a) \geq 0 \quad \forall (s, a), \quad \sum_{a \in \mathcal{A}} d(s, a) = (1 - \gamma)\mu(s) + \gamma \sum_{s', a'} P(s | s', a') d(s', a') \quad \forall s \in \mathcal{S} \right. \right\}.$$

Proposition 1.15. $\mathcal{K}_\mu$ is exactly the set of discounted occupancy measures induced by stationary policies: $d \in \mathcal{K}_\mu$ if and only if there exists a stationary (possibly randomized) policy π such that $d_\mu^\pi = d$.

The dual LP. Using the identity

$$V^\pi(\mu) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] = \frac{1}{1 - \gamma} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d_\mu^\pi(s, a) r(s, a),$$

we can write a dual LP in the variables $d \in \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|}$:

$$\begin{aligned} \max \quad & \frac{1}{1 - \gamma} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d(s, a) r(s, a) \\ \text{subject to} \quad & d \in \mathcal{K}_\mu. \end{aligned}$$

One can verify that this is dual to the primal LP above (and strong duality holds under standard conditions). This dual perspective provides an alternative route to computing an optimal policy.

If d^* is an optimal solution and μ has full support, then an optimal policy can be recovered via

$$\pi^*(a | s) = \frac{d^*(s, a)}{\sum_{a'} d^*(s, a')} \quad \text{whenever } \sum_{a'} d^*(s, a') > 0,$$

with $\pi^*(\cdot | s)$ defined arbitrarily on states with zero occupancy. An alternative optimal policy is $\arg\max_a d^*(s, a)$; these coincide when the optimizer is unique.

1.4 Sample Complexity and Sampling Models

Much of reinforcement learning is concerned with finding a near-optimal policy (or, equivalently, achieving near-optimal return) in settings where the MDP is *not* known to the learner. We will study these questions under several different *sampling models*, which specify how the agent obtains information about the unknown underlying MDP. In each model, we are interested in understanding the number of samples required to compute a near-optimal policy, i.e. the *sample complexity*. Ultimately, we are interested in results that remain meaningful when the number of states and actions is large (or even when $\mathcal{S}$ is countably or uncountably infinite). This is analogous in spirit to generalization in supervised learning, but, as we shall see, it is fundamentally more challenging in reinforcement learning because the data distribution depends on the learner’s behavior.

The episodic setting. In the episodic setting, the learner interacts with the environment in *episodes*. In each episode, the learner starts from an initial state $s_0 \sim \mu$, acts for a finite number of steps, observes the resulting trajectory, and then the environment resets and a new episode begins with a fresh draw $s_0 \sim \mu$. This episodic model of feedback is applicable to both finite-horizon and infinite-horizon MDPs:

- **Finite-horizon MDPs.** Each episode lasts exactly H steps, after which the environment resets to $s_0 \sim \mu$.
- **Infinite-horizon MDPs.** It is often natural to impose an episodic protocol even in the infinite-horizon discounted setting by ending each episode after a random (finite) number of steps. A standard model is to terminate the episode at each time step independently with probability $1 - \gamma$ (equivalently, the episode length has a geometric distribution). After termination, the state resets to $s_0 \sim \mu$. Under this protocol, the undiscounted cumulative reward observed in one episode is an unbiased estimate of the infinite-horizon discounted value $V^\pi(\mu)$.

In this setting, we are often interested either in the number of episodes needed to find a near-optimal policy (a *PAC* guarantee), or in *regret* guarantees (which we study in Chapter 6). Both are statements about the statistical (sample) complexity of learning.

The episodic setting is challenging because the agent must explore to obtain information about relevant state–action pairs. As we shall see in Chapter 6, this exploration must be strategic: simply behaving randomly can be far too inefficient. It is therefore often helpful to also study the statistical complexity of RL in an abstract sampling model that bypasses exploration concerns.

The generative model setting. A *generative model* (or simulator oracle) takes as input a state–action pair (s, a) and returns an independent sample $s' \sim P(\cdot \mid s, a)$ together with the reward $r(s, a)$ (or, in the stochastic-reward setting, an independent sample of the reward). This model allows the learner to sample transitions from any chosen (s, a) directly, and is a convenient abstraction for isolating statistical questions from exploration dynamics.

The offline RL setting. In *offline* RL, the agent is given a fixed dataset collected in advance, typically generated by some (unknown) behavior policy or a mixture of policies. In the simplest model, the dataset consists of tuples $\{(s, a, r, s')\}$, where r corresponds to $r(s, a)$ in the deterministic-reward case (or a reward sample otherwise) and $s' \sim P(\cdot \mid s, a)$. For simplicity, it is often helpful to assume that the state–action pairs (s, a) in the dataset are sampled i.i.d. from some distribution ν over $\mathcal{S} \times \mathcal{A}$, after which (r, s') are drawn from the conditional dynamics given (s, a) . This i.i.d. model captures the core statistical difficulties associated with distribution shift and coverage.

1.5 Bonus: Advantages and The Performance Difference Lemma

Throughout, we will overload notation where, for a distribution μ over $\mathcal{S}$, we write:

$$V^\pi(\mu) = \mathbb{E}_{s \sim \mu} [V^\pi(s)].$$

The *advantage* of a policy π is defined as

$$A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s).$$

Note that for an optimal policy π^* ,

$$A^{\pi^*}(s, a) = Q^{\pi^*}(s, a) - V^{\pi^*}(s) \leq 0$$

for all (s, a) .

Recall the discounted state-action visitation distribution d_μ^π from Equation 1.9. (Equivalently, d_μ^π is the distribution on $\mathcal{S} \times \mathcal{A}$ defined by the normalized discounted occupancy measure.) A basic identity we will repeatedly use is that for any function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$,

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \mid \pi, s_0 \sim \mu \right] = \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_\mu^\pi} [f(s, a)]. \quad (1.12)$$

This follows immediately from the definition of d_μ^π .

Lemma 1.16. (*The performance difference lemma*) For all policies π, π' and distributions μ over $\mathcal{S}$,

$$V^\pi(\mu) - V^{\pi'}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim d_\mu^\pi} [A^{\pi'}(s, a)].$$

Proof. We first prove the identity for a fixed start state $s_0 = s$, and then average over $s \sim \mu$.

Fix $s_0 = s$ and let $\tau = (s_0, a_0, s_1, a_1, \dots)$ denote the (infinite) trajectory generated by following π . Next, we use a telescoping decomposition. Since $V^{\pi'}$ is bounded and $\gamma \in [0, 1)$,

$$\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t)) = -V^{\pi'}(s_0) \quad (\text{almost surely}).$$

Therefore,

$$\begin{aligned} V^\pi(s) - V^{\pi'}(s) &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot | s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot | s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot | s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma \mathbb{E}[V^{\pi'}(s_{t+1}) \mid s_t, a_t] - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot | s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi'}(s_t, a_t) - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot | s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi'}(s_t, a_t) \right]. \end{aligned}$$

Applying (1.12) with $f = A^{\pi'}$ and averaging over $s_0 = s \sim \mu$ gives the stated result. $\square$

1.6 Bibliographic Remarks and Further Reading

We refer the reader to Puterman [1994] for a comprehensive treatment of dynamic programming and MDPs; this text also contains a thorough treatment of the dual LP formulation and a proof of Lemma 1.15.

Regarding the computational complexity of planning, Ye [2011] provides an excellent summary of various approaches. Notably, Ye [2011] showed that policy iteration is a strongly polynomial-time algorithm for a fixed discount rate.² Mansour and Singh [1999] proved that the number of iterations of policy iteration is bounded by roughly $|\mathcal{A}|^{|\mathcal{S}|}/|\mathcal{S}|$.

²The strongly polynomial runtime stated in Table 1.1 differs slightly from that in Ye [2011] because we assume the cost per iteration of policy iteration is $O(|\mathcal{S}|^3 + |\mathcal{S}|^2|\mathcal{A}|)$ (dominated by solving the linear system).

(improving on the trivial $|\mathcal{A}|^{|S|}$ bound). For general strongly polynomial algorithms (independent of γ), the CIPA algorithm proposed by Ye [2005] is an interior-point method achieving the runtime claimed in Table 1.1.

The error amplification bound (Lemma 1.11) is due to Singh and Yee [1994].

The performance difference lemma (Lemma 1.16) was explicitly stated and utilized by Kakade and Langford [2002], Kakade [2003], though the result (based on a telescoping expansion of advantages) was implicit in the analysis of several prior works.

Chapter 2

Sample Complexity with a Generative Model

This chapter begins our study of *sample complexity* in tabular reinforcement learning. We ask a minimax question: as a function of the problem dimensions $(|\mathcal{S}|, |\mathcal{A}|)$, an accuracy target ϵ , a failure probability δ , and an effective horizon (either H in finite-horizon MDPs or $(1 - \gamma)^{-1}$ in discounted MDPs), how many transition samples are required to (i) estimate the optimal action-value function Q^* , or (ii) output a policy whose performance is ϵ -close to optimal?

Throughout this chapter we work in the *generative model* setting: given any state–action pair (s, a) , we may query a simulator that returns an i.i.d. sample $s' \sim P(\cdot \mid s, a)$ together with the reward. For clarity, we assume rewards are deterministic; this assumption is often mild here because the main statistical difficulty in the tabular setting is learning the transition model P , and allowing stochastic rewards only changes constants (and can be absorbed into the variance terms that appear later).

Our first question is whether we must learn an accurate model of the world in order to act near-optimally. The transition kernel has $|\mathcal{S}|^2|\mathcal{A}|$ parameters, so a naive “learn- P -then-plan” intuition suggests that one might need on the order of $|\mathcal{S}|^2|\mathcal{A}|$ observed transitions. One of the main takeaways of this chapter is that this intuition is overly pessimistic: with a generative model, one can compute a near-optimal policy from a number of samples that is *sublinear* in the model size—scaling (up to horizon and logarithmic factors) like $|\mathcal{S}||\mathcal{A}|$ rather than $|\mathcal{S}|^2|\mathcal{A}|$.

A second, equally important question is how to think about accuracy when the horizon is long. In discounted problems, values naturally live on the scale $1/(1 - \gamma)$, so an additive error ϵ should be interpreted relative to this scale. More importantly, small one-step errors can compound over time: recall the matrix identity

$$Q^\pi = (I - \gamma P^\pi)^{-1} r,$$

which makes it clear that as $\gamma \uparrow 1$, the effective horizon grows and the mapping from one-step quantities to long-run values becomes increasingly sensitive. This raises a central question for the chapter: *what is the correct dependence on the effective horizon for obtaining a desired (relative) accuracy in Q^* or in the value of the learned policy?* And can we avoid paying extra “horizon amplification” factors that arise from worst-case error propagation arguments?

2.1 Warmup: a naive model-based approach

As a warmup, we analyze the most naive “model-based” strategy in the generative model setting. We sample enough transitions to build an accurate empirical transition kernel $\hat{P}$, form the empirical MDP $\hat{M}$ by replacing P with $\hat{P}$, and

then plan in $\widehat{M}$.

Concretely, we query the simulator N times for each state–action pair (s, a) . Let $\text{count}(s', s, a)$ be the number of times these N samples transition to s' . Define the empirical transition model

$$\widehat{P}(s' | s, a) := \frac{\text{count}(s', s, a)}{N}.$$

The total number of simulator calls is $|\mathcal{S}||\mathcal{A}|N$. As in Chapter 1, we will also view P and $\widehat{P}$ as matrices of size $|\mathcal{S}||\mathcal{A}| \times |\mathcal{S}|$.

Since the transition kernel has $|\mathcal{S}|^2|\mathcal{A}|$ parameters, one might expect that on the order of $|\mathcal{S}|^2|\mathcal{A}|$ observed transitions suffice to learn a globally accurate model. The next proposition confirms this intuition, and shows that such global model accuracy implies uniform accuracy of values (simultaneously for all policies), and hence near-optimal planning in $\widehat{M}$.

Proposition 2.1. *There exists an absolute constant c such that the following holds. Fix $\epsilon \in (0, \frac{1}{1-\gamma})$ and $\delta \in (0, 1)$, and suppose we collect N samples per state–action pair, with*

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N \geq \frac{c}{(1-\gamma)^4} \cdot \frac{|\mathcal{S}|^2|\mathcal{A}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2}.$$

(Equivalently, $N \geq \frac{c|\mathcal{S}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{(1-\gamma)^4 \epsilon^2}$.) Then, with probability at least $1 - \delta$, the following hold:

- **(Model accuracy)**

$$\max_{s,a} \|P(\cdot | s, a) - \widehat{P}(\cdot | s, a)\|_1 \leq (1-\gamma)^2 \epsilon.$$

- **(Uniform value accuracy)** For all stationary policies π ,

$$\|Q^\pi - \widehat{Q}^\pi\|_\infty \leq \epsilon.$$

- **(Near-optimal planning)** Let $\widehat{\pi}^*$ be an optimal policy in $\widehat{M}$. Then

$$\|\widehat{Q}^* - Q^*\|_\infty \leq \epsilon, \quad \text{and} \quad \|Q^{\widehat{\pi}^*} - Q^*\|_\infty \leq 2\epsilon.$$

Before proving Proposition 2.1, we record two basic lemmas.

Lemma 2.2. (Simulation lemma) For any stationary policy π ,

$$Q^\pi - \widehat{Q}^\pi = \gamma(I - \gamma\widehat{P}^\pi)^{-1}(P - \widehat{P})V^\pi.$$

Proof. Using $Q^\pi = (I - \gamma P^\pi)^{-1}r$ (Equation 1.2) and the analogous identity $\widehat{Q}^\pi = (I - \gamma\widehat{P}^\pi)^{-1}r$ in $\widehat{M}$, we have

$$\begin{aligned} Q^\pi - \widehat{Q}^\pi &= (I - \gamma P^\pi)^{-1}r - (I - \gamma\widehat{P}^\pi)^{-1}r \\ &= (I - \gamma\widehat{P}^\pi)^{-1} \left((I - \gamma\widehat{P}^\pi) - (I - \gamma P^\pi) \right) (I - \gamma P^\pi)^{-1}r \\ &= \gamma(I - \gamma\widehat{P}^\pi)^{-1}(P^\pi - \widehat{P}^\pi)Q^\pi. \end{aligned}$$

Finally, $(P^\pi Q^\pi)(s, a) = \mathbb{E}_{s' \sim P(\cdot | s, a)}[V^\pi(s')]$ and similarly for $\widehat{P}$, so $(P^\pi - \widehat{P}^\pi)Q^\pi = (P - \widehat{P})V^\pi$, which completes the proof. $\square$

Lemma 2.3. For any stationary policy π and any vector $v \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,

$$\|(I - \gamma P^\pi)^{-1}v\|_\infty \leq \frac{1}{1-\gamma} \|v\|_\infty.$$

Proof. Let $w = (I - \gamma P^\pi)^{-1}v$, so $v = (I - \gamma P^\pi)w = w - \gamma P^\pi w$. Therefore,

$$\|v\|_\infty = \|w - \gamma P^\pi w\|_\infty \geq \|w\|_\infty - \gamma \|P^\pi w\|_\infty \geq (1 - \gamma)\|w\|_\infty,$$

where $\|P^\pi w\|_\infty \leq \|w\|_\infty$ uses that P^π is row-stochastic. Rearranging gives the claim. $\square$

Proof. (Proof of Proposition 2.1) The first part of the proof is with regards to model accuracy. Fix (s, a) . By Proposition A.8 (applied with $d = |\mathcal{S}|$ and N samples), with probability at least $1 - \delta'$,

$$\|P(\cdot | s, a) - \hat{P}(\cdot | s, a)\|_1 \leq c \sqrt{\frac{|\mathcal{S}| \log(c/\delta')}{N}}.$$

Taking a union bound over all $|\mathcal{S}||\mathcal{A}|$ state-action pairs and setting $\delta' = \delta/(|\mathcal{S}||\mathcal{A}|)$ yields

$$\max_{s,a} \|P(\cdot | s, a) - \hat{P}(\cdot | s, a)\|_1 \leq c \sqrt{\frac{|\mathcal{S}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}}$$

with probability at least $1 - \delta$. Under the stated lower bound on N , the right-hand side is at most $(1 - \gamma)^2 \epsilon$, proving the first bullet.

Now we characterize the uniform value accuracy. Fix any stationary policy π . By Lemma 2.2 and Lemma 2.3,

$$\begin{aligned} \|Q^\pi - \hat{Q}^\pi\|_\infty &\leq \gamma \|(I - \gamma \hat{P}^\pi)^{-1}(P - \hat{P})V^\pi\|_\infty \\ &\leq \frac{\gamma}{1 - \gamma} \|(P - \hat{P})V^\pi\|_\infty. \end{aligned}$$

For each (s, a) , the quantity $[(P - \hat{P})V^\pi](s, a)$ is the difference of expectations of $V^\pi(s')$ under $P(\cdot | s, a)$ and $\hat{P}(\cdot | s, a)$, hence

$$\|(P - \hat{P})V^\pi\|_\infty \leq \left(\max_{s,a} \|P(\cdot | s, a) - \hat{P}(\cdot | s, a)\|_1 \right) \|V^\pi\|_\infty.$$

Since rewards lie in $[0, 1]$, we have $\|V^\pi\|_\infty \leq \frac{1}{1 - \gamma}$. Combining,

$$\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \frac{\gamma}{(1 - \gamma)^2} \max_{s,a} \|P(\cdot | s, a) - \hat{P}(\cdot | s, a)\|_1 \leq \frac{\gamma}{(1 - \gamma)^2} \cdot (1 - \gamma)^2 \epsilon \leq \epsilon,$$

which proves the second bullet (simultaneously for all π on the event from Step 1).

We bound the near-optimal planning accuracy. First, using $|\sup_x f(x) - \sup_x g(x)| \leq \sup_x |f(x) - g(x)|$,

$$|\hat{Q}^*(s, a) - Q^*(s, a)| = \left| \sup_{\pi} \hat{Q}^\pi(s, a) - \sup_{\pi} Q^\pi(s, a) \right| \leq \sup_{\pi} |\hat{Q}^\pi(s, a) - Q^\pi(s, a)| \leq \epsilon.$$

Taking the maximum over (s, a) yields $\|\hat{Q}^* - Q^*\|_\infty \leq \epsilon$.

Finally, since $\hat{\pi}^*$ is optimal in $\hat{M}$, we have $\hat{Q}^{\hat{\pi}^*} = \hat{Q}^*$. Therefore,

$$\|Q^{\hat{\pi}^*} - Q^*\|_\infty \leq \|Q^{\hat{\pi}^*} - \hat{Q}^{\hat{\pi}^*}\|_\infty + \|\hat{Q}^* - Q^*\|_\infty \leq \epsilon + \epsilon = 2\epsilon,$$

where the first term used the uniform value accuracy bound from Step 2. This proves the third bullet. $\square$

What our warmup does (and why it is not the full story). Proposition 2.1 formalizes a very strong guarantee: by learning $\hat{P}$ accurately everywhere, we obtain *uniform* value accuracy, meaning that $\hat{Q}^\pi$ is close to Q^π simultaneously for *every* policy π . This is more than is needed for planning—to act near-optimally, we only need to identify a single near-optimal policy (or, equivalently, approximate Q^* well enough to recover one). The uniform guarantee comes at a cost: it forces us to estimate each transition distribution in ℓ_1 to high accuracy, which leads to a sample requirement scaling like the number of parameters in the transition model, $|\mathcal{S}|^2|\mathcal{A}|$.

The rest of the chapter asks what happens if we target only what planning requires. With a generative model, it turns out that one can do substantially better: the minimax number of samples needed to estimate Q^* (and to find a near-optimal policy) scales, up to horizon and logarithmic factors, like $|\mathcal{S}||\mathcal{A}|$ rather than $|\mathcal{S}|^2|\mathcal{A}|$. This is the first concrete sense in which, in reinforcement learning, we can sometimes “act near-optimally without learning the world model.”

2.2 Sublinear Sample Complexity

In the warmup, we obtained *uniform* value accuracy for all policies, which effectively required learning each transition distribution $P(\cdot \mid s, a)$ in ℓ_1 —hence the $|\mathcal{S}|^2|\mathcal{A}|$ scaling. For planning, however, we do not need to approximate the value of *every* policy. Instead, it is natural to ask for guarantees tailored to the optimal value function Q^* (and to a near-optimal policy).

Throughout this section we measure sample size in terms of

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N,$$

where N is the number of simulator calls per state–action pair.

We begin with a crude but instructive bound which already yields a *sublinear* dependence on the number of parameters in P .

Proposition 2.4. (Crude value bounds) Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$,

$$\begin{aligned} \|Q^* - \hat{Q}^*\|_\infty &\leq \Delta_{\delta, N}, \\ \|Q^* - \hat{Q}^{\pi^*}\|_\infty &\leq \Delta_{\delta, N}, \end{aligned}$$

where

$$\Delta_{\delta, N} := \frac{\gamma}{(1 - \gamma)^2} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

The first inequality gives a sublinear rate for estimating Q^* itself. To translate value-estimation error into a policy guarantee, a generic (and somewhat pessimistic) route is to use the Q -error amplification lemma (Lemma 1.11): since $\hat{\pi}^*$ is greedy w.r.t. $\hat{Q}^*$ (ties broken deterministically),

$$\|V^* - V^{\hat{\pi}^*}\|_\infty \leq \frac{2}{1 - \gamma} \|\hat{Q}^* - Q^*\|_\infty \leq \frac{2}{1 - \gamma} \Delta_{\delta, N}.$$

Equivalently, using $Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}[V^\pi(s') \mid s, a]$, we also have

$$\|Q^* - Q^{\hat{\pi}^*}\|_\infty \leq \gamma \|V^* - V^{\hat{\pi}^*}\|_\infty \leq \frac{2\gamma}{1 - \gamma} \Delta_{\delta, N}.$$

A note on “horizon amplification” from our second approach. The crude bound already shows sublinear dependence on the model size, but its dependence on the effective horizon can be pessimistic. From Proposition 2.4,

achieving $\|Q^* - \hat{Q}^*\|_\infty \leq \epsilon$ requires

$$N \gtrsim \frac{1}{(1-\gamma)^4} \cdot \frac{\log(|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2}.$$

If we then translate estimation error into a policy guarantee using the generic Q -to-policy bound (Lemma 1.11), we lose an additional factor $1/(1-\gamma)$: to make the value of $\hat{\pi}^*$ ϵ -close to optimal, the same route suggests

$$N \gtrsim \frac{1}{(1-\gamma)^6} \cdot \frac{\log(|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2}.$$

This extra horizon cost is an artifact of worst-case error propagation, and one of the main goals of the sharper analysis later in the chapter is to avoid it: value estimation and near-optimal planning end up having the same effective-horizon dependence (up to constants and logarithmic factors).

We now provide the proof. We start with a planning-focused comparison lemma, which will be useful throughout the chapter.

Lemma 2.5. (Component-wise bounds) *Let π^* be an optimal policy in M , and let $\hat{\pi}^*$ be an optimal policy in $\widehat{M}$. Then*

$$\begin{aligned} Q^* - \hat{Q}^* &\leq \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P - \hat{P})V^*, \\ Q^* - \hat{Q}^* &\geq \gamma(I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}(P - \hat{P})V^*. \end{aligned}$$

Proof. Define the Bellman *evaluation* operator in the empirical MDP $\widehat{M}$ for a deterministic policy π by

$$(\hat{\mathcal{T}}^\pi Q)(s, a) := r(s, a) + \gamma \sum_{s'} \hat{P}(s' | s, a) Q(s', \pi(s')).$$

Its unique fixed point is $\hat{Q}^\pi$.

Let us first prove the upper bound. Since $\hat{\pi}^*$ is optimal in $\widehat{M}$, we have $\hat{Q}^* = \hat{Q}^{\hat{\pi}^*} \geq \hat{Q}^{\pi^*}$, and hence

$$Q^* - \hat{Q}^* \leq Q^* - \hat{Q}^{\pi^*}.$$

Because $\hat{Q}^{\pi^*}$ is the fixed point of $\hat{\mathcal{T}}^{\pi^*}$, we can write the exact identity

$$Q^* - \hat{Q}^{\pi^*} = (I - \gamma\hat{P}^{\pi^*})^{-1}(Q^* - \hat{\mathcal{T}}^{\pi^*} Q^*).$$

Using $Q^* = r + \gamma P V^*$ and the fact that π^* is greedy w.r.t. Q^* (so $Q^*(s, \pi^*(s)) = V^*(s)$), we have

$$(\hat{\mathcal{T}}^{\pi^*} Q^*)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot | s, a)}[Q^*(s', \pi^*(s'))] = r(s, a) + \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot | s, a)}[V^*(s')],$$

and therefore

$$Q^* - \hat{\mathcal{T}}^{\pi^*} Q^* = \gamma(P - \hat{P})V^*.$$

Combining yields the first inequality.

We now prove the lower bound. Since $\hat{Q}^* = \hat{Q}^{\hat{\pi}^*}$ is the fixed point of $\hat{\mathcal{T}}^{\hat{\pi}^*}$, we similarly have

$$Q^* - \hat{Q}^* = (I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}(Q^* - \hat{\mathcal{T}}^{\hat{\pi}^*} Q^*).$$

Now,

$$\begin{aligned} (Q^* - \hat{\mathcal{T}}^{\hat{\pi}^*} Q^*)(s, a) &= \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[V^*(s')] - \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot | s, a)}[Q^*(s', \hat{\pi}^*(s'))] \\ &\geq \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[V^*(s')] - \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot | s, a)}[V^*(s')] \\ &= \gamma((P - \hat{P})V^*)(s, a), \end{aligned}$$

where we used $Q^*(s', \hat{\pi}^*(s')) \leq \max_{a'} Q^*(s', a') = V^*(s')$. Since $(I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}$ has nonnegative entries, multiplying preserves the inequality, which gives the second claim. $\square$

Proof of Proposition 2.4. First, by Lemma 2.2 (applied to π^*) and Lemma 2.3,

$$\|Q^* - \hat{Q}^{\pi^*}\|_\infty = \|Q^{\pi^*} - \hat{Q}^{\pi^*}\|_\infty \leq \frac{\gamma}{1-\gamma} \|(P - \hat{P})V^*\|_\infty.$$

Second, Lemma 2.5 implies that

$$|Q^* - \hat{Q}^*| \leq \gamma \max\left\{(I - \gamma\hat{P}^{\pi^*})^{-1}, (I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}\right\} |(P - \hat{P})V^*|,$$

and therefore (using Lemma 2.3 again),

$$\|Q^* - \hat{Q}^*\|_\infty \leq \frac{\gamma}{1-\gamma} \|(P - \hat{P})V^*\|_\infty.$$

It remains to bound $\|(P - \hat{P})V^*\|_\infty$. Fix (s, a) and let $s'_1, \dots, s'_N \sim P(\cdot \mid s, a)$ be the N simulator samples used to form $\hat{P}(\cdot \mid s, a)$. Then $\mathbb{E}_{\hat{P}}[V^*(s')] = \frac{1}{N} \sum_{i=1}^N V^*(s'_i)$ and $\mathbb{E}_P[V^*(s')]$ is the true expectation. Since $0 \leq V^* \leq \frac{1}{1-\gamma}$, Hoeffding's inequality gives

$$\Pr\left(\left|\mathbb{E}_{\hat{P}(\cdot \mid s, a)}[V^*] - \mathbb{E}_{P(\cdot \mid s, a)}[V^*]\right| \geq \frac{1}{1-\gamma} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}\right) \leq \frac{\delta}{|\mathcal{S}||\mathcal{A}|}.$$

Taking a union bound over all (s, a) yields, with probability at least $1 - \delta$,

$$\|(P - \hat{P})V^*\|_\infty \leq \frac{1}{1-\gamma} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

Substituting into the previous displays proves both inequalities with the stated $\Delta_{\delta, N}$. $\square$

2.3 Minimax Optimal Sample Complexity (and the Model-Based Approach)

We now state the minimax-optimal sample complexity (up to logarithmic factors) in the generative model setting. The main message is that a simple model-based approach — estimate $\hat{P}$ from samples and then plan in the empirical MDP — achieves the optimal dependence on $|\mathcal{S}||\mathcal{A}|$, ϵ , and $(1 - \gamma)$.

2.3.1 The Discounted Case

Upper bounds. The next theorem packages the main quantitative consequence of the refined analysis from Section 2.4 into a clean sample-complexity statement.

Theorem 2.6 (Minimax upper bound: discounted case). *There exists an absolute constant c such that the following holds. Fix $\delta \in (0, 1)$ and suppose we collect N independent samples from the generative model for each state–action pair, so that the total number of samples is $|\mathcal{S}||\mathcal{A}|N$. Let $\hat{\pi}^*$ and $\hat{Q}^*$ denote an optimal policy and optimal Q -function in the empirical MDP $\hat{M}$.*

- **(Value estimation)** If $\epsilon \leq 1$ and

$$|\mathcal{S}||\mathcal{A}|N \geq \frac{c|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3} \cdot \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2},$$

then with probability at least $1 - \delta$,

$$\|\hat{Q}^* - Q^*\|_\infty \leq \epsilon.$$

- **(Near-optimal planning)** If $\epsilon \leq 1/\sqrt{1-\gamma}$ and

$$|\mathcal{S}||\mathcal{A}|N \geq \frac{c|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3} \cdot \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2},$$

then with probability at least $1 - \delta$,

$$\|Q^{\hat{\pi}^*} - Q^*\|_\infty \leq \epsilon.$$

We only prove the *value estimation* guarantee (the first bullet) in Theorem 2.6; the *planning* guarantee is more delicate. A black-box route using Lemma 1.11 would yield an additional $1/(1-\gamma)$ loss (the “horizon amplification” phenomenon discussed earlier). The second bullet shows that, with a sharper (variance-sensitive) argument, this amplification can be avoided. See Section 2.5 for further discussion, including results that improve the technical “burn-in” restriction on ϵ using different algorithms.

Lower bounds. We next state a matching minimax lower bound (again, up to logarithmic factors). We say that an estimator $\mathcal{A}$ is (ϵ, δ) -good on an MDP M if, given samples from the generative model, it outputs $\hat{Q}^*$ satisfying $\|\hat{Q}^* - Q^*\|_\infty \leq \epsilon$ with probability at least $1 - \delta$.

Theorem 2.7 (Minimax lower bound: discounted case). *There exist absolute constants $\epsilon_0 \in (0, 1)$, $\delta_0 \in (0, 1)$, and $c > 0$ and a class of MDPs $\mathcal{M}$ such that for all $\epsilon \in (0, \epsilon_0)$ and $\delta \in (0, \delta_0)$, any algorithm $\mathcal{A}$ that is (ϵ, δ) -good on every $M \in \mathcal{M}$ must use at least*

$$\# \text{ samples from generative model} \geq \frac{c|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3} \cdot \frac{\log(c/\delta)}{\epsilon^2}.$$

In other words, the model-based approach is minimax-optimal in its dependence on $|\mathcal{S}||\mathcal{A}|$, $(1-\gamma)$, and ϵ (up to logarithmic factors).

Interpreting the horizon dependence. It is useful to calibrate the target accuracy in discounted problems. Since rewards are bounded in $[0, 1]$, we have $\|Q^*\|_\infty \leq \frac{1}{1-\gamma}$, so an additive error of size $\frac{\epsilon}{1-\gamma}$ corresponds to *relative* accuracy on the scale of ϵ . Equivalently, if we define a normalized value function $\bar{Q}^\pi := (1-\gamma)Q^\pi$, then $\|\bar{Q}^\pi\|_\infty \leq 1$ and asking for additive error ϵ in $\bar{Q}$ is the same as asking for additive error $\epsilon/(1-\gamma)$ in Q .

Plugging this normalization into Theorem 2.6 (and ignoring logarithmic factors and the mild technical restrictions on ϵ), the minimax sample complexity for constant-accuracy estimation of $\bar{Q}^*$ scales like

$$\# \text{ samples} \approx \frac{|\mathcal{S}||\mathcal{A}|}{1-\gamma} \cdot \frac{1}{\epsilon^2}.$$

In this sense, after normalization, the dependence on the effective horizon is essentially linear: the extra powers of $(1-\gamma)^{-1}$ in the unnormalized bounds are largely accounting for the fact that Q^* itself grows like $(1-\gamma)^{-1}$.

2.3.2 Finite-Horizon Setting

Recall the finite-horizon (time-dependent) MDP model from Section 1.2, with horizon H and transition kernels $\{P_h\}_{h=0}^{H-1}$. In the generative-model setting, we can query any triple (s, a, h) and obtain an independent sample $s' \sim P_h(\cdot | s, a)$ (and the reward $r_h(s, a)$, which we treat as deterministic for simplicity).

We consider the natural plug-in model-based estimator: for each (s, a, h) we draw N independent samples and form the empirical transition kernel $\hat{P}_h(\cdot | s, a)$. The total number of samples is therefore

$$\# \text{ samples from generative model} = H|\mathcal{S}||\mathcal{A}|N.$$

We state the resulting performance guarantees without proof.

Upper bounds. The following theorem gives a high-probability error bound for planning in the empirical MDP.

Theorem 2.8 (Model-based planning: finite horizon). *There exists an absolute constant c such that the following holds. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

- (*Value estimation*)

$$\|Q_0^* - \hat{Q}_0^*\|_\infty \leq cH \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + cH \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

- (*Sub-optimality*) letting $\hat{\pi}^*$ be an optimal policy in the empirical MDP,

$$\|Q_0^* - Q_0^{\hat{\pi}^*}\|_\infty \leq cH \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + cH \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

Equivalently, ignoring lower-order terms, achieving $\|Q_0^* - \hat{Q}_0^*\|_\infty \leq \epsilon$ (or $\|Q_0^* - Q_0^{\hat{\pi}^*}\|_\infty \leq \epsilon$) is guaranteed once

$$N \gtrsim \frac{H^2}{\epsilon^2} \log\left(\frac{|\mathcal{S}||\mathcal{A}|}{\delta}\right), \quad \text{or equivalently} \quad \# \text{ samples from generative model} \gtrsim \frac{H^3 |\mathcal{S}||\mathcal{A}|}{\epsilon^2} \log\left(\frac{|\mathcal{S}||\mathcal{A}|}{\delta}\right).$$

The dependence on H looks different from the discounted case, but it is not directly comparable without specifying how one maps γ to an effective horizon. One rough correspondence is $H \asymp \frac{1}{1-\gamma}$; under this heuristic, the leading-order scalings match up to constants and logarithmic factors.

Lower bounds. In the same minimax sense as Theorem 2.7, the above model-based rate is optimal (up to constant and logarithmic factors) for finite-horizon MDPs under a generative model. We omit the proof.

2.4 Analysis

We now prove the *value-estimation* guarantee in Theorem 2.8 (i.e. the first bullet). For convenience, we restate it here.

Theorem 2.9 (Value estimation; restatement of Theorem 2.8). *Fix $\delta \in (0, 1)$. For an appropriately chosen absolute constant c , with probability at least $1 - \delta$,*

$$\|Q^* - \hat{Q}^*\|_\infty \leq \gamma \sqrt{\frac{c}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

2.4.1 Variance lemmas

The sharper dependence on $(1-\gamma)$ comes from replacing a worst-case (Hoeffding-style) deviation bound by a variance-sensitive one (Bernstein), and then exploiting a Bellman-type recursion for variances.

For a real-valued function f and distribution $\mathcal{D}$, define

$$\text{Var}_{\mathcal{D}}(f) := \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2] - \left(\mathbb{E}_{x \sim \mathcal{D}}[f(x)]\right)^2.$$

For a vector $V \in \mathbb{R}^{|\mathcal{S}|}$, define $\text{Var}_P(V) \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ componentwise by

$$\text{Var}_P(V)(s, a) := \text{Var}_{P(\cdot|s,a)}(V) = \mathbb{E}_{s' \sim P(\cdot|s,a)}[V(s')^2] - \left(\mathbb{E}_{s' \sim P(\cdot|s,a)}[V(s')]\right)^2.$$

Equivalently (with squares understood componentwise),

$$\text{Var}_P(V) = P(V^2) - (PV)^2.$$

A Bernstein deviation bound. We will repeatedly apply Bernstein's inequality to the empirical estimate $\hat{P}(\cdot \mid s, a)$.

Lemma 2.10. *Let $\delta \in (0, 1)$. With probability at least $1 - \delta$, the following holds simultaneously for all (s, a) :*

$$|(P - \hat{P})V^*| \leq \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}} \sqrt{\text{Var}_P(V^*)} + \frac{1}{1 - \gamma} \frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{3N} \mathbb{1}.$$

Here the absolute value and inequality are componentwise, and $\mathbb{1}$ denotes the all-ones vector in $\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$.

Proof. Fix (s, a) . Apply Bernstein's inequality to the i.i.d. random variables $V^*(s'_1), \dots, V^*(s'_N)$ with $s'_i \sim P(\cdot \mid s, a)$. Since $0 \leq V^* \leq \frac{1}{1 - \gamma}$, the range is at most $\frac{1}{1 - \gamma}$, and the variance term is exactly $\text{Var}_{P(\cdot \mid s, a)}(V^*)$. A union bound over $|\mathcal{S}||\mathcal{A}|$ state-action pairs yields the stated form. $\square$

The remaining work is to control how the (square-root) variance term propagates through the inverse Bellman operator, e.g. $\|(I - \gamma \hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_P(V^*)}\|_\infty$.

Variance of the discounted return. For any MDP M and policy π , define $\Sigma_M^\pi(s, a)$ as the conditional variance of the discounted return:

$$\Sigma_M^\pi(s, a) := \mathbb{E} \left[\left(\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) - Q_M^\pi(s, a) \right)^2 \middle| s_0 = s, a_0 = a \right],$$

where the expectation is over trajectories generated by π in M . Because we assume the one-step reward $r(s, a)$ is deterministic, all randomness in the return comes from future states, hence

$$\|\Sigma_M^\pi\|_\infty \leq \left(\frac{\gamma}{1 - \gamma} \right)^2 = \frac{\gamma^2}{(1 - \gamma)^2}.$$

The next lemma is the variance analogue of Bellman consistency.

Lemma 2.11 (Bellman consistency of Σ). *For any MDP M and policy π ,*

$$\Sigma_M^\pi = \gamma^2 \text{Var}_P(V_M^\pi) + \gamma^2 P^\pi \Sigma_M^\pi, \quad (2.1)$$

where P is the transition model of M .

The proof is left as an exercise to the reader (it follows by writing the return as $r(s, a) + \gamma(\text{future return})$ and applying the law of total variance).

We now prove a bound that turns square-root variances into the desired $(1 - \gamma)^{-3/2}$ scaling.

Lemma 2.12. (Weighted sum of deviations) *For any policy π and MDP M ,*

$$\left\| (I - \gamma P^\pi)^{-1} \sqrt{\text{Var}_P(V_M^\pi)} \right\|_\infty \leq \sqrt{\frac{2}{(1 - \gamma)^3}},$$

where P is the transition model of M .

Proof. Let $v := \text{Var}_P(V_M^\pi) \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, which is entrywise nonnegative. Observe that $(1 - \gamma)(I - \gamma P^\pi)^{-1}$ has nonnegative entries and each row sums to 1, so each row is a probability distribution over next state-action pairs.

Applying Jensen's inequality row-by-row (using concavity of $\sqrt{\cdot}$) gives

$$\|(I - \gamma P^\pi)^{-1} \sqrt{v}\|_\infty = \frac{1}{1-\gamma} \|(1-\gamma)(I - \gamma P^\pi)^{-1} \sqrt{v}\|_\infty \leq \sqrt{\left\| \frac{1}{1-\gamma} (I - \gamma P^\pi)^{-1} v \right\|_\infty}.$$

Next we relate $(I - \gamma P^\pi)^{-1}$ to $(I - \gamma^2 P^\pi)^{-1}$: for any entrywise nonnegative u ,

$$\|(I - \gamma P^\pi)^{-1} u\|_\infty \leq 2 \|(I - \gamma^2 P^\pi)^{-1} u\|_\infty.$$

(We verify this inequality at the end of the proof.) Applying this with $u = v$ yields

$$\left\| \frac{1}{1-\gamma} (I - \gamma P^\pi)^{-1} v \right\|_\infty \leq \left\| \frac{2}{1-\gamma} (I - \gamma^2 P^\pi)^{-1} v \right\|_\infty.$$

Finally, using the Bellman consistency equation (2.1) we have $\Sigma_M^\pi = \gamma^2 (I - \gamma^2 P^\pi)^{-1} v$, hence

$$(I - \gamma^2 P^\pi)^{-1} v = \frac{1}{\gamma^2} \Sigma_M^\pi, \quad \text{so} \quad \left\| \frac{2}{1-\gamma} (I - \gamma^2 P^\pi)^{-1} v \right\|_\infty \leq \frac{2}{1-\gamma} \cdot \frac{1}{\gamma^2} \|\Sigma_M^\pi\|_\infty \leq \frac{2}{(1-\gamma)^3}.$$

Taking square roots completes the proof.

Proof of $\|(I - \gamma P^\pi)^{-1} u\|_\infty \leq 2 \|(I - \gamma^2 P^\pi)^{-1} u\|_\infty$. Let $w = (I - \gamma^2 P^\pi)^{-1} u$. Then

$$(I - \gamma P^\pi)^{-1} u = (I - \gamma P^\pi)^{-1} (I - \gamma^2 P^\pi) w = (I - \gamma P^\pi)^{-1} ((1-\gamma)I + \gamma(I - \gamma P^\pi)) w = (1-\gamma)(I - \gamma P^\pi)^{-1} w + \gamma w.$$

Taking $\|\cdot\|_\infty$ norms and using Lemma 2.3 on the first term gives

$$\|(I - \gamma P^\pi)^{-1} u\|_\infty \leq (1-\gamma) \cdot \frac{1}{1-\gamma} \|w\|_\infty + \gamma \|w\|_\infty \leq 2 \|w\|_\infty = 2 \|(I - \gamma^2 P^\pi)^{-1} u\|_\infty,$$

as claimed. $\square$

2.4.2 Completing the proof

The next lemma relates the (unknown) one-step variance under P to an empirical variance under $\hat{P}$, up to lower-order estimation terms.

Lemma 2.13. Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$, we have the componentwise bounds

$$\text{Var}_P(V^*) \leq 2 \text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) + \Delta'_{\delta, N} \mathbf{1},$$

$$\text{Var}_P(V^*) \leq 2 \text{Var}_{\hat{P}}(\hat{V}^*) + \Delta'_{\delta, N} \mathbf{1},$$

where

$$\Delta'_{\delta, N} := \frac{1}{(1-\gamma)^2} \sqrt{\frac{18 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{1}{(1-\gamma)^4} \frac{4 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

Proof. We work componentwise in (s, a) , and write $(PV^*)(s, a) = \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^*(s')]$. By definition,

$$\begin{aligned} \text{Var}_P(V^*) &= P((V^*)^2) - (PV^*)^2 \\ &= \hat{P}((V^*)^2) - (\hat{P}V^*)^2 + (P - \hat{P})(V^*)^2 - ((PV^*)^2 - (\hat{P}V^*)^2) \\ &= \text{Var}_{\hat{P}}(V^*) + (P - \hat{P})(V^*)^2 - ((PV^*)^2 - (\hat{P}V^*)^2). \end{aligned}$$

We bound the last two terms via Hoeffding + a union bound over (s, a) .

We now handle term 1: $(P - \hat{P})(V^*)^2$. Since $0 \leq V^* \leq \frac{1}{1-\gamma}$, we have $0 \leq (V^*)^2 \leq \frac{1}{(1-\gamma)^2}$. Thus with probability at least $1 - \delta/3$,

$$\|(P - \hat{P})(V^*)^2\|_\infty \leq \frac{1}{(1-\gamma)^2} \sqrt{\frac{2 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

Next, we handle term 2: $(PV^*)^2 - (\hat{P}V^*)^2$. Using $(x^2 - y^2) = (x + y)(x - y)$ componentwise,

$$\begin{aligned} \|(PV^*)^2 - (\hat{P}V^*)^2\|_\infty &\leq \|PV^* + \hat{P}V^*\|_\infty \cdot \|PV^* - \hat{P}V^*\|_\infty \\ &\leq \frac{2}{1-\gamma} \cdot \|(P - \hat{P})V^*\|_\infty. \end{aligned}$$

Again $0 \leq V^* \leq \frac{1}{1-\gamma}$, so by Hoeffding + union bound, with probability at least $1 - \delta/3$,

$$\|(P - \hat{P})V^*\|_\infty \leq \frac{1}{1-\gamma} \sqrt{\frac{2 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

Combining gives, with probability at least $1 - \delta/3$,

$$\|(PV^*)^2 - (\hat{P}V^*)^2\|_\infty \leq \frac{2}{(1-\gamma)^2} \sqrt{\frac{2 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

We now handle term 3: $\text{Var}_{\hat{P}}(V^*)$. We compare $\text{Var}_{\hat{P}}(V^*)$ to an empirical variance of a nearby value function. Using $\text{Var}(X + Y) \leq 2 \text{Var}(X) + 2 \text{Var}(Y)$ and $\text{Var}_{\hat{P}}(f) \leq \|f\|_\infty^2$,

$$\begin{aligned} \text{Var}_{\hat{P}}(V^*) &= \text{Var}_{\hat{P}}((V^* - \hat{V}^{\pi^*}) + \hat{V}^{\pi^*}) \\ &\leq 2 \text{Var}_{\hat{P}}(V^* - \hat{V}^{\pi^*}) + 2 \text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) \\ &\leq 2\|V^* - \hat{V}^{\pi^*}\|_\infty^2 + 2 \text{Var}_{\hat{P}}(\hat{V}^{\pi^*}). \end{aligned}$$

Since π^* is optimal in M , we have $V^*(s) = Q^*(s, \pi^*(s))$ and $\hat{V}^{\pi^*}(s) = \hat{Q}^{\pi^*}(s, \pi^*(s))$, hence $\|V^* - \hat{V}^{\pi^*}\|_\infty \leq \|Q^* - \hat{Q}^{\pi^*}\|_\infty$. Applying Proposition 2.4 (with failure probability $\delta/3$) gives

$$\|V^* - \hat{V}^{\pi^*}\|_\infty \leq \Delta_{\delta/3, N} \Rightarrow 2\|V^* - \hat{V}^{\pi^*}\|_\infty^2 \leq \frac{4}{(1-\gamma)^4} \frac{\log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N},$$

where we used $\gamma \leq 1$ and the definition of $\Delta_{\delta, N}$.

Putting it together. With probability at least $1 - \delta$ (union bound over the three events),

$$\text{Var}_P(V^*) \leq 2 \text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) + \left(\frac{1}{(1-\gamma)^2} \sqrt{\frac{18 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{1}{(1-\gamma)^4} \frac{4 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right) \mathbf{1},$$

which is the first claim.

For the second claim, repeat the same argument but decompose $\text{Var}_{\hat{P}}(V^*) = \text{Var}_{\hat{P}}((V^* - \hat{V}^*) + \hat{V}^*)$ and use $\|V^* - \hat{V}^*\|_\infty \leq \|Q^* - \hat{Q}^*\|_\infty \leq \Delta_{\delta/3, N}$. $\square$

Combining the Bernstein deviation bound with the variance comparison above yields a variance-sensitive control of $(P - \hat{P})V^*$.

Corollary 2.14. Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$, we have the componentwise bounds

$$\begin{aligned} |(P - \hat{P})V^*| &\leq c\sqrt{\frac{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \Delta''_{\delta,N} \mathbf{1}, \\ |(P - \hat{P})V^*| &\leq c\sqrt{\frac{\text{Var}_{\hat{P}}(\hat{V}^*) \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \Delta''_{\delta,N} \mathbf{1}, \end{aligned}$$

where

$$\Delta''_{\delta,N} := c \frac{1}{1-\gamma} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} + \frac{c}{(1-\gamma)^2} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N},$$

and c is an absolute constant.

Proof. Apply Lemma 2.10 and substitute the upper bound on $\text{Var}_P(V^*)$ from Lemma 2.13. Then use $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ componentwise to separate the empirical-variance term from $\Delta'_{\delta,N}$. The term involving $\sqrt{\Delta'_{\delta,N}}$ contributes the $(\log/N)^{3/4}$ and $(\log/N)$ pieces, and the additive Bernstein remainder $\frac{1}{1-\gamma} \frac{\log(\cdot)}{N} \mathbf{1}$ is absorbed into the displayed $\Delta''_{\delta,N}$ by adjusting constants. $\square$

Proof of Theorem 2.9. We bound the two sides in Lemma 2.5.

We first handle the upper deviation. Using the first inequality in Lemma 2.5, monotonicity of $(I - \gamma\hat{P}^{\pi^*})^{-1}$, and Corollary 2.14 (first line), we obtain

$$\begin{aligned} \|Q^* - \hat{Q}^*\|_\infty &\leq \gamma \left\| (I - \gamma\hat{P}^{\pi^*})^{-1} |(P - \hat{P})V^*| \right\|_\infty \\ &\leq c\gamma \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} \left\| (I - \gamma\hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*})} \right\|_\infty + \gamma \Delta''_{\delta,N} \left\| (I - \gamma\hat{P}^{\pi^*})^{-1} \mathbf{1} \right\|_\infty \\ &\leq c\gamma \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} \left\| (I - \gamma\hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*})} \right\|_\infty \\ &\quad + \frac{c\gamma}{(1-\gamma)^2} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}, \end{aligned}$$

where we used $\|(I - \gamma\hat{P}^{\pi^*})^{-1} \mathbf{1}\|_\infty \leq \frac{1}{1-\gamma}$.

Now apply Lemma 2.12 to $\hat{M}$ and policy π^* (so $P = \hat{P}$ and $V_M^\pi = \hat{V}^{\pi^*}$) to get

$$\left\| (I - \gamma\hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*})} \right\|_\infty \leq \sqrt{\frac{2}{(1-\gamma)^3}}.$$

Substituting yields

$$\|Q^* - \hat{Q}^*\|_\infty \leq c\gamma \sqrt{\frac{2}{(1-\gamma)^3}} \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{c\gamma}{(1-\gamma)^2} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

The middle term can be absorbed into the other two terms using $2ab \leq a^2 + b^2$. Indeed, letting $L := \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}$,

$$\frac{1}{(1-\gamma)^2} L^{3/4} = \left(\frac{1}{(1-\gamma)^{3/2}} L^{1/2} \right) \left(\frac{1}{(1-\gamma)^{1/2}} L^{1/4} \right) \leq \frac{1}{(1-\gamma)^3} L + \frac{1}{1-\gamma} L^{1/2} \leq \frac{1}{(1-\gamma)^3} L + \frac{1}{(1-\gamma)^{3/2}} L^{1/2},$$

and adjusting constants gives the claimed form.

We now handle the lower deviation. Repeat the same argument using the second inequality in Lemma 2.5 (i.e. with $\hat{\pi}^*$ in place of π^*), which yields an identical bound for $\|\hat{Q}^* - Q^*\|_\infty$. Combining the two sides gives the stated ℓ_∞ error bound (with a possibly different absolute constant c). $\square$

2.5 Bibliographic Remarks and Further Reading

The generative model framework (also known as a simulator oracle) was introduced by Kearns and Singh [1999], who established that model-based and model-free methods have comparable sample complexity up to horizon and logarithmic factors. Kakade [2003] provided improved rates analogous to the “crude” bounds presented in Proposition 2.4.

The sharp value estimation bound (the first claim in Theorem 2.6) is due to Azar et al. [2013]; the analysis in Section 2.4 largely follows their technique. Regarding the sub-optimality of the learned policy $\hat{\pi}^*$, a naive analysis suggests an amplification of the error by $(1 - \gamma)^{-1}$. Sidford et al. [2018] provided the first proof that this amplification can be avoided, achieving the optimal rate using a variance-reduced value iteration algorithm. Subsequently, Agarwal et al. [2020c] showed that the naive model-based approach (the “plug-in” estimator) is also sufficient to achieve this optimal rate (the second claim in Theorem 2.6). The minimax lower bound (Theorem 2.7) is due to Azar et al. [2013].

We also note that the sub-optimality bounds presented here require ϵ to be sufficiently small (roughly $\epsilon \lesssim 1/\sqrt{1 - \gamma}$). Li et al. [2020] showed that the optimal rate is achievable for the full range of meaningful accuracy (up to $\epsilon \approx 1/(1 - \gamma)$) using a perturbed model-based algorithm. It remains an open question whether the standard (unperturbed) model-based approach achieves this optimal non-asymptotic behavior for large ϵ .

Finally, this chapter presented optimal sample complexity results for the finite-horizon setting (Section 2.3.2) without proof. These results follow from the line of reasoning in Azar et al. [2013], with the added simplification that the sub-optimality analysis is simpler in the finite-horizon setting with time-dependent transition matrices (see, e.g., Yin et al. [2021]).

Chapter 3

RL with Linear Features: When Does It Work, and When Does It Fail?

Up to now we have focused on tabular Markov decision processes, where both sample complexity and computation scale polynomially with $|\mathcal{S}|$ and $|\mathcal{A}|$. In many problems of interest, however, the state and action spaces are so large that tabular methods are not viable. A standard response is to introduce a feature map

$$\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$$

and to approximate value functions using linear predictors $f_\theta(s, a) = \theta^\top \phi(s, a)$.

A natural hope is that the relevant complexity parameter becomes d rather than $|\mathcal{S}||\mathcal{A}|$. This chapter asks a single organizing question:

When do linear features actually buy you poly(d) sample complexity — and when do they fundamentally not?

The answer is subtle. On the positive side, there are structural conditions under which linear methods reduce reinforcement learning to a sequence of well-conditioned regression problems, yielding $\text{poly}(d, H, 1/\epsilon)$ guarantees without explicit dependence on $|\mathcal{S}|$ or $|\mathcal{A}|$. On the negative side, several seemingly natural “linear realizability” assumptions are *not* enough: even with excellent coverage, and even when an algorithm is given the feature map as input, the sample complexity can be exponential in the horizon.

The chapter is organized around a short ladder of assumptions. We first introduce these assumptions and their relationships at a high level. We then present a sharp positive result under a completeness condition (Least-Squares Value Iteration with a well-conditioned design), followed by information-theoretic lower bounds showing that weaker notions of realizability do not suffice.

Why this is subtle (and why realizability feels tempting). Since most of our discussion so far has focused on the discounted setting, it is worth briefly recalling the finite-horizon dynamic programming picture. For any function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, the stage- h optimal Bellman operator is

$$(\mathcal{T}_h f)(s, a) := r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)} \left[\max_{a' \in \mathcal{A}} f(s', a') \right].$$

The optimal Q -functions satisfy $Q_H^* \equiv 0$ and the backward recursion $Q_h^* = \mathcal{T}_h Q_{h+1}^*$ for $h = H - 1, \dots, 0$.

This recursion suggests a very natural hope. If we can produce accurate estimates $\widehat{Q}_{h+1} \approx Q_{h+1}^*$, then we might plug $\widehat{Q}_{h+1}$ into the backup to form targets $\mathcal{T}_h \widehat{Q}_{h+1}$, fit $\widehat{Q}_h$ by regression, and expect errors to propagate in a controlled way across stages. From this perspective it is tempting to believe that a simple *realizability* assumption—for example, $Q_h^* \in \mathcal{F}$ for all h —should already be enough to obtain $\text{poly}(d)$ sample complexity.

The difficulty is that dynamic programming is not a single regression problem but a *composition* of backups: the algorithm repeatedly forms new regression targets using its *own* current estimates. Thus, what matters is not only whether Q^* lies in the hypothesis class, but whether the sequence of targets generated along the way remains learnable from the available data. The assumption ladder introduced next is a way of making precise which structural conditions guarantee this benign behavior—and which apparently natural linear conditions do not.

3.1 Setup and an Assumption Ladder

We work with a fixed feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$. For a parameter vector $\theta \in \mathbb{R}^d$, define the associated linear function $f_\theta(s, a) = \theta^\top \phi(s, a)$, and let

$$\mathcal{F} := \{f_\theta : \theta \in \mathbb{R}^d\}$$

denote the corresponding linear function class.

There are multiple distinct ways that “linear features” can enter reinforcement learning. The following assumptions form a useful ladder: as we move down the list, the assumptions become stronger and increasingly algorithmically useful.

(A) Agnostic linear approximation (no realizability). The weakest perspective is purely approximation-based: we do *not* assume that any relevant value function is linear. Instead, we measure the best approximation error of $\mathcal{F}$ to some target (e.g. Q^* , or the iterates of a fitted value iteration procedure) under a chosen norm or data distribution. This is the analogue of agnostic supervised learning. We defer this viewpoint—and its attendant distribution-shift / concentrability issues—to the chapter on fitted value iteration. However, the lower bounds presented later in this chapter for stronger assumptions (like realizability) imply that the agnostic viewpoint is prone to exponential sample complexity without further structural conditions.

(B) Linear Q^* realizability. A common structural assumption is that the *optimal* action-value function is linear: for each stage h there exists $\theta_h^* \in \mathbb{R}^d$ such that

$$Q_h^*(s, a) = (\theta_h^*)^\top \phi(s, a) \quad \forall (s, a).$$

This is the most direct analogue of realizable linear regression. A tempting intuition is that this should allow $\text{poly}(d)$ learning. We will see that this intuition is false in a strong information-theoretic sense.

(C) All-policies linear realizability. A stronger assumption requires linear realizability not just for Q^* , but for *every* policy: for each π and each stage h , there exists θ_h^π such that

$$Q_h^\pi(s, a) = (\theta_h^\pi)^\top \phi(s, a) \quad \forall (s, a).$$

This assumption is substantially stronger than (B). This assumption is subtle. It does permit polynomial sample complexity both with a generative model (via adaptive, on-policy rollouts) and in an episodic setting (see the discussion in Section 3.5). However, as we see in this chapter, (C) it still does *not* guarantee sample-efficient *offline* policy evaluation: even with extremely strong feature coverage, any estimator can require exponentially many samples (Section 3.4).

(D) Linear Bellman completeness (a closure property). The most algorithmically useful assumption in this chapter is a *closure* condition: roughly, applying the Bellman optimality operator to a linear function yields another linear function (with respect to the same feature map). One convenient way to state this is: for each stage h and each $\theta \in \mathbb{R}^d$, there exists $w \in \mathbb{R}^d$ such that the function

$$(s, a) \mapsto r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)} \left[\max_{a'} \theta^\top \phi(s', a') \right]$$

is exactly $w^\top \phi(s, a)$ for all (s, a) . Equivalently, letting $\mathcal{F} := \{(s, a) \mapsto w^\top \phi(s, a) \mid w \in \mathbb{R}^d\}$ denote the class of functions linear in ϕ , the Bellman completeness property can be written as

$$f \in \mathcal{F} \implies \mathcal{T}_h f \in \mathcal{F} \quad \text{for each } h \in [H], \quad (3.1)$$

where $\mathcal{T}_h$ is the stage- h Bellman optimality operator. This assumption turns dynamic programming into a sequence of regression problems, and it is what underlies the positive results in this chapter.

Relationships. A few simple implications are worth keeping in mind. First, linear Bellman completeness immediately implies that rewards are linear in ϕ (take $\theta = 0$ above). It also implies linear Q^* realizability (by backward induction / value iteration carried out within $\mathcal{F}$). In contrast, (C) and (D) are not comparable in general: (C) asserts realizability for *all* policies but does not impose closure under the *optimality* operator, while (D) enforces closure under the optimality operator but does not a priori assert that Q^π is linear for every policy π .

A remark on strength. Bellman completeness should be read as a structural condition on the MDP itself (via the transition kernel and reward), not merely a property of Q^* . We will give concrete examples of model classes satisfying Bellman completeness (e.g. linear MDPs and related low-complexity transition models) later in Chapters 7 and 8, once our algorithmic guarantees are on the table.

With our ladder in place, the remainder of the chapter answers our organizing question by pairing a sharp positive result under (D) with lower bounds showing that (B) and (C) can fail dramatically.

A remark on feature augmentation (non-monotonicity). A perhaps surprising subtlety is that Bellman completeness is *not* monotone in the feature map. If we enlarge the representation by appending an additional coordinate to ϕ (thus enlarging $\mathcal{F}$ from a d -dimensional linear class to a $(d + 1)$ -dimensional one), the closure property $f \in \mathcal{F} \implies \mathcal{T}_h f \in \mathcal{F}$ can *break*, even if it held before. Intuitively, completeness must hold for *every* linear function in the class; adding a new feature introduces new functions f whose Bellman backups may no longer be representable by the enlarged linear class. In contrast, realizability assumptions such as linear Q^* (or all-policies realizability) are monotone: adding features can only make it easier to represent the target functions. This fragility is another reason to view Bellman completeness as a structural assumption about the interaction between the MDP and the chosen feature map, rather than a mere approximation property of Q^* .

3.2 Positive results under Bellman completeness

This section gives the affirmative path for linear features: under the right structural condition, dynamic programming reduces to a sequence of (supervised) regression problems, yielding sample complexity polynomial in the feature dimension. We state the algorithm in a way that will be reused in both the generative-model and offline settings; the difference will be only in how the stage-wise datasets are obtained.

3.2.1 Setup and data model

Throughout this section we assume the Bellman completeness condition stated earlier (Section 3.1).

We work in the finite-horizon setting with stages $h \in [H] := \{0, 1, \dots, H-1\}$. An MDP is specified by a state space $\mathcal{S}$, an action space $\mathcal{A}$, an initial distribution μ over $\mathcal{S}$, stage-dependent transition kernels $\{P_h(\cdot | s, a)\}_{h=0}^{H-1}$, and stage-dependent rewards $\{r_h(s, a)\}_{h=0}^{H-1}$. Throughout we assume rewards are bounded, $r_h(s, a) \in [0, 1]$.¹

We are given a feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, and we consider the linear function class

$$\mathcal{F} := \{f_w : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R} \mid f_w(s, a) = w^\top \phi(s, a), w \in \mathbb{R}^d\}.$$

(As usual, one can rescale features so that $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$; we will be explicit about boundedness assumptions when they matter.)

Stage-wise datasets (offline view). To keep the algorithm reusable, we phrase everything in terms of per-stage datasets. For each $h \in [H]$, let

$$D_h = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^{n_h}$$

be a dataset of n_h transitions at stage h , where each $s'_i \sim P_h(\cdot | s_i, a_i)$ is sampled independently conditional on (s_i, a_i) and $r_i = r_h(s_i, a_i)$. In the generative-model setting we will *construct* these datasets; in the offline setting the datasets are given and we will impose an explicit coverage condition.

Bellman operators. Recall from Section 1.2, that for a function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, we have that our stage- h *optimal* Bellman operator $\mathcal{T}_h$ is defined by

$$(\mathcal{T}_h f)(s, a) := r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} f(s', a') \right].$$

In the finite-horizon problem, the optimal Q -functions satisfy the recursion $Q_H^* \equiv 0$ and $Q_h^* = \mathcal{T}_h Q_{h+1}^*$ for $h = H-1, \dots, 0$.

The remainder of Section 3.2 shows that, under Bellman completeness, ordinary least squares can drive Bellman residuals uniformly small, which in turn yields a near-optimal greedy policy.

3.2.2 Algorithm: Least-Squares Value Iteration (LSVI)

LSVI runs a backward dynamic program, where each Bellman backup is implemented by a least-squares regression onto the feature class $\mathcal{F}$.

If the empirical covariance at a stage is singular, one can take a minimum-norm solution (or add a small ridge). In the settings we analyze (D-optimal sampling from a generative model, or explicit coverage assumptions offline), the design is well-conditioned, and this technicality is inessential.

3.2.3 Main guarantee: LSVI with a generative model

We now prove the main positive result under Bellman completeness in the generative-model setting. The proof uses three ingredients: (i) a geometric sampling lemma (D-optimal design), (ii) a uniform generalization bound for ordinary least squares (Appendix), and (iii) a standard approximate dynamic programming lemma converting small Bellman residuals into near-optimality.

¹ Allowing stochastic rewards with bounded support is routine; to keep notation light we write rewards as deterministic.

Algorithm 1 Least-Squares Value Iteration (LSVI)

- 1: **Input:** datasets $D_0, \dots, D_{H-1}$, feature map ϕ
 - 2: Initialize $\hat{V}_H(s) \leftarrow 0$ for all $s \in \mathcal{S}$
 - 3: **for** $h = H - 1, H - 2, \dots, 0$ **do**
 - 4: $\hat{\theta}_h \in \arg \min_{\theta \in \mathbb{R}^d} \sum_{(s,a,r,s') \in D_h} \left(\theta^\top \phi(s,a) - r - \hat{V}_{h+1}(s') \right)^2$
 - 5: Define $\hat{Q}_h(s,a) \leftarrow \hat{\theta}_h^\top \phi(s,a)$ for all (s,a)
 - 6: Define $\hat{V}_h(s) \leftarrow \max_{a \in \mathcal{A}} \hat{Q}_h(s,a)$ for all s
 - 7: **end for**
 - 8: **Return:** greedy policy $\hat{\pi} = \{\hat{\pi}_h\}_{h=0}^{H-1}$ where $\hat{\pi}_h(s) \in \arg \max_{a \in \mathcal{A}} \hat{Q}_h(s,a)$
-

Tool lemma: D-optimal design

Let $\Phi := \{\phi(s,a) : (s,a) \in \mathcal{S} \times \mathcal{A}\} \subset \mathbb{R}^d$ denote the set of feature vectors. We assume Φ is bounded and full-dimensional (otherwise we may restrict to $\text{span}(\Phi)$).

Lemma 3.1 (D-optimal design). *There exists a distribution ρ supported on at most $d(d+1)/2$ state–action pairs such that, letting*

$$\Sigma := \mathbb{E}_{(s,a) \sim \rho} [\phi(s,a) \phi(s,a)^\top],$$

we have $\Sigma \succ 0$ and

$$\sup_{(s,a) \in \mathcal{S} \times \mathcal{A}} \phi(s,a)^\top \Sigma^{-1} \phi(s,a) \leq d.$$

One convenient characterization of ρ is as a maximizer of a log-determinant objective:

$$\rho \in \underset{\nu \in \Delta(\mathcal{S} \times \mathcal{A})}{\operatorname{argmax}} \ln \det \left(\mathbb{E}_{(s,a) \sim \nu} [\phi(s,a) \phi(s,a)^\top] \right), \quad (3.2)$$

(which is the Kiefer–Wolfowitz Theorem). Equivalently, the ellipsoid $\mathcal{E} = \{v : \|v\|_{\Sigma^{-1}}^2 \leq d\}$ is the unique minimum-volume centered ellipsoid containing Φ ; this result is known as John’s Theorem. We do not provide a proof of either claim here (both of which lead to the D-optimal design lemma). See Section 3.5 for references and further discussion.

Main theorem

We assume a generative model that can be queried at any stage h and any (s,a) to obtain an independent sample $s' \sim P_h(\cdot \mid s,a)$ (along with the reward $r_h(s,a)$).

Theorem 3.2 (LSVI under Bellman completeness; generative model). *Assume Bellman completeness and bounded rewards $r_h(s,a) \in [0,1]$. Let ρ be a D-optimal design as in Lemma 3.1. Fix $\epsilon \in (0,1)$ and $\delta \in (0,1)$, and set*

$$N := \left\lceil \frac{c H^6 d^2 \log(H/\delta)}{\epsilon^2} \right\rceil,$$

for a sufficiently large absolute constant c .

Construct datasets $\{D_h\}_{h=0}^{H-1}$ as follows: for each h and each support point $(s,a) \in \text{supp}(\rho)$, query the generative model $n_{s,a} = \lceil N \rho(s,a) \rceil$ times to obtain i.i.d. next-state samples, and include the corresponding tuples in D_h .

Then, with probability at least $1 - \delta$, running LSVI (Algorithm 1) on $\{D_h\}_{h=0}^{H-1}$ returns a greedy policy $\hat{\pi}$ satisfying

$$\mathbb{E}_{s \sim \mu} [V_0^*(s) - V_0^{\hat{\pi}}(s)] \leq \epsilon.$$

Moreover, the total number of samples is at most

$$\sum_{h=0}^{H-1} |D_h| \leq H \left(N + \frac{d(d+1)}{2} \right) = O \left(H d^2 + \frac{H^7 d^2 \log(H/\delta)}{\epsilon^2} \right).$$

3.2.4 Proof of Theorem 3.2

We begin with a standard lemma: small Bellman residuals imply near-optimality of the greedy policy.

For any sequence of functions $\{\hat{Q}_h\}_{h=0}^H$ with $\hat{Q}_H \equiv 0$, define $\hat{V}_h(s) := \max_{a \in \mathcal{A}} \hat{Q}_h(s, a)$ and the greedy policy $\hat{\pi}_h(s) \in \arg \max_a \hat{Q}_h(s, a)$. Define the stage- h Bellman residual

$$\text{Res}_h := \|\hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1}\|_\infty.$$

Lemma 3.3 (Small residual $\Rightarrow$ good policy). *Assume $\hat{Q}_H \equiv 0$ and that $\text{Res}_h \leq \eta$ for all $h \in [H]$. Then:*

1. For all $h \in [H]$, $\|\hat{Q}_h - Q_h^*\|_\infty \leq (H - h)\eta$.
2. For the greedy policy $\hat{\pi}$, for all $s \in \mathcal{S}$,

$$V_0^*(s) - V_0^{\hat{\pi}}(s) \leq 2H^2\eta.$$

Proof. For (1), we proceed by backward induction. Since $Q_H^* \equiv 0 = \hat{Q}_H$, the claim holds for $h = H$. Suppose it holds for $h + 1$. Then for any (s, a) ,

$$\begin{aligned} |\hat{Q}_h(s, a) - Q_h^*(s, a)| &\leq |\hat{Q}_h(s, a) - (\mathcal{T}_h \hat{Q}_{h+1})(s, a)| + |(\mathcal{T}_h \hat{Q}_{h+1})(s, a) - (\mathcal{T}_h Q_{h+1}^*)(s, a)| \\ &\leq \eta + \mathbb{E}_{s'} [\max_{a'} |\hat{Q}_{h+1}(s', a') - Q_{h+1}^*(s', a')|] \\ &\leq \eta + \|\hat{Q}_{h+1} - Q_{h+1}^*\|_\infty \leq (H - h)\eta, \end{aligned}$$

using that max is 1-Lipschitz in ℓ_∞ .

For (2), fix h and s . Let $a^* \in \arg \max_a Q_h^*(s, a)$ and $\hat{a} \in \arg \max_a \hat{Q}_h(s, a)$. Then

$$Q_h^*(s, a^*) - Q_h^*(s, \hat{a}) \leq (Q_h^*(s, a^*) - \hat{Q}_h(s, a^*)) + (\hat{Q}_h(s, \hat{a}) - Q_h^*(s, \hat{a})) \leq 2\|\hat{Q}_h - Q_h^*\|_\infty \leq 2(H - h)\eta.$$

Unrolling along the trajectory of $\hat{\pi}$ gives

$$V_0^*(s) - V_0^{\hat{\pi}}(s) \leq \sum_{h=0}^{H-1} 2(H - h)\eta \leq 2H^2\eta.$$

□

Proof of Theorem 3.2. We show that, with high probability, each stage regression in LSVI has small *uniform* prediction error, which implies small Bellman residuals; then we invoke Lemma 3.3.

The first step looks at one stage as fixed-design regression. Fix a stage $h \in [H]$ and condition on $\hat{Q}_{h+1}$ (equivalently, $\hat{V}_{h+1}$), which is measurable with respect to data from stages $h + 1, \dots, H - 1$.

Each sample has covariate $x_i := \phi(s_i, a_i)$ and response $y_i := r_h(s_i, a_i) + \hat{V}_{h+1}(s'_i)$. (Optionally clip $\hat{V}_{h+1}$ to $[0, H]$ to strengthen boundedness.)

By Bellman completeness, there exists $\tilde{\theta}_h \in \mathbb{R}^d$ such that for all (s, a) ,

$$\tilde{\theta}_h^\top \phi(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [\hat{V}_{h+1}(s')] = (\mathcal{T}_h \hat{Q}_{h+1})(s, a).$$

Hence $y_i = \tilde{\theta}_h^\top x_i + \xi_i$ where $\mathbb{E}[\xi_i | x_i] = 0$ and $\{\xi_i\}$ are independent across samples. Moreover $|\xi_i| \lesssim H$, so ξ_i is $O(H)$ -sub-Gaussian.

Let $n_h := |D_h|$. For notational consistency with the Appendix, we normalize all second-moment matrices by N (the nominal per-stage sample size used in the construction) and define

$$\Lambda_h := \frac{1}{N} \sum_{(s, a, r, s') \in D_h} \phi(s, a) \phi(s, a)^\top.$$

(Equivalently, if we also define the usual empirical covariance $\hat{\Sigma}_h := \frac{1}{n_h} \sum_{(s, a, r, s') \in D_h} \phi(s, a) \phi(s, a)^\top$, then $\Lambda_h = \frac{n_h}{N} \hat{\Sigma}_h$.)

Step 2: leverage control from D-optimal sampling (normalized). Let $\Sigma = \mathbb{E}_{(s, a) \sim \rho} [\phi(s, a) \phi(s, a)^\top]$. By construction of D_h ,

$$\sum_{(s, a, r, s') \in D_h} \phi(s, a) \phi(s, a)^\top = \sum_{(s, a) \in \text{supp}(\rho)} [N \rho(s, a)] \phi(s, a) \phi(s, a)^\top \succeq \sum_{(s, a) \in \text{supp}(\rho)} N \rho(s, a) \phi(s, a) \phi(s, a)^\top = N \Sigma.$$

Dividing by N yields $\Lambda_h \succeq \Sigma$, hence $\Lambda_h^{-1} \preceq \Sigma^{-1}$ and, by Lemma 3.1,

$$\sup_{(s, a)} \phi(s, a)^\top \Lambda_h^{-1} \phi(s, a) \leq \sup_{x \in \Phi} x^\top \Sigma^{-1} x \leq d.$$

Step 3: OLS generalizes uniformly (normalized). Apply the fixed-design OLS bound from the Appendix (Theorem A.10) to the stage- h regression with sample size n_h , failure probability $\delta' := \delta/H$, and noise scale $\sigma = O(H)$. Using the empirical covariance $\hat{\Sigma}_h$ (normalized by n_h), we obtain with probability at least $1 - \delta'$,

$$\|\hat{\theta}_h - \tilde{\theta}_h\|_{\hat{\Sigma}_h} \leq c_1 H \cdot \frac{\sqrt{d} + \sqrt{2 \log(1/\delta')}}{\sqrt{n_h}} = c_1 H \cdot \frac{\sqrt{d} + \sqrt{2 \log(H/\delta)}}{\sqrt{n_h}},$$

for an absolute constant c_1 . Since $\Lambda_h = \frac{n_h}{N} \hat{\Sigma}_h$, we have

$$\|\hat{\theta}_h - \tilde{\theta}_h\|_{\Lambda_h} = \sqrt{\frac{n_h}{N}} \|\hat{\theta}_h - \tilde{\theta}_h\|_{\hat{\Sigma}_h} \leq c_1 H \cdot \frac{\sqrt{d} + \sqrt{2 \log(H/\delta)}}{\sqrt{N}}.$$

Therefore, for any (s, a) ,

$$\begin{aligned} |(\hat{\theta}_h - \tilde{\theta}_h)^\top \phi(s, a)| &\leq \|\hat{\theta}_h - \tilde{\theta}_h\|_{\Lambda_h} \cdot \sqrt{\phi(s, a)^\top \Lambda_h^{-1} \phi(s, a)} \\ &\leq c_1 H \cdot \frac{\sqrt{d} + \sqrt{2 \log(H/\delta)}}{\sqrt{N}} \cdot \sqrt{d}. \end{aligned}$$

Equivalently,

$$\|\hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1}\|_\infty \leq c_1 H \cdot \frac{\sqrt{d}(\sqrt{d} + \sqrt{2 \log(H/\delta)})}{\sqrt{N}}.$$

Finally, apply the bound for all $h \in [H]$ and take a union bound: with probability at least $1 - \delta$,

$$\max_{h \in [H]} \|\hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1}\|_\infty \leq c_1 H \cdot \frac{\sqrt{d}(\sqrt{d} + \sqrt{2 \log(H/\delta)})}{\sqrt{N}}.$$

With our choice of N (and c large enough), the right-hand side is at most $\epsilon/(2H^2)$. Lemma 3.3 then yields, for all s , $V_0^*(s) - V_0^{\hat{\pi}}(s) \leq \epsilon$, and hence also $\mathbb{E}_{s \sim \mu}[V_0^*(s) - V_0^{\hat{\pi}}(s)] \leq \epsilon$.

Finally, for each h ,

$$|D_h| = \sum_{(s,a) \in \text{supp}(\rho)} \lceil N \rho(s,a) \rceil \leq N + |\text{supp}(\rho)| \leq N + \frac{d(d+1)}{2},$$

and summing over h gives the stated total sample bound. $\square$

3.3 Offline extensions under Bellman completeness

A notable feature of LSVI (Algorithm 1) is that it is non-adaptive: it operates on a fixed collection of observed transitions and rewards. Hence, it applies directly to the offline setting discussed in Chapter 1.

In this section we discuss two offline objectives: (i) learning a near-optimal policy, and (ii) policy evaluation (estimating the value of a given target policy π). The key additional requirement in the offline setting is *coverage*: the datasets must be sufficiently informative relative to the feature class.

3.3.1 Offline data model and a coverage condition

For each stage $h \in [H]$, we are given a dataset

$$D_h = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N,$$

where $r_i = r_h(s_i, a_i)$ and each $s'_i \sim P_h(\cdot \mid s_i, a_i)$ is sampled independently conditional on (s_i, a_i) . (The results below also extend to settings where the entire quadruple is i.i.d. from a fixed data-generating distribution; we focus on the above model to keep notation light.)

Let $\Phi := \{\phi(s, a) : (s, a) \in \mathcal{S} \times \mathcal{A}\}$, and recall the D-optimal covariance Σ induced by the maximizer in (3.2) (equivalently, Lemma 3.1). We impose coverage by requiring that the empirical covariance at each stage dominates Σ up to a factor κ .

Assumption 3.4 (Coverage). There exists $\kappa \geq 1$ such that for each stage $h \in [H]$,

$$\hat{\Sigma}_h := \frac{1}{N} \sum_{(s,a,r,s') \in D_h} \phi(s,a) \phi(s,a)^\top \succeq \frac{1}{\kappa} \Sigma,$$

where $\Sigma = \mathbb{E}_{(s,a) \sim \rho}[\phi(s,a) \phi(s,a)^\top]$ for a D-optimal design ρ (see (3.2)).

Assumption 3.4 is a concise way to ensure that the offline design has uniformly bounded leverage scores relative to the geometry of Φ . Indeed, since $\hat{\Sigma}_h \succeq \Sigma/\kappa$,

$$\sup_{(s,a)} \phi(s,a)^\top \hat{\Sigma}_h^{-1} \phi(s,a) \leq \kappa \cdot \sup_{(s,a)} \phi(s,a)^\top \Sigma^{-1} \phi(s,a) \leq \kappa d,$$

using Lemma 3.1.

3.3.2 Offline policy optimization via LSVI

We first consider learning a near-optimal policy from offline data. Under Bellman completeness and coverage, the proof of the generative-model guarantee carries over with only minor changes.

Theorem 3.5 (Offline LSVI under Bellman completeness). *Assume Bellman completeness, bounded rewards $r_h(s, a) \in [0, 1]$, and the coverage condition (Assumption 3.4) with parameter κ . Fix $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, and suppose each stage dataset has size*

$$N \geq \frac{c \kappa H^6 d^2 \log(H/\delta)}{\epsilon^2},$$

for a sufficiently large absolute constant c . Then with probability at least $1 - \delta$, running LSVI (Algorithm 1) on $\{D_h\}_{h=0}^{H-1}$ returns a greedy policy $\hat{\pi}$ satisfying

$$\mathbb{E}_{s \sim \mu} [V_0^*(s) - V_0^{\hat{\pi}}(s)] \leq \epsilon.$$

Proof sketch. The argument is identical to the proof of Theorem 3.2, except that leverage control now comes from coverage rather than D-optimal sampling. Concretely, at stage h , define the normalized covariance

$$\Lambda_h := \frac{1}{N} \sum_{(s,a,r,s') \in D_h} \phi(s,a) \phi(s,a)^\top = \hat{\Sigma}_h.$$

Then Assumption 3.4 implies $\Lambda_h \succeq \Sigma/\kappa$, hence

$$\sup_{(s,a)} \phi(s,a)^\top \Lambda_h^{-1} \phi(s,a) \leq \kappa \cdot \sup_{(s,a)} \phi(s,a)^\top \Sigma^{-1} \phi(s,a) \leq \kappa d.$$

Applying the same fixed-design OLS generalization bound (Appendix, Theorem A.10) and the same residual-to-policy lemma (Lemma 3.3), and union bounding over stages, yields the stated sample complexity. $\square$

3.3.3 Offline policy evaluation via LSPE

We now consider policy evaluation: estimating the value of a *given* policy $\pi = \{\pi_h\}_{h=0}^{H-1}$. For simplicity we assume π is deterministic; the extension to stochastic policies is routine by replacing $\pi_h(s)$ with an action drawn from $\pi_h(\cdot | s)$ (or by working with $\mathbb{E}_{a \sim \pi_h(\cdot | s)} \phi(s, a)$).

LSPE algorithm. LSPE mirrors LSVI but replaces the optimality backup by the π -backup.

Given parameters θ_h , define the induced value estimate $\hat{V}_h^\pi(s) := \theta_h^\top \phi(s, \pi_h(s))$.

A π -completeness assumption. For policy evaluation, we only need closure under the *policy* Bellman operator.

Let $(\mathcal{T}_h^\pi f)(s, a) := r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} f(s', \pi_{h+1}(s'))$.

Definition 3.6 (Linear completeness for π). We say ϕ satisfies *policy completeness* for π if for each $h \in [H]$ and each $\theta \in \mathbb{R}^d$, there exists $w \in \mathbb{R}^d$ such that

$$(s, a) \mapsto r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} \theta^\top \phi(s', \pi_{h+1}(s'))$$

is exactly $w^\top \phi(s, a)$ for all (s, a) ; equivalently,

$$f \in \mathcal{F} \implies \mathcal{T}_h^\pi f \in \mathcal{F} \quad \text{for each } h \in [H].$$

Algorithm 2 Least-Squares Policy Evaluation (LSPE)

- 1: **Input:** policy $\pi = \{\pi_h\}_{h=0}^{H-1}$, datasets $D_0, \dots, D_{H-1}$, feature map ϕ
 - 2: Initialize $\widehat{V}_H^\pi(s) \leftarrow 0$ for all $s \in \mathcal{S}$
 - 3: **for** $h = H-1, H-2, \dots, 0$ **do**
 - 4: $\widehat{\theta}_h \in \arg \min_{\theta \in \mathbb{R}^d} \sum_{(s,a,r,s') \in D_h} \left(\theta^\top \phi(s,a) - r - \widehat{V}_{h+1}^\pi(s') \right)^2$
 - 5: Define $\widehat{V}_h^\pi(s) \leftarrow \widehat{\theta}_h^\top \phi(s, \pi_h(s))$ for all s
 - 6: **end for**
 - 7: **Return:** $\{\widehat{\theta}_h\}_{h=0}^{H-1}$ (equivalently, $\{\widehat{V}_h^\pi\}_{h=0}^H$)
-

Residual-to-estimation lemma (policy evaluation). For evaluation, small Bellman residuals control the value error directly (no greedy policy mismatch), yielding a milder horizon dependence.

Lemma 3.7 (Small π -residual $\Rightarrow$ accurate evaluation). *Let $\{\widehat{V}_h^\pi\}_{h=0}^H$ be any functions with $\widehat{V}_H^\pi \equiv 0$. Define the stage- h π -residual*

$$\text{Res}_h^\pi := \sup_{(s,a)} \left| \widehat{Q}_h(s,a) - (\mathcal{T}_h^\pi \widehat{V}_{h+1}^\pi)(s,a) \right|, \quad \widehat{Q}_h(s,a) := \widehat{\theta}_h^\top \phi(s,a).$$

If $\text{Res}_h^\pi \leq \eta$ for all $h \in [H]$, then for all $s \in \mathcal{S}$,

$$|V_0^\pi(s) - \widehat{V}_0^\pi(s)| \leq H\eta.$$

Proof. The proof is the same backward-induction argument as in Lemma 3.3(1), but with $\max$ replaced by following π , so there is no additional greedy suboptimality term. $\square$

Theorem 3.8 (Offline LSPE under policy completeness). *Assume bounded rewards $r_h(s,a) \in [0,1]$, coverage (Assumption 3.4) with parameter κ , and policy completeness for π (Definition 3.6). Fix $\epsilon \in (0,1)$ and $\delta \in (0,1)$, and suppose each stage dataset has size*

$$N \geq \frac{c \kappa H^4 d^2 \log(H/\delta)}{\epsilon^2},$$

for a sufficiently large absolute constant c . Then with probability at least $1 - \delta$, the LSPE estimate satisfies

$$\sup_{s \in \mathcal{S}} |V_0^\pi(s) - \widehat{V}_0^\pi(s)| \leq \epsilon.$$

Consequently, $|\mathbb{E}_{s \sim \mu}[V_0^\pi(s)] - \mathbb{E}_{s \sim \mu}[\widehat{V}_0^\pi(s)]| \leq \epsilon$.

Proof sketch. Fix a stage h and condition on $\widehat{V}_{h+1}^\pi$. By policy completeness, the regression target has the form

$$r_h(s,a) + \mathbb{E}[\widehat{V}_{h+1}^\pi(s') \mid s,a] = (\tilde{\theta}_h)^\top \phi(s,a)$$

for some $\tilde{\theta}_h \in \mathbb{R}^d$. Thus each stage is a fixed-design linear regression with $O(H)$ -sub-Gaussian noise. Define the normalized covariance $\Lambda_h := \frac{1}{N} \sum_{(s,a,r,s') \in D_h} \phi(s,a) \phi(s,a)^\top = \widehat{\Sigma}_h$. Coverage implies $\Lambda_h \succeq \Sigma/\kappa$, hence

$$\sup_{(s,a)} \phi(s,a)^\top \Lambda_h^{-1} \phi(s,a) \leq \kappa d.$$

Applying the fixed-design OLS bound (Appendix, Theorem A.10) and union bounding over h yields a uniform residual bound of order

$$\max_h \text{Res}_h^\pi \lesssim Hd \sqrt{\frac{\kappa \log(H/\delta)}{N}}.$$

Choosing N as in the theorem makes this at most ϵ/H , and Lemma 3.7 gives $\sup_s |V_0^\pi(s) - \widehat{V}_0^\pi(s)| \leq \epsilon$. $\square$

Remark on horizon dependence. The improvement from H^6 (offline LSVI) to H^4 (offline LSPE) reflects that for evaluation we only need value accuracy under a *fixed* policy, so we use Lemma 3.7 (which incurs an H factor) rather than the greedy-policy performance bound in Lemma 3.3 (which incurs an additional H factor). As elsewhere in this chapter, we have not optimized the dependence on H .

3.4 Negative Results with Weaker Linear Notions

The positive results in Section 3.2 rely on a closure condition (Bellman completeness) that keeps the algorithm’s regression targets inside the hypothesis class. This section explains why weaker “linear” conditions can be misleading.

We present two complementary impossibility results. First, we show that even the seemingly strong assumption that Q^* is linear in a known feature map does *not* guarantee $\text{poly}(d, H)$ sample complexity, even with a generative model. Second, we give an offline lower bound under an even stronger realizability assumption (linearity of Q^π for *all* policies) and a maximally well-conditioned feature covariance. The offline construction is particularly revealing: it exhibits a concrete mechanism by which errors (or variance) can compound exponentially with the horizon despite excellent “coverage” in the usual linear-regression sense.

3.4.1 Information-theoretic lower bounds with linearly realizable Q^*

We start with the most basic realizability assumption: the optimal Q -function itself lies in the linear span of the features.

Assumption 3.9 (Linear Q^* realizability). For all $h \in [H]$, there exists $\theta_h^* \in \mathbb{R}^d$ such that for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$Q_h^*(s, a) = (\theta_h^*)^\top \phi(s, a).$$

One might hope that Assumption 3.9 is already sufficient for sample-efficient RL when the feature map is known to the learner. The next theorem rules this out in a strong sense.

Theorem 3.10 (Linear Q^* is not enough, even with a generative model). *Consider any algorithm $\mathcal{A}$ with access to a generative model, which takes as input the feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$. There exists a finite-horizon MDP and a feature map ϕ satisfying Assumption 3.9 such that the action space size is*

$$|\mathcal{A}| = c_1 \left\lceil \min\{d^{1/4}, H^{1/2}\} \right\rceil,$$

and if $\mathcal{A}$ outputs a policy π satisfying

$$\mathbb{E}_{s_0 \sim \mu} [V_0^\pi(s_0)] \geq \mathbb{E}_{s_0 \sim \mu} [V_0^*(s_0)] - 0.05$$

with probability at least 0.1, then $\mathcal{A}$ requires at least $\min\{2^{c_2 d}, 2^{c_2 H}\}$ generative-model samples, where $c_1, c_2 > 0$ are absolute constants.

Theorem 3.10 shows that linear Q^* realizability alone does not prevent exponential sample complexity. Necessarily, such lower bounds rely on state spaces that are exponential in d or H (otherwise one could revert to a tabular algorithm with polynomial dependence on $|\mathcal{S}||\mathcal{A}|$). In Appendix B, we state and prove a weaker version of this theorem based on a simpler construction with $|\mathcal{A}| = \exp(d)$ (though, even for this weaker statement, the construction is quite involved).

3.4.2 Offline impossibility under all-policies linear realizability

We now turn to the offline setting. In Section 3.2 we saw that a closure condition (Bellman completeness) yields polynomial guarantees for LSVI and LSPE given suitable data. Here we ask: what can be guaranteed under the *strictly weaker* condition of realizability, even if we strengthen it to hold for *all* policies and even if we assume essentially optimal coverage?

Offline data model. We work in a finite-horizon MDP with horizon H . The learner does not interact with the MDP; instead, for each stage $h \in \{0, \dots, H-1\}$ it receives a dataset D_h of n i.i.d. transition samples

$$(s, a, r, s') \in \mathcal{S}_h \times \mathcal{A} \times \mathbb{R} \times \mathcal{S}_{h+1}, \quad (s, a) \sim \mu_h, \quad r \sim r_h(s, a), \quad s' \sim P_h(\cdot | s, a),$$

for some (unknown) data distribution μ_h over $\mathcal{S}_h \times \mathcal{A}$. (The results can also be stated with stage-dependent sample sizes $\{n_h\}$; we use n for simplicity.)

Offline policy evaluation goal. Given a target policy π (assumed known) and the feature map ϕ , the goal is to estimate $V_0^\pi(s_0)$ (or $\mathbb{E}_{s_0 \sim \mu}[V_0^\pi(s_0)]$) from the offline datasets with as few samples as possible.

Realizability (for all policies). We assume that Q^π is linear in the features *for every* policy.

Assumption 3.11 (All-policies linear realizability). For every policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, there exist parameter vectors $\theta_0^\pi, \dots, \theta_{H-1}^\pi \in \mathbb{R}^d$ such that for all $h \in \{0, \dots, H-1\}$ and all $(s, a) \in \mathcal{S}_h \times \mathcal{A}$,

$$Q_h^\pi(s, a) = (\theta_h^\pi)^\top \phi(s, a).$$

This is substantially stronger than realizability for a single target policy: it postulates a linear representation simultaneously over *all* policies.

Coverage (maximally well-conditioned design). To isolate the effect of compounding across stages, we impose a very strong coverage condition: the feature covariance under the data distribution is as well-conditioned as possible under $\|\phi\|_2 \leq 1$.

Assumption 3.12 (Coverage). Assume $\|\phi(s, a)\|_2 \leq 1$ for all (s, a) . For each stage $h \in \{0, \dots, H-1\}$, the data distribution μ_h satisfies

$$\mathbb{E}_{(s,a) \sim \mu_h} [\phi(s, a) \phi(s, a)^\top] = \frac{1}{d} I.$$

For $H = 1$, Assumptions 3.11 and 3.12 reduce to classical linear regression. The next theorem shows that this intuition fails dramatically once H is large.

Theorem 3.13 (Offline policy evaluation can be exponentially hard). *Suppose Assumption 3.12 holds. Fix any algorithm that, given (π, ϕ) and the offline datasets $\{D_h\}_{h=0}^{H-1}$, outputs an estimate of $V_0^\pi(s_0)$. There exists a finite-horizon MDP satisfying Assumption 3.11 such that for every policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, the algorithm requires*

$$\Omega((d/2)^H)$$

samples to obtain a constant additive approximation to $V_0^\pi(s_0)$ with probability at least 0.9.

Note that this theorem says it not possible to estimate the value of *any* given policy, from the starting state s_0 .

Implication for offline control. Although stated for policy evaluation, the hardness also applies to offline control under Assumption 3.11. Indeed, add a new initial state s_{new} in which action a_1 yields reward 0.5 and terminates, while action a_2 transitions into the hard instance from Theorem 3.13. Any algorithm that returns a policy with suboptimality at most 0.5 must distinguish whether entering the hard instance is better than taking a_1 , and therefore inherits the same exponential lower bound.

LSPE (and LSVI) can have exponential variance. Theorem 3.13 is a minimax lower bound: it implies that no offline estimator can avoid an exponential dependence on H under realizability alone. In particular, any unbiased estimator (when it exists) must have variance growing exponentially with H , so Least-Squares Policy Evaluation inherits this pathology despite the strong coverage in Assumption 3.12.

A hard instance for offline policy evaluation

We now describe the hard family used in Theorem 3.13. The key idea is that although the offline data have essentially perfect coverage over the *observed* feature directions, the value of the policy depends on *unobserved* “aggregator” states. Linearity then forces the value at these aggregator states to be an amplified combination of quantities that can only be estimated from the last layer, causing an exponential dependence on the horizon. Figure 3.1 shows this instance.

Let $m \geq 1$ be an integer and set

$$d := 2m, \quad \mathcal{A} := \{a_1, a_2\}.$$

The family is indexed by a scalar parameter $r_\infty \in [0, m^{-H/2}]$; in the lower bound we only need two values, $r_\infty \in \{0, m^{-H/2}\}$.

State space and transitions. For each layer $h \in \{0, 1, \dots, H-1\}$, define

$$\mathcal{S}_h := \{s_h^1, \dots, s_h^m\} \cup \{g_h\},$$

where g_h is an “aggregator” state. The initial state is $s_0 = g_0$. For $h \in \{0, 1, \dots, H-2\}$, transitions are deterministic:

$$\begin{aligned} P(g_{h+1} \mid s, a_1) &= 1 && \text{for all } s \in \mathcal{S}_h, \\ P(s_{h+1}^i \mid s_h^i, a_2) &= 1 && \text{for all } i \in [m], \\ P(g_{h+1} \mid g_h, a_2) &= 1. \end{aligned}$$

Equivalently: action a_1 always jumps to the aggregator chain, while a_2 preserves the index i on the $\{s_h^i\}$ states (and keeps the g -chain on g).

Rewards. Define the shorthand

$$\alpha_h := r_\infty m^{(H-h)/2}, \quad h \in \{0, 1, \dots, H-1\}.$$

(Note $\alpha_0 \leq 1$ since $r_\infty \leq m^{-H/2}$.) For layers $h = 0, 1, \dots, H-2$, rewards are deterministic:

$$\begin{aligned} r_h(s_h^i, a) &= 0 && \text{for all } i \in [m], a \in \{a_1, a_2\}, \\ r_h(g_h, a) &= \alpha_h - \alpha_{h+1} && \text{for all } a \in \{a_1, a_2\}. \end{aligned}$$

At the last layer $h = H-1$, rewards depend only on r_∞ :

$$\begin{aligned} r_{H-1}(s_{H-1}^i, a) &= \begin{cases} 1 & \text{with prob. } (1 + r_\infty)/2, \\ -1 & \text{with prob. } (1 - r_\infty)/2, \end{cases} && \text{for all } i \in [m], a \in \{a_1, a_2\}, \\ r_{H-1}(g_{H-1}, a) &= \alpha_{H-1} = r_\infty \sqrt{m} && \text{for all } a \in \{a_1, a_2\}. \end{aligned}$$

reward params:
 $r_\infty \in \{0, m^{-H/2}\}$
 $\alpha_h := r_\infty m^{(H-h)/2}$

features:
 $\phi(s_h^c, a_1) = e_c$
 $\phi(s_h^c, a_2) = e_{m+c}$
 $\phi(g_h, a) = \frac{1}{\sqrt{m}}(e_1 + e_2 + \dots + e_m)$

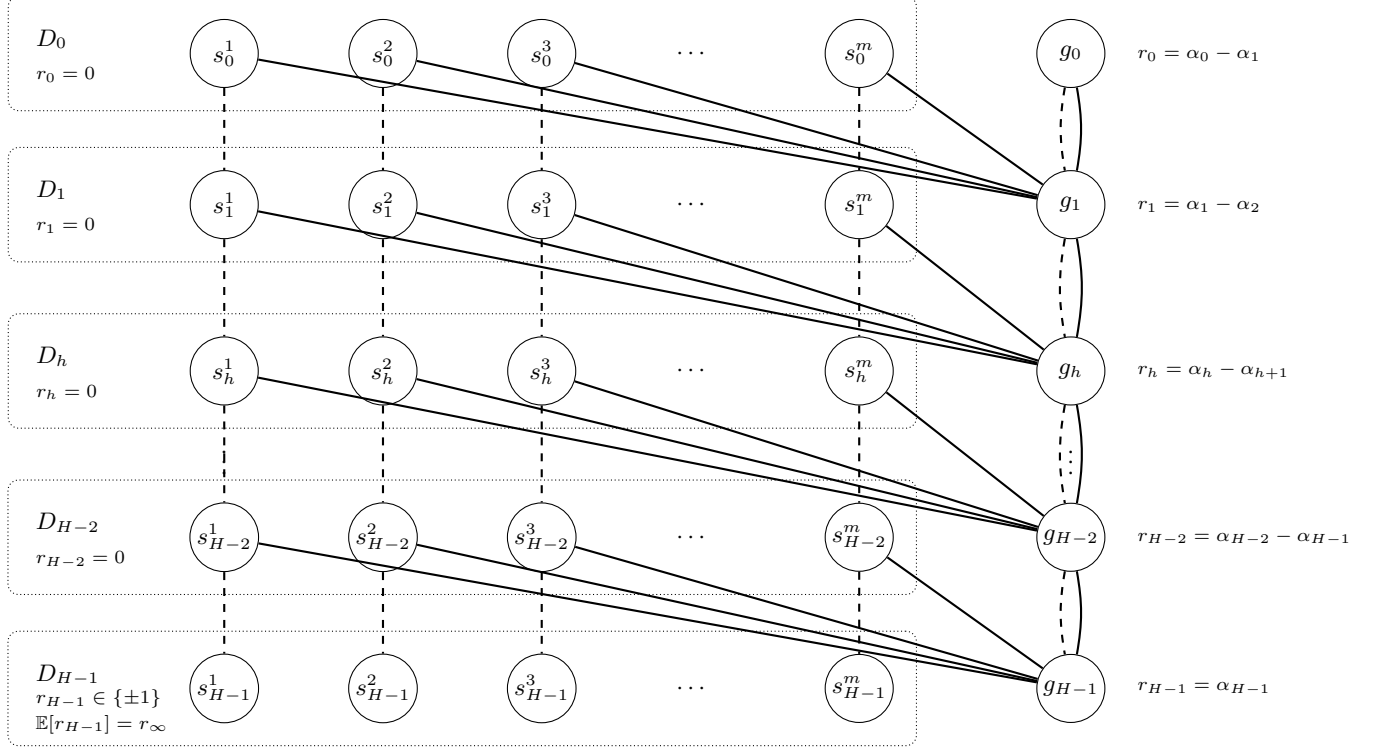
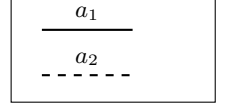


Figure 3.1: Offline policy-evaluation hard instance (schematic). Each layer h has m “observed” states $\{s_h^i\}_{i=1}^m$ (boxed: states covered by the offline dataset D_h) and one “aggregator” state g_h (never covered by the offline data). Action a_2 (dashed) keeps the trajectory in the observed chain $s_h^i \rightarrow s_{h+1}^i$, while action a_1 (solid) jumps to the aggregator chain $s_h^i \rightarrow g_{h+1}$; from aggregator states, both actions transition to the next aggregator state. Rewards are stage-dependent. On observed states, $r_h = 0$ for $h \leq H-2$ and $r_{H-1} \in \{\pm 1\}$ with $\mathbb{E}[r_{H-1}] = r_\infty$. On aggregator states, $r_h(g_h, \cdot) = \alpha_h - \alpha_{h+1}$ for $h \leq H-2$ and $r_{H-1}(g_{H-1}, \cdot) = \alpha_{H-1} = r_\infty \sqrt{m}$. (Note that $\alpha_h := r_\infty m^{(H-h)/2}$). The lower bound compares two instances with $r_\infty = 0$ versus $r_\infty = m^{-H/2}$, which induce identical offline data on the boxed support but different values at g_0 . See text for further description.

Thus $\mathbb{E}[r_{H-1}(s_{H-1}^i, a)] = r_\infty$ while $r_{H-1}(g_{H-1}, a)$ is deterministic.

Feature map. Let $e_1, \dots, e_{2m}$ be an orthonormal basis of $\mathbb{R}^{2m}$. Define $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^{2m}$ by, for each layer h ,

$$\begin{aligned} \phi(s_h^i, a_1) &= e_i, & \phi(s_h^i, a_2) &= e_{m+i} & \text{for all } i \in [m], \\ \phi(g_h, a_1) &= \phi(g_h, a_2) = \frac{1}{\sqrt{m}} \sum_{i=1}^m e_i. \end{aligned}$$

In particular, $\|\phi(s, a)\|_2 \leq 1$ for all (s, a) .

Offline data distributions (isotropic, but miss the aggregators). For each layer $h \in \{0, 1, \dots, H-1\}$, let μ_h be the uniform distribution over

$$\{(s_h^i, a_1), (s_h^i, a_2) : i \in [m]\}.$$

In particular, (g_h, a) is *not* in the support of μ_h for any h or a . A direct calculation gives

$$\mathbb{E}_{(s,a) \sim \mu_h} [\phi(s, a) \phi(s, a)^\top] = \frac{1}{2m} \sum_{j=1}^{2m} e_j e_j^\top = \frac{1}{d} I,$$

so Assumption 3.12 holds.

Where the exponential dependence comes from (informal). The policy value at the initial state is

$$V_0^\pi(g_0) = \alpha_0 = r_\infty m^{H/2},$$

which equals 0 when $r_\infty = 0$ and equals 1 when $r_\infty = m^{-H/2}$. However, the offline data *never* visit g_h , so the only information about r_∞ comes from the last layer rewards at the observed states s_{H-1}^i , whose mean differs by only $m^{-H/2}$. Estimating that tiny mean to constant confidence requires $\Omega(m^H)$ samples, yielding the claimed lower bound.

We now verify Assumption 3.11.

Lemma 3.14. *For every policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$, for each $h \in [H]$, there exists $\theta_h^\pi \in \mathbb{R}^{2m}$ such that for all $(s, a) \in \mathcal{S}_h \times \mathcal{A}$,*

$$Q_h^\pi(s, a) = \langle \theta_h^\pi, \phi(s, a) \rangle.$$

Proof. First note that from any aggregator state g_h , the next state is always g_{h+1} (regardless of the action), and rewards at g_h do not depend on the action. Hence $V_h^\pi(g_h) = Q_h^\pi(g_h, a_1) = Q_h^\pi(g_h, a_2)$ for all h and π .

We compute $V_h^\pi(g_h)$ by backward induction. At $h = H-1$,

$$V_{H-1}^\pi(g_{H-1}) = r_{H-1}(g_{H-1}, \cdot) = \alpha_{H-1}.$$

For $h \leq H-2$,

$$V_h^\pi(g_h) = r_h(g_h, \cdot) + V_{h+1}^\pi(g_{h+1}) = (\alpha_h - \alpha_{h+1}) + \alpha_{h+1} = \alpha_h.$$

In particular, for $h \leq H-2$ and any $i \in [m]$,

$$Q_h^\pi(s_h^i, a_1) = r_h(s_h^i, a_1) + V_{h+1}^\pi(g_{h+1}) = \alpha_{h+1}.$$

Also, for any $i \in [m]$ and $h \leq H-2$,

$$Q_h^\pi(s_h^i, a_2) = r_h(s_h^i, a_2) + V_{h+1}^\pi(s_{h+1}^i) = V_{h+1}^\pi(s_{h+1}^i),$$

which may depend on π but is a scalar that we can represent directly.

Now define, for $h \leq H-2$,

$$\theta_h^\pi := \sum_{i=1}^m \alpha_{h+1} e_i + \sum_{i=1}^m Q_h^\pi(s_h^i, a_2) e_{m+i}.$$

Then for $i \in [m]$,

$$\langle \theta_h^\pi, \phi(s_h^i, a_1) \rangle = \alpha_{h+1} = Q_h^\pi(s_h^i, a_1), \quad \langle \theta_h^\pi, \phi(s_h^i, a_2) \rangle = Q_h^\pi(s_h^i, a_2).$$

Moreover,

$$\langle \theta_h^\pi, \phi(g_h, \cdot) \rangle = \left\langle \sum_{i=1}^m \alpha_{h+1} e_i, \frac{1}{\sqrt{m}} \sum_{i=1}^m e_i \right\rangle = \alpha_{h+1} \sqrt{m} = r_\infty m^{(H-h-1)/2} \sqrt{m} = \alpha_h = Q_h^\pi(g_h, \cdot).$$

At the last layer $h = H-1$, we have $Q_{H-1}^\pi(s, a) = \mathbb{E}[r_{H-1}(s, a)]$. For the observed states $\{s_{H-1}^i\}_{i=1}^m$, $\mathbb{E}[r_{H-1}(s_{H-1}^i, a)] = r_\infty$ for both actions; for g_{H-1} , $Q_{H-1}^\pi(g_{H-1}, \cdot) = \alpha_{H-1} = r_\infty \sqrt{m}$. Setting

$$\theta_{H-1}^\pi := \sum_{j=1}^{2m} r_\infty e_j$$

gives $Q_{H-1}^\pi(s, a) = \langle \theta_{H-1}^\pi, \phi(s, a) \rangle$ for all (s, a) , since $\langle \theta_{H-1}^\pi, e_i \rangle = r_\infty$ and $\left\langle \theta_{H-1}^\pi, \frac{1}{\sqrt{m}} \sum_{i=1}^m e_i \right\rangle = r_\infty \sqrt{m}$. $\square$

Proof sketch of Theorem 3.13 (information-theoretic argument)

We reduce policy evaluation to a two-point hypothesis test. Fix any policy π . In this construction, the value at the initial state does not depend on π :

$$V_0^\pi(g_0) = \alpha_0 = r_\infty m^{H/2}.$$

Consider two instances, identical in transitions, features, and data distributions, but with different parameters:

$$r_\infty^{(0)} := 0, \quad r_\infty^{(1)} := m^{-H/2}.$$

Then $V_0^\pi(g_0) = 0$ under $r_\infty^{(0)}$ and $V_0^\pi(g_0) = 1$ under $r_\infty^{(1)}$. Therefore, any algorithm that outputs an estimate $\hat{V}$ satisfying $|\hat{V} - V_0^\pi(g_0)| \leq 1/2$ with probability at least 0.9 must distinguish these two instances with probability at least 0.9.

Under both instances, the offline distributions $\{\mu_h\}_{h=0}^{H-1}$ are the same. Moreover, for layers $h \leq H-2$, all rewards on the support of μ_h are identically 0 in both instances. Hence *all information about r_∞ available in the offline data comes from the last layer samples D_{H-1} .*

On the support of μ_{H-1} , rewards are $\{\pm 1\}$ -valued with mean r_∞ : under $r_\infty^{(0)}$,

$$\Pr(r = 1) = 1/2, \quad \Pr(r = -1) = 1/2,$$

while under $r_\infty^{(1)}$,

$$\Pr(r = 1) = (1 + m^{-H/2})/2, \quad \Pr(r = -1) = (1 - m^{-H/2})/2.$$

Let $\Delta := m^{-H/2}$. A direct calculation shows that the KL divergence between a single sample under these two reward distributions satisfies

$$\text{KL}(P_\Delta \| P_0) \leq -\frac{1}{2} \log(1 - \Delta^2) \leq 2\Delta^2 \quad (\text{for } \Delta \leq 1/2).$$

By independence, the KL divergence between n last-layer samples is at most $2n\Delta^2$. Standard two-point testing inequalities (e.g. Le Cam's method) imply that any test that distinguishes the two instances with probability at least 0.9 must have total KL at least a constant, and thus requires

$$n\Delta^2 = \Omega(1) \implies n = \Omega(\Delta^{-2}) = \Omega(m^H).$$

Recalling $d = 2m$, this is $\Omega((d/2)^H)$, establishing Theorem 3.13.

3.5 Bibliographic Remarks and Further Readings

The closure idea behind *Bellman completeness* (for general function classes) appears already in early work on error propagation in approximate dynamic programming and batch RL; see, for example, Munos [2005]. In episodic online RL, Zanette et al. [2020] gave statistically efficient algorithms under *linear* Bellman completeness, and Jin et al. [2021] extended the picture to statistically efficient algorithms under Bellman completeness with general function approximation.

The geometric viewpoint on D -optimal design is classical and goes back to John’s theorem on minimum-volume ellipsoids [John, 1948]. The Kiefer–Wolfowitz equivalence between the D -optimal (log-determinant) and G -optimal (maximum variance) formulations is due to Kiefer and Wolfowitz [1960]. For an accessible modern proof and additional context (including connections to bandits and experimental design), see Lattimore and Szepesvári [2020].

The theme that, under the right structural assumptions, reinforcement learning can be reduced to a sequence of supervised learning problems goes back at least to Kearns et al. [2000], which analyzed a “trajectory tree” approach and connected sample complexity to the VC dimension of a policy class. We examine this in the next chapter with importance sampling.

Linear methods for dynamic programming have a long history, going back at least to Shannon [1959], Bellman and Dreyfus [1959]. The modern formulation of the *linear Q^* realizability* assumption appears (at least) in Wen and Van Roy [2017] and is highlighted as a central learnability question in Du et al. [2019]. In the offline (policy evaluation) setting, exponential lower bounds under strong linear realizability assumptions are due to Wang et al. [2020a], Zanette [2021], and we present a variant as Theorem 3.13. With a generative model, the breakthrough impossibility result under linear Q^* realizability is due to Weisz et al. [2021a]. Subsequent work [Weisz et al., 2021b] extends these ideas to yield lower bounds that continue to hold even when the action space is only polynomial in (d, H) ; this is the form we invoke for Theorem 3.10. In Appendix B we also include a simpler (but weaker) proof in the spirit of the original construction of Weisz et al. [2021a], using exponentially many actions.

A note on (C) with a generative model. Our exponential lower bound under all-policies linear realizability (C) is an *offline* phenomenon: the fixed dataset can systematically miss the state–action pairs whose values drive V^π , forcing repeated extrapolation through Bellman backups and allowing estimation error to amplify across stages. With a generative model, this particular obstruction disappears because the learner can collect *fresh* samples under (or targeted to) the current candidate policy, producing unbiased estimates of the relevant Bellman targets and avoiding the worst-case compounding that occurs with fixed data (see Du et al. [2019]). Furthermore, polynomial-sample PAC guarantees under (C) are also available in the online episodic setting. Here, Weisz et al. [2023] shows that linearly Q^π -realizable MDPs are PAC learnable with sample complexity polynomial in $(d, H, 1/\epsilon)$. Related polynomial-sample guarantees for online learning under Q^π -realizability (with computational efficiency and with more restrictions) are obtained by Mhammedi [2024]. This makes the (C) model subtle: naive least-squares dynamic programming can fail offline, yet the model is still learnable.

Finally, Wang et al. [2021] studies the *episodic* (online) setting under linear Q^* realizability, together with an additional *constant suboptimality gap* assumption. As we discuss below, this gap assumption permits sample-efficient learning with a generative model. However, the main result of Wang et al. [2021] is negative: even with a constant gap, *episodic* interaction alone does not guarantee sample-efficient learning under these two assumptions — they prove an $\exp(\Omega(\min\{d, H\}))$ sample lower bound for obtaining a constant-accuracy guarantee in the online model.

This stands in sharp contrast to the *generative-model* setting, where the same gap assumption can be exploited for *identification*. A backward-induction-style elimination argument can compare actions locally and rule out suboptimal ones using only polynomially many queries, since each is separated from the optimum by a constant margin. Once the optimal action is identified stage by stage, one can propagate *unbiased* estimates of Q^* backward through the horizon, because the Bellman targets are evaluated under the true optimal continuation rather than a biased surrogate—precisely the kind of targeted identification that episodic data do not reliably support.

Chapter 4

Approximate Dynamic Programming in Large MDPs: Offline Learning and Coverage

This chapter studies a complementary regime to Chapter 3. In Chapter 3 we obtained strong (worst-case) sample complexity guarantees in large MDPs by imposing a *structural closure* assumption: the relevant Bellman updates remain within a prescribed (typically low-dimensional) function class (e.g., linear Bellman completeness / linear MDP structure). Under such closure, one can propagate estimation error cleanly and avoid explicit dependence on the size of the state space.

Here, we consider a more classical approach based on *approximate dynamic programming*, but with using *supervised regression* subroutines. Concretely, we will study *fitted* methods that repeatedly regress to Bellman targets and then act greedily with respect to the resulting approximations. This style of algorithm is attractive because it is modular — it reduces reinforcement learning to a sequence of standard prediction problems — and it allows substantial flexibility in the choice of function class. We study this question in the discounted MDP framework in this chapter. Again, we seek guarantees which do not depend on the size of the state and action spaces.

However, this flexibility comes at a price. In the offline setting, the learner does not control data collection: it is restricted to a fixed dataset drawn from some behavior distribution. In such settings, purely worst-case guarantees cannot hold without additional assumptions: the data distribution must provide *coverage* of the state-action pairs that matter for planning. Moreover, even with good coverage, the function class need not be closed under Bellman updates, leading to an irreducible approximation term.

The main message of this chapter is that fitted methods yield meaningful guarantees under two explicit requirements:

1. **Distributional coverage (concentrability).** The offline data distribution must dominate the state-action distributions induced by policies the algorithm might implicitly reason about. This is quantified by a *concentrability coefficient*.
2. **Bellman approximation (inherent Bellman error).** The function class must be able to approximate Bellman updates in an average-case sense under the data distribution.

Under these conditions, fitted Q -iteration (FQI) can be analyzed as a stable approximate value iteration procedure with regression error at each step, propagated through the MDP dynamics via concentrability.

4.1 Fitted Q -Iteration (FQI) in Offline RL

We work with an infinite-horizon discounted MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$ where $\gamma \in (0, 1)$, the reward is bounded $r(s, a) \in [0, 1]$, and μ is an initial-state distribution. Let

$$V_{\max} := \frac{1}{1 - \gamma}.$$

For any function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, recall the Bellman optimality operator

$$(\mathcal{T}f)(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in \mathcal{A}} f(s', a').$$

We write Q^* for the unique fixed point of $\mathcal{T}$ and $V^*(s) := \max_a Q^*(s, a)$, with overall value $V^* := \mathbb{E}_{s_0 \sim \mu}[V^*(s_0)]$. For a policy π , we similarly denote $V^\pi(s_0)$ and $V^\pi := \mathbb{E}_{s_0 \sim \mu}[V^\pi(s_0)]$.

Offline sampling model. We are given a fixed distribution $\nu \in \Delta(\mathcal{S} \times \mathcal{A})$ and an i.i.d. dataset

$$\mathcal{D} := \{(s_i, a_i, r_i, s'_i)\}_{i=1}^n, \quad (s_i, a_i) \sim \nu, \quad r_i = r(s_i, a_i), \quad s'_i \sim P(\cdot | s_i, a_i).$$

The learner must output a policy $\hat{\pi}$ using only $\mathcal{D}$. For any distribution ρ over $\mathcal{S} \times \mathcal{A}$ we use $\|f\|_{2, \rho}^2 := \mathbb{E}_{(s, a) \sim \rho}[f(s, a)^2]$.

4.1.1 Two explicit requirements: coverage and Bellman approximation

Coverage via concentrability. Let d_μ^π denote the discounted state-action occupancy measure of policy π , i.e. the distribution over (s, a) obtained by starting at $s_0 \sim \mu$, then following π , and sampling a time step according to the geometric weights $(1 - \gamma)\gamma^t$. The dataset distribution ν must cover the occupancy measures of the policies that matter for evaluation and improvement.

Assumption 4.1 (Concentrability / coverage). There exists a constant $C \geq 1$ such that for every (possibly non-stationary) policy π ,

$$\sup_{s, a} \frac{d_\mu^\pi(s, a)}{\nu(s, a)} \leq C,$$

with the convention that the ratio is $+\infty$ when $\nu(s, a) = 0$ and $d_\mu^\pi(s, a) > 0$.

This assumption ensures that average-case errors measured under ν control the same errors under the visitation distributions that appear in performance comparisons. Informally: *if the data never visits a region that an evaluated policy might visit, then no offline method can reliably reason about that region.*

Bellman approximation via inherent Bellman error. Fix a bounded function class

$$\mathcal{F} \subseteq \{f : \mathcal{S} \times \mathcal{A} \rightarrow [0, V_{\max}]\}.$$

In fitted methods, regression targets are Bellman updates $\mathcal{T}f$ of previous iterates $f \in \mathcal{F}$. Without closure of $\mathcal{F}$ under $\mathcal{T}$, there is an irreducible approximation error. We quantify this by the standard *inherent Bellman error* under ν .

Assumption 4.2 (Inherent Bellman error under ν). Define

$$\epsilon_{\text{approx}, \nu} := \max_{f \in \mathcal{F}} \min_{f' \in \mathcal{F}} \|f' - \mathcal{T}f\|_{2, \nu}^2.$$

We refer to $\epsilon_{\text{approx}, \nu}$ as the inherent Bellman error (with respect to ν).

When $\epsilon_{\text{approx}, \nu} = 0$, the class is (average-case) Bellman complete under ν ; otherwise, this term will appear additively in the final suboptimality bound.

4.1.2 The FQI algorithm

Fitted Q -iteration (FQI) constructs a sequence $\{f_t\}_{t \geq 0} \subset \mathcal{F}$ by repeatedly regressing to Bellman targets computed from the previous iterate. Starting from an arbitrary $f_0 \in \mathcal{F}$, iterate for $t = 1, 2, \dots, K$:

$$f_t \in \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n \left(f(s_i, a_i) - [r_i + \gamma \max_{a' \in \mathcal{A}} f_{t-1}(s'_i, a')] \right)^2. \quad (4.1)$$

After K iterations, output the greedy policy

$$\pi^K(s) \in \operatorname{argmax}_{a \in \mathcal{A}} f_K(s, a).$$

The interpretation is as follows: At each iteration, the regression problem in (4.1) has a well-defined conditional expectation target:

$$\mathbb{E} \left[r + \gamma \max_{a'} f_{t-1}(s', a') \mid s, a \right] = (\mathcal{T} f_{t-1})(s, a).$$

Moreover, since $r \in [0, 1]$ and $f_{t-1} \in [0, V_{\max}]$,

$$0 \leq r + \gamma \max_{a'} f_{t-1}(s', a') \leq 1 + \gamma V_{\max} = V_{\max}, \quad \text{and} \quad 0 \leq (\mathcal{T} f_{t-1})(s, a) \leq V_{\max}.$$

Thus, if the regression step returns f_t close to $\mathcal{T} f_{t-1}$ in $\|\cdot\|_{2,\nu}$, and if ν provides coverage so that $\|\cdot\|_{2,\nu}$ controls errors under relevant visitation distributions, then the iterates behave like an approximate value iteration procedure.

4.1.3 Main guarantee: what you get, and what you pay

The following theorem bounds the performance loss by three terms: (i) a statistical term scaling as $n^{-1/2}$, (ii) an approximation term governed by $\epsilon_{\text{approx},\nu}$, and (iii) a finite-iteration term that decays geometrically in K .

Theorem 4.3 (Performance guarantee for FQI). *Fix $K \in \mathbb{N}^+$. Under Assumptions 4.1 and 4.2, and for a finite function class $\mathcal{F}$, FQI (4.1) satisfies the following: with probability at least $1 - \delta$,*

$$V^* - V^{\pi^K} \leq \frac{2}{(1-\gamma)^2} \left(\sqrt{\frac{20 C V_{\max}^2 \ln(2|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{3 C \epsilon_{\text{approx},\nu}} \right) + \frac{2\gamma^K V_{\max}}{1-\gamma}.$$

The dependence on C is the explicit coverage price: regression error under ν must transfer to the visitation distributions induced by the greedy policies. The term $\epsilon_{\text{approx},\nu}$ is the Bellman approximation price: even with infinite data, lack of (average-case) Bellman closure creates an error floor. The final term is purely algorithmic, from running only K fitted iterations.

Proofs for Theorem 4.3

We combine a propagation bound (Bellman residuals $\Rightarrow$ performance loss) with a uniform regression bound for each fitted step.

Lemma 4.4 (Bellman residuals $\Rightarrow$ performance loss). *Assume that for all $t = 0, 1, \dots, K-1$ we have*

$$\|f_{t+1} - \mathcal{T} f_t\|_{2,\nu} \leq \varepsilon.$$

Let π^t be greedy w.r.t. f_t . Then for every $k \in \{0, 1, \dots, K\}$,

$$V^* - V^{\pi^k} \leq \frac{2\sqrt{C} \varepsilon}{(1-\gamma)^2} + \frac{2\gamma^k V_{\max}}{1-\gamma}.$$

Proof. A standard approximate value-iteration bound (see, e.g., the fitted value-iteration analyses in Munos [2005], Antos et al. [2008]) implies that for greedy policies $\{\pi^t\}$,

$$V^* - V^{\pi^K} \leq \frac{2}{1-\gamma} \sum_{t=0}^{K-1} \gamma^{k-1-t} \|f_{t+1} - \mathcal{T}f_t\|_{1,d_\mu^{\pi^t}} + \frac{2\gamma^K V_{\max}}{1-\gamma}.$$

By Assumption 4.1 and Cauchy–Schwarz,

$$\|f_{t+1} - \mathcal{T}f_t\|_{1,d_\mu^{\pi^t}} \leq \|f_{t+1} - \mathcal{T}f_t\|_{2,d_\mu^{\pi^t}} \leq \sqrt{C} \|f_{t+1} - \mathcal{T}f_t\|_{2,\nu} \leq \sqrt{C} \varepsilon.$$

Using $\sum_{t=0}^{K-1} \gamma^{k-1-t} \leq (1-\gamma)^{-1}$ yields the claim. $\square$

Lemma 4.5 (Uniform regression (Bellman error) bound). *With probability at least $1 - \delta$, for all $t = 0, 1, \dots, K - 1$,*

$$\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu}^2 \leq \frac{20V_{\max}^2 \ln(2|\mathcal{F}|^2 K/\delta)}{n} + 3\epsilon_{\text{approx},\nu}.$$

Proof. Fix $g \in \mathcal{F}$ and consider the regression problem with inputs (s_i, a_i) and targets

$$y_i := r_i + \gamma \max_{a'} g(s'_i, a').$$

Then $\mathbb{E}[y_i \mid s_i, a_i] = (\mathcal{T}g)(s_i, a_i)$. Since $r \in [0, 1]$ and $g \in [0, V_{\max}]$,

$$0 \leq y_i \leq 1 + \gamma V_{\max} = V_{\max}, \quad 0 \leq (\mathcal{T}g)(s, a) \leq V_{\max}.$$

By Assumption 4.2, $\min_{f \in \mathcal{F}} \|f - \mathcal{T}g\|_{2,\nu}^2 \leq \epsilon_{\text{approx},\nu}$. Applying Lemma A.11 with $B = V_{\max}$ gives that, with probability at least $1 - \delta$, for this fixed g ,

$$\|\hat{f}_g - \mathcal{T}g\|_{2,\nu}^2 \leq \frac{20V_{\max}^2 \ln(2|\mathcal{F}|/\delta)}{n} + 3\epsilon_{\text{approx},\nu},$$

where $\hat{f}_g$ is the ERM over $\mathcal{F}$ for targets y_i . A union bound over $g \in \mathcal{F}$ and $t = 0, \dots, K - 1$ (equivalently, replace δ by $\delta/(|\mathcal{F}|K)$) yields the stated bound, and $f_{t+1} = \hat{f}_{f_t}$ by construction. $\square$

Proof of Theorem 4.3. By Lemma 4.5, with probability at least $1 - \delta$, for all $t \leq K - 1$,

$$\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu} \leq \sqrt{\frac{20V_{\max}^2 \ln(2|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{3\epsilon_{\text{approx},\nu}}.$$

Set

$$\varepsilon := \sqrt{\frac{20V_{\max}^2 \ln(2|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{3\epsilon_{\text{approx},\nu}}.$$

Applying Lemma 4.4 with this ε and $k = K$ yields

$$V^* - V^{\pi^K} \leq \frac{2\sqrt{C}}{(1-\gamma)^2} \left(\sqrt{\frac{20V_{\max}^2 \ln(2|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{3\epsilon_{\text{approx},\nu}} \right) + \frac{2\gamma^K V_{\max}}{1-\gamma},$$

which is exactly the theorem statement. $\square$

4.1.4 What breaks without coverage: a minimal impossibility

Assumption 4.1 is not a technical artifact of the analysis: without some form of *coverage* (overlap between the data distribution ν and the occupancy measures of relevant policies), offline RL is information-theoretically impossible. The simplest obstruction is *support mismatch*: if ν assigns zero probability to a state–action pair that an optimal (or even just a competitive) policy may visit, then the offline dataset contains *no information* about what would happen there.

We give a crisp two-action counterexample showing that without concentrability, even the most basic offline learning goal can fail.

Proposition 4.6 (No coverage $\Rightarrow$ offline RL is impossible). *There exists an offline data distribution ν and two discounted MDPs $\mathcal{M}_+$ and $\mathcal{M}_-$ such that:*

1. $\mathcal{M}_+$ and $\mathcal{M}_-$ induce exactly the same distribution over datasets $\mathcal{D}$ drawn i.i.d. from ν (hence no offline algorithm can distinguish them from $\mathcal{D}$), but
2. the optimal policies (and optimal values) differ substantially, so any offline algorithm must incur nontrivial suboptimality on at least one of the two MDPs.

In particular, for any (possibly randomized) offline algorithm $\mathcal{A}$ that maps $\mathcal{D}$ to a policy $\hat{\pi}$, we have

$$\max \left\{ \mathbb{E}_{\mathcal{D} \sim \mathcal{M}_+} [V^* - V^{\hat{\pi}}], \mathbb{E}_{\mathcal{D} \sim \mathcal{M}_-} [V^* - V^{\hat{\pi}}] \right\} \geq \frac{1}{4} V_{\max}.$$

Proof. Consider the one-state MDP with $\mathcal{S} = \{s\}$, two actions $\mathcal{A} = \{a_0, a_1\}$, deterministic self-loop dynamics $P(s \mid s, a) = 1$, and initial distribution $\mu(s) = 1$. Define the offline data distribution ν to put all of its mass on the single pair (s, a_0) , i.e. $\nu(s, a_0) = 1$ and $\nu(s, a_1) = 0$. Hence every offline dataset $\mathcal{D}$ consists only of samples of (s, a_0) .

Now define two reward functions (keeping the same dynamics):

$$\mathcal{M}_+ : \quad r(s, a_0) = \frac{1}{2}, \quad r(s, a_1) = 1, \quad \mathcal{M}_- : \quad r(s, a_0) = \frac{1}{2}, \quad r(s, a_1) = 0.$$

Under ν , the algorithm only ever observes rewards for (s, a_0) , which are identically $\frac{1}{2}$ in both MDPs. Therefore the distribution of $\mathcal{D}$ is *exactly the same* under $\mathcal{M}_+$ and $\mathcal{M}_-$, and any algorithm $\mathcal{A}$ induces the same distribution over outputs $\hat{\pi}$ under both MDPs.

Since $\mathcal{D}$ has the same distribution under $\mathcal{M}_+$ and $\mathcal{M}_-$, no offline algorithm can reliably tell which MDP it is in, and therefore it cannot reliably choose between a_0 and a_1 . In $\mathcal{M}_+$, choosing a_0 is suboptimal by $V_{\max}/2$ (since $V^* = V_{\max}$ but $V^{a_0} = V_{\max}/2$), while in $\mathcal{M}_-$, choosing a_1 is suboptimal by $V_{\max}/2$ (since $V^* = V_{\max}/2$ but $V^{a_1} = 0$). Consequently, any decision rule that outputs some action (possibly randomized) must incur expected suboptimality at least $V_{\max}/4$ on at least one of the two MDPs (the minimax choice is to randomize 50–50, yielding $V_{\max}/4$ in either case). $\square$

Let us connect this proposition back to concentrability. In Proposition 4.6, the policy that plays a_1 induces occupancy $d_{\mu}^{\pi}(s, a_1) = 1$, yet $\nu(s, a_1) = 0$, so the ratio $d_{\mu}^{\pi}(s, a_1)/\nu(s, a_1)$ is infinite. This is exactly the failure mode that Assumption 4.1 rules out: it enforces that whenever a policy can place mass on (s, a) , the dataset distribution must also place mass there (with a bounded density ratio).

4.2 Optional: Agnostic policy search and the horizon barrier

The previous sections focused on *discounted* offline learning with concentrability. Here we briefly switch to a *finite-horizon* lens to make a complementary point: without additional structure, even the most basic goal of selecting a near-best policy from a class can require sample sizes that are exponential in the horizon. This “horizon barrier” is the finite-horizon analogue of what goes wrong in offline RL when coverage deteriorates.

Consider a horizon- H MDP with initial distribution μ and finite action set $\mathcal{A}$. Let Π be a finite set of deterministic policies. The *agnostic objective* is

$$\max_{\pi \in \Pi} V_0^\pi(\mu),$$

i.e. compete with the best policy in the class without assuming realizability or Bellman closure.

Importance sampling gives an unbiased estimator, with $|\mathcal{A}|^H$ variance. Collect N trajectories under the behavior policy that acts uniformly at random at each step, denoted $\text{Unif}_{\mathcal{A}}$. For any deterministic π , the standard importance-sampling (IS) estimator is

$$\widehat{V}_0^\pi(\mu) := \frac{1}{N} \sum_{n=1}^N w(\tau^{(n)}; \pi) \sum_{t=0}^{H-1} r_t^{(n)}, \quad w(\tau; \pi) = |\mathcal{A}|^H \mathbf{1}(\pi(s_t) = a_t \forall t),$$

which is unbiased: $\mathbb{E}[\widehat{V}_0^\pi(\mu)] = V_0^\pi(\mu)$. The difficulty is that $w(\tau; \pi)$ is nonzero only when the behavior trajectory happens to match π ’s entire length- H action sequence, an event of probability $|\mathcal{A}|^{-H}$. Thus the estimator has variance that scales like $|\mathcal{A}|^H$ (up to polynomial factors in H), so concentration requires exponentially many samples.

A simple union-bound argument yields the following representative statement: with probability at least $1 - \delta$, simultaneously for all $\pi \in \Pi$,

$$\left| \widehat{V}_0^\pi(\mu) - V_0^\pi(\mu) \right| \lesssim H |\mathcal{A}|^H \sqrt{\frac{\log(|\Pi|/\delta)}{N}}.$$

In particular, selecting a near-best policy in Π from this shared data requires N that is exponential in H in the worst case.

A matching lower bound (needle in a haystack). This exponential dependence is not an artifact of IS. There exist families of horizon- H MDPs (e.g. a balanced $|\mathcal{A}|$ -ary tree in which only one leaf is rewarding) for which any algorithm—even with access to a generative model—must use $\Omega(|\mathcal{A}|^H)$ samples to obtain even a constant additive optimality guarantee with nontrivial probability. Intuitively, unless the algorithm samples the rewarding path, it receives no information about where reward lies.

Taken together, these facts motivate why offline RL analyses must assume some form of overlap/coverage: without it, the effective importance weights needed to reason about candidate policies can be exponentially large, mirroring the $|\mathcal{A}|^H$ barrier in agnostic horizon- H problems.

4.3 Bibliographic Remarks and Further Readings

The notion of *concentrability* (or coverage via occupancy-measure density ratios) was introduced by Munos [2003, 2005] to obtain sharp *average-case* approximation bounds for approximate dynamic programming, under the assumption that the relevant concentrability coefficients are bounded. This line of work also supports sample-based fitted methods, with explicit finite-sample bounds, provided that the data-collection policy induces adequate coverage; see,

e.g., Munos [2005], Szepesvári and Munos [2005], Antos et al. [2008], Lazaric et al. [2016]. For further discussion of concentrability-type quantities and their role in offline RL, see Chen and Jiang [2019].

The reduction of reinforcement learning to supervised learning via reusing data across policies was developed early in Kearns et al. [2000]. That work analyzed a related evaluation/selection scheme based on a “trajectory tree” view of policy behavior, and explicitly connected learnability to capacity measures of the policy class (including VC-type dimensions). Our optional agnostic-learning discussion in Section 4.2 is a stripped-down version of this theme, using importance sampling to make the exponential-in- H barrier transparent. The fundamental sample complexity tradeoff — avoiding explicit dependence on the size of the state space at the cost of exponential dependence on the horizon — is explored further in Kakade [2003].

Part II

Strategic Exploration

Chapter 5

Multi-Armed & Linear Bandits

When $\gamma = 0$ (equivalently $H = 1$ in the finite-horizon setting), learning in an unknown MDP reduces to the multi-armed bandit problem. The basic algorithms and proof techniques in this chapter are important in their own right, and we will later extend these ideas to handle exploration–exploitation tradeoffs in more general reinforcement learning problems.

Throughout this chapter, rewards are stochastic. In the case of linear bandits, we will be interested in a setting where the number of possible decisions is large (or uncountably infinite).

5.1 The K -Armed Bandit Problem

There are K actions (“arms”). Pulling arm $a \in [K]$ produces a random reward $r_a \in [-1, 1]$ distributed according to an unknown distribution $R(a) \in \Delta([-1, 1])$, with mean

$$\mu_a := \mathbb{E}_{r_a \sim R(a)}[r_a] \in [-1, 1].$$

The interaction proceeds for T rounds, indexed by $t = 0, 1, \dots, T-1$. At round t , the learner selects an arm $I_t \in [K]$ and observes a reward

$$r_t \sim R(I_t), \quad \mathbb{E}[r_t \mid I_t = a] = \mu_a.$$

Regret. The (cumulative) regret compares our total reward to that of the best fixed arm:

$$R_T := T\mu_\star - \sum_{t=0}^{T-1} r_t, \quad \text{where} \quad \mu_\star := \max_{a \in [K]} \mu_a.$$

For the analysis it is convenient to work with the (pseudo) regret

$$R_T := T\mu_\star - \sum_{t=0}^{T-1} \mu_{I_t},$$

and we will refer to this quantity as the regret throughout. (One can relate $\sum_{t=0}^{T-1} r_t$ to $\sum_{t=0}^{T-1} \mu_{I_t}$ via Hoeffding–Azuma, since the reward noise forms a martingale difference sequence.)

Let $a^\star \in \operatorname{argmax}_{a \in [K]} \mu_a$ be an optimal arm and define the gap

$$\Delta_a := \mu_{a^\star} - \mu_a \geq 0.$$

Algorithm 3 Explore–Then–Commit (ETC)

Input: exploration length $m \in \mathbb{N}$

- 1: Pull each arm $a \in [K]$ exactly m times (in any order), and compute the empirical mean $\hat{\mu}(a)$ from these m samples.
 - 2: Let $\hat{a} \in \operatorname{argmax}_{a \in [K]} \hat{\mu}(a)$.
 - 3: For all remaining rounds, play $I_t = \hat{a}$.
-

5.1.1 A baseline: explore–then–commit

A natural first idea is to *explore* uniformly to estimate the arm means, then *commit* to the empirically best arm.

ETC is easy to analyze and motivates why *adaptive* exploration (e.g., UCB) is needed: if we knew the gaps, we could choose m so that the best arm is identified with high probability, giving gap-dependent regret. Without gap knowledge, fixed exploration can be suboptimal in the worst case.

Proposition 5.1 (ETC, gap-dependent bound). *Fix $\delta \in (0, 1)$. Suppose ETC explores each arm m times, and then commits. If*

$$m \geq \frac{8}{\Delta_{\min}^2} \ln\left(\frac{2K}{\delta}\right), \quad \text{where } \Delta_{\min} := \min_{a \neq a^*} \Delta_a,$$

then with probability at least $1 - \delta$, ETC selects $\hat{a} = a^$ and*

$$R_T \leq m \sum_{a \neq a^*} \Delta_a = \frac{8 \sum_{a \neq a^*} \Delta_a}{\Delta_{\min}^2} \ln\left(\frac{2K}{\delta}\right).$$

Proof sketch. After m pulls of each arm, Hoeffding’s inequality (Lemma A.1) implies that simultaneously for all $a \in [K]$, $|\hat{\mu}(a) - \mu_a| \leq \sqrt{2 \ln(2K/\delta)/m}$ with probability $\geq 1 - \delta$. Using our choice of m ensures that this deviation is at most $\Delta_{\min}/2$. With this, we have that $\hat{\mu}(a^*) > \hat{\mu}(a)$ for all $a \neq a^*$, so the algorithm commits to a^* and regret comes only from the exploration pulls. $\square$

We now turn to an adaptive method (UCB) which automatically balances exploration and exploitation and achieves both a distribution-free $\tilde{O}(\sqrt{KT})$ regret bound and a gap-dependent $O(\sum_{a \neq a^*} \frac{\log T}{\Delta_a})$ bound. Note that, for the gap dependent case, this bound improves upon ECT by, roughly, a $1/\Delta_{\min}$ factor.

5.1.2 The Upper Confidence Bound (UCB) Algorithm

We now present and analyze the upper confidence bound (UCB) algorithm (Algorithm 4). For simplicity, we ensure every arm has been pulled at least once by allocating the first K rounds to pulling each arm once. Here $N_t(a)$ is the number of times arm a was pulled prior to round t :

$$N_t(a) := \sum_{s=0}^{t-1} \mathbf{1}\{I_s = a\},$$

and the empirical mean is

$$\hat{\mu}_t(a) := \frac{1}{N_t(a)} \sum_{s=0}^{t-1} \mathbf{1}\{I_s = a\} r_s.$$

(After the initialization phase, $N_t(a) \geq 1$ for all a and $t \geq K$.)

The algorithm enjoys the following guarantee:

Algorithm 4 UCB

Input: horizon T , failure probability δ

```
1: for  $a = 1, 2, \dots, K$  do
2:   Play arm  $I_{a-1} = a$  and observe reward  $r_{a-1}$ .
3: end for
4: for  $t = K, K + 1, \dots, T - 1$  do
5:   Set  $I_t \in \operatorname{argmax}_{a \in [K]} \left( \hat{\mu}_t(a) + \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(a)}} \right)$ .
6:   Play arm  $I_t$  and observe reward  $r_t$ .
7: end for
```

Theorem 5.2. *With probability at least $1 - \delta$, the UCB algorithm obtains the following regret guarantee:*

$$R_T = O \left(\min \left\{ \sqrt{KT \ln \left(\frac{TK}{\delta} \right)}, \sum_{a \neq a^*} \frac{\ln \left(\frac{TK}{\delta} \right)}{\Delta_a} \right\} + K \right).$$

We define the UCB index

$$\text{UCB}_t(a) := \hat{\mu}_t(a) + \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(a)}}.$$

The following lemma shows that, uniformly over all arms and all times, this is a valid confidence bound with high probability.

Lemma 5.3 (Uniform concentration of empirical means). *With probability at least $1 - \delta$, simultaneously for all rounds $t \in \{K, \dots, T - 1\}$ and all arms $a \in [K]$,*

$$|\hat{\mu}_t(a) - \mu_a| \leq \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(a)}}.$$

Proof. For each arm a and each pull-count $n \in \{1, \dots, T\}$, let $\hat{\mu}_{a,n}$ denote the empirical mean of the first n rewards obtained from arm a . Hoeffding's inequality gives $\Pr(|\hat{\mu}_{a,n} - \mu_a| > \sqrt{2 \ln(2TK/\delta)/n}) \leq \delta/(TK)$. A union bound over all $(a, n) \in [K] \times [T]$ yields the claim, and at time t we simply take $n = N_t(a)$. $\square$

Now provide the proof of UCB.

Proof. (Proof of Theorem 5.2.) Let

$$\text{rad}_t(a) := \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(a)}}, \quad \text{UCB}_t(a) := \hat{\mu}_t(a) + \text{rad}_t(a).$$

Define the high-probability event from Lemma 5.3:

$$\mathcal{E} := \{\forall t \in \{K, \dots, T - 1\}, \forall a \in [K], |\hat{\mu}_t(a) - \mu_a| \leq \text{rad}_t(a)\}.$$

By Lemma 5.3, we have $\Pr(\mathcal{E}) \geq 1 - \delta$.

Condition on $\mathcal{E}$. First note the *optimism* property: for every $t \geq K$ and every $a \in [K]$,

$$\mu_a \leq \hat{\mu}_t(a) + \text{rad}_t(a) = \text{UCB}_t(a).$$

In particular, $\mu_\star = \mu_{a^\star} \leq \text{UCB}_t(a^\star)$.

Since UCB selects $I_t \in \operatorname{argmax}_a \text{UCB}_t(a)$, we have $\text{UCB}_t(I_t) \geq \text{UCB}_t(a^*) \geq \mu_*$. Therefore,

$$\begin{aligned} \mu_* - \mu_{I_t} &\leq \text{UCB}_t(I_t) - \mu_{I_t} = \hat{\mu}_t(I_t) + \text{rad}_t(I_t) - \mu_{I_t} \\ &\leq |\hat{\mu}_t(I_t) - \mu_{I_t}| + \text{rad}_t(I_t) \leq 2 \text{rad}_t(I_t) = 2 \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(I_t)}}. \end{aligned}$$

First, let us provide a distribution-free bound. Summing over $t = K, \dots, T-1$ gives

$$R_T = \sum_{t=0}^{T-1} (\mu_* - \mu_{I_t}) \leq 2K + 2\sqrt{2 \ln(2TK/\delta)} \sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(I_t)}},$$

where we used the trivial bound $\mu_* - \mu_{I_t} \leq 2$ for the first K rounds. Now use the standard counting argument: for each arm a , each time we play a the quantity $N_t(a)$ increases by 1, hence

$$\sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(I_t)}} = \sum_{a \in [K]} \sum_{i=1}^{N_T(a)} \frac{1}{\sqrt{i}} \leq \sum_{a \in [K]} 2\sqrt{N_T(a)} \leq 2\sqrt{K \sum_a N_T(a)} = 2\sqrt{KT}.$$

(We used $\sum_{i=1}^n i^{-1/2} \leq 2\sqrt{n}$ and Cauchy–Schwarz.) Thus,

$$R_T \leq 2K + 8\sqrt{KT \ln(2TK/\delta)}.$$

Now, we provide a gap-dependent bound. Fix a suboptimal arm $a \neq a^*$ with gap $\Delta_a > 0$. Suppose that at some round $t \geq K$ the algorithm selects $I_t = a$. Then

$$\text{UCB}_t(a) \geq \text{UCB}_t(a^*) \geq \mu_*.$$

On $\mathcal{E}$, we also have $\hat{\mu}_t(a) \leq \mu_a + \text{rad}_t(a)$, hence

$$\text{UCB}_t(a) = \hat{\mu}_t(a) + \text{rad}_t(a) \leq \mu_a + 2 \text{rad}_t(a).$$

Combining the two displays yields

$$\mu_* \leq \mu_a + 2 \text{rad}_t(a) \quad \Rightarrow \quad \Delta_a \leq 2 \text{rad}_t(a) = 2 \sqrt{\frac{2 \ln(2TK/\delta)}{N_t(a)}},$$

so necessarily

$$N_t(a) \leq \frac{8 \ln(2TK/\delta)}{\Delta_a^2}.$$

Therefore, the total number of pulls of arm a up to time T satisfies

$$N_T(a) \leq 1 + \frac{8 \ln(2TK/\delta)}{\Delta_a^2},$$

and hence the regret can be bounded as

$$R_T = \sum_{t=0}^{T-1} (\mu_* - \mu_{I_t}) = \sum_{a \neq a^*} \Delta_a N_T(a) \leq 2K + \sum_{a \neq a^*} \left(\Delta_a + \frac{8 \ln(2TK/\delta)}{\Delta_a} \right) \leq O \left(K + \sum_{a \neq a^*} \frac{\ln(2TK/\delta)}{\Delta_a} \right).$$

To conclude the proof, note that on the event $\mathcal{E}$ (which holds with probability at least $1 - \delta$), we have both bounds, so we may take their minimum. This proves Theorem 5.2. $\square$

Algorithm 5 LinUCB (linear bandits with optimistic action selection)

Input: regularization $\lambda > 0$; confidence radii $(\beta_t)_{t \geq 0}$

1: Initialize $\Sigma_0 \leftarrow \lambda I$ and $b_0 \leftarrow 0$.

2: **for** $t = 0, 1, 2, \dots$ **do**

3: $\hat{\mu}_t \leftarrow \Sigma_t^{-1} b_t$

4: Choose

$$x_t \in \operatorname{argmax}_{x \in D} \left(\hat{\mu}_t \cdot x + \sqrt{\beta_t} \sqrt{x^\top \Sigma_t^{-1} x} \right)$$

and observe reward r_t .

5: Update $b_{t+1} \leftarrow b_t + r_t x_t$ and $\Sigma_{t+1} \leftarrow \Sigma_t + x_t x_t^\top$.

6: **end for**

5.2 Linear Bandits: Handling Large Action Spaces

Let $D \subset \mathbb{R}^d$ be a compact (but otherwise arbitrary) set of decisions. On each round $t = 0, 1, 2, \dots$, the learner chooses an action $x_t \in D$ and observes a reward $r_t \in [-1, 1]$.

Linear reward model. We assume that, for every history $\mathcal{H}_{<t}$ and every $x \in D$, the conditional expectation of r_t is linear:

$$\mathbb{E}[r_t \mid \mathcal{H}_{<t}, x_t = x] = \mu^* \cdot x \in [-1, 1],$$

for some unknown parameter $\mu^* \in \mathbb{R}^d$. Equivalently, the noise sequence

$$\eta_t := r_t - \mu^* \cdot x_t$$

is a martingale difference sequence (with respect to the natural filtration).

Regret. Let

$$x^* \in \operatorname{argmax}_{x \in D} \mu^* \cdot x,$$

which exists since D is compact. We measure performance via the (pseudo-)regret

$$R_T := T \mu^* \cdot x^* - \sum_{t=0}^{T-1} \mu^* \cdot x_t.$$

(Again, one can relate $\sum_{t=0}^{T-1} r_t$ to $\sum_{t=0}^{T-1} \mu^* \cdot x_t$ via Hoeffding–Azuma since (η_t) is a martingale difference sequence.)

5.2.1 The LinUCB algorithm

LinUCB is an “optimism in the face of uncertainty” algorithm. At time t , it forms a confidence region (an ellipsoid) around the regularized least-squares estimate $\hat{\mu}_t$:

$$\hat{\mu}_t = \arg \min_{\mu \in \mathbb{R}^d} \sum_{\tau=0}^{t-1} (r_\tau - \mu \cdot x_\tau)^2 + \lambda \|\mu\|_2^2 = \Sigma_t^{-1} \sum_{\tau=0}^{t-1} r_\tau x_\tau,$$

where

$$\Sigma_t = \lambda I + \sum_{\tau=0}^{t-1} x_\tau x_\tau^\top, \quad \text{with } \Sigma_0 = \lambda I.$$

The confidence set is

$$\text{BALL}_t := \left\{ \mu \in \mathbb{R}^d \mid (\hat{\mu}_t - \mu)^\top \Sigma_t (\hat{\mu}_t - \mu) \leq \beta_t \right\}. \quad (5.1)$$

The action rule in Step 4 is exactly

$$x_t \in \operatorname{argmax}_{x \in D} \max_{\mu \in \text{BALL}_t} \mu \cdot x,$$

since $\max_{\mu \in \text{BALL}_t} \mu \cdot x = \hat{\mu}_t \cdot x + \sqrt{\beta_t} \sqrt{x^\top \Sigma_t^{-1} x}$.

Computation. Even if we can optimize linear functions over D (i.e., solve $\max_{x \in D} \nu \cdot x$ for any ν), the objective in Step 4 includes the (convex) term $\sqrt{x^\top \Sigma_t^{-1} x}$, so maximizing it over an arbitrary set D need not be computationally tractable. We return to computational issues and common relaxations in Section 5.4.

5.2.2 Upper and Lower Bounds

Our main message is that one can achieve sublinear regret with only a polynomial dependence on the dimension d , and *no* dependence on $|D|$.

Theorem 5.4. Assume $|\mu^* \cdot x| \leq 1$ for all $x \in D$, $\|\mu^*\|_2 \leq W$, $\|x\|_2 \leq B$ for all $x \in D$, and that the noise η_t is conditionally σ^2 -sub-Gaussian.¹ Set

$$\lambda = \sigma^2 / W^2, \quad \beta_t := \sigma^2 \left(2 + 4d \log \left(1 + \frac{tB^2W^2}{d\sigma^2} \right) + 8 \log(4/\delta) \right).$$

Then with probability at least $1 - \delta$, simultaneously for all $T \geq 1$,

$$R_T \leq c \sigma \sqrt{T} \left(d \log \left(1 + \frac{TB^2W^2}{d\sigma^2} \right) + \log(4/\delta) \right),$$

where c is an absolute constant. In particular, $R_T = \tilde{O}(d\sqrt{T})$ with high probability.

Theorem 5.5 (Lower bound). There exists a distribution over linear bandit instances (i.e., over μ^*) with rewards bounded in $[-1, 1]$ and $\sigma^2 \leq 1$ such that for every (randomized) algorithm and all $T \geq \max\{256, d^2/16\}$,

$$\mathbb{E}_{\mu^*} \mathbb{E}[R_T] \geq \frac{1}{2500} d\sqrt{T},$$

where the inner expectation is over the randomness in the interaction (rewards) and the algorithm.

We will only prove the upper bound (see Section 5.4 for pointers).

Remark: D -optimal preconditioning. The dependence on the a priori bounds B and W in Theorem 5.4 is not intrinsic. One can remove it (up to dimension factors) by working in a well-chosen coordinate system.

Let Σ_D be the D -optimal design matrix from Theorem 3.1, and define the change of variables

$$\tilde{x} := \Sigma_D^{-1/2} x, \quad \tilde{\mu}^* := \Sigma_D^{1/2} \mu^*.$$

Then the expected reward is preserved: $\tilde{\mu}^* \cdot \tilde{x} = \mu^* \cdot x$. Moreover, by properties of Σ_D ,

$$\|\tilde{x}\| = \|x\|_{\Sigma_D^{-1}} \leq \sqrt{d} \quad \text{and} \quad \|\tilde{\mu}^*\|^2 = (\mu^*)^\top \Sigma_D \mu^* = \mathbb{E}_{x \sim \rho} [(\mu^* \cdot x)^2] \leq 1,$$

where the last inequality uses the assumption $|\mu^* \cdot x| \leq 1$ for all $x \in D$.

¹ See Definition A.2.

Corollary 5.6. Assume $|\mu^* \cdot x| \leq 1$ for all $x \in D$ and that the noise η_t is σ^2 sub-Gaussian. Run LinUCB in the preconditioned coordinates $\tilde{x}_t = \Sigma_D^{-1/2} x_t$ with

$$\lambda = \sigma^2, \quad \beta_t := \sigma^2 \left(2 + 4d \log(1+t) + 8 \log(4/\delta) \right).$$

Then with probability at least $1 - \delta$, simultaneously for all $T \geq 1$,

$$R_T \leq c \sigma \sqrt{T} \left(d \log \left(1 + \frac{T}{\sigma^2} \right) + \log(4/\delta) \right),$$

for a universal constant c .

5.3 LinUCB Analysis

The proof of Theorem 5.4 decomposes into two steps. First, we show that the confidence region BALL_t contains the true parameter μ^* for all times t with high probability. Second, we bound the regret on this event via a potential-function argument based on $\log \det(\Sigma_t)$.

Proposition 5.7 (Confidence). *Let $\delta \in (0, 1)$. Under the assumptions of Theorem 5.4 (in particular, η_t is conditionally σ -sub-Gaussian), we have*

$$\Pr(\forall t \geq 0, \mu^* \in \text{BALL}_t) \geq 1 - \delta.$$

Section 5.3.2 proves this by controlling $\|\hat{\mu}_t - \mu^*\|_{\Sigma_t}^2 := (\hat{\mu}_t - \mu^*)^\top \Sigma_t (\hat{\mu}_t - \mu^*)$.

The second step bounds the regret assuming the above high-probability event. Define the instantaneous regret

$$\text{regret}_t := (\mu^*)^\top x^* - (\mu^*)^\top x_t, \quad \text{so that} \quad R_T = \sum_{t=0}^{T-1} \text{regret}_t.$$

Proposition 5.8 (Sum of squares regret bound). *Assume $\|x\| \leq B$ for all $x \in D$, and that $(\beta_t)_{t \geq 0}$ is nondecreasing with $\beta_t \geq 1$. For LinUCB, if $\mu^* \in \text{BALL}_t$ for all $t = 0, 1, \dots, T-1$, then*

$$\sum_{t=0}^{T-1} \text{regret}_t^2 \leq 8 \beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right).$$

This is proven in Section 5.3.1 using a potential function based on the log-determinant of the “design” matrix Σ_t , which tracks how accurately we have learned μ^* in each direction.

Proof of Theorem 5.4. On the event $\{\forall t, \mu^* \in \text{BALL}_t\}$, Proposition 5.8 and Cauchy–Schwarz give

$$R_T = \sum_{t=0}^{T-1} \text{regret}_t \leq \sqrt{T \sum_{t=0}^{T-1} \text{regret}_t^2} \leq \sqrt{8T\beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right)}.$$

Proposition 5.7 shows this event holds with probability at least $1 - \delta$. Plugging in the chosen β_T and simplifying yields the stated bound. $\square$

We now prove Propositions 5.8 and 5.7.

5.3.1 Regret Analysis

Throughout this section, assume $\mu^* \in \text{BALL}_t$ for all $t = 0, 1, \dots, T-1$. Unless explicitly stated, $\|\cdot\|$ denotes the Euclidean norm. For a positive definite matrix A , we write $\|v\|_A^2 := v^\top A v$.

Lemma 5.9 (Ellipsoid width). *For any $x \in D$ and any $\mu \in \text{BALL}_t$,*

$$|(\mu - \hat{\mu}_t)^\top x| \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x}.$$

Proof. By Cauchy–Schwarz,

$$|(\mu - \hat{\mu}_t)^\top x| = |(\Sigma_t^{1/2}(\mu - \hat{\mu}_t))^\top (\Sigma_t^{-1/2}x)| \leq \|\mu - \hat{\mu}_t\|_{\Sigma_t} \|x\|_{\Sigma_t^{-1}}.$$

Since $\mu \in \text{BALL}_t$, we have $\|\mu - \hat{\mu}_t\|_{\Sigma_t} \leq \sqrt{\beta_t}$, and $\|x\|_{\Sigma_t^{-1}} = \sqrt{x^\top \Sigma_t^{-1} x}$. □

Define the normalized width in the chosen direction

$$w_t := \sqrt{x_t^\top \Sigma_t^{-1} x_t}.$$

Lemma 5.10 (Regret is controlled by width). *Fix $t \leq T-1$. If $\mu^* \in \text{BALL}_t$ and $\beta_t \geq 1$, then*

$$\text{regret}_t \leq \min\{2\sqrt{\beta_t} w_t, 2\} = 2 \min\{\sqrt{\beta_t} w_t, 1\} \leq 2\sqrt{\beta_T} \min\{w_t, 1\}.$$

Proof. Let $\mu_t^+ \in \arg \max_{\mu \in \text{BALL}_t} \mu^\top x_t$. By the choice of x_t in LinUCB,

$$(\mu_t^+)^\top x_t = \max_{x \in D} \max_{\mu \in \text{BALL}_t} \mu^\top x \geq (\mu^*)^\top x^*,$$

where the inequality uses $\mu^* \in \text{BALL}_t$. Therefore,

$$\text{regret}_t = (\mu^*)^\top x^* - (\mu^*)^\top x_t \leq (\mu_t^+ - \mu^*)^\top x_t \leq |(\mu_t^+ - \hat{\mu}_t)^\top x_t| + |(\hat{\mu}_t - \mu^*)^\top x_t|.$$

Applying Lemma 5.9 to μ_t^+ and to μ^* (both in BALL_t) yields $\text{regret}_t \leq 2\sqrt{\beta_t} w_t$. Finally, since $|(\mu^*)^\top x| \leq 1$ for all $x \in D$, we always have $\text{regret}_t \leq 2$, giving the stated bound. □

We now relate the sum of widths to a log-determinant potential.

Lemma 5.11 (Determinant growth). *We have*

$$\det(\Sigma_T) = \det(\Sigma_0) \prod_{t=0}^{T-1} (1 + w_t^2).$$

Proof. Using $\Sigma_{t+1} = \Sigma_t + x_t x_t^\top$ and the matrix determinant lemma,

$$\det(\Sigma_{t+1}) = \det(\Sigma_t) \left(1 + x_t^\top \Sigma_t^{-1} x_t\right) = \det(\Sigma_t) (1 + w_t^2).$$

Iterating from $t = 0$ to $T-1$ gives the claim. □

Lemma 5.12 (Elliptical potential function bound). *For any sequence $x_0, \dots, x_{T-1}$ with $\|x_t\| \leq B$,*

$$\log \frac{\det(\Sigma_T)}{\det(\Sigma_0)} = \log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right).$$

Consequently,

$$\sum_{t=0}^{T-1} \min\{w_t^2, 1\} \leq 2d \log \left(1 + \frac{TB^2}{d\lambda} \right).$$

Proof. Let $\sigma_1, \dots, \sigma_d$ be the eigenvalues of $\sum_{t=0}^{T-1} x_t x_t^\top$. Then $\sum_{i=1}^d \sigma_i = \text{Tr}(\sum_t x_t x_t^\top) = \sum_t \|x_t\|^2 \leq TB^2$. By AM-GM,

$$\log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) = \sum_{i=1}^d \log \left(1 + \frac{\sigma_i}{\lambda} \right) \leq d \log \left(1 + \frac{1}{d\lambda} \sum_{i=1}^d \sigma_i \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right).$$

For the second claim, note for all $u \geq 0$ one has $\min\{u, 1\} \leq 2 \log(1 + u)$, so

$$\sum_{t=0}^{T-1} \min\{w_t^2, 1\} \leq 2 \sum_{t=0}^{T-1} \log(1 + w_t^2) = 2 \log \frac{\det(\Sigma_T)}{\det(\Sigma_0)},$$

where the last equality uses Lemma 5.11. Combining with the first claim in the lemma yields the second claim. $\square$

Proof of Proposition 5.8. Using Lemma 5.10 and that β_t is nondecreasing,

$$\sum_{t=0}^{T-1} \text{regret}_t^2 \leq \sum_{t=0}^{T-1} 4\beta_t \min\{w_t^2, 1\} \leq 4\beta_T \sum_{t=0}^{T-1} \min\{w_t^2, 1\} \leq 8\beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right),$$

where the last step uses Lemma 5.12. $\square$

5.3.2 Confidence Analysis

Proof of Proposition 5.7. Recall $\Sigma_t = \lambda I + \sum_{\tau=0}^{t-1} x_\tau x_\tau^\top$ and $\hat{\mu}_t = \Sigma_t^{-1} \sum_{\tau=0}^{t-1} r_\tau x_\tau$. Since $r_\tau = (\mu^*)^\top x_\tau + \eta_\tau$, we have

$$\hat{\mu}_t - \mu^* = \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau - \lambda \Sigma_t^{-1} \mu^*.$$

Therefore, by the triangle inequality,

$$\begin{aligned} \|\hat{\mu}_t - \mu^*\|_{\Sigma_t} &\leq \left\| \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t^{-1}} + \lambda \|\mu^*\|_{\Sigma_t^{-1}} \\ &\leq \left\| \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t^{-1}} + \sqrt{\lambda} \|\mu^*\|, \end{aligned}$$

where we used $\Sigma_t \succeq \lambda I$, hence $\|\mu^*\|_{\Sigma_t^{-1}} \leq \|\mu^*\|/\lambda^{1/2}$.

Now apply the self-normalized bound (Lemma A.9) with $V_0 = \Sigma_0 = \lambda I$ and $X_{\tau+1} = x_\tau$. For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $t \geq 0$,

$$\left\| \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\|_{\Sigma_t^{-1}} \leq \sigma \sqrt{2 \log \left(\frac{\det(\Sigma_t)^{1/2} \det(\Sigma_0)^{-1/2}}{\delta} \right)}.$$

Combining the last two displays yields that, with probability at least $1 - \delta$, simultaneously for all $t \geq 0$,

$$\|\hat{\mu}_t - \mu^*\|_{\Sigma_t} \leq \sqrt{\lambda} \|\mu^*\| + \sigma \sqrt{2 \log \left(\frac{\det(\Sigma_t)^{1/2} \det(\Sigma_0)^{-1/2}}{\delta} \right)}.$$

With our choice of β_t (as in Theorem 5.4) and using the log determinant bound (Lemma 5.12), this is equivalent to $\mu^* \in \text{BALL}_t$ for all t , proving the proposition. $\square$

5.4 Bibliographic Remarks and Further Readings

A comprehensive modern reference for multi-armed and linear bandits is the textbook of Lattimore and Szepesvári [2020].

The stochastic K -armed bandit model dates back to early work such as Robbins [1952]. The first sharp regret guarantees for simple algorithms such as UCB were developed in Auer et al. [2002] (see also the exposition in Lattimore and Szepesvári [2020]).

The stochastic linear bandit model was originally introduced by Abe and Long [1999]. For stochastic linear bandits, our presentation of LinUCB follows the original $\sqrt{T}$ -regret analysis of Dani et al. [2008]. The confidence-set argument we use is based on the self-normalized concentration framework of Abbasi-Yadkori et al. [2011], which streamlines the proof and yields clean high-probability bounds. The minimax lower bound stated here is also due to Dani et al. [2008].

Computation. LinUCB requires solving, at each round, an optimistic optimization problem over a confidence ellipsoid. Even when one has an efficient linear optimization oracle over D , this ellipsoidal optimization can be computationally hard for general action sets D (e.g., for polytopes given by linear inequalities). A standard workaround (as presented in Dani et al. [2008]) is to replace the ℓ_2 ellipsoid by a containing polyhedral (e.g. ℓ_1 -type) confidence set, which makes the optimistic step reducible to finitely many calls to the linear oracle (e.g., by checking extreme points). This typically incurs an extra $\sqrt{d}$ factor in regret but yields an implementable algorithm (see Dani et al. [2008], Lattimore and Szepesvári [2020]).

Chapter 6

Strategic Exploration in Tabular MDPs

In the generative-model setting, we could query the dynamics at any state–action pair and reduce reinforcement learning to estimating a collection of local quantities. In an *online* MDP, the learner has a much more limited interface: it can only interact with the environment by executing trajectories. As a result, data are not obtained uniformly over the state space—they are biased toward states the learner can *reach*.

This makes exploration a central algorithmic challenge. To learn a near-optimal policy, the agent must sometimes take actions that are not immediately rewarding in order to gather information about parts of the MDP that are currently uncertain. The goal is to balance this exploration–exploitation tradeoff so that performance approaches that of an optimal policy as the number of episodes grows.

In this chapter we focus on *tabular* finite-horizon MDPs and present *UCB Value Iteration (UCBVI)*, an optimism-based algorithm that achieves (sublinear) regret. The algorithm combines (i) an empirical transition model built from trajectory data with (ii) an explicit exploration bonus, and then plans greedily in the resulting optimistic model.

6.1 Setup and regret

We consider a finite-horizon episodic MDP with horizon H , discrete state space $\mathcal{S}$ and action space $\mathcal{A}$, and a fixed start state s_0 . Let $S := |\mathcal{S}|$ and $A := |\mathcal{A}|$. The (stage-dependent) transition kernels are $\{P_h^*(\cdot \mid s, a)\}_{h=0}^{H-1}$. Rewards are bounded as $r_h(s, a) \in [0, 1]$. For simplicity, we treat rewards as deterministic and known; the extension to unknown and/or stochastic rewards estimates r_h by an empirical mean and adds an additional (Hoeffding-style) bonus term.

A learning algorithm runs for K episodes. In episode k , the learner selects a (possibly history-dependent) policy $\pi^k = \{\pi_h^k\}_{h=0}^{H-1}$ based on the past history $\mathcal{H}_{<k}$, then executes it to generate a trajectory

$$s_0^k = s_0, \quad a_h^k = \pi_h^k(s_h^k), \quad s_{h+1}^k \sim P_h^*(\cdot \mid s_h^k, a_h^k), \quad h = 0, \dots, H-1.$$

For any policy π , define the value functions in the true MDP as

$$V_h^\pi(s) := \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(s_t, a_t) \mid s_h = s, \pi, P^* \right], \quad V_h^*(s) := \max_{\pi} V_h^\pi(s).$$

Since rewards lie in $[0, 1]$, we have $0 \leq V_h^\pi(s) \leq H - h$ for all (h, s, π) .

It will be useful later to write expectations along a trajectory using the *stage- h occupancy measure* of a policy π :

$$d_h^\pi(s, a) := \Pr(s_h = s, a_h = a \mid s_0, \pi, P^*), \quad h = 0, \dots, H-1.$$

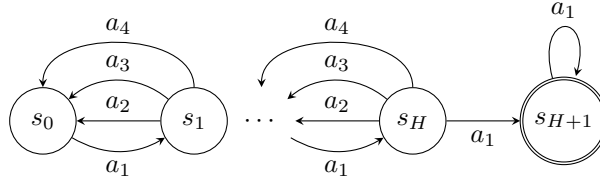


Figure 6.1: A deterministic, chain MDP of length $H + 2$. Rewards are 0 everywhere other than $r(s_{H+1}, a_1) = 1$. In this example we have $A = 4$.

Regret. We measure performance using the (pseudo-)regret over K episodes:

$$R_K := \mathbb{E} \left[\sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right].$$

Equivalently, since $\mathbb{E}[\sum_{h=0}^{H-1} r_h(s_h^k, a_h^k) \mid \pi^k] = V_0^{\pi^k}(s_0)$, we may write

$$R_K = \mathbb{E} \left[KV_0^*(s_0) - \sum_{k=0}^{K-1} \sum_{h=0}^{H-1} r_h(s_h^k, a_h^k) \right].$$

(When rewards are stochastic, one can relate pseudo-regret to realized cumulative regret via Hoeffding–Azuma; see Appendix A.)

6.2 Why exploration must be strategic

Consider the chain MDP in Figure 10.1. From the start state s_0 , there is a unique action that moves one step to the right; all other actions move left or stay. A uniformly random policy that picks actions uniformly from $\mathcal{A}$ at every step reaches the rewarding state s_{H+1} only if it chooses the “move right” action at each of the H steps. This happens with probability $(1/A)^H$, which is exponentially small in the horizon.

In particular, the expected number of episodes before such a policy reaches the rewarding state even once is on the order of A^H . This illustrates a key point: in online RL, the learner must sometimes *plan* to reach informative states. Exploration is not only about adding random noise to actions—it must be driven by which parts of the MDP are currently uncertain and how to reach them.

6.3 Algorithm: UCB Value Iteration (UCBVI)

UCBVI (see Algorithm 6) is an “optimism in the face of uncertainty” algorithm, directly analogous to UCB for multi-armed bandits (Chapter 5). The learner:

1. builds an empirical transition model from the trajectory data collected so far;
2. adds an exploration bonus to encourage actions with high uncertainty;
3. plans optimistically in this modified model via value iteration; and
4. executes the resulting greedy policy for one episode.

Algorithm 6 UCBVI (episodic, tabular)

Input: failure probability δ

- 1: Initialize $N_h^0(s, a) \leftarrow 0$ and $N_h^0(s, a, s') \leftarrow 0$ for all h, s, a, s' .
 - 2: **for** $k = 0, 1, \dots, K - 1$ **do**
 - 3: Form $\hat{P}_h^k(\cdot | s, a)$ from counts (arbitrary if $N_h^k(s, a) = 0$).
 - 4: Set bonuses $b_h^k(s, a)$ by (6.1).
 - 5: Compute $(\hat{V}_h^k, \hat{Q}_h^k, \pi_h^k)_{h=0}^{H-1}$ by (6.2).
 - 6: Execute π^k : $s_0^k = s_0$; for $h = 0, \dots, H - 1$ take $a_h^k = \pi_h^k(s_h^k)$ and sample $s_{h+1}^k \sim P_h^*(\cdot | s_h^k, a_h^k)$.
 - 7: Update counts: $N_h^{k+1}(\cdot) \leftarrow N_h^k(\cdot)$ then increment the visited $(s_h^k, a_h^k, s_{h+1}^k)$.
 - 8: **end for**
-

Counts and empirical transitions. For each stage $h \in \{0, \dots, H - 1\}$ and each $(s, a, s') \in \mathcal{S} \times \mathcal{A} \times \mathcal{S}$, define the pre-episode- k counts

$$N_h^k(s, a) := \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}, \quad N_h^k(s, a, s') := \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\}.$$

Define the empirical transition model $\hat{P}_h^k(\cdot | s, a)$ by

$$\hat{P}_h^k(s' | s, a) := \begin{cases} \frac{N_h^k(s, a, s')}{N_h^k(s, a)} & \text{if } N_h^k(s, a) > 0, \\ \text{any distribution on } \mathcal{S} & \text{if } N_h^k(s, a) = 0. \end{cases}$$

When $N_h^k(s, a) = 0$, the model $\hat{P}_h^k(\cdot | s, a)$ is arbitrary; the bonus term defined next ensures that such state-action pairs are treated as highly uncertain.

Exploration bonus. Fix a failure probability $\delta \in (0, 1)$ and define the log factor

$$L := \ln\left(\frac{2SAHK}{\delta}\right).$$

For each episode k , stage h , and (s, a) , define

$$b_h^k(s, a) := \min \left\{ H - h, 2H \sqrt{\frac{L}{N_h^k(s, a) + 1}} \right\}. \quad (6.1)$$

The bonus is large when (s, a) has been visited only a few times and decays like $1/\sqrt{N_h^k(s, a)}$ as more data are collected. The clipping by $H - h$ matches the maximum possible remaining return from stage h .

Planning (optimistic value iteration). Given the empirical model $\{\hat{P}_h^k\}$ and bonuses $\{b_h^k\}$, UCBVI performs a backward dynamic program. Initialize $\hat{V}_H^k(\cdot) \equiv 0$ and for $h = H - 1, \dots, 0$ compute

$$\begin{aligned} \hat{Q}_h^k(s, a) &:= r_h(s, a) + b_h^k(s, a) + \sum_{s' \in \mathcal{S}} \hat{P}_h^k(s' | s, a) \hat{V}_{h+1}^k(s'), \\ \hat{V}_h^k(s) &:= \min \left\{ H - h, \max_{a \in \mathcal{A}} \hat{Q}_h^k(s, a) \right\}, \quad \pi_h^k(s) \in \arg \max_{a \in \mathcal{A}} \hat{Q}_h^k(s, a). \end{aligned} \quad (6.2)$$

Thus π^k is greedy with respect to the optimistic Q -values. The truncation $\hat{V}_h^k(s) \leq H - h$ simply enforces the natural range of values under bounded rewards.

Execution and updates. Finally, execute π^k for one episode to obtain $(s_h^k, a_h^k, s_{h+1}^k)_{h=0}^{H-1}$, then update the counts and repeat.

6.4 A simple regret analysis

This section presents a warm-up regret analysis of UCBVI that yields a bound of order $\tilde{O}(H^2 S \sqrt{AK})$. The proof follows a recurring pattern in regret analyses (as in UCB in Chapter 5) for optimistic reinforcement learning algorithms: we first establish a time-uniform, high-probability concentration event controlling the empirical transition estimates; we then use this event to show *optimism*, namely that the value functions produced by optimistic planning dominate the optimal value function V^* ; next we derive a one-episode *decomposition* that upper bounds the suboptimality of the executed policy by a sum of exploration bonuses and transition-estimation error terms along the episode; finally, we apply a simple counting (potential) argument to control the total contribution of these terms over time via a bound on $\sum_{k,h} 1/\sqrt{\text{visits}}$ along the realized trajectories.

The dependence on S in Theorem 6.1 is not optimal: the leading term scales like $S\sqrt{AK}$ rather than $\sqrt{SAK}$. A refined analysis in a Section 6.5 removes this extra factor by controlling transition errors more sharply than the crude bound $\|\hat{P} - P^*\|_1 \cdot \|V\|_\infty$.

Theorem 6.1 (UCBVI regret bound (simple analysis)). *Run UCBVI (Algorithm 6) with bonuses (6.1). Then for any $\delta \in (0, 1)$, letting $L := \ln(\frac{2SAHK}{\delta})$,*

$$R_K \leq O\left(H^2 S \sqrt{AK \cdot L}\right) + \delta KH.$$

In particular, setting $\delta = 1/(KH)$ yields

$$R_K = \tilde{O}\left(H^2 S \sqrt{AK}\right).$$

When is this nontrivial? Since $0 \leq V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq H$, we always have the trivial bound $R_K \leq KH$. Theorem 6.1 becomes smaller than KH once $K \gtrsim H^2 S^2 A$ (up to logs), reflecting that this warm-up argument is still “model-size” limited.

6.4.1 A time-uniform transition concentration event

We first record a time-uniform ℓ_1 concentration bound for empirical transitions. The key point is that $N_h^k(s, a)$ is random and adaptively chosen; to avoid stopping-time issues we union bound over all sample sizes $n \in \{1, \dots, K\}$.

SK: really don't like using Xi (feel people forget the name of the letter). so just using c instead.

Lemma 6.2 (Uniform ℓ_1 transition concentration). *Fix $\delta \in (0, 1)$ and let $L = \ln(2SAHK/\delta)$. With probability at least $1 - \delta/2$, simultaneously for all $h \in [H]$, all $(s, a) \in \mathcal{S} \times \mathcal{A}$, and all episodes k with $N_h^k(s, a) \geq 1$,*

$$\|\hat{P}_h^k(\cdot \mid s, a) - P_h^*(\cdot \mid s, a)\|_1 \leq 2\sqrt{\frac{SL}{N_h^k(s, a)}}.$$

Consequently, on the same event, for any (possibly data-dependent) $f : \mathcal{S} \rightarrow [0, H]$,

$$\left| \left(\hat{P}_h^k(\cdot \mid s, a) - P_h^*(\cdot \mid s, a) \right)^\top f \right| \leq 2H\sqrt{\frac{SL}{N_h^k(s, a)}}.$$

Proof. Fix (h, s, a) and consider the empirical distribution after exactly n i.i.d. samples from $P_h^*(\cdot \mid s, a)$; denote it by $\hat{P}_{h,n}(\cdot \mid s, a)$. (One can couple the online interaction to an infinite i.i.d. sequence for each (h, s, a) , revealing the n -th sample on the n -th visit; this makes the “fixed- n ” view rigorous.)

By Proposition A.8 (Appendix), with probability at least $1 - \delta'$,

$$\|\hat{P}_{h,n}(\cdot \mid s, a) - P_h^*(\cdot \mid s, a)\|_1 \leq \sqrt{S} \left(\frac{1}{\sqrt{n}} + \sqrt{\frac{\ln(1/\delta')}{2n}} \right) \leq 2\sqrt{\frac{S \ln(1/\delta')}{n}},$$

where the last inequality uses $\ln(1/\delta') \geq \ln 2$ (since $\delta' \leq 1/2$ below). Choose $\delta' := \delta/(2SAHK)$ and union bound over all (h, s, a) and all $n \in \{1, \dots, K\}$. If $N_h^k(s, a) = n$, then $\hat{P}_h^k(\cdot \mid s, a) = \hat{P}_{h,n}(\cdot \mid s, a)$, so the stated bound holds simultaneously for all episodes k .

The final display follows from Hölder: $|(\hat{P} - P)^\top f| \leq \|\hat{P} - P\|_1 \|f\|_\infty \leq H \|\hat{P} - P\|_1$. $\square$

Next we need a sharper bound against the *deterministic* optimal value function (to prove optimism without paying a $\sqrt{S}$ factor).

Lemma 6.3 (Model error against V^*). *Fix $\delta \in (0, 1)$ and let $L = \ln(2SAHK/\delta)$. With probability at least $1 - \delta/2$, simultaneously for all $h \in [H]$, all (s, a) , and all episodes k with $N_h^k(s, a) \geq 1$,*

$$\left| (\hat{P}_h^k(\cdot \mid s, a) - P_h^*(\cdot \mid s, a))^\top V_{h+1}^* \right| \leq H \sqrt{\frac{L}{N_h^k(s, a)}}.$$

Proof. Fix (h, s, a) and consider n i.i.d. samples $s'_1, \dots, s'_n \sim P_h^*(\cdot \mid s, a)$. Let $Y_i := V_{h+1}^*(s'_i) \in [0, H]$, so $\mathbb{E}[Y_i] = P_h^*(\cdot \mid s, a)^\top V_{h+1}^*$. By Hoeffding’s inequality (Lemma A.1),

$$\Pr \left(\left| \frac{1}{n} \sum_{i=1}^n Y_i - \mathbb{E}[Y_i] \right| \geq \epsilon \right) \leq 2 \exp \left(-\frac{2n\epsilon^2}{H^2} \right).$$

Choosing $\epsilon = H \sqrt{\ln(4SAHK/\delta)/(2n)}$ makes the RHS at most $\delta/(2SAHK)$. Since $\ln(4SAHK/\delta) \leq 2 \ln(2SAHK/\delta) = 2L$ (using $L \geq \ln 2$), this ϵ is at most $H \sqrt{L/n}$. Union bound over all (h, s, a) and $n \in \{1, \dots, K\}$, and translate from n to the random count $N_h^k(s, a)$ as in Lemma 6.2. $\square$

Let $\mathcal{E}_{\text{model}}$ denote the intersection of the events in Lemmas 6.2 and 6.3. Then $\Pr(\mathcal{E}_{\text{model}}) \geq 1 - \delta$.

6.4.2 Optimism

Lemma 6.4 (Optimism). *On $\mathcal{E}_{\text{model}}$, for every episode k , every stage h , and every state s ,*

$$\hat{V}_h^k(s) \geq V_h^*(s).$$

In particular, $\hat{V}_0^k(s_0) \geq V_0^(s_0)$ for all k .*

Proof. We prove by backward induction on h . At $h = H$, $\hat{V}_H^k \equiv 0 = V_H^*$.

Assume $\hat{V}_{h+1}^k(s) \geq V_{h+1}^*(s)$ for all s . Fix a state s at stage h .

If $\hat{V}_h^k(s) = H - h$, then $\hat{V}_h^k(s) \geq V_h^*(s)$ holds since $V_h^*(s) \leq H - h$.

Otherwise, $\widehat{V}_h^k(s) < H - h$. In this case the truncation in (6.2) is inactive, so $\widehat{V}_h^k(s) = \max_a \widehat{Q}_h^k(s, a)$ and therefore $\widehat{Q}_h^k(s, a) < H - h$ for all a . In particular, no action has a clipped bonus at (h, s) , so for every a we have

$$b_h^k(s, a) = 2H \sqrt{\frac{L}{N_h^k(s, a) + 1}} \quad \text{and hence } N_h^k(s, a) \geq 1.$$

Let $a \in \mathcal{A}$ be arbitrary. Using Lemma 6.3 and the induction hypothesis,

$$\begin{aligned} Q_h^*(s, a) &= r_h(s, a) + P_h^*(\cdot | s, a)^\top V_{h+1}^* \\ &\leq r_h(s, a) + \widehat{P}_h^k(\cdot | s, a)^\top V_{h+1}^* + H \sqrt{\frac{L}{N_h^k(s, a)}} \\ &\leq r_h(s, a) + \widehat{P}_h^k(\cdot | s, a)^\top \widehat{V}_{h+1}^k + H \sqrt{\frac{L}{N_h^k(s, a)}} \\ &\leq r_h(s, a) + \widehat{P}_h^k(\cdot | s, a)^\top \widehat{V}_{h+1}^k + b_h^k(s, a) = \widehat{Q}_h^k(s, a), \end{aligned}$$

where the third line uses $\widehat{V}_{h+1}^k \geq V_{h+1}^*$ and $\widehat{P}_h^k$ being a distribution, and the fourth line uses $b_h^k(s, a) \geq H \sqrt{L/N_h^k(s, a)}$ since $N_h^k(s, a) + 1 \leq 2N_h^k(s, a)$.

Taking maxima over a yields

$$V_h^*(s) = \max_a Q_h^*(s, a) \leq \max_a \widehat{Q}_h^k(s, a) = \widehat{V}_h^k(s),$$

completing the induction. $\square$

6.4.3 A one-episode decomposition

Recall the stage- h occupancy measure of a policy π in the true MDP:

$$d_h^\pi(s, a) := \Pr(s_h = s, a_h = a | s_0, \pi, P^*).$$

Lemma 6.5 (One-episode decomposition). *For any episode k , the policy π^k returned by UCBVI satisfies*

$$\widehat{V}_0^k(s_0) - V_0^{\pi^k}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + (\widehat{P}_h^k(\cdot | s_h, a_h) - P_h^*(\cdot | s_h, a_h))^\top \widehat{V}_{h+1}^k \right].$$

Proof. Fix episode k and abbreviate $\pi = \pi^k$, $\widehat{V}_h = \widehat{V}_h^k$, $\widehat{P}_h = \widehat{P}_h^k$, $b_h = b_h^k$. Define $\Delta_h(s) := \widehat{V}_h(s) - V_h^\pi(s)$, so $\Delta_H \equiv 0$.

For any state s at stage h , since $\pi_h(s) \in \arg \max_a \widehat{Q}_h(s, a)$ and $\widehat{V}_h(s) = \min\{H-h, \max_a \widehat{Q}_h(s, a)\} \leq \widehat{Q}_h(s, \pi_h(s))$, we have

$$\widehat{V}_h(s) \leq r_h(s, \pi_h(s)) + b_h(s, \pi_h(s)) + \widehat{P}_h(\cdot | s, \pi_h(s))^\top \widehat{V}_{h+1}.$$

On the other hand,

$$V_h^\pi(s) = r_h(s, \pi_h(s)) + P_h^*(\cdot | s, \pi_h(s))^\top V_{h+1}^\pi.$$

Subtracting gives

$$\Delta_h(s) \leq b_h(s, \pi_h(s)) + (\widehat{P}_h(\cdot | s, \pi_h(s)) - P_h^*(\cdot | s, \pi_h(s)))^\top \widehat{V}_{h+1} + P_h^*(\cdot | s, \pi_h(s))^\top \Delta_{h+1}.$$

Taking expectations under the trajectory distribution induced by π in the true MDP and unrolling (tower property) from $h = 0$ to $H - 1$, using $\Delta_H \equiv 0$, yields the stated bound for $\Delta_0(s_0) = \widehat{V}_0(s_0) - V_0^\pi(s_0)$. $\square$

6.4.4 A counting lemma

Lemma 6.6 (Counting / potential bound). *For any sequence of K trajectories $\{(s_h^k, a_h^k)\}_{h=0}^{H-1}$, we have*

$$\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k) + 1}} \leq 2H\sqrt{SAK}.$$

Proof. Fix a stage h and a state–action pair (s, a) . Let $n := N_h^K(s, a)$ be the total number of visits to (s, a) at stage h over all K episodes. Each time (s, a) is visited, the denominator takes values $\sqrt{1}, \sqrt{2}, \dots, \sqrt{n}$, so

$$\sum_{k: (s_h^k, a_h^k) = (s, a)} \frac{1}{\sqrt{N_h^k(s, a) + 1}} \leq \sum_{i=1}^n \frac{1}{\sqrt{i}} \leq 2\sqrt{n}.$$

Summing over all (s, a) and using Cauchy–Schwarz,

$$\sum_{k=0}^{K-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k) + 1}} \leq 2 \sum_{s, a} \sqrt{N_h^K(s, a)} \leq 2 \sqrt{SA \sum_{s, a} N_h^K(s, a)} = 2\sqrt{SA \cdot K}.$$

Finally sum over $h = 0, \dots, H - 1$ to get $2H\sqrt{SAK}$. □

6.4.5 Proof of Theorem 6.1

Proof. Fix $\delta \in (0, 1)$ and let $\mathcal{E}_{\text{model}}$ be as above, with $\Pr(\mathcal{E}_{\text{model}}) \geq 1 - \delta$.

First, we will reduce the regret to the optimistic planning gap. On $\mathcal{E}_{\text{model}}$, optimism (Lemma 6.4) implies

$$V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq \widehat{V}_0^k(s_0) - V_0^{\pi^k}(s_0).$$

We now decompose one episode and bound the model error term. Apply Lemma 6.5 and upper bound the model error term by absolute value:

$$\widehat{V}_0^k(s_0) - V_0^{\pi^k}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + \left| (\widehat{P}_h^k(\cdot | s_h, a_h) - P_h^*(\cdot | s_h, a_h))^\top \widehat{V}_{h+1}^k \right| \right].$$

Since $\|\widehat{V}_{h+1}^k\|_\infty \leq H$, Lemma 6.2 gives (for $N_h^k(s_h, a_h) \geq 1$)

$$\left| (\widehat{P}_h^k - P_h^*)^\top \widehat{V}_{h+1}^k \right| \leq 2H \sqrt{\frac{SL}{N_h^k(s_h, a_h)}} \leq 2\sqrt{2} H \sqrt{\frac{SL}{N_h^k(s_h, a_h) + 1}},$$

where we used $N + 1 \leq 2N$ for $N \geq 1$. If $N_h^k(s_h, a_h) = 0$, then trivially $\left| (\widehat{P}_h^k - P_h^*)^\top \widehat{V}_{h+1}^k \right| \leq \|\widehat{V}_{h+1}^k\|_\infty \leq H$, and since $L \geq \ln 2$ we still have $H \leq 2\sqrt{2} H \sqrt{SL/(0 + 1)}$. Thus, in all cases,

$$\left| (\widehat{P}_h^k(\cdot | s_h, a_h) - P_h^*(\cdot | s_h, a_h))^\top \widehat{V}_{h+1}^k \right| \leq 2\sqrt{2} H \sqrt{\frac{SL}{N_h^k(s_h, a_h) + 1}}.$$

Also, by the bonus definition (6.1) and $S \geq 1$,

$$b_h^k(s_h, a_h) \leq 2H \sqrt{\frac{L}{N_h^k(s_h, a_h) + 1}} \leq 2H \sqrt{\frac{SL}{N_h^k(s_h, a_h) + 1}}.$$

Combining the last two displays yields, on $\mathcal{E}_{\text{model}}$,

$$V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq (2 + 2\sqrt{2})H\sqrt{SL} \cdot \mathbb{E} \left[\sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k) + 1}} \middle| \mathcal{H}_{<k} \right].$$

Next, we sum over episodes using the counting lemma. Summing over k and taking total expectation, then applying Lemma 6.6, gives

$$\mathbb{E} \left[\mathbf{1}\{\mathcal{E}_{\text{model}}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right] \leq (2 + 2\sqrt{2})H\sqrt{SL} \cdot \mathbb{E} \left[\sum_{k,h} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k) + 1}} \right] \leq O(H^2 S \sqrt{AK \cdot L}).$$

On the complement $\bar{\mathcal{E}}_{\text{model}}$, use $0 \leq V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq H$ to obtain

$$\mathbb{E} \left[\mathbf{1}\{\bar{\mathcal{E}}_{\text{model}}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right] \leq \Pr(\bar{\mathcal{E}}_{\text{model}}) \cdot KH \leq \delta KH.$$

Combining completes the proof. $\square$

6.5 An Improved Regret Bound

The simple analysis in Theorem 6.1 bounds the model error term $(\hat{P}_h^k - P_h^*)^\top \hat{V}_{h+1}^k$ by $\|\hat{P}_h^k - P_h^*\|_1 \cdot \|\hat{V}_{h+1}^k\|_\infty$, which incurs an extraneous $\sqrt{S}$ factor. This section refines the argument and improves the leading dependence to $\tilde{O}(H^2 \sqrt{SAK})$, at the cost of an additional lower-order term. The leading dependence on (S, A, K) is optimal up to factors depending on H (see Chapter 2 for related sample-complexity results with a generative model). We discuss minimax regret rates and sharper bounds in Section 6.6.

Theorem 6.7 (UCBVI regret bound (refined analysis)). *Run UCBVI (Algorithm 6) with bonuses (6.1). There exists a universal constant $C > 0$ such that for any $\delta \in (0, 1)$, letting*

$$L := \ln\left(\frac{2S^2 AHK}{\delta}\right),$$

the expected regret satisfies

$$R_K \leq C \left(H^2 \sqrt{SAK \cdot L} + H^3 S^2 A \cdot L \cdot \ln(1 + K) \right) + \delta KH.$$

In particular, setting $\delta = 1/(KH)$ yields $R_K = \tilde{O}(H^2 \sqrt{SAK} + H^3 S^2 A)$.

The proof follows the same template as in Section 6.4: optimism, a one-episode decomposition, and counting. The main new ingredient is a Bernstein/Freedman-style bound that controls $(\hat{P}_h^k - P_h^*)^\top (\hat{V}_{h+1}^k - V_{h+1}^*)$ without paying $\sqrt{S}$ in the leading term.

6.5.1 A Bernstein-style transition bound

We begin with a pointwise empirical Bernstein bound for each next-state probability. It can be proved by applying Freedman's inequality (Lemma A.7) to the martingale differences associated with indicator variables and then taking a union bound over (h, s, a, s', k) .

Lemma 6.8 (Pointwise empirical Bernstein bound). *Fix $\delta \in (0, 1)$ and let $L = \ln(2S^2 AHK/\delta)$. With probability at least $1 - \delta$, simultaneously for all stages $h \in \{0, \dots, H - 1\}$, all $(s, a) \in \mathcal{S} \times \mathcal{A}$, all $s' \in \mathcal{S}$, and all episodes $k \in \{0, \dots, K - 1\}$,*

$$|\hat{P}_h^k(s' | s, a) - P_h^*(s' | s, a)| \leq 2\sqrt{\frac{P_h^*(s' | s, a) L}{N_h^k(s, a) + 1}} + \frac{4L}{N_h^k(s, a) + 1}.$$

Proof. Fix (h, s, a, s') and abbreviate $p := P_h^*(s' | s, a)$. For episode index $i \in \{0, \dots, K - 1\}$ define

$$X_i := \mathbf{1}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\} - \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} p.$$

Then $(X_i)_{i \geq 0}$ is a martingale difference sequence with $|X_i| \leq 1$, and

$$\sum_{i=0}^{k-1} X_i = N_h^k(s, a, s') - p N_h^k(s, a).$$

Moreover,

$$\sum_{i=0}^{k-1} \mathbb{E}[X_i^2 | \mathcal{H}_{<i}] = p(1-p) N_h^k(s, a) \leq p N_h^k(s, a).$$

Freedman's inequality (Lemma A.7) and a standard inversion imply that with probability at least $1 - \delta'$, simultaneously for all k ,

$$|N_h^k(s, a, s') - p N_h^k(s, a)| \leq \sqrt{2p N_h^k(s, a) \ln(1/\delta')} + \frac{2}{3} \ln(1/\delta').$$

Divide by $N_h^k(s, a) \vee 1$ and use $N_h^k(s, a) \vee 1 \geq (N_h^k(s, a) + 1)/2$ to obtain the stated bound (with slightly loosened constants). Finally, set $\delta' = \delta/(S^2 AHK)$ and take a union bound over all (h, s, a, s') . $\square$

A convenient consequence is a *self-bounding* inequality for linear functionals of the form $(\hat{P} - P^*)^\top f$ when f is nonnegative. This is the key step that ultimately yields the improved $\sqrt{SAK}$ dependence.

Lemma 6.9 (Self-bounding transition error). *On the event of Lemma 6.8, for all h, k, s, a and all functions $f : \mathcal{S} \rightarrow [0, H]$,*

$$\left| (\hat{P}_h^k(\cdot | s, a) - P_h^*(\cdot | s, a))^\top f \right| \leq \frac{P_h^*(\cdot | s, a)^\top f}{H} + c_h^k(s, a), \quad c_h^k(s, a) := \frac{c H^2 S L}{N_h^k(s, a) + 1},$$

for a universal constant $c > 0$.

Proof. Fix (h, k, s, a) and abbreviate $\hat{P} := \hat{P}_h^k(\cdot | s, a)$, $P := P_h^*(\cdot | s, a)$, and $N := N_h^k(s, a)$. Using Lemma 6.8 and $f \geq 0$,

$$|(\hat{P} - P)^\top f| \leq \sum_{s'} f(s') |\hat{P}(s') - P(s')| \leq T_1 + T_2,$$

where

$$T_1 := \sum_{s'} f(s') \cdot 2\sqrt{\frac{P(s') L}{N + 1}}, \quad T_2 := \sum_{s'} f(s') \cdot \frac{4L}{N + 1}.$$

For the linear term, $T_2 \leq \frac{4SHL}{N+1} \leq \frac{4H^2SL}{N+1}$ (since $H \geq 1$). For the square-root term, by Cauchy–Schwarz,

$$T_1 \leq 2\sqrt{\frac{L}{N+1}} \sum_{s'} \sqrt{P(s') f(s')^2} \leq 2\sqrt{\frac{SL}{N+1}} \sqrt{\sum_{s'} P(s') f(s')^2} = 2\sqrt{\frac{SL}{N+1}} \sqrt{P f^2}.$$

Since $0 \leq f \leq H$, we have $f^2 \leq Hf$ and thus $Pf^2 \leq H Pf$, giving

$$T_1 \leq 2\sqrt{\frac{HSL}{N+1}}\sqrt{Pf} = \sqrt{\frac{4H^2SL}{N+1} \cdot \frac{Pf}{H}}.$$

Applying $\sqrt{uv} \leq \frac{u+v}{2}$ with $u = (Pf)/H$ and $v = 4H^2SL/(N+1)$ yields

$$T_1 \leq \frac{Pf}{H} + \frac{2H^2SL}{N+1}.$$

Combining the bounds on T_1 and T_2 proves the claim (absorbing constants into c). $\square$

6.5.2 A recursive control of the optimistic gap

For each episode k define the optimistic gap

$$\Delta_h^k(s) := \widehat{V}_h^k(s) - V_h^*(s) \geq 0,$$

where nonnegativity holds on $\mathcal{E}_{\text{model}}$ by Lemma 6.4. The next lemma shows that Δ_h^k satisfies a useful recursion.

Lemma 6.10 (Gap recursion). *Assume $\mathcal{E}_{\text{model}}$ holds (so $\widehat{V}^k \geq V^*$) and Lemma 6.9 holds. Then for every episode k , stage h , and state s , letting $a = \pi_h^k(s)$,*

$$\Delta_h^k(s) \leq 2b_h^k(s, a) + c_h^k(s, a) + \left(1 + \frac{1}{H}\right) \mathbb{E}_{s' \sim P_h^*(\cdot | s, a)}[\Delta_{h+1}^k(s')].$$

Consequently, for every h and s ,

$$\begin{aligned} \Delta_h^k(s) &\leq \mathbb{E} \left[\sum_{\tau=h}^{H-1} \left(1 + \frac{1}{H}\right)^{\tau-h} (2b_\tau^k(s_\tau, a_\tau) + c_\tau^k(s_\tau, a_\tau)) \mid s_h = s, \pi^k \right] \\ &\leq e \cdot \mathbb{E} \left[\sum_{\tau=h}^{H-1} (2b_\tau^k(s_\tau, a_\tau) + c_\tau^k(s_\tau, a_\tau)) \mid s_h = s, \pi^k \right]. \end{aligned}$$

Proof. Fix (k, h, s) and let $a = \pi_h^k(s)$. Using $\widehat{V}_h^k(s) \leq \widehat{Q}_h^k(s, a)$ and $V_h^*(s) \geq Q_h^*(s, a)$,

$$\Delta_h^k(s) \leq \widehat{Q}_h^k(s, a) - Q_h^*(s, a) = b_h^k(s, a) + \widehat{P}_h^k(\cdot | s, a)^\top \widehat{V}_{h+1}^k - P_h^*(\cdot | s, a)^\top V_{h+1}^*.$$

Write $\widehat{V}_{h+1}^k = V_{h+1}^* + \Delta_{h+1}^k$ and expand:

$$\Delta_h^k(s) \leq b_h^k(s, a) + (\widehat{P}_h^k - P_h^*)^\top V_{h+1}^* + \widehat{P}_h^k^\top \Delta_{h+1}^k.$$

On $\mathcal{E}_{\text{model}}$, Lemma 6.3 bounds the middle term, and by the choice of bonus (and the fact that $\ln(2S^2 AHK/\delta) \leq 2 \ln(2SAHK/\delta)$), it can be absorbed into $b_h^k(s, a)$ up to a constant factor. This yields an additional $+b_h^k(s, a)$ term.

For the remaining term, write

$$\widehat{P}_h^k^\top \Delta_{h+1}^k = P_h^{*\top} \Delta_{h+1}^k + (\widehat{P}_h^k - P_h^*)^\top \Delta_{h+1}^k,$$

and bound the last inner product by Lemma 6.9 with $f = \Delta_{h+1}^k$. Since $P_h^{*\top} \Delta_{h+1}^k = \mathbb{E}_{s' \sim P_h^*(\cdot | s, a)}[\Delta_{h+1}^k(s')]$, this gives the stated one-step recursion.

Unrolling the recursion gives the weighted sum, and the final inequality uses $(1 + \frac{1}{H})^m \leq e$ for all integers $m \geq 0$. $\square$

6.5.3 Per-episode regret bound and summation

The refined recursion allows us to bound the one-episode regret by the same terms $2b + c$ (up to a constant factor), without ever using $\|\hat{P} - P^*\|_1 \cdot \|\hat{V}\|_\infty$.

Lemma 6.11 (Per-episode refined bound). *Assume $\mathcal{E}_{\text{model}}$ and Lemmas 6.9–6.10 hold. Then for every episode k ,*

$$V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq e \sum_{h=0}^{H-1} \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi^k}} [2b_h^k(s_h, a_h) + c_h^k(s_h, a_h)].$$

Proof. Fix episode k and abbreviate $\pi = \pi^k$. On $\mathcal{E}_{\text{model}}$, optimism (Lemma 6.4) implies $V_0^*(s_0) - V_0^\pi(s_0) \leq \hat{V}_0^k(s_0) - V_0^\pi(s_0)$. Apply the one-episode decomposition (Lemma 6.5):

$$\hat{V}_0^k(s_0) - V_0^\pi(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [b_h^k + (\hat{P}_h^k - P_h^*)^\top \hat{V}_{h+1}^k].$$

Write $\hat{V}_{h+1}^k = V_{h+1}^* + \Delta_{h+1}^k$ to obtain

$$(\hat{P}_h^k - P_h^*)^\top \hat{V}_{h+1}^k = (\hat{P}_h^k - P_h^*)^\top V_{h+1}^* + (\hat{P}_h^k - P_h^*)^\top \Delta_{h+1}^k.$$

The first term is bounded by b_h^k using Lemma 6.3 (and the bonus definition), so

$$\hat{V}_0^k(s_0) - V_0^\pi(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [2b_h^k + (\hat{P}_h^k - P_h^*)^\top \Delta_{h+1}^k].$$

Next, use Lemma 6.9 with $f = \Delta_{h+1}^k$:

$$(\hat{P}_h^k - P_h^*)^\top \Delta_{h+1}^k \leq \frac{P_h^{*\top} \Delta_{h+1}^k}{H} + c_h^k.$$

Let $\alpha_h^k(s, a) := 2b_h^k(s, a) + c_h^k(s, a)$. Combining the last two displays yields

$$\hat{V}_0^k(s_0) - V_0^\pi(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [\alpha_h^k] + \frac{1}{H} \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [P_h^{*\top} \Delta_{h+1}^k].$$

By the tower property, $\mathbb{E}_{d_h^\pi} [P_h^{*\top} \Delta_{h+1}^k] = \mathbb{E}[\Delta_{h+1}^k(s_{h+1})]$ under π . Using the unrolled bound in Lemma 6.10 and exchanging sums,

$$\begin{aligned} \frac{1}{H} \sum_{h=0}^{H-1} \mathbb{E}[\Delta_{h+1}^k(s_{h+1})] &\leq \frac{1}{H} \sum_{h=0}^{H-1} \sum_{\tau=h+1}^{H-1} \left(1 + \frac{1}{H}\right)^{\tau-h-1} \mathbb{E}_{d_\tau^\pi} [\alpha_\tau^k] \\ &= \sum_{\tau=0}^{H-1} \mathbb{E}_{d_\tau^\pi} [\alpha_\tau^k] \cdot \frac{1}{H} \sum_{j=0}^{\tau-1} \left(1 + \frac{1}{H}\right)^j \leq (e-1) \sum_{\tau=0}^{H-1} \mathbb{E}_{d_\tau^\pi} [\alpha_\tau^k], \end{aligned}$$

where we used the geometric-series identity and $(1 + 1/H)^\tau \leq e$. Substituting back gives

$$\hat{V}_0^k(s_0) - V_0^\pi(s_0) \leq e \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [\alpha_h^k],$$

which is the stated bound. $\square$

Proof of Theorem 6.7. Let $\mathcal{E}$ be the intersection of $\mathcal{E}_{\text{model}}$ and the event in Lemma 6.8, each invoked with failure probability $\delta/2$. Then $\Pr(\mathcal{E}) \geq 1 - \delta$, and on $\mathcal{E}$ all assumptions of Lemma 6.11 hold.

Decompose the expected regret as

$$R_K = \mathbb{E} \left[\mathbf{1}\{\mathcal{E}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right] + \mathbb{E} \left[\mathbf{1}\{\bar{\mathcal{E}}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right].$$

On $\mathcal{E}$, Lemma 6.11 and the law of total expectation yield

$$\mathbb{E} \left[\mathbf{1}\{\mathcal{E}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right] \leq e \mathbb{E} \left[\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} (2b_h^k(s_h^k, a_h^k) + c_h^k(s_h^k, a_h^k)) \right].$$

For the bonus term, using $b_h^k(s, a) \leq 2H\sqrt{L/(N_h^k(s, a) + 1)}$ and the counting lemma (Lemma 6.6),

$$\sum_{k,h} 2b_h^k(s_h^k, a_h^k) \leq 4H\sqrt{L} \sum_{k,h} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k) + 1}} \leq 8H^2\sqrt{SAK \cdot L}.$$

For the c term, since $c_h^k(s, a) = \frac{cH^2SL}{N_h^k(s, a) + 1}$,

$$\sum_{k,h} c_h^k(s_h^k, a_h^k) = cH^2SL \sum_{h=0}^{H-1} \sum_{s,a} \sum_{i=1}^{N_h^K(s,a)} \frac{1}{i} \leq cH^2SL \cdot (HSA) \cdot \ln(1 + K) = O(H^3S^2A \cdot L \cdot \ln(1 + K)).$$

On $\bar{\mathcal{E}}$, we use the trivial bound $0 \leq V_0^*(s_0) - V_0^{\pi^k}(s_0) \leq H$ to obtain

$$\mathbb{E} \left[\mathbf{1}\{\bar{\mathcal{E}}\} \sum_{k=0}^{K-1} (V_0^*(s_0) - V_0^{\pi^k}(s_0)) \right] \leq \Pr(\bar{\mathcal{E}}) \cdot KH \leq \delta KH.$$

Combining the bounds proves the theorem. $\square$

6.6 Bibliographic Remarks and Further Readings

This chapter studies strategic exploration in finite-horizon episodic tabular MDPs under the regret criterion. The main algorithmic template is *optimism + planning*: construct an empirical MDP from trajectory data, add an exploration bonus to encourage visiting uncertain state–action pairs, and plan in the resulting optimistic model. Our presentation and proofs follow Azar et al. [2017] (see also the related treatment of Dann et al. [2017]). We first give a simplified analysis of UCBVI yielding a regret bound of order $H^2S\sqrt{AK}$ (up to logarithmic factors), and then sketch a refined analysis that improves the leading dependence on S to obtain $H^2\sqrt{SAK}$ (again up to logarithmic factors), at the cost of an additional lower-order term.

The first provably sample-efficient exploration algorithms for finite MDPs were developed under the PAC criterion. Classic examples include E^3 [Kearns and Singh, 2002] and R-max [Brafman and Tennenholtz, 2002], which formalize optimism via either explicit “known/unknown” reasoning or optimistic models. The thesis of [Kakade, 2003] gives a broader treatment and refinements, including improved sample complexity bounds in the tabular setting.

In the regret framework, an influential starting point for continuing (average-reward) MDPs is UCRL2 [Jaksch et al., 2010], which established $\tilde{O}(\sqrt{T})$ regret as a function of the time horizon T (with additional dependence on structural parameters such as the diameter). The $\sqrt{T}$ scaling is minimax-optimal in T , while sharpening the dependence on other problem parameters has motivated substantial follow-up work.

For episodic finite-horizon MDPs (the setting of this chapter), Azar et al. [2017] gives near-minimax regret guarantees for UCBVI-type algorithms. A complementary perspective that unifies PAC and regret analyses via *Uniform-PAC* guarantees is developed by Dann et al. [2017].

There is also a long line of work on model-free learning. Early finite-sample analyses of Q-learning and related “indirect” methods appear in Kearns and Singh [1998] (in a generative model setting). On the PAC side, Strehl et al. [2006] provides a model-free exploration algorithm with provable sample-efficiency guarantees. For regret guarantees in episodic MDPs with model-free learning, Jin et al. [2018] shows that Q-learning with UCB-style exploration achieves $\tilde{O}(\sqrt{H^3 SAT})$ regret (where $T = KH$), matching the optimal $\sqrt{T}$ dependence and coming within a $\sqrt{H}$ factor (up to logs) of the best-known model-based rates.

An alternative to optimism is posterior sampling (Thompson sampling) for MDPs. Posterior Sampling for RL (PSRL) was studied by Osband et al. [2013], and further finite-horizon analyses and comparisons between posterior sampling and optimism are given by Osband and Van Roy [2017].

Lower bounds for regret in tabular reinforcement learning, and further discussion of the minimax benchmark, can be found in Azar et al. [2017], Osband and Van Roy [2016]. We also refer to Dann and Brunskill [2015] for lower bounds in closely related finite-horizon settings.

Chapter 7

Linearly Parameterized MDPs

This chapter studies strategic exploration in episodic finite-horizon MDPs under a *linear transition model*. The goal is to obtain regret bounds that scale polynomially in the feature dimension d (and the horizon H), without explicit dependence on the number of states.

The algorithmic template mirrors tabular UCBVI (Chapter 6): build an optimistic model from the data, plan via value iteration in that model, execute the resulting policy for one episode, and repeat. The main new ingredient is that we cannot estimate a full transition table; instead, we must estimate the one-step lookahead terms needed by dynamic programming through a regression primitive based on known features.

A recurring theme from Chapter 3 is that linear features lead to $\text{poly}(d)$ guarantees only under a *closure* condition: the regression targets created by dynamic programming must remain learnable from the available data (Bellman completeness). The linear transition model we assume here enforces a *stronger* version of this idea for the transition component: regardless of how nonlinear the value function may be, its one-step lookahead under the true dynamics can still be represented linearly in the features. This is what makes the least-squares steps inside optimistic value iteration genuinely well-specified, even though the targets depend on the algorithm's own value estimates.

A common strengthening of the model additionally assumes that rewards are also linear in the same features. Under that assumption, the *entire* Bellman backup (reward plus transition) is closed over a linear class, aligning exactly with the linear Bellman completeness condition from Chapter 3. In this chapter we keep rewards known to isolate the main statistical difficulty: learning transition effects from trajectory data.

7.1 Setting: online episodic learning and regret

We adopt the episodic protocol and regret criterion from Chapter 6. The MDP has horizon H , state space $\mathcal{S}$ (possibly very large but finite, so matrix notation is well-defined), action space $\mathcal{A}$, and a fixed start state s_0 . Transitions are stage-dependent: $\{P_h^*(\cdot \mid s, a)\}_{h=0}^{H-1}$.

Rewards are deterministic, known, and bounded:

$$r_h(s, a) \in [0, 1] \quad \text{for all } h \in \{0, \dots, H-1\}, (s, a) \in \mathcal{S} \times \mathcal{A}.$$

(Unknown and/or stochastic rewards can be handled by empirical means plus an additional Hoeffding-style bonus; we suppress this to keep the focus on learning the transitions.)

A learning algorithm runs for K episodes. In episode k , it chooses a (possibly history-dependent) nonstationary policy

$\pi^k = \{\pi_h^k\}_{h=0}^{H-1}$, then executes it to generate a trajectory

$$s_0^k = s_0, \quad a_h^k = \pi_h^k(s_h^k), \quad s_{h+1}^k \sim P_h^*(\cdot \mid s_h^k, a_h^k), \quad h = 0, \dots, H-1.$$

For any policy π , define the stage- h value in the true MDP as

$$V_h^\pi(s) := \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(s_t, a_t) \mid s_h = s, \pi, P^* \right], \quad V_h^*(s) := \max_{\pi} V_h^\pi(s).$$

We abbreviate $V^\pi := V_0^\pi(s_0)$ and $V^* := V_0^*(s_0)$.

The (pseudo-)regret over K episodes is

$$R_K := \mathbb{E} \left[\sum_{k=0}^{K-1} (V^* - V^{\pi^k}) \right].$$

We will also use the stage- h occupancy measure (defined as in Chapter 6): for a policy π ,

$$d_h^\pi(s, a) := \Pr(s_h = s, a_h = a \mid s_0, \pi, P^*).$$

7.2 Linear MDP model

We assume the transition kernel is *linear in known features*. Let $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ be a known feature map.

Assumption 7.1 (Linear transition model). There exists a feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ with $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$, and unknown parameters $\{\mu_h^*\}_{h=0}^{H-1}$, where each $\mu_h^* \in \mathbb{R}^{|\mathcal{S}| \times d}$, such that for all h and (s, a) ,

$$P_h^*(\cdot \mid s, a) = \mu_h^* \phi(s, a),$$

and the right-hand side is a valid probability distribution on $\mathcal{S}$.

Moreover, we assume the (normalization) bound

$$\|(\mu_h^*)^\top f\|_2 \leq \sqrt{d} \|f\|_\infty \quad \text{for all } f : \mathcal{S} \rightarrow \mathbb{R} \text{ and all } h \in \{0, \dots, H-1\}.$$

Assumption 7.1 induces a low-rank structure: viewing the transition model as a matrix mapping state-action pairs to next-state distributions, its rank is at most d . Tabular MDPs are recovered by taking ϕ to be the one-hot encoding of (s, a) , in which case $d = |\mathcal{S}||\mathcal{A}|$ and the model becomes unconstrained.

The only role of the operator-norm condition above is to control the size of certain linear coefficients that represent Bellman backups; it is a convenient normalization and can be replaced by a generic constant if desired.

7.3 A key property: Bellman backups are linear in features

A basic consequence of Assumption 7.1 is that, for any bounded $f : \mathcal{S} \rightarrow \mathbb{R}$, the conditional expectation $(P_h^* f)(s, a)$ is linear in $\phi(s, a)$.

Lemma 7.2 (Linearization of $P_h^* f$). *Fix a stage h and a function $f : \mathcal{S} \rightarrow \mathbb{R}$. Define*

$$w_{h,f}^* := (\mu_h^*)^\top f \in \mathbb{R}^d.$$

Then for all (s, a) ,

$$(P_h^* f)(s, a) := \mathbb{E}_{s' \sim P_h^*(\cdot | s, a)}[f(s')] = \phi(s, a)^\top w_{h,f}^*.$$

Moreover, if $\|f\|_\infty \leq H$ then $\|w_{h,f}^\|_2 \leq H\sqrt{d}$.*

Proof. By Assumption 7.1, $P_h^*(\cdot | s, a) = \mu_h^* \phi(s, a)$, hence

$$(P_h^* f)(s, a) = \sum_{s' \in \mathcal{S}} f(s') P_h^*(s' | s, a) = f^\top \mu_h^* \phi(s, a) = \phi(s, a)^\top (\mu_h^*)^\top f = \phi(s, a)^\top w_{h,f}^*.$$

The norm bound follows from the second part of Assumption 7.1: $\|w_{h,f}^*\|_2 = \|(\mu_h^*)^\top f\|_2 \leq \sqrt{d} \|f\|_\infty \leq H\sqrt{d}$. $\square$

If the transition parameters $\{\mu_h^*\}$ were known, the optimal values could be computed by backward dynamic programming:

$$V_H^* \equiv 0, \quad Q_h^*(s, a) = r_h(s, a) + (P_h^* V_{h+1}^*)(s, a), \quad V_h^*(s) = \max_{a \in \mathcal{A}} Q_h^*(s, a).$$

Lemma 7.2 shows that, even though V_{h+1}^* need not be linear, the function $(s, a) \mapsto (P_h^* V_{h+1}^*)(s, a)$ is always linear in $\phi(s, a)$. Thus, the central statistical problem is to estimate linear functionals $(P_h^* f)(s, a)$ accurately for the value functions f encountered during planning.

Remark (a strong form of completeness: all functions back up into the span). Lemma 7.2 can be read as the key “completeness” property behind this model. In Chapter 3, Bellman completeness was phrased as a closure requirement of the form $f \in \mathcal{F} \Rightarrow \mathcal{T}_h f \in \mathcal{F}$ for a chosen linear class $\mathcal{F}$. Here we assume something even stronger for the transition component: for each stage h , the map

$$f : \mathcal{S} \rightarrow \mathbb{R} \quad \mapsto \quad ((s, a) \mapsto (P_h^* f)(s, a))$$

takes every bounded f (not just linear ones) into $\text{span}(\phi)$. This is exactly what makes least-squares estimation viable even when the value functions we plug in — such as V_{h+1}^* or the optimistic iterates $\widehat{V}_{h+1}^k$ — are highly nonlinear (they involve max over actions and clipping). The algorithm never needs f itself to be linear; it only needs the one-step conditional expectation $P_h^* f$ to admit a linear representation in $\phi(s, a)$.

7.4 Lin-UCBVI: ridge regression and optimistic planning

We now define a ridge-regression primitive for estimating $(P_h^* f)(s, a)$ and use it to construct an optimistic value-iteration algorithm in the spirit of UCBVI.

7.4.1 Stage-wise ridge regression for $(P_h^* f)(s, a)$

Fix a regularization parameter $\lambda > 0$. For each episode k and stage h , define the pre-episode- k dataset at stage h :

$$D_h^k := \{(s_h^i, a_h^i, s_{h+1}^i)\}_{i=0}^{k-1}.$$

Let $x_h^i := \phi(s_h^i, a_h^i) \in \mathbb{R}^d$ and define the regularized design matrix

$$\Lambda_h^k := \lambda I + \sum_{i=0}^{k-1} x_h^i (x_h^i)^\top.$$

Given any function $f : \mathcal{S} \rightarrow \mathbb{R}$, define the ridge estimator of the linear coefficient $w_{h,f}^* = (\mu_h^*)^\top f$ by

$$\hat{w}_{h,f}^k := (\Lambda_h^k)^{-1} \sum_{i=0}^{k-1} x_h^i f(s_{h+1}^i), \quad (\hat{P}_h^k f)(s, a) := \phi(s, a)^\top \hat{w}_{h,f}^k.$$

The algorithm will only use $\hat{P}_h^k$ through such linear functionals $(\hat{P}_h^k f)(s, a)$.

7.4.2 Optimistic planning and policy selection

Fix a bonus scale $\beta > 0$ and define the (UCB-style) exploration bonus

$$b_h^k(s, a) := \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}} \quad \text{where} \quad \|u\|_{(\Lambda_h^k)^{-1}} := \sqrt{u^\top (\Lambda_h^k)^{-1} u}.$$

(We will later choose β using a uniform confidence bound over the random value functions produced by the algorithm.)

Define the stage-dependent clipping operator

$$\text{clip}_{[0, H-h]}(x) := \min\{H-h, \max\{0, x\}\}.$$

At the start of episode k , Lin-UCBVI computes optimistic Q - and value-functions by a backward recursion. Initialize $\hat{V}_H^k(\cdot) \equiv 0$, and for $h = H-1, \dots, 0$ define

$$\hat{Q}_h^k(s, a) := \text{clip}_{[0, H-h]}(r_h(s, a) + b_h^k(s, a) + (\hat{P}_h^k \hat{V}_{h+1}^k)(s, a)), \quad (7.1)$$

$$\hat{V}_h^k(s) := \max_{a \in \mathcal{A}} \hat{Q}_h^k(s, a), \quad \pi_h^k(s) \in \arg \max_{a \in \mathcal{A}} \hat{Q}_h^k(s, a). \quad (7.2)$$

The policy $\pi^k = \{\pi_h^k\}_{h=0}^{H-1}$ is then executed for one episode and the design matrices are updated.

We assume the maximization over $a \in \mathcal{A}$ in (7.2) is tractable (e.g., finite $\mathcal{A}$ or access to an appropriate optimization oracle).

7.5 Regret guarantee and analysis

We now bound the expected regret of Lin-UCBVI. The argument mirrors tabular UCBVI: a uniform confidence statement for the value-dependent regression targets, an optimism lemma, a one-episode performance decomposition, and an elliptical-potential summation.

Theorem 7.3 (Regret of Lin-UCBVI). *Assume the linear transition model (Assumption 7.1) and bounded rewards $r_h(s, a) \in [0, 1]$. Run Lin-UCBVI (Algorithm 7) with regularization $\lambda \geq 1$ and bonus*

$$b_h^k(s, a) = \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

Fix $\delta \in (0, 1)$. Suppose β is chosen so that the uniform confidence event in Proposition 7.4 holds with probability at least $1 - \delta$. Then the expected regret satisfies

$$R_K = \mathbb{E} \left[\sum_{k=0}^{K-1} (V^* - V^{\pi^k}) \right] \leq c \beta H \sqrt{K d \log \left(1 + \frac{K}{\lambda} \right)} + \delta K H,$$

Algorithm 7 Lin-UCBVI (least-squares value iteration with UCB bonus)

Input: horizon H , episodes K , regularization $\lambda > 0$, bonus scale $\beta > 0$

```
1: Initialize  $\Lambda_h^0 \leftarrow \lambda I$  for all  $h \in \{0, \dots, H-1\}$ .
2: for  $k = 0, 1, \dots, K-1$  do
3:   Set  $\widehat{V}_H^k(\cdot) \equiv 0$ .
4:   for  $h = H-1, H-2, \dots, 0$  do
5:     Compute  $\widehat{w}_{h, \widehat{V}_{h+1}^k}^k \leftarrow (\Lambda_h^k)^{-1} \sum_{i=0}^{k-1} \phi(s_h^i, a_h^i) \widehat{V}_{h+1}^k(s_{h+1}^i)$ .
6:     Define  $(\widehat{P}_h^k \widehat{V}_{h+1}^k)(s, a) = \phi(s, a)^\top \widehat{w}_{h, \widehat{V}_{h+1}^k}^k$ .
7:     Define  $\widehat{Q}_h^k$  by (7.1), and then  $\widehat{V}_h^k, \pi_h^k$  by (7.2).
8:   end for
9:   Execute  $\pi^k$ :  $s_0^k = s_0$ ; for  $h = 0, \dots, H-1$ , play  $a_h^k = \pi_h^k(s_h^k)$  and sample  $s_{h+1}^k \sim P_h^*(\cdot \mid s_h^k, a_h^k)$ .
10:  for  $h = 0, 1, \dots, H-1$  do
11:    Update  $\Lambda_h^{k+1} \leftarrow \Lambda_h^k + \phi(s_h^k, a_h^k) \phi(s_h^k, a_h^k)^\top$ .
12:  end for
13: end for
```

for a universal constant $c > 0$.

In particular, one may take $\beta = \beta_{K, \delta/H}$ as in Corollary A.17 (applied stage-wise and union-bounded over h), which yields $\beta = \widetilde{O}(Hd)$ and hence $R_K = \widetilde{O}(H^2 \sqrt{d^3 K})$.

7.5.1 Uniform confidence for the value-dependent targets

For each stage h , the algorithm fits ridge regression using samples $\{(s_h^i, a_h^i, s_{h+1}^i)\}_{i < k}$ and then queries the fitted model at the *random* value function $f = \widehat{V}_{h+1}^k$. The next proposition is exactly the uniformization step that justifies the UCB bonus.

Proposition 7.4 (Uniform model confidence). *Fix $\delta \in (0, 1)$. With probability at least $1 - \delta$, simultaneously for all episodes $k \in \{0, \dots, K-1\}$, all stages $h \in \{0, \dots, H-1\}$, all $(s, a) \in \mathcal{S} \times \mathcal{A}$, and for $f = \widehat{V}_{h+1}^k$ produced by Lin-UCBVI,*

$$|(\widehat{P}_h^k f)(s, a) - (P_h^* f)(s, a)| \leq \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}.$$

Proof. Fix a stage h . Consider the optimistic value-function class $\mathcal{F}$ from Lemma A.15 (Appendix A.6.1), with parameters chosen large enough to contain the algorithm's iterates:

- the known reward offset $r_h(s, a)$ is handled by the remark in Lemma A.15;
- max and clipping commute and clipping is a $\|\cdot\|_\infty$ contraction (Lemma A.14);
- and for all k, h one has the deterministic bound $\|\widehat{w}_{h, \widehat{V}_{h+1}^k}^k\|_2 \leq \frac{1}{\lambda} \sum_{i < k} \|\phi(s_h^i, a_h^i)\|_2 \|\widehat{V}_{h+1}^k\|_\infty \leq HK/\lambda$, so we may take $L = HK/\lambda$.

Under Assumption 7.1, the linear conditional expectation condition (A.3) in Corollary A.17 holds with $w_f^* = (\mu_h^*)^\top f$ and $\|w_f^*\|_2 \leq H\sqrt{d}$ (Lemma 7.2).

Apply Corollary A.17 at this stage with failure probability δ/H , then take a union bound over $h \in \{0, \dots, H-1\}$. This yields the stated event. $\square$

For the remainder of the analysis we condition on the event in Proposition 7.4.

7.5.2 Optimism and a one-episode bound

Lemma 7.5 (Optimism). *On the event of Proposition 7.4, for every episode k , stage h , and state s ,*

$$\widehat{V}_h^k(s) \geq V_h^*(s).$$

Proof. Fix episode k and argue by backward induction on h . At $h = H$, $\widehat{V}_H^k \equiv 0 = V_H^*$. Assume $\widehat{V}_{h+1}^k \geq V_{h+1}^*$ pointwise.

Fix (s, a) . Since $P_h^*(\cdot \mid s, a)$ is a distribution and $\widehat{V}_{h+1}^k \geq V_{h+1}^*$, we have $(P_h^* V_{h+1}^*)(s, a) \leq (P_h^* \widehat{V}_{h+1}^k)(s, a)$. By Proposition 7.4 applied to $f = \widehat{V}_{h+1}^k$,

$$(P_h^* \widehat{V}_{h+1}^k)(s, a) \leq (\widehat{P}_h \widehat{V}_{h+1}^k)(s, a) + b_h^k(s, a).$$

Therefore,

$$Q_h^*(s, a) = r_h(s, a) + (P_h^* V_{h+1}^*)(s, a) \leq r_h(s, a) + b_h^k(s, a) + (\widehat{P}_h \widehat{V}_{h+1}^k)(s, a).$$

Since $Q_h^*(s, a) \in [0, H - h]$, clipping to $[0, H - h]$ preserves the inequality and gives $Q_h^*(s, a) \leq \widehat{Q}_h^k(s, a)$. Taking maxima over a yields $V_h^*(s) \leq \widehat{V}_h^k(s)$. $\square$

Lemma 7.6 (Per-episode bonus bound). *On the event of Proposition 7.4, for every episode k ,*

$$V^* - V^{\pi^k} \leq 2 \sum_{h=0}^{H-1} \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi^k}} [b_h^k(s_h, a_h)].$$

Proof. Fix episode k and abbreviate $\pi = \pi^k$, $\widehat{V}_h = \widehat{V}_h^k$, $\widehat{P}_h = \widehat{P}_h^k$, $b_h = b_h^k$. Optimism (Lemma 7.5) gives $V^* - V^\pi \leq \widehat{V}_0(s_0) - V^\pi$. Define $\Delta_h(s) := \widehat{V}_h(s) - V_h^\pi(s)$, so $\Delta_H \equiv 0$.

On the confidence event, $(\widehat{P}_h \widehat{V}_{h+1})(s, a) \geq (P_h^* \widehat{V}_{h+1})(s, a) - b_h(s, a) \geq -b_h(s, a)$, so $r_h(s, a) + b_h(s, a) + (\widehat{P}_h \widehat{V}_{h+1})(s, a) \geq r_h(s, a) \geq 0$ and the clipping can only decrease the expression. Now, for any state s at stage h , using that $\pi_h(s) \in \arg \max_a \widehat{Q}_h(s, a)$ and that $\widehat{Q}_h$ is obtained by clipping the Bellman-style expression, we have

$$\widehat{V}_h(s) = \widehat{Q}_h(s, \pi_h(s)) \leq r_h(s, \pi_h(s)) + b_h(s, \pi_h(s)) + (\widehat{P}_h \widehat{V}_{h+1})(s, \pi_h(s)).$$

On the other hand,

$$V_h^\pi(s) = r_h(s, \pi_h(s)) + (P_h^* V_{h+1}^\pi)(s, \pi_h(s)).$$

Subtracting and writing $\widehat{V}_{h+1} = V_{h+1}^\pi + \Delta_{h+1}$ yields

$$\Delta_h(s) \leq b_h(s, \pi_h(s)) + ((\widehat{P}_h - P_h^*) \widehat{V}_{h+1})(s, \pi_h(s)) + (P_h^* \Delta_{h+1})(s, \pi_h(s)).$$

Take expectations along a trajectory of π in the true MDP and unroll from $h = 0$ to $H - 1$ (using $\Delta_H \equiv 0$) to obtain

$$\widehat{V}_0(s_0) - V^\pi \leq \sum_{h=0}^{H-1} \mathbb{E}_{d_h^\pi} [b_h(s_h, a_h) + ((\widehat{P}_h - P_h^*) \widehat{V}_{h+1})(s_h, a_h)].$$

Finally, Proposition 7.4 implies $|((\widehat{P}_h - P_h^*) \widehat{V}_{h+1})(s, a)| \leq b_h(s, a)$, and the claim follows. $\square$

7.5.3 Summation via an elliptical potential

Proof of Theorem 7.3. Let $\mathcal{E}$ denote the event in Proposition 7.4. On $\mathcal{E}$, Lemma 7.6 implies that for each episode k ,

$$V^* - V^{\pi^k} \leq 2 \sum_{h=0}^{H-1} \mathbb{E}[b_h^k(s_h^k, a_h^k) \mid \mathcal{H}_{<k}],$$

since b_h^k is $\mathcal{H}_{<k}$ -measurable and (s_h^k, a_h^k) has occupancy distribution under π^k in the true MDP. Therefore, using $\mathbf{1}\{\mathcal{E}\} \leq 1$ and the tower property,

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}\} \sum_{k=0}^{K-1} (V^* - V^{\pi^k})\right] \leq 2 \mathbb{E}\left[\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} b_h^k(s_h^k, a_h^k)\right].$$

Substituting $b_h^k(s, a) = \beta \|\phi(s, a)\|_{(\Lambda_h^k)^{-1}}$ gives

$$\mathbb{E}\left[\mathbf{1}\{\mathcal{E}\} \sum_{k=0}^{K-1} (V^* - V^{\pi^k})\right] \leq 2\beta \sum_{h=0}^{H-1} \mathbb{E}\left[\sum_{k=0}^{K-1} \|\phi(s_h^k, a_h^k)\|_{(\Lambda_h^k)^{-1}}\right].$$

Fix a stage h and write $x_k := \phi(s_h^k, a_h^k)$ and $\Lambda_k := \Lambda_h^k$, so $\Lambda_{k+1} = \Lambda_k + x_k x_k^\top$ and $\Lambda_0 = \lambda I$. Since $\lambda \geq 1$ and $\|x_k\|_2 \leq 1$, we have $\|x_k\|_{\Lambda_k^{-1}}^2 \leq 1$. The standard log-determinant argument (matrix determinant lemma; see Lemma 5.11 in Chapter 5) gives

$$\sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}}^2 \leq 2 \log \frac{\det(\Lambda_K)}{\det(\Lambda_0)}.$$

Moreover, $\Lambda_K = \lambda I + \sum_{k=0}^{K-1} x_k x_k^\top \preceq (\lambda + K)I$, hence $\det(\Lambda_K)/\det(\Lambda_0) \leq (1 + K/\lambda)^d$ and therefore

$$\sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}}^2 \leq 2d \log\left(1 + \frac{K}{\lambda}\right).$$

By Cauchy–Schwarz,

$$\sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}} \leq \sqrt{K \sum_{k=0}^{K-1} \|x_k\|_{\Lambda_k^{-1}}^2} \leq \sqrt{2K d \log\left(1 + \frac{K}{\lambda}\right)}.$$

Summing this bound over $h = 0, \dots, H-1$ yields, on $\mathcal{E}$,

$$\sum_{k=0}^{K-1} (V^* - V^{\pi^k}) \leq c\beta H \sqrt{K d \log\left(1 + \frac{K}{\lambda}\right)}$$

for a universal constant $c > 0$.

On the complement $\bar{\mathcal{E}}$ we use $0 \leq V^* - V^{\pi^k} \leq H$ to obtain

$$\mathbb{E}\left[\mathbf{1}\{\bar{\mathcal{E}}\} \sum_{k=0}^{K-1} (V^* - V^{\pi^k})\right] \leq \Pr(\bar{\mathcal{E}}) \cdot KH \leq \delta KH.$$

Combining completes the proof. $\square$

7.6 Bibliographic Remarks and Further Readings

There are several ways to impose linear structure on an MDP so that learning can be statistically and computationally efficient. A useful organizing viewpoint, developed in Jiang et al. [2017], is that many such models have low Bellman rank (and related complexity measures such as witness rank [Sun et al., 2019a]); we return to these notions in Chapter 8.

The specific linear transition model used in this chapter, in which $P_h(\cdot \mid s, a) = \mu_h^* \phi(s, a)$, was introduced and analyzed in Jin et al. [2020], building on earlier algorithmic developments for linearly parameterized MDPs in Yang and Wang [2019a,b]. A notable feature of this model is that the unknown parameter μ_h^* has ambient size $S \times d$, yet regret bounds can depend polynomially on d (and H) without explicit dependence on S . In particular, the algorithm never needs to estimate the full transition model; it only estimates the linear functionals $(P_h^* f)(s, a)$ required by planning.

A different family of models parameterizes the transition probabilities directly as linear combinations of known features $\phi(s, a, s')$; see, for example, Modi et al. [2020b], Jia et al. [2020], Ayoub et al. [2020b], Zhou et al. [2020]. These models typically have $O(d)$ free parameters in the transition model, while the model $P_h(\cdot \mid s, a) = \mu_h^* \phi(s, a)$ has $S \cdot d$ free parameters. The two viewpoints are therefore not directly comparable: smaller ambient parameter count does not by itself determine learnability or the sharp dependence on (d, H) in regret.

The style of model-based implementation we use here—estimating only the Bellman-relevant linear predictions rather than a full model—appears in related forms in other settings; see Lykouris et al. [2019] for an example in the context of adversarial corruption.

Finally, extensions beyond finite-dimensional features (e.g. RKHS settings) typically replace the dimension parameter d by an information-gain quantity; see Agarwal et al. [2020a] for one such development.

Chapter 8

Generalization with Bounded Bellman Rank

In the tabular setting, statistical efficiency — and strategic exploration with UCBVI — comes from directly estimating a finite transition table. This idea appears explicitly in the generative-model analysis of Chapter 2, and in the online episodic setting of Chapter 6, where optimism drives the learner to visit uncertain state–action pairs.

Moving beyond tabular representations requires structural assumptions. Chapter 3 studied linear function approximation under closure conditions such as linear Bellman completeness, which reduce dynamic programming to regression; however, the clean learnability results there were obtained under a generative-model interface, and therefore sidestepped the exploration problem. Chapter 7 imposed a stronger transition model that makes the Bellman-relevant predictions linear, enabling optimistic exploration and regret guarantees that scale polynomially in the feature dimension.

This chapter develops a more general lens for *generalization* in reinforcement learning. We seek conditions under which a learner can find a near-optimal policy using a number of episodes that scales with an intrinsic complexity parameter rather than with $|S|$ (and sometimes rather than with $|\mathcal{A}|$). The key object is the *average Bellman error* of one hypothesis evaluated under the state distributions induced by another. When these average Bellman errors admit a low-rank factorization (the *Bellman rank* condition), trajectory data collected under one roll-in policy can be reused to test many candidate hypotheses, yielding Probably Approximately Correct guarantees in large state spaces.

Of central importance is the data interface: here the learner has only episodic interaction and must explore in order to obtain informative data. Our focus is therefore on the *PAC-RL* question — how to return an ϵ -optimal policy with high probability from trajectory access alone — which is typically more challenging than planning with a simulator. This is a weaker requirement than regret: we aim to output a single good policy, rather than control cumulative suboptimality online. We discuss the relationship between PAC and regret herein.

8.1 Setting

We work in an episodic finite-horizon MDP with horizon H , state space $\mathcal{S}$, action space $\mathcal{A}$, and a fixed start state s_0 .¹ Rewards satisfy $r_h(s, a) \in [0, 1]$.

For any nonstationary policy $\pi = \{\pi_h\}_{h=0}^{H-1}$, define the stage- h occupancy measure

$$d_h^\pi(s, a) := \Pr(s_h = s, a_h = a \mid s_0, \pi, P^*), \quad d_h^\pi(s) := \sum_{a \in \mathcal{A}} d_h^\pi(s, a).$$

¹Extending to an initial distribution μ is straightforward; we use a fixed s_0 to keep notation light.

Our goal is PAC-RL: given $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, output a policy $\hat{\pi}$ such that

$$V_0^*(s_0) - V_0^{\hat{\pi}}(s_0) \leq \epsilon \quad \text{with probability at least } 1 - \delta.$$

PAC versus regret. This is a *fixed-confidence* guarantee: we aim to return one good policy, rather than control cumulative suboptimality online (regret). Regret bounds typically require finer control of how errors evolve along the interaction; we discuss regret and related work in the bibliographic notes. As in bandits (Chapter 5), one can convert PAC guarantees to regret by an explore-then-commit schedule, yielding sublinear regret rates of the familiar $T^{2/3}$ form.

Hypothesis class and realizability. We assume access to a hypothesis class

$$\mathcal{H} = \mathcal{H}_0 \times \cdots \times \mathcal{H}_{H-1}.$$

Each hypothesis $f \in \mathcal{H}$ induces functions $\{Q_{h,f}\}_{h=0}^{H-1}$ and $\{V_{h,f}\}_{h=0}^{H-1}$ with the consistency constraint

$$V_{h,f}(s) = \max_{a \in \mathcal{A}} Q_{h,f}(s, a) \quad \forall (h, s, f), \quad V_{H,f} \equiv 0.$$

Let $\pi_{h,f}(s) \in \arg \max_a Q_{h,f}(s, a)$ and $\pi_f = \{\pi_{h,f}\}_{h=0}^{H-1}$ be the greedy policy induced by f (ties broken in some fixed manner).

We assume $\mathcal{H}$ is realizable: there exists $f^* \in \mathcal{H}$ such that $Q_{h,f^*} = Q_h^*$ for all h (equivalently, π_{f^*} is optimal).

8.2 The Bellman rank

For a fixed stage $h \in \{0, \dots, H-1\}$ and hypothesis $g \in \mathcal{H}$, define the one-step Bellman residual integrand

$$\ell_h(s, a, s'; g) := Q_{h,g}(s, a) - r_h(s, a) - V_{h+1,g}(s').$$

8.2.1 V -Bellman rank

Fix hypotheses $f, g \in \mathcal{H}$. The V -average Bellman error of g under roll-in $d_h^{\pi_f}$ is

$$\mathcal{E}_h^V(f, g) := \mathbb{E}_{s \sim d_h^{\pi_f}, a = \pi_{h,g}(s), s' \sim P_h^*(\cdot | s, a)} [\ell_h(s, a, s'; g)]. \quad (8.1)$$

Here the state s is drawn from the distribution induced by π_f , while the action is the greedy action of g at s . Since $V_{h,g}(s) = Q_{h,g}(s, \pi_{h,g}(s))$, $\mathcal{E}_h^V(f, g)$ is the expected V -Bellman residual of g on states visited by π_f .

Definition 8.1 (V -Bellman rank). We say $(\mathcal{M}, \mathcal{H})$ has V -Bellman rank at most d if for each stage h there exist maps

$$X_h : \mathcal{H} \rightarrow \mathbb{R}^d, \quad W_h : \mathcal{H} \rightarrow \mathbb{R}^d$$

and constants $B_X, B_W \geq 1$ such that

$$\mathcal{E}_h^V(f, g) = \langle X_h(f), W_h(g) \rangle \quad \forall f, g \in \mathcal{H},$$

and $\sup_{f \in \mathcal{H}} \|X_h(f)\|_2 \leq B_X, \sup_{g \in \mathcal{H}} \|W_h(g)\|_2 \leq B_W$.

8.2.2 Q -Bellman rank

The Q -version averages the same residual under the full *state–action* occupancy of π_f :

$$\mathcal{E}_h^Q(f, g) := \mathbb{E}_{(s,a) \sim d_h^{\pi_f}, s' \sim P_h^*(\cdot|s,a)} [\ell_h(s, a, s'; g)]. \quad (8.2)$$

Definition 8.2 (Q -Bellman rank). We say $(\mathcal{M}, \mathcal{H})$ has Q -Bellman rank at most d if for each stage h there exist maps

$$X_h : \mathcal{H} \rightarrow \mathbb{R}^d, \quad W_h : \mathcal{H} \rightarrow \mathbb{R}^d$$

and constants $B_X, B_W \geq 1$ such that

$$\mathcal{E}_h^Q(f, g) = \langle X_h(f), W_h(g) \rangle \quad \forall f, g \in \mathcal{H},$$

and $\sup_{f \in \mathcal{H}} \|X_h(f)\|_2 \leq B_X, \sup_{g \in \mathcal{H}} \|W_h(g)\|_2 \leq B_W$.

Shifting W_h so that $W_h(f^*) = 0$. Under realizability, $\mathcal{E}_h^V(f, f^*) = \mathcal{E}_h^Q(f, f^*) = 0$ for all f and h . Thus, for either definition, we may assume without loss of generality that $W_h(f^*) = 0$ by replacing $W_h(g)$ with $W_h(g) - W_h(f^*)$ (absorbing any constant-factor change into B_W).

Why there are two versions. The Q -version is “on-policy” at stage h in the sense that $(s_h, a_h) \sim d_h^{\pi_f}$; consequently, $\mathcal{E}_h^Q(f, g)$ can be estimated directly from data collected by following π_f . The V -version evaluates the residual at the *action chosen by g* on states visited by π_f . To estimate $\mathcal{E}_h^V(f, g)$ from data that does not depend on g , one typically injects exploration at time h (e.g. uniform actions) and uses importance weighting. This allows data collected under one roll-in policy to be reused across many candidate g , at the cost of explicit $|\mathcal{A}|$ -dependence in the estimation error.

8.2.3 Two anchor examples

Contextual linear bandits ($H = 1$): small Q -Bellman rank

Let $H = 1$ and suppose the initial state is drawn from a distribution μ . Assume rewards satisfy $\mathbb{E}[r(s, a) \mid s, a] = \langle \theta^*, \phi(s, a) \rangle$ with $\phi(s, a) \in \mathbb{R}^d$ and $\|\phi(s, a)\|_2 \leq 1$. Take the hypothesis class

$$\mathcal{H}_0 = \{Q_\theta(s, a) = \langle \theta, \phi(s, a) \rangle : \|\theta\|_2 \leq W\},$$

with $V_\theta(s) = \max_a Q_\theta(s, a)$ and greedy π_θ .

Proposition 8.3 (Contextual linear bandits have Q -Bellman rank at most d). *The pair $(\mathcal{M}, \mathcal{H})$ above has Q -Bellman rank at most d .*

Proof. Since $H = 1$, $V_{1,g} \equiv 0$ and

$$\mathcal{E}_0^Q(f, g) = \mathbb{E}_{s \sim \mu, a = \pi_f(s)} [Q_g(s, a) - r(s, a)] = \mathbb{E}_{s \sim \mu, a = \pi_f(s)} \langle \theta_g - \theta^*, \phi(s, a) \rangle.$$

Define $X_0(f) := \mathbb{E}_{s \sim \mu, a = \pi_f(s)} [\phi(s, a)] \in \mathbb{R}^d$ and $W_0(g) := \theta_g - \theta^* \in \mathbb{R}^d$. Then $\mathcal{E}_0^Q(f, g) = \langle X_0(f), W_0(g) \rangle$. $\square$

Linear Bellman completeness: small Q -Bellman rank

We phrase this in the value-based linear Bellman completeness setting (Definition 3.1). Fix features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ with $\|\phi(s, a)\|_2 \leq 1$ and take a linear class

$$\mathcal{H}_h = \left\{ Q_{h,w}(s, a) = \langle w, \phi(s, a) \rangle : \|w\|_2 \leq W, \|Q_{h,w}\|_\infty \leq H \right\}, \quad V_{h,w}(s) = \max_a Q_{h,w}(s, a).$$

Assume linear Bellman completeness: for each h there exists an operator $\mathcal{T}_h : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that for all $w \in \mathbb{R}^d$ and all (s, a) ,

$$(\mathcal{T}_h(w))^\top \phi(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h^*(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} w^\top \phi(s', a') \right],$$

and $\sup_{h, \|w\|_2 \leq W} \|\mathcal{T}_h(w)\|_2 \leq W$.

Proposition 8.4 (Linear Bellman completeness implies Q -Bellman rank at most d). *Under the above assumptions, $(\mathcal{M}, \mathcal{H})$ has Q -Bellman rank at most d . One valid choice is*

$$X_h(f) = \mathbb{E}_{(s_h, a_h) \sim d_h^{\pi_f}} [\phi(s_h, a_h)], \quad W_h(g) = w_{h,g} - \mathcal{T}_h(w_{h+1,g}).$$

Proof. Fix f, g and stage h . Conditioning on (s_h, a_h) and using the defining identity of $\mathcal{T}_h$,

$$\begin{aligned} \mathbb{E}[\ell_h(s_h, a_h, s_{h+1}; g) \mid s_h, a_h] &= w_{h,g}^\top \phi(s_h, a_h) - r_h(s_h, a_h) - \mathbb{E}[V_{h+1,g}(s_{h+1}) \mid s_h, a_h] \\ &= w_{h,g}^\top \phi(s_h, a_h) - (\mathcal{T}_h(w_{h+1,g}))^\top \phi(s_h, a_h) \\ &= (w_{h,g} - \mathcal{T}_h(w_{h+1,g}))^\top \phi(s_h, a_h). \end{aligned}$$

Taking expectation over $(s_h, a_h) \sim d_h^{\pi_f}$ yields $\mathcal{E}_h^Q(f, g) = \langle X_h(f), W_h(g) \rangle$. □

8.3 PAC RL under bounded Bellman rank

We now give an optimistic-elimination template that works for either the Q - or the V -version. The two versions differ only in how we estimate the average Bellman error.

8.3.1 Uniform convergence primitive

For a dataset $\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^m$, we write

$$\mathbb{E}_{\mathcal{D}}[Z(s, a, s')] := \frac{1}{m} \sum_{i=1}^m Z(s_i, a_i, s'_i)$$

for the empirical average.

Estimating $\mathcal{E}_h^Q(f, g)$. If (s, a, s') are sampled with $(s, a) \sim d_h^{\pi_f}$ and $s' \sim P_h^*(\cdot \mid s, a)$, then $\mathbb{E}_{\mathcal{D}}[\ell_h(s, a, s'; g)]$ is an unbiased estimate of $\mathcal{E}_h^Q(f, g)$.

Algorithm 8 Bellman-rank optimistic elimination (Q/V versions)

Input: iterations T , batch size m , radius R , mode $\text{mode} \in \{Q, V\}$

1: **for** $t = 0, 1, \dots, T - 1$ **do**

2: Choose

$$f_t \in \operatorname{argmax}_{g \in \mathcal{H}} V_{0,g}(s_0) \quad \text{s.t.} \quad \sum_{i=0}^{t-1} \left(\widehat{\mathcal{E}}_{i,h}(g) \right)^2 \leq R^2 \quad \forall h \in \{0, \dots, H-1\},$$

where $\widehat{\mathcal{E}}_{i,h}(g) := \mathbb{E}_{\mathcal{D}_{i,h}} [\tilde{\ell}(s, a, s'; g)]$ and $\tilde{\ell}$ is chosen according to mode.

3: **for** $h = 0, 1, \dots, H - 1$ **do**

4: Collect a batch $\mathcal{D}_{t,h} = \{(s^j, a^j, (s')^j)\}_{j=1}^m$ by rolling in with π_{f_t} to time h to obtain $s^j \sim d_h^{\pi_{f_t}}$, then

$$a^j = \begin{cases} \pi_{h,f_t}(s^j), & \text{mode} = Q, \\ \text{Unif}(\mathcal{A}), & \text{mode} = V, \end{cases} \quad (s')^j \sim P_h^*(\cdot \mid s^j, a^j).$$

5: **end for**

6: **end for**

7: Let $\hat{t} \in \arg \min_{t \in \{0, \dots, T-1\}} \sum_{h=0}^{H-1} |\widehat{\mathcal{E}}_{t,h}(f_t)|$.

8: **Output:** return $\hat{\pi} := \pi_{f_{\hat{t}}}$.

Estimating $\mathcal{E}_h^V(f, g)$. To estimate $\mathcal{E}_h^V(f, g)$ from data that does not depend on g , we use the importance-weighted loss

$$\tilde{\ell}_h^V(s, a, s'; g) := \frac{\mathbf{1}\{a = \pi_{h,g}(s)\}}{1/A} \ell_h(s, a, s'; g). \quad (8.3)$$

If $s \sim d_h^{\pi_f}$, $a \sim \text{Unif}(\mathcal{A})$ and $s' \sim P_h^*(\cdot \mid s, a)$, then $\mathbb{E}[\tilde{\ell}_h^V(s, a, s'; g)] = \mathcal{E}_h^V(f, g)$.

Assumption 8.5 (Uniform convergence of Bellman-error estimates). There exists a function $\varepsilon_{\text{gen}}(m, \mathcal{H}, \delta)$ such that for any stage h , any distribution ν over (s, a, s') , if $\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^m$ are i.i.d. from ν , then with probability at least $1 - \delta$,

$$\sup_{g \in \mathcal{H}} \left| \mathbb{E}_{\mathcal{D}} [\tilde{\ell}(s, a, s'; g)] - \mathbb{E}_{(s,a,s') \sim \nu} [\tilde{\ell}(s, a, s'; g)] \right| \leq \varepsilon_{\text{gen}}(m, \mathcal{H}, \delta),$$

where $\tilde{\ell}$ is either $\ell_h(\cdot; g)$ (for the Q -version) or $\tilde{\ell}_h^V(\cdot; g)$ (for the V -version).

In Appendix A.6.2 we give explicit ε_{gen} bounds for both choices of $\tilde{\ell}$.

8.3.2 Algorithm: Bellman-rank optimistic elimination

Algorithm 8 maintains a set of hypotheses that are consistent with all previously collected Bellman-error tests. At iteration t it selects an *optimistic* hypothesis f_t , maximizing the predicted value $V_{0,g}(s_0)$ subject to the constraint that, at each stage h , the cumulative squared estimated Bellman errors against g over the past batches are small. It then rolls in with π_{f_t} to generate a fresh batch of transition samples at each stage and repeats.

The only difference between mode = Q and mode = V is how we obtain an unbiased estimate of the average Bellman error at stage h . In the Q -version we take the on-policy action $a_h = \pi_{h,f_t}(s_h)$. In the V -version we inject a uniform action at stage h and use the importance-weighted loss (8.3), which permits testing many candidates g using data whose action choices do not depend on g .

8.3.3 Main PAC guarantee

Theorem 8.6 (PAC RL under bounded Bellman rank). *Fix $\text{mode} \in \{\mathbf{Q}, \mathbf{V}\}$. Assume realizability and that $(\mathcal{M}, \mathcal{H})$ has Bellman rank at most d in the corresponding sense: Q -Bellman rank if $\text{mode} = \mathbf{Q}$, and V -Bellman rank if $\text{mode} = \mathbf{V}$, with bounds B_X, B_W . Run Algorithm 8 with batch size m and define*

$$\varepsilon_{\text{gen}} := \varepsilon_{\text{gen}}\left(m, \mathcal{H}, \frac{\delta}{TH}\right).$$

Set

$$T := \left\lceil c H d \ln\left(e H d \left(\frac{B_X^2 B_W^2}{\varepsilon_{\text{gen}}^2} + 1\right)\right) \right\rceil, \quad R := \sqrt{T} \varepsilon_{\text{gen}},$$

for universal constants $c > 0$. Then with probability at least $1 - \delta$, the output policy $\hat{\pi}$ satisfies

$$V_0^*(s_0) - V_0^{\hat{\pi}}(s_0) \leq C \varepsilon_{\text{gen}} \sqrt{d H^3 \ln\left(e H d \left(\frac{B_X^2 B_W^2}{\varepsilon_{\text{gen}}^2} + 1\right)\right)},$$

for a universal constant $C > 0$.

Where the $|\mathcal{A}|$ dependence enters. The theorem is stated in terms of ε_{gen} . For $\text{mode} = \mathbf{Q}$, the loss $\tilde{\ell} = \ell_h$ is uniformly $O(H)$ -bounded under the usual range assumptions on $Q_{h,g}$, so standard concentration does not introduce an explicit $|\mathcal{A}|$ factor. For $\text{mode} = \mathbf{V}$, $\tilde{\ell} = \tilde{\ell}_h^V$ includes an importance weight of size A , and ε_{gen} typically scales linearly with A .

8.3.4 Instantiations

Theorem 8.6 is stated in terms of ε_{gen} . This section records a few standard plug-ins from Appendix A.6.2.

Corollary 8.7 (Finite hypothesis classes). *Assume $|\mathcal{H}| < \infty$ and $\|Q_{h,g}\|_\infty \leq H$ for all h, g . Fix T and m and apply Lemma A.18 with failure probability $\delta/(TH)$. Then one may take*

$$\varepsilon_{\text{gen}} \leq 3H \sqrt{\frac{2 \ln(2|\mathcal{H}|TH/\delta)}{m}} \quad \text{for } \text{mode} = \mathbf{Q},$$

and

$$\varepsilon_{\text{gen}} \leq 3AH \sqrt{\frac{2 \ln(2|\mathcal{H}|TH/\delta)}{m}} \quad \text{for } \text{mode} = \mathbf{V}.$$

Substituting into Theorem 8.6 yields an explicit PAC guarantee after choosing m so that the right-hand side is at most ϵ .

Beyond finite classes. For infinite hypothesis classes, one replaces the finite-class union bound by a standard complexity measure (covering numbers, VC-type dimensions, Rademacher complexity, etc.). Appendix A.6.2 records one convenient covering-number instantiation for linear Q -classes, which is the one used next. We use this below and in the next section.

Linear Bellman completeness. In the linear Bellman completeness setting (Chapter 3), the one-step Bellman residuals can be written as linear functionals of a d -dimensional feature vector. Proposition 8.4 shows this implies bounded Q -Bellman rank. Algorithm 8 therefore yields PAC guarantees for linear Bellman completeness from trajectory access alone, complementing the earlier generative-model guarantees for linear Bellman completeness.

Corollary 8.8 (Linear Bellman completeness). *Consider the linear Bellman completeness setting of Proposition 8.4, with feature dimension d and weight bound W (so $\|w_{h,g}\|_2 \leq W$ for all h, g). Take $\mathcal{H}$ to be the corresponding product of stage-wise linear classes and run Algorithm 8 in `mode = Q`. Then $(\mathcal{M}, \mathcal{H})$ has Q -Bellman rank at most d and Lemma A.19 yields*

$$\varepsilon_{\text{gen}} \leq 3H \sqrt{\frac{2d \ln(1 + 4Wm) + \ln(TH/\delta)}{m}}.$$

In particular, there is a choice of m and T for which Theorem 8.6 guarantees $V_0^*(s_0) - V_0^{\hat{\pi}}(s_0) \leq \epsilon$ with probability at least $1 - \delta$, using

$$m = \tilde{O}\left(\frac{d^2 H^5}{\epsilon^2}\right) \quad \text{and} \quad T = \tilde{O}(Hd),$$

where $\tilde{O}(\cdot)$ hides logarithmic factors in $(H, d, W, 1/\epsilon, 1/\delta)$. In particular, the sample complexity is polynomial in $(H, d, 1/\epsilon)$ and does not depend on $|\mathcal{S}|$.

As written, Algorithm 8 collects m independent roll-ins for each stage, for a total of $O(mHT)$ trajectories of length at most H . In `mode = Q` one can form all batches $\{\mathcal{D}_{t,h}\}_{h=0}^{H-1}$ from the same m full episodes of π_{f_t} , reducing the interaction cost to $O(mT)$ full episodes.

8.4 Model zoo: more bounded Bellman rank examples

Theorem 8.6 converts two ingredients into a PAC-RL guarantee: a low-rank factorization of the average Bellman errors (bounded Bellman rank) and a uniform convergence bound for the corresponding empirical losses. In this section we focus on the first ingredient and record structural assumptions that imply bounded Bellman rank. To obtain explicit trajectory/sample complexity statements one then plugs in an appropriate $\varepsilon_{\text{gen}}(m, \mathcal{H}, \delta)$ bound from Appendix A.6.2. In particular, the appendix includes covering-number bounds for infinite linear classes and a $\log |\Phi|$ overhead for unions of candidate representations, which is the basic mechanism behind the feature-selection corollary below.

8.4.1 Linear Q^* and linear V^* (a restricted consistent class)

Assume there are known feature maps $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ and $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ such that $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$ and $\sup_s \|\psi(s)\|_2 \leq 1$, and for each stage h ,

$$Q_h^*(s, a) = \langle \theta_h^*, \phi(s, a) \rangle, \quad V_h^*(s) = \langle w_h^*, \psi(s) \rangle.$$

Restrict the hypothesis class to pairs (θ_h, w_h) that are internally consistent:

$$\mathcal{H}_h = \left\{ (\theta, w) : \|\theta\|_2 \leq W_1, \|w\|_2 \leq W_2, \max_a \langle \theta, \phi(s, a) \rangle = \langle w, \psi(s) \rangle \forall s \right\}.$$

Proposition 8.9 (Linear Q^* & V^* imply bounded Q -Bellman rank). *Under the assumptions above, $(\mathcal{M}, \mathcal{H})$ has Q -Bellman rank at most $2d$.*

Proof. Fix f, g and stage h , and write g 's parameters as (θ_h, w_{h+1}) . Using $Q_h^*(s, a) = r_h(s, a) + \mathbb{E}[V_{h+1}^*(s_{h+1}) \mid s_h = s, a_h = a]$,

$$\begin{aligned} \mathcal{E}_h^Q(f, g) &= \mathbb{E}_{(s, a) \sim d_h^{\pi_f}} \left[\langle \theta_h - \theta_h^*, \phi(s, a) \rangle + \left\langle w_{h+1} - w_{h+1}^*, \mathbb{E}[\psi(s_{h+1}) \mid s_h = s, a_h = a] \right\rangle \right] \\ &= \left\langle \begin{bmatrix} \theta_h - \theta_h^* \\ w_{h+1} - w_{h+1}^* \end{bmatrix}, \mathbb{E}_{(s, a) \sim d_h^{\pi_f}} \left[\begin{bmatrix} \phi(s, a) \\ \mathbb{E}[\psi(s_{h+1}) \mid s_h = s, a_h = a] \end{bmatrix} \right] \right\rangle. \end{aligned}$$

□

PAC-RL sample complexity. Combining Proposition 8.9 with Theorem 8.6 and the linear-class uniform convergence bound in Lemma A.19 yields a PAC guarantee with sample complexity polynomial in $(d, H, 1/\epsilon)$ (up to logarithmic factors). We omit the algebra.

8.4.2 Q^* -state abstraction

Suppose there is an abstraction map $\xi : \mathcal{S} \rightarrow \mathcal{Z}$ with $|\mathcal{Z}| \ll |\mathcal{S}|$ such that for all h ,

$$\xi(s) = \xi(s') \Rightarrow Q_h^*(s, a) = Q_h^*(s', a) \forall a, \quad \text{and} \quad V_h^*(s) = V_h^*(s').$$

This can be viewed as a special case of Proposition 8.9 by taking one-hot features on $\mathcal{Z}$ (and $\mathcal{Z} \times \mathcal{A}$ for Q). In particular, the relevant complexity parameters depend on $|\mathcal{Z}|$ rather than $|\mathcal{S}|$.

8.4.3 Low-occupancy structure

Proposition 8.10 (Low state-action occupancy implies bounded Q -Bellman rank). *Assume that for each stage h there exist maps $\beta_h : \mathcal{H} \rightarrow \mathbb{R}^d$ and $\psi_h : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that for all $f \in \mathcal{H}$, $d_h^{\pi_f}(s, a) = \langle \beta_h(f), \psi_h(s, a) \rangle$. Then any realizable $\mathcal{H}$ has Q -Bellman rank at most d .*

Proof. For any f, g ,

$$\mathcal{E}_h^Q(f, g) = \sum_{s, a} d_h^{\pi_f}(s, a) \mathbb{E}[\ell_h(s, a, s'; g) \mid s, a] = \left\langle \beta_h(f), \sum_{s, a} \psi_h(s, a) \mathbb{E}[\ell_h(s, a, s'; g) \mid s, a] \right\rangle.$$

□

Proposition 8.11 (Low state occupancy implies bounded V -Bellman rank). *Assume that for each stage h there exist maps $\beta_h : \mathcal{H} \rightarrow \mathbb{R}^d$ and $\psi_h : \mathcal{S} \rightarrow \mathbb{R}^d$ such that for all $f \in \mathcal{H}$, $d_h^{\pi_f}(s) = \langle \beta_h(f), \psi_h(s) \rangle$. Then any realizable $\mathcal{H}$ has V -Bellman rank at most d .*

Proof. For any f, g ,

$$\mathcal{E}_h^V(f, g) = \sum_s d_h^{\pi_f}(s) \mathbb{E}[\ell_h(s, \pi_{h,g}(s), s'; g) \mid s] = \left\langle \beta_h(f), \sum_s \psi_h(s) \mathbb{E}[\ell_h(s, \pi_{h,g}(s), s'; g) \mid s] \right\rangle.$$

□

PAC-RL sample complexity. If the family of roll-in occupancies $\{d_h^{\pi_f} : f \in \mathcal{H}\}$ lies in a low-dimensional linear span (in either the state-action or state sense), then Theorem 8.6 yields PAC guarantees with complexity governed by that dimension. The statistical term is obtained by selecting an appropriate ε_{gen} bound for the hypothesis class $\mathcal{H}$ (Appendix A.6.2).

8.4.4 Low-rank transitions and representation learning

Assume a low-rank transition factorization: there exist unknown maps $\mu_h^* : \mathcal{S} \rightarrow \mathbb{R}^d$ and $\phi_h^* : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that for all h ,

$$P_h^*(s' | s, a) = \langle \mu_h^*(s'), \phi_h^*(s, a) \rangle, \quad \sup_{s,a} \|\phi_h^*(s, a)\|_2 \leq 1.$$

We also assume a mild normalization: there exists $B_\mu \geq 1$ such that for all bounded $f : \mathcal{S} \rightarrow \mathbb{R}$ and all h ,

$$\left\| \sum_{s \in \mathcal{S}} \mu_h^*(s) f(s) \right\|_2 \leq B_\mu \|f\|_\infty.$$

Proposition 8.12 (Low-rank transitions imply bounded V -Bellman rank). *Assume the low-rank transition factorization above and that $\|Q_{h,g}\|_\infty \leq H$ for all $h, g \in \mathcal{H}$ (and hence $\|V_{h,g}\|_\infty \leq H$). Then $(\mathcal{M}, \mathcal{H})$ has V -Bellman rank at most d . One valid choice satisfies $\sup_f \|X_h(f)\|_2 \leq 1$ and $\sup_g \|W_h(g)\|_2 \leq B_\mu H$.*

Proof. Fix a stage h and hypotheses f, g .

For $h = 0$, the roll-in state is always s_0 , so $\mathcal{E}_0^V(f, g)$ depends only on g . Let $u \in \mathbb{R}^d$ be any fixed unit vector. Define $X_0(\cdot) \equiv u$ and $W_0(g) := \mathcal{E}_0^V(\cdot, g) u$. Then $\mathcal{E}_0^V(f, g) = \langle X_0(f), W_0(g) \rangle$ and $\|W_0(g)\|_2 \leq H$.

Now assume $h \geq 1$. Define

$$X_h(f) := \mathbb{E}_{(s,a) \sim d_{h-1}^{\pi_f}} [\phi_{h-1}^*(s, a)] \in \mathbb{R}^d.$$

By $\|\phi_{h-1}^*\|_2 \leq 1$, we have $\|X_h(f)\|_2 \leq 1$. Let $p_h^{\pi_f}$ denote the stage- h state distribution under π_f . Using the low-rank factorization and the definition of occupancy,

$$\begin{aligned} p_h^{\pi_f}(s) &= \sum_{s', a'} d_{h-1}^{\pi_f}(s', a') P_{h-1}^*(s | s', a') \\ &= \sum_{s', a'} d_{h-1}^{\pi_f}(s', a') \langle \mu_{h-1}^*(s), \phi_{h-1}^*(s', a') \rangle \\ &= \left\langle \mu_{h-1}^*(s), \mathbb{E}_{(s', a') \sim d_{h-1}^{\pi_f}} [\phi_{h-1}^*(s', a')] \right\rangle = \langle \mu_{h-1}^*(s), X_h(f) \rangle. \end{aligned}$$

Next define the state-wise expected residual under g at stage h :

$$b_h(s; g) := \mathbb{E}_{a=\pi_{h,g}(s), s' \sim P_h^*(\cdot | s, a)} [\ell_h(s, a, s'; g)].$$

Since $\ell_h(\cdot; g)$ is bounded in $[-H, H]$ under the range assumptions on Q and V , we have $\|b_h(\cdot; g)\|_\infty \leq H$. Using the definition of $\mathcal{E}_h^V$ and the identity for $p_h^{\pi_f}$,

$$\begin{aligned} \mathcal{E}_h^V(f, g) &= \sum_{s \in \mathcal{S}} p_h^{\pi_f}(s) b_h(s; g) = \sum_{s \in \mathcal{S}} \langle \mu_{h-1}^*(s), X_h(f) \rangle b_h(s; g) \\ &= \left\langle X_h(f), \sum_{s \in \mathcal{S}} \mu_{h-1}^*(s) b_h(s; g) \right\rangle. \end{aligned}$$

Define

$$W_h(g) := \sum_{s \in \mathcal{S}} \mu_{h-1}^*(s) b_h(s; g) \in \mathbb{R}^d.$$

Then $\mathcal{E}_h^V(f, g) = \langle X_h(f), W_h(g) \rangle$. Moreover, by the normalization assumption and $\|b_h(\cdot; g)\|_\infty \leq H$, $\|W_h(g)\|_2 \leq B_\mu H$. $\square$

Why this suggests representation learning. Proposition 8.12 shows that bounded Bellman rank can arise from a *latent* low-dimensional transition structure, even when the feature map ϕ_h^* is not known. If the learner searches over a family Φ of candidate representations, the Bellman-rank parameter can remain d while the statistical error pays only a $\log |\Phi|$ overhead via uniform convergence (Lemma A.20), which is the basic mechanism behind sparse feature selection.

8.4.5 Sparse linear MDPs and feature selection

A sparse linear MDP is a special case of the low-rank model in which the ambient feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^D$ is known but only an unknown subset of coordinates matters. Let $K^* \subset [D]$ be an unknown support with $|K^*| = s \ll D$, and write $\phi_K(s, a) \in \mathbb{R}^s$ for the restriction to coordinates in K . Assume the true transitions admit a rank- s factorization in these coordinates (so the latent dimension is s). Then Proposition 8.12 implies bounded V -Bellman rank with parameter $d = s$ (up to the normalization constant B_μ).

PAC-RL sample complexity. If K^* is unknown, a natural strategy is to take a union over candidate supports K . The Bellman-rank parameter is governed by the *latent* dimension s , while the statistical term depends on the candidate family only through $\log |\Phi|$ in the union bound (Appendix A.6.2).

Corollary 8.13 (Feature selection: polynomial in s , polylogarithmic in D). *Fix $s \in \{1, \dots, D\}$ and a finite family Φ of candidate supports $K \subset [D]$ with $|K| = s$ (e.g. all s -subsets). For each $K \in \Phi$, let $\mathcal{H}(\phi_K)$ be a stage-wise linear Q -class as in the appendix, and take the union hypothesis class $\mathcal{H} := \bigcup_{K \in \Phi} \mathcal{H}(\phi_K)$.*

Assume realizability and that $(\mathcal{M}, \mathcal{H})$ has V -Bellman rank at most s (with some bounds B_X, B_W), and run Algorithm 8 in `mode = V`. Then ε_{gen} depends on Φ only through $\ln |\Phi|$ (Lemma A.20). In particular, when Φ is the family of all s -subsets of $[D]$, $\ln |\Phi| \leq s \ln(eD/s)$, so the resulting PAC sample complexity is polynomial in s and polylogarithmic in D (up to additional logarithmic factors in $(H, A, 1/\epsilon, 1/\delta)$).

Proof. Lemma A.20 bounds $\varepsilon_{\text{gen}}(m, \mathcal{H}, \delta)$ for the union class $\mathcal{H}$ with $\ln(1/\delta)$ replaced by $\ln(|\Phi|/\delta)$. Substituting this ε_{gen} into Theorem 8.6 yields a PAC bound whose dependence on the candidate family enters only through $\ln |\Phi|$. For $\Phi = \{K \subset [D] : |K| = s\}$, we use $\ln |\Phi| = \ln \binom{D}{s} \leq s \ln(eD/s)$. $\square$

8.5 Analysis

We prove Theorem 8.6. Fix `mode` $\in \{\text{Q}, \text{V}\}$. For each stage h , let $\tilde{\ell}_h(\cdot; g)$ denote the loss used in the constraints and in data collection: $\tilde{\ell}_h = \ell_h$ for `mode` = Q, and $\tilde{\ell}_h = \tilde{\ell}_h^V$ for `mode` = V.

For each iteration t and stage h , the algorithm collects an i.i.d. batch $\mathcal{D}_{t,h} = \{(s_i, a_i, s'_i)\}_{i=1}^m$ by rolling in with π_{f_t} to time h and then choosing actions according to the mode-dependent rule at time h . Define the empirical Bellman-error estimate

$$\hat{\mathcal{E}}_{t,h}(g) := \mathbb{E}_{\mathcal{D}_{t,h}} [\tilde{\ell}_h(s, a, s'; g)] = \frac{1}{m} \sum_{i=1}^m \tilde{\ell}_h(s_i, a_i, s'_i; g).$$

By construction (and by the importance-weighting identity in the V -case),

$$\mathbb{E}[\hat{\mathcal{E}}_{t,h}(g) \mid \mathcal{H}_{<t}] = \mathcal{E}_h^{\text{mode}}(f_t, g),$$

where $\mathcal{E}_h^{\text{mode}}$ denotes $\mathcal{E}_h^Q$ if `mode` = Q and $\mathcal{E}_h^V$ if `mode` = V.

Lemma 8.14 (Uniform estimation event). *Let $\varepsilon_{\text{gen}} := \varepsilon_{\text{gen}}(m, \mathcal{H}, \delta/(TH))$. With probability at least $1 - \delta$, simultaneously for all $t \in \{0, \dots, T-1\}$, all $h \in \{0, \dots, H-1\}$, and all $g \in \mathcal{H}$,*

$$|\widehat{\mathcal{E}}_{t,h}(g) - \mathcal{E}_h^{\text{mode}}(f_t, g)| \leq \varepsilon_{\text{gen}}.$$

Proof. Condition on $\mathcal{H}_{<t}$. Then $\mathcal{D}_{t,h}$ is i.i.d. from some distribution $\nu_{t,h}$ over (s, a, s') . Assumption 8.5 applied with failure probability $\delta/(TH)$ gives the desired bound for this fixed pair (t, h) . Union bound over t and h completes the proof. $\square$

In the rest of the proof we condition on the event of Lemma 8.14. We also apply the standard shift so that $W_h(f^*) = 0$ for all h , absorbing any constant-factor change into B_W .

Lemma 8.15 (Feasibility and optimism). *On the event of Lemma 8.14, if $R = \sqrt{T} \varepsilon_{\text{gen}}$ then f^* is feasible in every iteration t , and hence $V_{0,f_t}(s_0) \geq V_0^*(s_0)$ for all t .*

Proof. Fix t and h . Realizability implies $Q_{h,f^*} = Q_h^*$ and $V_{h+1,f^*} = V_{h+1}^*$. Therefore, for any (s, a) ,

$$\mathbb{E}_{s' \sim P_h^*(\cdot|s,a)}[\ell_h(s, a, s'; f^*)] = Q_h^*(s, a) - r_h(s, a) - \mathbb{E}[V_{h+1}^*(s')] = 0.$$

It follows that $\mathcal{E}_h^{\text{mode}}(f, f^*) = 0$ for every roll-in f (both Q and V). By Lemma 8.14, $|\widehat{\mathcal{E}}_{i,h}(f^*)| \leq \varepsilon_{\text{gen}}$ for all $i \in \{0, \dots, t-1\}$. Hence

$$\sum_{i=0}^{t-1} (\widehat{\mathcal{E}}_{i,h}(f^*))^2 \leq t \varepsilon_{\text{gen}}^2 \leq T \varepsilon_{\text{gen}}^2 = R^2,$$

so f^* is feasible at iteration t . Since f_t maximizes $V_{0,g}(s_0)$ over the feasible set and $V_{0,f^*}(s_0) = V_0^*(s_0)$, we have $V_{0,f_t}(s_0) \geq V_0^*(s_0)$. $\square$

Lemma 8.16 (Performance bound via self Bellman error). *For every t ,*

$$V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} \left| \mathcal{E}_h^{\text{mode}}(f_t, f_t) \right| = \sum_{h=0}^{H-1} \left| \langle X_h(f_t), W_h(f_t) \rangle \right|.$$

Proof. By Lemma 8.15, $V_0^*(s_0) \leq V_{0,f_t}(s_0)$, so

$$V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) \leq V_{0,f_t}(s_0) - V_0^{\pi_{f_t}}(s_0).$$

Let $(s_h, a_h)_{h=0}^{H-1}$ be the trajectory generated by executing π_{f_t} in the true MDP. Since $a_h = \pi_{h,f_t}(s_h)$ and $V_{h,f_t}(s_h) = Q_{h,f_t}(s_h, a_h)$,

$$V_{0,f_t}(s_0) - \sum_{h=0}^{H-1} r_h(s_h, a_h) = \sum_{h=0}^{H-1} (Q_{h,f_t}(s_h, a_h) - r_h(s_h, a_h) - V_{h+1,f_t}(s_{h+1})).$$

Taking expectation over the trajectory gives

$$V_{0,f_t}(s_0) - V_0^{\pi_{f_t}}(s_0) = \sum_{h=0}^{H-1} \mathbb{E}[\ell_h(s_h, a_h, s_{h+1}; f_t)] = \sum_{h=0}^{H-1} \mathcal{E}_h^{\text{mode}}(f_t, f_t),$$

and the absolute-value bound follows. Finally, Bellman rank gives $\mathcal{E}_h^{\text{mode}}(f_t, f_t) = \langle X_h(f_t), W_h(f_t) \rangle$. $\square$

Proof of Theorem 8.6. Fix a regularization parameter $\lambda > 0$ (chosen at the end) and define, for each stage h ,

$$\Sigma_{t,h} := \lambda I + \sum_{i=0}^{t-1} X_h(f_i) X_h(f_i)^\top, \quad \Sigma_{0,h} = \lambda I.$$

First, we put bound on $\|W_h(f_t)\|_{\Sigma_{t,h}}$. Since f_t is feasible, $\sum_{i=0}^{t-1} (\widehat{\mathcal{E}}_{i,h}(f_t))^2 \leq R^2$. On the good event, for each $i \in \{0, \dots, t-1\}$,

$$|\langle X_h(f_i), W_h(f_t) \rangle| = |\mathcal{E}_h^{\text{mode}}(f_i, f_t)| \leq |\widehat{\mathcal{E}}_{i,h}(f_t)| + \varepsilon_{\text{gen}}.$$

Thus,

$$\sum_{i=0}^{t-1} \langle X_h(f_i), W_h(f_t) \rangle^2 \leq 2 \sum_{i=0}^{t-1} (\widehat{\mathcal{E}}_{i,h}(f_t))^2 + 2t \varepsilon_{\text{gen}}^2 \leq 2R^2 + 2T \varepsilon_{\text{gen}}^2 = 4T \varepsilon_{\text{gen}}^2.$$

Therefore,

$$\|W_h(f_t)\|_{\Sigma_{t,h}}^2 = \lambda \|W_h(f_t)\|_2^2 + \sum_{i=0}^{t-1} \langle X_h(f_i), W_h(f_t) \rangle^2 \leq \lambda B_W^2 + 4T \varepsilon_{\text{gen}}^2. \quad (8.4)$$

Now, we bound $\|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}$ for a given t . For each fixed h , the matrix determinant lemma gives

$$\det(\Sigma_{t+1,h}) = \det(\Sigma_{t,h}) \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2\right),$$

so summing over $t = 0, \dots, T-1$ yields

$$\sum_{t=0}^{T-1} \ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2\right) = \ln \frac{\det(\Sigma_{T,h})}{\det(\lambda I)}.$$

Since $\|X_h(f_t)\|_2 \leq B_X$,

$$\text{tr}(\Sigma_{T,h}) \leq d\lambda + \sum_{t=0}^{T-1} \|X_h(f_t)\|_2^2 \leq d\lambda + TB_X^2.$$

Using $\det(M) \leq (\text{tr}(M)/d)^d$ for PSD M gives

$$\ln \frac{\det(\Sigma_{T,h})}{\det(\lambda I)} \leq d \ln \left(1 + \frac{TB_X^2}{d\lambda}\right).$$

Summing over h and averaging over t yields

$$\frac{1}{T} \sum_{t=0}^{T-1} \sum_{h=0}^{H-1} \ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2\right) \leq \frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda}\right).$$

Hence there exists $t^* \in \{0, \dots, T-1\}$ such that

$$\sum_{h=0}^{H-1} \ln \left(1 + \|X_h(f_{t^*})\|_{\Sigma_{t^*,h}^{-1}}^2\right) \leq \frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda}\right).$$

Since each summand is nonnegative, this implies that for every h ,

$$\|X_h(f_{t^*})\|_{\Sigma_{t^*,h}^{-1}}^2 \leq \exp \left(\frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda}\right) \right) - 1. \quad (8.5)$$

By Lemma 8.16 and Cauchy–Schwarz,

$$\begin{aligned} V_0^*(s_0) - V_0^{\pi_{f_{t^*}}}(s_0) &\leq \sum_{h=0}^{H-1} \|X_h(f_{t^*})\|_{\Sigma_{t^*,h}^{-1}} \|W_h(f_{t^*})\|_{\Sigma_{t^*,h}} \\ &\leq H \sqrt{\lambda B_W^2 + 4T \varepsilon_{\text{gen}}^2} \sqrt{\exp\left(\frac{Hd}{T} \ln\left(1 + \frac{TB_X^2}{d\lambda}\right)\right) - 1}, \end{aligned}$$

using (8.4) and (8.5).

Now choose

$$\lambda := \frac{B_X^2}{\frac{B_X^2 B_W^2}{\varepsilon_{\text{gen}}^2} + 1}, \quad L := \ln\left(eHd\left(\frac{B_X^2 B_W^2}{\varepsilon_{\text{gen}}^2} + 1\right)\right), \quad T := \lceil cHdL \rceil,$$

with $c > 0$ a sufficiently large universal constant. Then $\lambda B_W^2 \leq \varepsilon_{\text{gen}}^2$ and, since $T \asymp HdL$,

$$\ln\left(1 + \frac{TB_X^2}{d\lambda}\right) = \ln\left(1 + \frac{T}{d}\left(\frac{B_X^2 B_W^2}{\varepsilon_{\text{gen}}^2} + 1\right)\right) \leq L + \ln(3cL),$$

which implies (for c large enough) that

$$\frac{Hd}{T} \ln\left(1 + \frac{TB_X^2}{d\lambda}\right) \leq 1.$$

Using $\exp(x) - 1 \leq 2x$ for $x \in [0, 1]$ yields

$$V_0^*(s_0) - V_0^{\pi_{f_{t^*}}}(s_0) \leq C_0 \varepsilon_{\text{gen}} \sqrt{dH^3 L}$$

for a universal constant $C_0 > 0$.

Finally, we select the output iterate. Algorithm 8 returns $\hat{\pi} = \pi_{f_{\hat{t}}}$ where $\hat{t} \in \arg \min_{t \in \{0, \dots, T-1\}} \sum_{h=0}^{H-1} |\hat{\mathcal{E}}_{t,h}(f_t)|$. On the good event, for every t and h ,

$$|\mathcal{E}_h^{\text{mode}}(f_t, f_t)| \leq |\hat{\mathcal{E}}_{t,h}(f_t)| + \varepsilon_{\text{gen}}, \quad |\hat{\mathcal{E}}_{t,h}(f_t)| \leq |\mathcal{E}_h^{\text{mode}}(f_t, f_t)| + \varepsilon_{\text{gen}}.$$

Therefore,

$$\begin{aligned} \sum_{h=0}^{H-1} |\mathcal{E}_h^{\text{mode}}(f_{\hat{t}}, f_{\hat{t}})| &\leq \sum_{h=0}^{H-1} |\hat{\mathcal{E}}_{\hat{t},h}(f_{\hat{t}})| + H\varepsilon_{\text{gen}} \\ &\leq \sum_{h=0}^{H-1} |\hat{\mathcal{E}}_{t^*,h}(f_{t^*})| + H\varepsilon_{\text{gen}} \\ &\leq \sum_{h=0}^{H-1} |\mathcal{E}_h^{\text{mode}}(f_{t^*}, f_{t^*})| + 2H\varepsilon_{\text{gen}}. \end{aligned}$$

Combining with Lemma 8.16 and the bound for t^* gives

$$V_0^*(s_0) - V_0^{\hat{\pi}}(s_0) \leq C_0 \varepsilon_{\text{gen}} \sqrt{dH^3 L} + 2H\varepsilon_{\text{gen}}.$$

Since $L = \ln(eHd(\dots)) \geq 1$ and $d \geq 1$, the additive term $2H\varepsilon_{\text{gen}}$ is dominated by the leading term up to a constant factor, yielding the stated bound of Theorem 8.6. $\square$

8.6 Bibliographic Remarks and Further Readings

The Bellman rank was introduced by Jiang et al. [2017] as a structural condition under which data collected under one policy can be reused to test many candidate value functions. The original formulation corresponds to the *V-version* used in this chapter. Subsequent work clarified that there are two natural variants, depending on whether the average Bellman error is defined under the state–action occupancy of the roll-in policy (the *Q-version*) or under roll-in states paired with the greedy action of the tested hypothesis (the *V-version*); see Du et al. [2021], Jin et al. [2021]. We focus on PAC-RL guarantees (returning one near-optimal policy). Regret guarantees are subtler, particularly for the *V-version*, because estimating $\mathcal{E}_h^V(f, g)$ typically requires explicit exploration or importance weighting rather than a purely on-policy estimate. In the regret setting, Dong et al. [2020] obtained $\tilde{O}(\sqrt{T})$ regret under a *V*-Bellman-rank-type assumption using a careful sampling and elimination procedure. For the *Q-version*, on-policy estimation enables more direct optimistic algorithms: Jin et al. [2021] established $\tilde{O}(\sqrt{T})$ regret guarantees in this framework. Related regret bounds under linear Bellman completeness were obtained earlier by Zanette et al. [2020].

A closely related perspective is the *bilinear class* framework of Du et al. [2021]. The bilinear class can be viewed as a natural extension of Bellman rank that unifies a number of low-complexity models through low-rank factorizations of average Bellman errors. While we do not define the full bilinear class in this chapter, our definitions of *Q*- and *V*-Bellman rank follow the variants emphasized in Du et al. [2021], Jin et al. [2021], and many of the examples here are drawn from Jiang et al. [2017] and Du et al. [2021]. The analysis in Du et al. [2021] also motivates the elliptical-potential style argument used in our PAC proof.

Another complexity measure is the *Eluder dimension*, introduced by Russo and Van Roy [2013] for bandits and extended to reinforcement learning in, e.g., Osband and Roy [2014], Wang et al. [2020b], Ayoub et al. [2020a], Jin et al. [2021]. The Bellman–Eluder dimension and bilinear-rank-type conditions are distinct. At a high level, bilinear rank is often straightforward to verify for standard parametric models, and it captures many settings where RL is known to be statistically efficient. The Bellman–Eluder dimension can be viewed as a different way to control exploration by quantifying how quickly the learner can rule out hypotheses. We refer the reader to Jin et al. [2021] for a detailed treatment and comparisons. The bilinear class also subsumes witness-rank-type conditions [Sun et al., 2019a], yielding coverage of models such as factored MDPs [Kearns and Koller, 1999]; see Du et al. [2021] for a discussion of separations where Bellman rank or Bellman–Eluder dimension can be exponentially larger than bilinear-rank-based measures.

We conclude with pointers on several special cases highlighted in this chapter. Feature selection and representation learning in low-rank models appear already in Jiang et al. [2017]. Later work developed oracle-efficient approaches and refined statistical guarantees; see Agarwal et al. [2020b], Uehara et al. [2021]. Model-free approaches were also studied, for example by Modi et al. [2021]. Sparse linear MDP models were proposed in Hao et al. [2021] (see also related work on feature selection and sparsity in structured MDPs). The linear mixture model was introduced and analyzed in Modi et al. [2020a], Ayoub et al. [2020a], Zhou et al. [2021b,a]. The linear Q^*-V^* model and the low-occupancy model appear explicitly in Du et al. [2021].

Finally, Foster et al. [2023] studies an asymptotic minimax view of interactive learning and proposes the *decision–estimation coefficient* (DEC), a complexity measure characterized via a Le Cam notion of statistical estimation and shown to be both necessary and sufficient (in that asymptotic sense) for sample-efficient interactive learning.

Part III

Policy Optimization

Chapter 9

Policy Gradient Methods and Non-Convex Optimization

For a starting-state distribution $\mu \in \Delta(\mathcal{S})$, define

$$J_\mu(\theta) := V^{\pi_\theta}(\mu) = \mathbb{E}_{s_0 \sim \mu} [V^{\pi_\theta}(s_0)].$$

We study differentiable stochastic policy classes

$$\Pi = \{\pi_\theta : \theta \in \Theta \subseteq \mathbb{R}^d\}$$

and the optimization problem

$$\max_{\theta \in \Theta} J_\mu(\theta).$$

We suppress the MDP subscript throughout the chapter.

A direct optimization over deterministic policies is typically not differentiable, which motivates the use of stochastic parameterizations.

Example 9.1 (Softmax policies). For the tabular softmax parameterization,

$$\pi_\theta(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta_{s,a'})}, \quad \theta \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}. \quad (9.1)$$

The closure of this policy class contains all stationary deterministic policies.

Example 9.2 (Log-linear policies). Given features $\phi_{s,a} \in \mathbb{R}^d$, consider

$$\pi_\theta(a|s) = \frac{\exp(\theta \cdot \phi_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta \cdot \phi_{s,a'})}, \quad \theta \in \mathbb{R}^d.$$

Example 9.3 (Neural softmax policies). Let $f_\theta(s, a)$ be a differentiable scalar-valued function, for example a neural network. The induced softmax policy class is

$$\pi_\theta(a|s) = \frac{\exp(f_\theta(s, a))}{\sum_{a' \in \mathcal{A}} \exp(f_\theta(s, a'))}, \quad \theta \in \mathbb{R}^d.$$

9.1 Policy Gradient Expressions

Let $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots)$ denote a trajectory. For a fixed policy π , the trajectory distribution under start distribution μ is

$$\Pr_{\mu}^{\pi}(\tau) = \mu(s_0)\pi(a_0|s_0)P(s_1|s_0, a_0)\pi(a_1|s_1)P(s_2|s_1, a_1) \cdots \quad (9.2)$$

We write $\Pr_{\mu}^{\pi_{\theta}}$ when the policy is parameterized by θ .

Define the discounted return of a trajectory by

$$R(\tau) := \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t).$$

Then

$$J_{\mu}(\theta) = V^{\pi_{\theta}}(\mu) = \mathbb{E}_{\tau \sim \Pr_{\mu}^{\pi_{\theta}}} [R(\tau)].$$

It will be convenient to also write the discounted visitation distributions

$$d_{\mu}^{\pi}(s) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t \mathbf{1}\{s_t = s\} \right], \quad d_{\mu}^{\pi}(s, a) := d_{\mu}^{\pi}(s) \pi(a|s).$$

The next theorem records the basic policy gradient identities. We assume throughout that $\theta \mapsto \pi_{\theta}(a|s)$ is differentiable and that differentiation may be interchanged with the relevant infinite sums / expectations. This is standard for the policy classes above in the tabular discounted setting.

Theorem 9.4. *For any differentiable policy class $\{\pi_{\theta}\}$ and any start distribution μ ,*

$$\nabla_{\theta} J_{\mu}(\theta) = \mathbb{E}_{\tau \sim \Pr_{\mu}^{\pi_{\theta}}} \left[R(\tau) \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \right].$$

Moreover,

$$\nabla_{\theta} J_{\mu}(\theta) = \mathbb{E}_{\tau \sim \Pr_{\mu}^{\pi_{\theta}}} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t|s_t) \right],$$

and therefore

$$\nabla_{\theta} J_{\mu}(\theta) = \frac{1}{1 - \gamma} \mathbb{E}_{(s,a) \sim d_{\mu}^{\pi_{\theta}}} \left[Q^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a|s) \right].$$

Finally,

$$\nabla_{\theta} J_{\mu}(\theta) = \frac{1}{1 - \gamma} \mathbb{E}_{(s,a) \sim d_{\mu}^{\pi_{\theta}}} \left[A^{\pi_{\theta}}(s, a) \nabla_{\theta} \log \pi_{\theta}(a|s) \right].$$

Proof. For the first identity,

$$\begin{aligned} \nabla_{\theta} J_{\mu}(\theta) &= \nabla_{\theta} \sum_{\tau} R(\tau) \Pr_{\mu}^{\pi_{\theta}}(\tau) \\ &= \sum_{\tau} R(\tau) \nabla_{\theta} \Pr_{\mu}^{\pi_{\theta}}(\tau) \\ &= \sum_{\tau} R(\tau) \Pr_{\mu}^{\pi_{\theta}}(\tau) \nabla_{\theta} \log \Pr_{\mu}^{\pi_{\theta}}(\tau). \end{aligned}$$

Since neither the start distribution μ nor the transition kernel depends on θ ,

$$\nabla_{\theta} \log \Pr_{\mu}^{\pi_{\theta}}(\tau) = \sum_{t=0}^{\infty} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t),$$

which proves the REINFORCE identity.

For the second identity, define

$$G_t := \sum_{u=t}^{\infty} \gamma^{u-t} r(s_u, a_u),$$

so that

$$R(\tau) = \sum_{u=0}^{t-1} \gamma^u r(s_u, a_u) + \gamma^t G_t.$$

Fix t , and let

$$\mathcal{F}_t := \sigma(s_0, a_0, r_0, \dots, s_{t-1}, a_{t-1}, r_{t-1}, s_t)$$

be the history up to time t , before sampling a_t . Then

$$\mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \mid \mathcal{F}_t] = \sum_{a \in \mathcal{A}} \pi_{\theta}(a | s_t) \nabla_{\theta} \log \pi_{\theta}(a | s_t) = \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s_t) = 0.$$

Hence the past rewards do not contribute:

$$\mathbb{E} \left[\left(\sum_{u=0}^{t-1} \gamma^u r(s_u, a_u) \right) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] = 0.$$

Also, by the Markov property,

$$\mathbb{E}[G_t \mid s_t, a_t] = Q^{\pi_{\theta}}(s_t, a_t).$$

Therefore

$$\mathbb{E}[R(\tau) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)] = \mathbb{E}[\gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)].$$

Summing over t gives

$$\nabla_{\theta} J_{\mu}(\theta) = \mathbb{E}_{\tau \sim \Pr_{\mu}^{\pi_{\theta}}} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right].$$

The discounted occupancy form follows from the definition of $d_{\mu}^{\pi_{\theta}}(s, a)$.

For the advantage expression, write

$$Q^{\pi_{\theta}}(s, a) = A^{\pi_{\theta}}(s, a) + V^{\pi_{\theta}}(s)$$

and note that

$$\mathbb{E}_{a \sim \pi_{\theta}(\cdot | s)} [V^{\pi_{\theta}}(s) \nabla_{\theta} \log \pi_{\theta}(a | s)] = V^{\pi_{\theta}}(s) \nabla_{\theta} \sum_a \pi_{\theta}(a | s) = 0.$$

Substituting proves the final identity. □

The action-value and advantage forms are typically more useful for Monte Carlo estimation.

9.2 Monte Carlo Gradient Estimation

The previous theorem suggests a simulation-based estimator that only requires sampled trajectories. For a single trajectory τ , define

$$\widehat{Q}_t := \sum_{u=t}^{\infty} \gamma^{u-t} r(s_u, a_u),$$

and

$$\widehat{\nabla_{\theta} J_{\mu}}(\tau) := \sum_{t=0}^{\infty} \gamma^t \widehat{Q}_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t).$$

Lemma 9.5. *We have*

$$\mathbb{E}_{\tau \sim \text{Pr}_{\mu}^{\pi_{\theta}}} [\widehat{\nabla_{\theta} J_{\mu}}(\tau)] = \nabla_{\theta} J_{\mu}(\theta).$$

Proof. By the tower property and the Markov property,

$$\mathbb{E}[\widehat{Q}_t | s_t, a_t] = Q^{\pi_{\theta}}(s_t, a_t).$$

Therefore

$$\begin{aligned} \mathbb{E}[\widehat{\nabla_{\theta} J_{\mu}}(\tau)] &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \widehat{Q}_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \\ &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_{\theta}}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right], \end{aligned}$$

which is exactly the action-value expression in Theorem 9.4. $\square$

In practice, one typically uses finite-horizon episodes or truncates the infinite-horizon trajectory. In the discounted setting, truncating after T_{tr} steps introduces a bias of order $O(\gamma^{T_{\text{tr}}}/(1-\gamma))$.

9.2.1 Baselines

A basic and extremely useful variance-reduction device is to subtract a baseline.

Lemma 9.6 (Baselines preserve unbiasedness). *Let $(b_t)_{t \geq 0}$ be any sequence such that, for each t , b_t is measurable with respect to $\mathcal{F}_t$ (the history up to time t , before sampling a_t). Assume the expectations below are well-defined. Then*

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t b_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] = 0.$$

Consequently,

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t (\widehat{Q}_t - b_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] = \nabla_{\theta} J_{\mu}(\theta).$$

Proof. For each t ,

$$\begin{aligned} \mathbb{E}[b_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)] &= \mathbb{E}[\mathbb{E}[b_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) | \mathcal{F}_t]] \\ &= \mathbb{E}[b_t \mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) | \mathcal{F}_t]] \\ &= 0, \end{aligned}$$

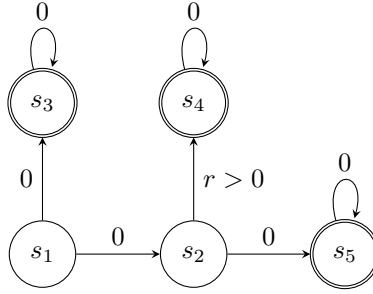


Figure 9.1: (Non-concavity example) A deterministic MDP corresponding to Lemma 9.7 where $V^{\pi_\theta}(s)$ is not concave. Numbers on arrows represent the rewards for each action.

using the same identity as in the proof of Theorem 9.4. Summing over t gives the first claim, and the second follows from Lemma 9.5. $\square$

The most common choice is a state-dependent baseline $b_t = b(s_t)$, often intended to approximate $V^{\pi_\theta}(s_t)$. This leads to the familiar advantage-style estimator. More generally, any baseline measurable with respect to the history before sampling a_t is valid.

9.3 Non-Convex Optimization

Even in very small MDPs, the map $\theta \mapsto J_\mu(\theta)$ is generally non-concave.

Proposition 9.7 (Non-concavity of the softmax parameterization). *There exists an MDP such that the optimization problem*

$$\max_{\theta} V^{\pi_\theta}(s_1)$$

is non-concave under the softmax parameterization.

Proof. Consider the MDP in Figure 9.1. Only the actions at states s_1 and s_2 affect the value. Let a_1 denote the “up” action and a_2 the “right” action. Then

$$V^\pi(s_1) = \pi(a_2|s_1) \pi(a_1|s_2) r.$$

Write the parameters in the order

$$(\theta_{s_1, a_1}, \theta_{s_1, a_2}, \theta_{s_2, a_1}, \theta_{s_2, a_2}).$$

Consider

$$\theta^{(1)} = (0, \log 3, \log 3, 0), \quad \theta^{(2)} = (0, -\log 3, -\log 3, 0).$$

Under the softmax parameterization,

$$\pi^{(1)}(a_2|s_1) = \frac{3}{4}, \quad \pi^{(1)}(a_1|s_2) = \frac{3}{4}, \quad V^{(1)}(s_1) = \frac{9}{16}r,$$

while

$$\pi^{(2)}(a_2|s_1) = \frac{1}{4}, \quad \pi^{(2)}(a_1|s_2) = \frac{1}{4}, \quad V^{(2)}(s_1) = \frac{1}{16}r.$$

The midpoint is

$$\theta^{(\text{mid})} = \frac{\theta^{(1)} + \theta^{(2)}}{2} = (0, 0, 0, 0),$$

for which

$$\pi^{(\text{mid})}(a_2|s_1) = \frac{1}{2}, \quad \pi^{(\text{mid})}(a_1|s_2) = \frac{1}{2}, \quad V^{(\text{mid})}(s_1) = \frac{1}{4}r.$$

Thus

$$V^{(\text{mid})}(s_1) < \frac{1}{2}V^{(1)}(s_1) + \frac{1}{2}V^{(2)}(s_1),$$

so the objective is not concave. $\square$

For this reason, the standard tools of convex optimization do not directly apply. Nevertheless, smooth non-convex optimization still provides useful guarantees: if the gradient is small, then the iterate is close to a first-order stationary point.

Let $J : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable. We say J is β -smooth if

$$\|\nabla J(\theta) - \nabla J(\theta')\|_2 \leq \beta \|\theta - \theta'\|_2 \quad \forall \theta, \theta' \in \mathbb{R}^d. \quad (9.3)$$

We now record two standard consequences.

Lemma 9.8. *Let $J : \mathbb{R}^d \rightarrow \mathbb{R}$ be β -smooth and bounded above by*

$$J^* := \sup_{\theta} J(\theta) < \infty.$$

Consider gradient ascent

$$\theta_{t+1} = \theta_t + \eta \nabla J(\theta_t)$$

with stepsize $\eta = 1/\beta$. Then for every $T \geq 1$,

$$\min_{0 \leq t < T} \|\nabla J(\theta_t)\|_2^2 \leq \frac{2\beta(J^* - J(\theta_0))}{T}.$$

Proof. By β -smoothness,

$$J(\theta_{t+1}) \geq J(\theta_t) + \eta \|\nabla J(\theta_t)\|_2^2 - \frac{\beta\eta^2}{2} \|\nabla J(\theta_t)\|_2^2.$$

With $\eta = 1/\beta$, this becomes

$$J(\theta_{t+1}) \geq J(\theta_t) + \frac{1}{2\beta} \|\nabla J(\theta_t)\|_2^2.$$

Summing from $t = 0$ to $T - 1$ gives

$$\frac{1}{2\beta} \sum_{t=0}^{T-1} \|\nabla J(\theta_t)\|_2^2 \leq J^* - J(\theta_0).$$

The claim follows by taking the minimum over $t < T$. $\square$

Lemma 9.9. *Let $J : \mathbb{R}^d \rightarrow \mathbb{R}$ be β -smooth and bounded above by J^* . Suppose we run the stochastic gradient ascent updates*

$$\theta_{t+1} = \theta_t + \eta g_t,$$

where

$$\mathbb{E}[g_t \mid \theta_t] = \nabla J(\theta_t), \quad \mathbb{E}[\|g_t - \nabla J(\theta_t)\|_2^2 \mid \theta_t] \leq \sigma^2,$$

and $\eta \in (0, 1/\beta]$ is constant. Then for every $T \geq 1$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla J(\theta_t)\|_2^2 \leq \frac{2(J^* - J(\theta_0))}{\eta T} + \beta\eta\sigma^2.$$

In particular,

$$\min_{0 \leq t < T} \mathbb{E} \|\nabla J(\theta_t)\|_2^2 \leq \frac{2(J^* - J(\theta_0))}{\eta T} + \beta\eta\sigma^2.$$

Proof. Taking conditional expectation in the smoothness inequality gives

$$\begin{aligned} \mathbb{E}[J(\theta_{t+1}) \mid \theta_t] &\geq J(\theta_t) + \eta \nabla J(\theta_t) \cdot \mathbb{E}[g_t \mid \theta_t] - \frac{\beta\eta^2}{2} \mathbb{E}[\|g_t\|_2^2 \mid \theta_t] \\ &= J(\theta_t) + \eta \|\nabla J(\theta_t)\|_2^2 - \frac{\beta\eta^2}{2} \left(\|\nabla J(\theta_t)\|_2^2 + \mathbb{E}[\|g_t - \nabla J(\theta_t)\|_2^2 \mid \theta_t] \right) \\ &\geq J(\theta_t) + \left(\eta - \frac{\beta\eta^2}{2} \right) \|\nabla J(\theta_t)\|_2^2 - \frac{\beta\eta^2}{2} \sigma^2. \end{aligned}$$

Since $\eta \leq 1/\beta$,

$$\eta - \frac{\beta\eta^2}{2} \geq \frac{\eta}{2}.$$

Therefore,

$$\frac{\eta}{2} \mathbb{E} \|\nabla J(\theta_t)\|_2^2 \leq \mathbb{E}[J(\theta_{t+1})] - \mathbb{E}[J(\theta_t)] + \frac{\beta\eta^2}{2} \sigma^2.$$

Summing over $t = 0, \dots, T-1$ and using $\mathbb{E}[J(\theta_T)] \leq J^*$ yields

$$\frac{\eta}{2} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla J(\theta_t)\|_2^2 \leq J^* - J(\theta_0) + \frac{\beta\eta^2 T}{2} \sigma^2.$$

Divide by $\eta T/2$. □

In the policy-gradient setting, these lemmas apply to $J_\mu(\theta) = V^{\pi_\theta}(\mu)$ whenever the policy class is sufficiently regular to make J_μ smooth and the Monte Carlo gradient estimator has controlled variance. Later chapters will exploit additional structure beyond generic stationarity in order to obtain global guarantees.

Global optimality vs. first-order stationarity. Obtaining an approximate first-order stationary point is, in general, a fairly weak guarantee. As we will see in the next chapter, policy gradient methods can become trapped at first-order stationary points that are still far from globally optimal, and in some cases far from any meaningful local optimum.

9.4 Bibliographic Remarks and Further Readings

The REINFORCE algorithm is due to Williams [1992], and is an example of the likelihood-ratio method for gradient estimation Glynn [1990].

For the optimization facts used in this chapter, including smooth non-convex gradient and stochastic gradient methods, we refer the reader to Beck [2017]; see also Ghadimi and Lan [2013] for standard stochastic-gradient convergence bounds to approximate stationary points.

Chapter 10

Optimality

In this chapter we study global convergence properties of policy gradient methods when the gradients are computed exactly. We largely restrict attention to the tabular softmax policy class from Example 9.1.

Given a start-state distribution $\rho \in \Delta(\mathcal{S})$, our ultimate objective is

$$\max_{\theta \in \Theta} V^{\pi_\theta}(\rho).$$

As we will see, it is often helpful to optimize under a different start-state distribution μ , even when performance is ultimately measured under ρ :

$$\max_{\theta \in \Theta} V^{\pi_\theta}(\mu).$$

Recall the normalized discounted state-visitation distribution

$$d_\mu^\pi(s) := (1 - \gamma) \mathbb{E}_{s_0 \sim \mu} \left[\sum_{t=0}^{\infty} \gamma^t \mathbf{1}\{s_t = s\} \right].$$

We will frequently use the fact that

$$d_\mu^\pi(s) \geq (1 - \gamma) \mu(s) \quad \forall s \in \mathcal{S}. \quad (10.1)$$

We study three algorithms for the softmax policy class: the vanilla policy gradient method, a log-barrier regularized variant, and the natural policy gradient method. The main message is that first-order stationarity alone is not enough for global optimality, but additional structure can restore strong guarantees.

10.1 Vanishing gradients and the need for coverage

Before turning to the softmax policy class, it is helpful to isolate the basic difficulty: small gradients need not imply near-optimality. If a policy rarely visits rewarding states, then the gradient can be very small simply because the learning signal almost never appears.

This issue already arises in the chain MDP of Figure 10.1. The starting state is s_0 , the discount factor is $\gamma = H/(H + 1)$, and we use the direct parameterization

$$\pi_\theta(a_i|s) = \theta_{s,a_i}, \quad i = 1, 2, 3, \quad \pi_\theta(a_4|s) = 1 - \theta_{s,a_1} - \theta_{s,a_2} - \theta_{s,a_3}.$$

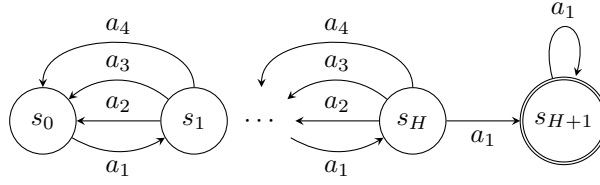


Figure 10.1: (Vanishing gradient example) A deterministic, chain MDP of length $H + 2$. We consider a policy where $\pi(a|s_i) = \theta_{s_i,a}$ for $i = 1, 2, \dots, H$. Rewards are 0 everywhere other than $r(s_{H+1}, a_1) = 1$. See Proposition 10.1.

Thus the parameter vector has dimension $3|\mathcal{S}|$ and we do not over-parameterize the policy.

If the action corresponding to moving right is assigned probability less than $1/4$ at every state, then the probability of reaching the rewarding end of the chain is exponentially small in H . The next proposition records a quantitative form of this phenomenon.

Proposition 10.1 (Vanishing gradients at suboptimal parameters). *Consider the chain MDP of Figure 10.1, with $H + 2$ states, discount factor $\gamma = H/(H + 1)$, and the direct policy parameterization above. Suppose θ satisfies $0 < \theta < 1$ componentwise and*

$$\theta_{s,a_1} < \frac{1}{4} \quad \text{for all } s \in \mathcal{S}.$$

Then for all

$$k \leq \frac{H}{40 \log(2H)} - 1,$$

we have

$$\|\nabla_{\theta}^k V^{\pi_{\theta}}(s_0)\| \leq (1/3)^{H/4},$$

where $\nabla_{\theta}^k V^{\pi_{\theta}}(s_0)$ is the k th derivative tensor and the norm is the operator norm. Furthermore,

$$V^{\star}(s_0) - V^{\pi_{\theta}}(s_0) \geq \frac{H+1}{8} - \frac{(H+1)^2}{3^H}.$$

We omit the proof; see the bibliographic notes. The proposition shows that even exact gradients, and indeed many higher-order derivatives, can be exponentially small at highly suboptimal parameters. Thus generic non-convex optimization guarantees do not by themselves imply global convergence.

10.2 Policy gradient ascent with softmax policies

We now return to the softmax parameterization

$$\pi_{\theta}(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a'} \exp(\theta_{s,a'})}.$$

The following coordinate formula is immediate from differentiation:

$$\frac{\partial \log \pi_{\theta}(a|s)}{\partial \theta_{s',a'}} = \mathbf{1}\{s = s'\} \left(\mathbf{1}\{a = a'\} - \pi_{\theta}(a'|s) \right).$$

Combining this with the advantage form of the policy gradient from Theorem 9.4 gives a particularly useful identity.

Lemma 10.2. *For the softmax policy class,*

$$\frac{\partial V^{\pi_\theta}(\mu)}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s) \pi_\theta(a|s) A^{\pi_\theta}(s, a). \quad (10.2)$$

Proof. Using the advantage expression in Theorem 9.4,

$$\begin{aligned} \frac{\partial V^{\pi_\theta}(\mu)}{\partial \theta_{s',a'}} &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[A^{\pi_\theta}(s, a) \frac{\partial \log \pi_\theta(a|s)}{\partial \theta_{s',a'}} \right] \\ &= \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s') \mathbb{E}_{a \sim \pi_\theta(\cdot|s')} \left[A^{\pi_\theta}(s', a) (\mathbf{1}\{a = a'\} - \pi_\theta(a'|s')) \right]. \end{aligned}$$

The first term in the expectation is

$$\pi_\theta(a'|s') A^{\pi_\theta}(s', a'),$$

and the second is

$$\pi_\theta(a'|s') \sum_a \pi_\theta(a|s') A^{\pi_\theta}(s', a) = 0.$$

This proves the claim. $\square$

The identity (10.2) makes the coverage issue explicit: the effective learning rate of coordinate (s, a) is scaled by $d_\mu^{\pi_\theta}(s) \pi_\theta(a|s)$. If a state is rarely visited, then its parameters move slowly.

The vanilla policy gradient ascent update is

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta V^{\pi_{\theta^{(t)}}}(\mu). \quad (10.3)$$

We write

$$\pi^{(t)} := \pi_{\theta^{(t)}}, \quad V^{(t)} := V^{\pi_{\theta^{(t)}}}.$$

Although the problem is non-convex, the following theorem shows that the vanilla policy gradient method is asymptotically globally convergent under a full-support assumption on μ .

Theorem 10.3 (Global convergence for the softmax parameterization). *Assume the update rule (10.3) and suppose $\mu(s) > 0$ for all $s \in \mathcal{S}$. If*

$$\eta \leq \frac{(1-\gamma)^3}{8},$$

then for every state s ,

$$V^{(t)}(s) \rightarrow V^*(s) \quad \text{as } t \rightarrow \infty.$$

We do not prove this result here. The theorem is asymptotic, and the worst-case convergence rate is believed to be extremely slow. The full-support condition on μ is essential: without it, states outside the support of $d_\mu^{\pi_\theta}$ may receive vanishing updates even when they are crucial for performance under another start-state distribution ρ .

10.3 Log barrier regularization

A major source of difficulty with softmax policies is that the parameters can grow large in magnitude, driving the policy toward determinism and causing some action probabilities to become extremely small. This slows learning and complicates global analysis.

To prevent this, we consider a barrier that penalizes small probabilities. Recall that

$$\text{KL}(p\|q) := \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right].$$

Let $\text{Unif}_{\mathcal{X}}$ denote the uniform distribution over a finite set $\mathcal{X}$. Define the log-barrier regularized objective

$$\begin{aligned} L_{\lambda}(\theta) &:= V^{\pi_{\theta}}(\mu) - \lambda \mathbb{E}_{s \sim \text{Unif}_{\mathcal{S}}} \left[\text{KL}(\text{Unif}_{\mathcal{A}} \parallel \pi_{\theta}(\cdot|s)) \right] \\ &= V^{\pi_{\theta}}(\mu) + \frac{\lambda}{|\mathcal{S}||\mathcal{A}|} \sum_{s,a} \log \pi_{\theta}(a|s) + \lambda \log |\mathcal{A}|. \end{aligned} \quad (10.4)$$

The additive constant is irrelevant for optimization.

The regularized gradient ascent update is

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_{\theta} L_{\lambda}(\theta^{(t)}). \quad (10.5)$$

The key fact is that approximate first-order stationarity for L_{λ} implies near-optimality for the original value function, with a dependence on the mismatch between the optimization distribution μ and the target distribution ρ .

Theorem 10.4 (Log barrier regularization). *Suppose θ satisfies*

$$\|\nabla_{\theta} L_{\lambda}(\theta)\|_2 \leq \varepsilon_{\text{opt}} \quad \text{and} \quad \varepsilon_{\text{opt}} \leq \frac{\lambda}{2|\mathcal{S}||\mathcal{A}|}.$$

Then for every start-state distribution ρ ,

$$V^{\pi_{\theta}}(\rho) \geq V^{\star}(\rho) - \frac{2\lambda}{1-\gamma} \left\| \frac{d_{\rho}^{\pi^{\star}}}{\mu} \right\|_{\infty}.$$

Proof. By the performance difference lemma,

$$\begin{aligned} V^{\star}(\rho) - V^{\pi_{\theta}}(\rho) &= \frac{1}{1-\gamma} \sum_{s,a} d_{\rho}^{\pi^{\star}}(s) \pi^{\star}(a|s) A^{\pi_{\theta}}(s, a) \\ &\leq \frac{1}{1-\gamma} \sum_s d_{\rho}^{\pi^{\star}}(s) \max_a A^{\pi_{\theta}}(s, a). \end{aligned}$$

Thus it is enough to show that for every state s ,

$$\max_a A^{\pi_{\theta}}(s, a) \leq \frac{2\lambda}{|\mathcal{S}|\mu(s)}.$$

Fix (s, a) such that $A^{\pi_{\theta}}(s, a) \geq 0$. Differentiating (10.4) and using (10.2),

$$\frac{\partial L_{\lambda}(\theta)}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_{\mu}^{\pi_{\theta}}(s) \pi_{\theta}(a|s) A^{\pi_{\theta}}(s, a) + \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_{\theta}(a|s) \right). \quad (10.6)$$

Since $A^{\pi_{\theta}}(s, a) \geq 0$, we have

$$\frac{\partial L_{\lambda}(\theta)}{\partial \theta_{s,a}} \geq \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_{\theta}(a|s) \right).$$

Also,

$$\left| \frac{\partial L_{\lambda}(\theta)}{\partial \theta_{s,a}} \right| \leq \|\nabla_{\theta} L_{\lambda}(\theta)\|_2 \leq \varepsilon_{\text{opt}},$$

so

$$\varepsilon_{\text{opt}} \geq \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_{\theta}(a|s) \right).$$

Using $\varepsilon_{\text{opt}} \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$ gives

$$\pi_{\theta}(a|s) \geq \frac{1}{2|\mathcal{A}|}.$$

Now solve (10.6) for the advantage:

$$\begin{aligned} A^{\pi_{\theta}}(s, a) &= \frac{1 - \gamma}{d_{\mu}^{\pi_{\theta}}(s) \pi_{\theta}(a|s)} \left(\frac{\partial L_{\lambda}(\theta)}{\partial \theta_{s,a}} - \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_{\theta}(a|s) \right) \right) \\ &= \frac{1 - \gamma}{d_{\mu}^{\pi_{\theta}}(s)} \left(\frac{1}{\pi_{\theta}(a|s)} \frac{\partial L_{\lambda}(\theta)}{\partial \theta_{s,a}} + \frac{\lambda}{|\mathcal{S}|} \left(1 - \frac{1}{\pi_{\theta}(a|s)|\mathcal{A}|} \right) \right). \end{aligned}$$

Using $\pi_{\theta}(a|s) \geq 1/(2|\mathcal{A}|)$ and $\left| \frac{\partial L_{\lambda}}{\partial \theta_{s,a}} \right| \leq \varepsilon_{\text{opt}}$,

$$\begin{aligned} A^{\pi_{\theta}}(s, a) &\leq \frac{1 - \gamma}{d_{\mu}^{\pi_{\theta}}(s)} \left(2|\mathcal{A}| \varepsilon_{\text{opt}} + \frac{\lambda}{|\mathcal{S}|} \right) \\ &\leq \frac{2(1 - \gamma)\lambda}{|\mathcal{S}| d_{\mu}^{\pi_{\theta}}(s)}, \end{aligned}$$

where the last step uses $\varepsilon_{\text{opt}} \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$. Finally, by (10.1),

$$d_{\mu}^{\pi_{\theta}}(s) \geq (1 - \gamma)\mu(s),$$

so

$$A^{\pi_{\theta}}(s, a) \leq \frac{2\lambda}{|\mathcal{S}|\mu(s)}.$$

Since this holds for every (s, a) with nonnegative advantage, the same bound holds for $\max_a A^{\pi_{\theta}}(s, a)$. Substituting into the performance difference bound gives

$$\begin{aligned} V^{\star}(\rho) - V^{\pi_{\theta}}(\rho) &\leq \frac{1}{1 - \gamma} \sum_s d_{\rho}^{\pi^{\star}}(s) \frac{2\lambda}{|\mathcal{S}|\mu(s)} \\ &\leq \frac{2\lambda}{1 - \gamma} \left\| \frac{d_{\rho}^{\pi^{\star}}}{\mu} \right\|_{\infty}, \end{aligned}$$

since $\sum_s 1/|\mathcal{S}| = 1$ after factoring out the maximum ratio. □

We refer to

$$\left\| \frac{d_{\rho}^{\pi^{\star}}}{\mu} \right\|_{\infty}$$

as the *distribution mismatch coefficient*. The theorem shows that first-order stationarity becomes globally meaningful once the barrier prevents vanishing action probabilities and the optimization distribution μ provides adequate coverage of the states relevant for ρ .

Combining Theorem 10.4 with Lemma 9.8 yields an explicit iteration bound.

Corollary 10.5 (Iteration complexity with log barrier regularization). *Assume the softmax policy class and suppose L_{λ} is β_{λ} -smooth with*

$$\beta_{\lambda} \leq \frac{8\gamma}{(1 - \gamma)^3} + \frac{2\lambda}{|\mathcal{S}|}.$$

Initialize at the uniform policy, e.g. $\theta^{(0)} = 0$ in the tabular softmax parameterization, and run the updates (10.5) with step size $\eta = 1/\beta_\lambda$.

Let

$$\lambda := \frac{\epsilon(1-\gamma)}{2 \left\| \frac{d_{\rho}^{\pi^*}}{\mu} \right\|_{\infty}}.$$

Then

$$\min_{0 \leq t < T} \{V^*(\rho) - V^{(t)}(\rho)\} \leq \epsilon$$

whenever

$$T \geq \frac{320|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)^6 \epsilon^2} \left\| \frac{d_{\rho}^{\pi^*}}{\mu} \right\|_{\infty}^2.$$

Proof. By Theorem 10.4, it is enough to ensure

$$\|\nabla_{\theta} L_{\lambda}(\theta^{(t)})\|_2 \leq \frac{\lambda}{2|\mathcal{S}||\mathcal{A}|}.$$

Since $\theta^{(0)} = 0$ corresponds to the uniform policy, the regularization term vanishes initially, so

$$L_{\lambda}(\theta^{(0)}) = V^{(0)}(\mu) \geq 0.$$

Also $L_{\lambda}(\theta) \leq V^{\pi_{\theta}}(\mu) \leq 1/(1-\gamma)$ for every θ , hence

$$\sup_{\theta} L_{\lambda}(\theta) - L_{\lambda}(\theta^{(0)}) \leq \frac{1}{1-\gamma}.$$

Applying Lemma 9.8 to the objective L_{λ} gives

$$\min_{0 \leq t < T} \|\nabla_{\theta} L_{\lambda}(\theta^{(t)})\|_2^2 \leq \frac{2\beta_{\lambda}}{(1-\gamma)T}.$$

Thus it suffices that

$$\sqrt{\frac{2\beta_{\lambda}}{(1-\gamma)T}} \leq \frac{\lambda}{2|\mathcal{S}||\mathcal{A}|},$$

or equivalently,

$$T \geq \frac{8\beta_{\lambda}|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)\lambda^2}.$$

Using the bound on β_{λ} ,

$$\frac{8\beta_{\lambda}|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)\lambda^2} \leq \frac{64|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)^4\lambda^2} + \frac{16|\mathcal{S}||\mathcal{A}|^2}{(1-\gamma)\lambda}.$$

Since $\left\| \frac{d_{\rho}^{\pi^*}}{\mu} \right\|_{\infty} \geq 1$ and $\epsilon \leq 1/(1-\gamma)$ is the only meaningful regime, the chosen λ satisfies $\lambda \leq 1$, so the second term is dominated by the first. Hence

$$T \geq \frac{80|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)^4\lambda^2}$$

is sufficient. Substituting the chosen λ yields the stated bound. $\square$

The corollary gives a polynomial iteration complexity bound, but it depends on the distribution mismatch coefficient and on a strong barrier that keeps the policy away from the boundary of the simplex.

It is worth contrasting this log barrier with the more common entropy regularizer

$$\frac{1}{|S|} \sum_s H(\pi_\theta(\cdot|s)) = \frac{1}{|S|} \sum_s \sum_a -\pi_\theta(a|s) \log \pi_\theta(a|s).$$

The entropy regularizer is bounded, whereas the barrier

$$\text{KL}(\text{Unif}_{\mathcal{A}} \parallel \pi_\theta(\cdot|s))$$

diverges as any action probability tends to zero. The proof above crucially uses this aggressive penalty against vanishing probabilities.

10.4 The natural policy gradient

We now turn to a preconditioned gradient method based on the information geometry of the policy class.

For a parametric density $p_\theta(x)$, the Fisher information matrix is

$$\mathbb{E}_{x \sim p_\theta} [\nabla_\theta \log p_\theta(x) \nabla_\theta \log p_\theta(x)^\top].$$

For policies, we average this quantity over the discounted state-visitation distribution:

$$F_\rho(\theta) := \mathbb{E}_{s \sim d_\rho^\pi} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)^\top].$$

The natural policy gradient (NPG) update is

$$\theta^{(t+1)} = \theta^{(t)} + \eta F_\rho(\theta^{(t)})^\dagger \nabla_\theta V^{(t)}(\rho), \quad (10.7)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse.

In this section we optimize directly with respect to the performance distribution ρ , so there is no separate optimization measure μ .

For the softmax parameterization, the NPG update takes a particularly simple form.

Lemma 10.6 (Softmax NPG as soft policy iteration). *For the softmax parameterization (9.1), the NPG updates (10.7) satisfy*

$$\theta^{(t+1)} = \theta^{(t)} + \frac{\eta}{1-\gamma} A^{(t)} + \eta v,$$

where v is a state-dependent offset, i.e. $v_{s,a} = c_s$ for some constants $\{c_s\}_{s \in S}$. Consequently,

$$\pi^{(t+1)}(a|s) = \pi^{(t)}(a|s) \frac{\exp(\eta A^{(t)}(s,a)/(1-\gamma))}{Z_t(s)},$$

where

$$Z_t(s) := \sum_{a \in \mathcal{A}} \pi^{(t)}(a|s) \exp(\eta A^{(t)}(s,a)/(1-\gamma)).$$

Proof. Let $\pi = \pi_\theta$ for brevity. By Lemma 10.2,

$$[\nabla_\theta V^\pi(\rho)]_{s,a} = \frac{1}{1-\gamma} d_\rho^\pi(s) \pi(a|s) A^\pi(s,a).$$

For any vector $w \in \mathbb{R}^{|S||\mathcal{A}|}$, define

$$\bar{w}_s := \sum_{a \in \mathcal{A}} \pi(a|s) w_{s,a}.$$

Using the softmax derivative,

$$w^\top \nabla_\theta \log \pi(a|s) = w_{s,a} - \bar{w}_s.$$

Hence

$$[F_\rho(\theta)w]_{s,a} = d_\rho^\pi(s) \pi(a|s) (w_{s,a} - \bar{w}_s).$$

Comparing with the gradient formula above shows that any solution of

$$F_\rho(\theta)w = \nabla_\theta V^\pi(\rho)$$

must satisfy

$$w_{s,a} - \bar{w}_s = \frac{1}{1-\gamma} A^\pi(s, a).$$

Equivalently, every solution is of the form

$$w = \frac{1}{1-\gamma} A^\pi + v,$$

where v is state-dependent: $v_{s,a} = c_s$. The pseudoinverse selects the minimum-norm solution among these.

Substituting this into the update of θ proves the first claim. Since the softmax policy is invariant to adding the same constant to all action logits at a state, the state-dependent offset v disappears in the induced policy update, yielding the multiplicative-weights form. $\square$

A notable feature of this formula is that, although the ascent direction in (10.7) is defined using the gradient of $V^\pi(\rho)$, the closed-form update for the softmax policy depends only on the advantages and not explicitly on the visitation distribution d_ρ^π .

The following improvement lemma is the key quantitative step.

Lemma 10.7 (Improvement lower bound for NPG). *For the iterates generated by (10.7), for every start-state distribution μ ,*

$$V^{(t+1)}(\mu) - V^{(t)}(\mu) \geq \frac{1-\gamma}{\eta} \mathbb{E}_{s \sim \mu} [\log Z_t(s)] \geq 0.$$

Proof. First, Jensen's inequality and the identity

$$\mathbb{E}_{a \sim \pi^{(t)}(\cdot|s)} [A^{(t)}(s, a)] = 0$$

imply

$$\begin{aligned} \log Z_t(s) &= \log \sum_a \pi^{(t)}(a|s) \exp(\eta A^{(t)}(s, a)/(1-\gamma)) \\ &\geq \sum_a \pi^{(t)}(a|s) \frac{\eta}{1-\gamma} A^{(t)}(s, a) = 0. \end{aligned}$$

By the performance difference lemma,

$$V^{(t+1)}(\mu) - V^{(t)}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \mathbb{E}_{a \sim \pi^{(t+1)}(\cdot|s)} [A^{(t)}(s, a)].$$

Using the update formula from Lemma 10.6,

$$\frac{\eta}{1-\gamma} A^{(t)}(s, a) = \log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)}.$$

Therefore

$$\begin{aligned} V^{(t+1)}(\mu) - V^{(t)}(\mu) &= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \mathbb{E}_{a \sim \pi^{(t+1)}(\cdot|s)} \left[\log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)} \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \left[\text{KL}(\pi_s^{(t+1)} \parallel \pi_s^{(t)}) + \log Z_t(s) \right] \\ &\geq \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} [\log Z_t(s)]. \end{aligned}$$

Since $\log Z_t(s) \geq 0$ and $d_\mu^{(t+1)}(s) \geq (1-\gamma)\mu(s)$ by (10.1),

$$\mathbb{E}_{s \sim d_\mu^{(t+1)}} [\log Z_t(s)] \geq (1-\gamma) \mathbb{E}_{s \sim \mu} [\log Z_t(s)].$$

This proves the claim. $\square$

We can now show that NPG enjoys a dimension-free global convergence rate.

Theorem 10.8 (Global convergence for NPG). *Suppose we run the NPG updates (10.7) with start distribution ρ , initialization $\theta^{(0)} = 0$, and stepsize $\eta > 0$. Then for every $T \geq 1$,*

$$V^{(T)}(\rho) \geq V^*(\rho) - \frac{\log |\mathcal{A}|}{\eta T} - \frac{1}{(1-\gamma)^2 T}.$$

Proof. Let

$$d^* := d_\rho^*.$$

Since $\theta^{(0)} = 0$, the initial policy is uniform at every state, so

$$\text{KL}(\pi_s^* \parallel \pi_s^{(0)}) \leq \log |\mathcal{A}| \quad \forall s \in \mathcal{S}.$$

By the performance difference lemma,

$$V^*(\rho) - V^{(t)}(\rho) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^*} \mathbb{E}_{a \sim \pi^*(\cdot|s)} [A^{(t)}(s, a)].$$

Using the update formula from Lemma 10.6,

$$\frac{\eta}{1-\gamma} A^{(t)}(s, a) = \log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)}.$$

Hence

$$\begin{aligned} V^*(\rho) - V^{(t)}(\rho) &= \frac{1}{\eta} \mathbb{E}_{s \sim d^*} \mathbb{E}_{a \sim \pi^*(\cdot|s)} \left[\log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)} \right] \\ &= \frac{1}{\eta} \mathbb{E}_{s \sim d^*} \left[\text{KL}(\pi_s^* \parallel \pi_s^{(t)}) - \text{KL}(\pi_s^* \parallel \pi_s^{(t+1)}) + \log Z_t(s) \right]. \end{aligned}$$

Next, apply Lemma 10.7 with start distribution $d^\star$:

$$\frac{1}{\eta} \mathbb{E}_{s \sim d^\star} [\log Z_t(s)] \leq \frac{1}{1-\gamma} (V^{(t+1)}(d^\star) - V^{(t)}(d^\star)).$$

Therefore

$$\begin{aligned} V^\star(\rho) - V^{(t)}(\rho) &\leq \frac{1}{\eta} \mathbb{E}_{s \sim d^\star} [\text{KL}(\pi_s^\star \| \pi_s^{(t)}) - \text{KL}(\pi_s^\star \| \pi_s^{(t+1)})] \\ &\quad + \frac{1}{1-\gamma} (V^{(t+1)}(d^\star) - V^{(t)}(d^\star)). \end{aligned}$$

Now average over $t = 0, \dots, T-1$. Since Lemma 10.7 implies $V^{(t+1)}(\rho) \geq V^{(t)}(\rho)$, we have

$$V^\star(\rho) - V^{(T)}(\rho) \leq \frac{1}{T} \sum_{t=0}^{T-1} (V^\star(\rho) - V^{(t)}(\rho)).$$

Using the previous display and telescoping,

$$\begin{aligned} V^\star(\rho) - V^{(T)}(\rho) &\leq \frac{1}{\eta T} \mathbb{E}_{s \sim d^\star} \text{KL}(\pi_s^\star \| \pi_s^{(0)}) + \frac{1}{(1-\gamma)T} (V^{(T)}(d^\star) - V^{(0)}(d^\star)) \\ &\leq \frac{\log |\mathcal{A}|}{\eta T} + \frac{1}{(1-\gamma)^2 T}, \end{aligned}$$

since values lie in $[0, 1/(1-\gamma)]$. □

Thus, by choosing

$$\eta \geq (1-\gamma)^2 \log |\mathcal{A}|,$$

we obtain

$$V^{(T)}(\rho) \geq V^\star(\rho) - \frac{2}{(1-\gamma)^2 T}.$$

In particular, NPG reaches an ϵ -optimal policy after

$$T \leq \frac{2}{(1-\gamma)^2 \epsilon}$$

iterations, with no explicit dependence on $|\mathcal{S}|$.

10.5 Bibliographic Remarks and Further Readings

The natural policy gradient method was introduced in Kakade [2001]; see also Bagnell and Schneider [2003], Peters and Schaal [2008] for information-geometric perspectives.

The convergence results in this chapter are largely based on Agarwal et al. [2020d]. The proof strategy behind the NPG rate has connections to online-learning analyses in changing MDPs Even-Dar et al. [2009]; the sharper $O(1/T)$ rate stated here is due to Agarwal et al. [2020d]. Closely related $O(1/T)$ guarantees also follow from the analysis of mirror-descent modified policy iteration in Geist et al. [2019]. The observation that the online-learning proof strategy yields a convergence guarantee for NPG was made in Neu et al. [2017].

Building on Agarwal et al. [2020d], Mei et al. [2020] showed that both vanilla policy gradient ascent and entropy-regularized policy gradient ascent for the softmax class converge asymptotically at a $O(1/t)$ rate. It is not expected

that these methods admit finite-time polynomial guarantees in the relevant structural quantities (such as $|\mathcal{S}|$, $|\mathcal{A}|$, $1/(1 - \gamma)$, and the distribution mismatch coefficient).

Entropy regularization is standard in practical policy-gradient methods; see Mnih et al. [2016] and the more detailed empirical discussion in Ahmed et al. [2019].

Chapter 11

Function Approximation and the Natural Policy Gradient

In Chapter 10.5, we analyzed the natural policy gradient in the tabular softmax policy class and obtained global convergence guarantees. We now turn to the setting of *function approximation*, where the policy class

$$\Pi = \{\pi_\theta : \theta \in \mathbb{R}^d\}$$

may be far smaller than the set of all stochastic policies. In particular, we no longer assume that Π contains an optimal policy.

Accordingly, we adopt an *agnostic* viewpoint: given a start-state distribution ρ , we seek guarantees relative to a comparator policy $\tilde{\pi}$ (often the best policy in the class, but not necessarily). As in the previous chapter, it will be useful to distinguish the distribution with respect to which performance is measured from the distribution under which gradients or regression problems are formed.

Our focus is on the natural policy gradient (NPG) update

$$\theta \leftarrow \theta + \eta F_\rho(\theta)^\dagger \nabla_\theta V^{\pi_\theta}(\rho), \quad (11.1)$$

and on a closely related variant, which we call *Q-NPG*. The key theme of the chapter is that the effectiveness of these updates is governed by a precise approximation–estimation decomposition based on *compatible function approximation*.

11.1 Compatible function approximation and the NPG

We begin by recalling the geometric meaning of the natural policy gradient.

Fix a start-state distribution μ and a parameter θ . Let $\pi = \pi_\theta$ for brevity, and define the Fisher matrix

$$F_\mu(\theta) := \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\nabla_\theta \log \pi(a|s) \nabla_\theta \log \pi(a|s)^\top \right].$$

The compatible function approximation problem is the regression problem of approximating the advantage function using the score features $\nabla_\theta \log \pi(a|s)$.

Lemma 11.1 (Compatible function approximation). *Fix μ and θ , and let $\pi = \pi_\theta$. Let $w^\star$ be any minimizer of*

$$\min_{w \in \mathbb{R}^d} \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\left(A^\pi(s, a) - w^\top \nabla_\theta \log \pi(a|s) \right)^2 \right].$$

Define the predictor

$$\hat{A}^\pi(s, a) := (w^\star)^\top \nabla_\theta \log \pi(a|s).$$

Then

$$\nabla_\theta V^\pi(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\nabla_\theta \log \pi(a|s) \hat{A}^\pi(s, a) \right].$$

Proof. The first-order optimality condition for $w^\star$ is

$$\mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\left(A^\pi(s, a) - (w^\star)^\top \nabla_\theta \log \pi(a|s) \right) \nabla_\theta \log \pi(a|s) \right] = 0. \quad (11.2)$$

Rearranging and using the advantage form of the policy gradient from Theorem 9.4 yields

$$\nabla_\theta V^\pi(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\nabla_\theta \log \pi(a|s) \hat{A}^\pi(s, a) \right].$$

□

The next lemma shows that the NPG update direction is exactly the solution of this regression problem.

Lemma 11.2. *Fix μ and θ , and let $\pi = \pi_\theta$. Let $w^\star$ denote the minimum-Euclidean-norm minimizer of*

$$\min_{w \in \mathbb{R}^d} \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\left(A^\pi(s, a) - w^\top \nabla_\theta \log \pi(a|s) \right)^2 \right].$$

Then

$$F_\mu(\theta)^\dagger \nabla_\theta V^\pi(\mu) = \frac{1}{1-\gamma} w^\star.$$

Proof. By (11.2),

$$\mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} \left[\nabla_\theta \log \pi(a|s) A^\pi(s, a) \right] = F_\mu(\theta) w^\star.$$

Using the advantage expression from Theorem 9.4,

$$\nabla_\theta V^\pi(\mu) = \frac{1}{1-\gamma} F_\mu(\theta) w^\star.$$

Thus $w^\star$ is a minimum-norm solution of the linear system

$$F_\mu(\theta) w = (1-\gamma) \nabla_\theta V^\pi(\mu),$$

which is exactly the Moore–Penrose pseudoinverse solution. □

Lemma 11.2 shows that the NPG update can be viewed as

$$\theta \leftarrow \theta + \frac{\eta}{1-\gamma} w^\star,$$

where $w^\star$ is the compatible regression solution. This viewpoint is central in what follows.

11.2 Examples: NPG and Q-NPG

We now specialize the compatible approximation viewpoint to the policy classes most often used in practice. Throughout this section, we consider policy classes of the form

$$\Pi = \left\{ \pi_\theta(a|s) = \frac{\exp(f_\theta(s, a))}{\sum_{a' \in \mathcal{A}} \exp(f_\theta(s, a'))} \mid \theta \in \mathbb{R}^d \right\}, \quad (11.3)$$

where $f_\theta(s, a)$ is differentiable.

11.2.1 Log-linear policy classes

Suppose $f_\theta(s, a) = \theta^\top \phi_{s,a}$ for features $\phi_{s,a} \in \mathbb{R}^d$. Then

$$\pi_\theta(a|s) = \frac{\exp(\theta^\top \phi_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta^\top \phi_{s,a'})}.$$

Define the centered features

$$\bar{\phi}_{s,a}^\pi := \phi_{s,a} - \mathbb{E}_{a' \sim \pi(\cdot|s)}[\phi_{s,a'}].$$

A direct calculation shows

$$\nabla_\theta \log \pi_\theta(a|s) = \bar{\phi}_{s,a}^{\pi_\theta}.$$

Hence, by Lemma 11.2, the NPG update (11.1) is equivalent, up to rescaling of the learning rate, to

$$\theta \leftarrow \theta + \eta w^*, \quad w^* \in \underset{w}{\operatorname{argmin}} \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[(A^{\pi_\theta}(s, a) - w^\top \bar{\phi}_{s,a}^{\pi_\theta})^2 \right].$$

We will also consider a variant based on regressing Q rather than A :

$$\theta \leftarrow \theta + \eta w^*, \quad w^* \in \underset{w}{\operatorname{argmin}} \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[(Q^{\pi_\theta}(s, a) - w^\top \phi_{s,a})^2 \right].$$

We refer to this update rule as Q -NPG.

For log-linear policies, these updates can also be written directly in policy space. Indeed, an update of the form $\theta \leftarrow \theta + \eta w$ induces

$$\pi(a|s) \leftarrow \frac{\pi(a|s) \exp(\eta w^\top \phi_{s,a})}{Z_s},$$

where Z_s is the normalization constant. In the NPG case one may equivalently replace $\phi_{s,a}$ by the centered feature $\bar{\phi}_{s,a}^\pi$ inside the exponential, since the difference is a state-dependent shift that is absorbed by normalization.

11.2.2 Neural policy classes

Now let $f_\theta(s, a)$ be a differentiable neural network. Define

$$g_\theta(s, a) := \nabla_\theta f_\theta(s, a) - \mathbb{E}_{a' \sim \pi_\theta(\cdot|s)}[\nabla_\theta f_\theta(s, a')].$$

Then

$$\nabla_\theta \log \pi_\theta(a|s) = g_\theta(s, a).$$

Accordingly, the NPG update takes the form

$$\theta \leftarrow \theta + \eta w^*, \quad w^* \in \underset{w}{\operatorname{argmin}} \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[\left(A^{\pi_\theta}(s, a) - w^\top g_\theta(s, a) \right)^2 \right].$$

The Q -NPG variant is

$$\theta \leftarrow \theta + \eta w^*, \quad w^* \in \underset{w}{\operatorname{argmin}} \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[\left(Q^{\pi_\theta}(s, a) - w^\top \nabla_\theta f_\theta(s, a) \right)^2 \right].$$

11.3 A regret lemma for approximate NPG updates

We now give a generic mirror-descent-style inequality for updates of the form

$$\theta^{(t+1)} = \theta^{(t)} + \eta w^{(t)}. \quad (11.4)$$

The sequence $w^{(0)}, \dots, w^{(T-1)}$ is arbitrary; later we will specialize it to approximate regression solutions.

Let $\tilde{\pi}$ be an arbitrary comparator policy and write

$$\tilde{d} := d_\rho^{\tilde{\pi}}.$$

Lemma 11.3 (NPG regret lemma). *Assume that for every state s and action a , the map*

$$\theta \mapsto \log \pi_\theta(a|s)$$

is β -smooth. Consider the update rule (11.4), initialized at the uniform policy $\pi^{(0)}(\cdot|s) = \operatorname{Unif}_{\mathcal{A}}(\cdot)$ for all s , and assume

$$\|w^{(t)}\|_2 \leq W \quad \text{for all } t < T.$$

Define

$$\operatorname{err}_t := \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[A^{(t)}(s, a) - (w^{(t)})^\top \nabla_\theta \log \pi^{(t)}(a|s) \right].$$

If

$$\eta = \sqrt{\frac{2 \log |\mathcal{A}|}{\beta W^2 T}},$$

then

$$\min_{0 \leq t < T} \{V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho)\} \leq \frac{1}{1-\gamma} \left(W \sqrt{\frac{2\beta \log |\mathcal{A}|}{T}} + \frac{1}{T} \sum_{t=0}^{T-1} \operatorname{err}_t \right).$$

Proof. By smoothness of $\log \pi_\theta(a|s)$,

$$\log \pi^{(t+1)}(a|s) - \log \pi^{(t)}(a|s) \geq \eta \nabla_\theta \log \pi^{(t)}(a|s)^\top w^{(t)} - \frac{\beta \eta^2}{2} \|w^{(t)}\|_2^2.$$

Taking expectation over $s \sim \tilde{d}$, $a \sim \tilde{\pi}(\cdot|s)$ gives

$$\begin{aligned} & \mathbb{E}_{s \sim \tilde{d}} \left[\operatorname{KL}(\tilde{\pi}_s \| \pi_s^{(t)}) - \operatorname{KL}(\tilde{\pi}_s \| \pi_s^{(t+1)}) \right] \\ & \geq \eta \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[\nabla_\theta \log \pi^{(t)}(a|s)^\top w^{(t)} \right] - \frac{\beta \eta^2}{2} W^2. \end{aligned}$$

Add and subtract $A^{(t)}(s, a)$ inside the expectation, and use the performance difference lemma:

$$\mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} [A^{(t)}(s, a)] = (1 - \gamma)(V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho)).$$

This yields

$$(1 - \gamma)\eta(V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho)) \leq \mathbb{E}_{s \sim \tilde{d}} \left[\text{KL}(\tilde{\pi}_s \| \pi_s^{(t)}) - \text{KL}(\tilde{\pi}_s \| \pi_s^{(t+1)}) \right] + \frac{\beta\eta^2}{2} W^2 + \eta \text{err}_t.$$

Summing over $t = 0, \dots, T - 1$ and telescoping,

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} (V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho)) &\leq \frac{1}{(1 - \gamma)\eta T} \mathbb{E}_{s \sim \tilde{d}} \text{KL}(\tilde{\pi}_s \| \pi_s^{(0)}) \\ &\quad + \frac{\eta\beta W^2}{2(1 - \gamma)} + \frac{1}{(1 - \gamma)T} \sum_{t=0}^{T-1} \text{err}_t. \end{aligned}$$

Since $\pi_s^{(0)}$ is uniform, $\text{KL}(\tilde{\pi}_s \| \pi_s^{(0)}) \leq \log |\mathcal{A}|$ for every s . Substituting the chosen η gives the claim. $\square$

When $\text{err}_t = 0$ for all t , the lemma yields an $O(T^{-1/2})$ rate. This is slower than the sharper $O(1/T)$ rate obtained in Theorem 10.8 for exact NPG in the tabular softmax class. The point of Lemma 11.3 is not sharpness, but robustness: it cleanly isolates the effect of approximation and estimation error.

11.4 Q -NPG with function approximation

We now specialize to the log-linear policy class from Section 11.2.1 and analyze an approximate Q -NPG update.

For a state-action distribution ν , define the squared loss

$$L(w; \theta, \nu) := \mathbb{E}_{(s,a) \sim \nu} \left[(Q^{\pi_\theta}(s, a) - w^\top \phi_{s,a})^2 \right].$$

To discuss coverage, we allow the update to be based on a state-action distribution ν rather than only a state distribution. For a starting pair (s_0, a_0) , define

$$d_{s_0, a_0}^\pi(s, a) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr(s_t = s, a_t = a \mid s_0, a_0). \quad (11.5)$$

For a distribution ν over $\mathcal{S} \times \mathcal{A}$, define

$$d_\nu^\pi(s, a) := \mathbb{E}_{(s_0, a_0) \sim \nu} [d_{s_0, a_0}^\pi(s, a)].$$

We abbreviate

$$d^{(t)} := d_\nu^{\pi^{(t)}}.$$

The approximate Q -NPG update is

$$\theta^{(t+1)} = \theta^{(t)} + \eta w^{(t)}, \quad w^{(t)} \in \mathcal{W} := \{w : \|w\|_2 \leq W\}, \quad (11.6)$$

where $w^{(t)}$ is intended to approximately minimize $L(w; \theta^{(t)}, d^{(t)})$ over $\mathcal{W}$. We denote the exact constrained minimizer by

$$w_\star^{(t)} \in \underset{w \in \mathcal{W}}{\text{argmin}} L(w; \theta^{(t)}, d^{(t)}).$$

We now separate *excess risk* from *approximation error*.

Assumption 11.4 (Approximation and estimation error). For all $t < T$, the iterates satisfy

$$L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_{\star}^{(t)}; \theta^{(t)}, d^{(t)}) \leq \epsilon_{\text{stat}}$$

and

$$L(w_{\star}^{(t)}; \theta^{(t)}, d^{(t)}) \leq \epsilon_{\text{approx}}.$$

The estimation term ϵ_{stat} reflects how accurately the regression problem is solved. The approximation term ϵ_{approx} reflects model misspecification: even the best linear predictor of $Q^{\pi^{(t)}}$ under the fitting distribution $d^{(t)}$ may not be exact.

We also need a coverage assumption on the feature covariance. For any state-action distribution ν , let

$$\Sigma_{\nu} := \mathbb{E}_{(s,a) \sim \nu} [\phi_{s,a} \phi_{s,a}^{\top}].$$

Fix a comparator policy $\pi^{\star}$ and define the associated state-action distribution

$$d^{\star}(s, a) := d_{\rho}^{\pi^{\star}}(s) \cdot \text{Unif}_{\mathcal{A}}(a).$$

Assumption 11.5 (Relative condition number). The quantity

$$\kappa := \sup_{w \in \mathbb{R}^d \setminus \{0\}} \frac{w^{\top} \Sigma_{d^{\star}} w}{w^{\top} \Sigma_{\nu} w}$$

is finite.

Thus ν covers all feature directions that matter under the comparator distribution $d^{\star}$. As in linear bandits and linearly parameterized MDPs, this is a feature-space coverage assumption rather than a state-space one.

We can now state the main result.

Theorem 11.6. Fix a start-state distribution ρ , a start state-action distribution ν , and an arbitrary comparator policy $\pi^{\star}$. Assume the log-linear policy class, Assumptions 11.4 and 11.5, and that $\|\phi_{s,a}\|_2 \leq B$ for all (s, a) . Suppose $\theta^{(0)} = 0$ and

$$\eta = \sqrt{\frac{2 \log |\mathcal{A}|}{B^2 W^2 T}}.$$

Then

$$\begin{aligned} \mathbb{E} \left[\min_{0 \leq t < T} \{V^{\pi^{\star}}(\rho) - V^{(t)}(\rho)\} \right] &\leq \frac{BW}{1-\gamma} \sqrt{\frac{2 \log |\mathcal{A}|}{T}} \\ &\quad + \frac{2}{1-\gamma} \sqrt{\frac{|\mathcal{A}|}{1-\gamma} \left\| \frac{d^{\star}}{\nu} \right\|_{\infty} \epsilon_{\text{approx}}} \\ &\quad + \frac{2}{1-\gamma} \sqrt{\frac{|\mathcal{A}| \kappa}{1-\gamma} \epsilon_{\text{stat}}}. \end{aligned}$$

The theorem makes the distinction between approximation and estimation especially clear. The approximation error is transferred to the comparator distribution through the mismatch factor $\|d^{\star}/\nu\|_{\infty}$, while the estimation term depends only on the relative conditioning parameter κ .

Proof. For the log-linear policy class, $\log \pi_\theta(a|s)$ is B^2 -smooth as a function of θ . Applying Lemma 11.3 with comparator $\tilde{\pi} = \pi^*$,

$$\min_{0 \leq t < T} \{V^{\pi^*}(\rho) - V^{(t)}(\rho)\} \leq \frac{BW}{1-\gamma} \sqrt{\frac{2 \log |\mathcal{A}|}{T}} + \frac{1}{(1-\gamma)T} \sum_{t=0}^{T-1} \text{err}_t,$$

where

$$\text{err}_t = \mathbb{E}_{s \sim d_\rho^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot|s)} \left[A^{(t)}(s, a) - (w^{(t)})^\top \bar{\phi}_{s,a}^{\pi^{(t)}} \right].$$

We first decompose

$$\text{err}_t = T_{1,t} + T_{2,t},$$

where

$$\begin{aligned} T_{1,t} &:= \mathbb{E}_{s \sim d_\rho^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot|s)} \left[A^{(t)}(s, a) - (w_\star^{(t)})^\top \bar{\phi}_{s,a}^{\pi^{(t)}} \right], \\ T_{2,t} &:= \mathbb{E}_{s \sim d_\rho^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)})^\top \bar{\phi}_{s,a}^{\pi^{(t)}} \right]. \end{aligned}$$

For the first term, write

$$e_t(s, a) := Q^{(t)}(s, a) - (w_\star^{(t)})^\top \phi_{s,a}.$$

Since

$$A^{(t)}(s, a) - (w_\star^{(t)})^\top \bar{\phi}_{s,a}^{\pi^{(t)}} = e_t(s, a) - \mathbb{E}_{a' \sim \pi^{(t)}(\cdot|s)} [e_t(s, a')],$$

we have

$$\begin{aligned} |T_{1,t}| &\leq \mathbb{E}_{s \sim d_\rho^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot|s)} [|e_t(s, a)|] + \mathbb{E}_{s \sim d_\rho^{\pi^*}} \mathbb{E}_{a \sim \pi^{(t)}(\cdot|s)} [|e_t(s, a)|] \\ &\leq \sqrt{\mathbb{E}_{s \sim d_\rho^{\pi^*}, a \sim \pi^*(\cdot|s)} [e_t(s, a)^2]} + \sqrt{\mathbb{E}_{s \sim d_\rho^{\pi^*}, a \sim \pi^{(t)}(\cdot|s)} [e_t(s, a)^2]}. \end{aligned}$$

By the definition of $d^\star(s, a) = d_\rho^{\pi^*}(s) \text{Unif}_{\mathcal{A}}(a)$, for any conditional action distribution $\alpha(\cdot|s)$,

$$\mathbb{E}_{s \sim d_\rho^{\pi^*}, a \sim \alpha(\cdot|s)} [e_t(s, a)^2] \leq |\mathcal{A}| \mathbb{E}_{(s,a) \sim d^\star} [e_t(s, a)^2].$$

Therefore,

$$|T_{1,t}| \leq 2 \sqrt{|\mathcal{A}| L(w_\star^{(t)}; \theta^{(t)}, d^\star)}.$$

Next,

$$L(w_\star^{(t)}; \theta^{(t)}, d^\star) \leq \left\| \frac{d^\star}{d^{(t)}} \right\|_\infty L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}).$$

Since $d^{(t)} = d_\nu^{\pi^{(t)}}$ and (11.5) implies

$$d^{(t)}(s, a) \geq (1-\gamma) \nu(s, a),$$

we have

$$\left\| \frac{d^\star}{d^{(t)}} \right\|_\infty \leq \frac{1}{1-\gamma} \left\| \frac{d^\star}{\nu} \right\|_\infty.$$

Using Assumption 11.4,

$$|T_{1,t}| \leq 2 \sqrt{\frac{|\mathcal{A}|}{1-\gamma} \left\| \frac{d^\star}{\nu} \right\|_\infty} \epsilon_{\text{approx}}.$$

For the second term, let

$$u_t := w_\star^{(t)} - w^{(t)}.$$

Since

$$u_t^\top \bar{\phi}_{s,a}^{\pi^{(t)}} = u_t^\top \phi_{s,a} - \mathbb{E}_{a' \sim \pi^{(t)}(\cdot|s)}[u_t^\top \phi_{s,a'}],$$

the same argument as above gives

$$|T_{2,t}| \leq 2\sqrt{|\mathcal{A}| u_t^\top \Sigma_{d^*} u_t}.$$

By Assumption 11.5,

$$u_t^\top \Sigma_{d^*} u_t \leq \kappa u_t^\top \Sigma_\nu u_t.$$

Again using $d^{(t)} \geq (1 - \gamma)\nu$,

$$u_t^\top \Sigma_\nu u_t \leq \frac{1}{1 - \gamma} u_t^\top \Sigma_{d^{(t)}} u_t.$$

It remains to control $u_t^\top \Sigma_{d^{(t)}} u_t$. Since $w_\star^{(t)}$ minimizes $L(\cdot; \theta^{(t)}, d^{(t)})$ over the convex set $\mathcal{W}$ and $w^{(t)} \in \mathcal{W}$, the first-order optimality condition implies

$$L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \geq \|w^{(t)} - w_\star^{(t)}\|_{\Sigma_{d^{(t)}}}^2 = u_t^\top \Sigma_{d^{(t)}} u_t.$$

By Assumption 11.4,

$$u_t^\top \Sigma_{d^{(t)}} u_t \leq \epsilon_{\text{stat}}.$$

Combining,

$$|T_{2,t}| \leq 2\sqrt{\frac{|\mathcal{A}|\kappa}{1 - \gamma} \epsilon_{\text{stat}}}.$$

Hence

$$\text{err}_t \leq 2\sqrt{\frac{|\mathcal{A}|}{1 - \gamma} \left\| \frac{d^*}{\nu} \right\|_\infty \epsilon_{\text{approx}}} + 2\sqrt{\frac{|\mathcal{A}|\kappa}{1 - \gamma} \epsilon_{\text{stat}}}.$$

This bound is uniform in t , so substituting into Lemma 11.3 gives the theorem. Taking expectation preserves the bound. $\square$

The theorem shows that the two sources of error enter in qualitatively different ways.

First, the approximation term is multiplied by the density ratio

$$\left\| \frac{d^*}{\nu} \right\|_\infty,$$

which is a transfer term from the fitting distribution to the comparator distribution. This is the natural price of model misspecification.

Second, the estimation term is controlled instead by the relative condition number κ , which is a much milder coverage requirement. Intuitively, to estimate a linear predictor it is enough that ν cover the relevant feature directions, rather than all state-action pairs uniformly.

Finally, although κ may depend on the feature geometry, it need not depend directly on the size of the state space. In particular, one can often choose ν using only the feature set so that κ is controlled by a dimension-dependent quantity; we do not pursue this here.

11.5 Bibliographic Remarks and Further Readings

The notion of *compatible function approximation* is due to Sutton et al. [1999], which also proved the basic projection identity in Lemma 11.1. The close connection between compatible approximation and the natural policy gradient, as in Lemma 11.2, was noted in Kakade [2001].

The regret lemma (Lemma 11.3) is closely related to mirror-descent and online-learning analyses, going back to Even-Dar et al. [2009]. The specific performance guarantees in this chapter are derived from Agarwal et al. [2020d]. The Q -NPG update for log-linear policies is closely related to POLITEX Abbasi-Yadkori et al. [2019]. The same general perspective also applies to neural policy classes; see Agarwal et al. [2020d]. Related analyses of trust-region and natural gradient methods for neural policies appear in Liu et al. [2019], Schulman et al. [2015].

One of the main conceptual messages of this chapter is that direct policy optimization can tolerate approximation error in a comparatively mild way. The approximation term enters through a transfer factor to a *fixed* comparator distribution, whereas approximate value-function-based methods typically require much stronger control over errors under a far larger family of distributions.

Chapter 12

Conservative Policy Optimization

In this chapter, we consider a family of policy optimization algorithms that perform conservative policy updates, i.e., updates where the new policy stays close to the old policy. We start with conservative policy iteration (CPI) and trust-region constrained policy optimization (TRPO). Both CPI and TRPO can be understood as making small, incremental updates to the policy by forcing the new policy's action distribution to be not far from the old policy's. We will see that CPI achieves that by forming a new policy that is a mixture of the current policy and a local greedy policy, while TRPO forces that by explicitly adding a trust-region constraint in the optimization procedure. We will show that the practical version of TRPO gives an equivalent update procedure as Natural Policy Gradient. Finally, we present a new policy optimization algorithm called *Orthogonalized Policy Gradient* (OPG). OPG performs orthogonalization on the policy gradient before taking a gradient ascent step. We show that the orthogonalization step induces a conservative policy update, which leads to monotonic policy improvement.

The algorithm and analysis of CPI is adapted from the original one in Kakade and Langford [2002], and the we follow the presentation of TRPO from Schulman et al. [2015], while making a connection to the NGP algorithm.

12.1 Conservative Policy Iteration

We consider the discounted MDP here $\{\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho\}$ where ρ is the initial state distribution. Similar to Policy Gradient Methods, we assume that we have a restart distribution μ (i.e., the μ -restart setting). Throughout this section, for any policy π , we denote d_μ^π as the state-action visitation starting from $s_0 \sim \mu$ instead of ρ , and d^π the state-action visitation starting from the true initial state distribution ρ , i.e., $s_0 \sim \rho$. Similarly, we denote V_μ^π as expected discounted total reward of policy π starting at μ , while V^π as the expected discounted total reward of π with ρ as the initial state distribution. We assume $\mathcal{A}$ is discrete but $\mathcal{S}$ could be continuous.

CPI is based on the concept of *Reduction to Supervised Learning*. Specifically we will use the Approximate Greedy Policy Selector. We recall the definition of the ε -approximate Greedy Policy Selector $\mathcal{G}_\varepsilon(\pi, \Pi, \mu)$ below. Given a policy π , policy class Π , and a restart distribution μ , denote $\hat{\pi} = \mathcal{G}_\varepsilon(\pi, \Pi, \mu)$, we have that:

$$\mathbb{E}_{s \sim d_\mu^\pi} A^\pi(s, \hat{\pi}(s)) \geq \max_{\tilde{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^\pi} A^\pi(s, \tilde{\pi}(s)) - \varepsilon.$$

12.1.1 The CPI Algorithm

CPI, summarized in Alg. 9, will iteratively generate a sequence of policies π_t . Note we use $\pi_\alpha = (1-\alpha)\pi + \alpha\pi'$ to refer to a randomized policy which at any state s , chooses an action according to π with probability $1-\alpha$ and according to π'

with probability α . The greedy policy π' is computed using the ε -approximate greedy policy selector $\mathcal{G}_\varepsilon(\pi_t, \Pi, \mu)$. The algorithm is terminated when there is no significant one-step improvement over π_t , i.e., $\mathbb{E}_{s \sim d_\mu^{\pi_t}} A^{\pi_t}(s, \pi'(s)) \leq \varepsilon$.

Algorithm 9 Conservative Policy Iteration (CPI)

Input: Initial policy $\pi_0 \in \Pi$, accuracy parameter ε , mixing rate α .

```

1: for  $t = 0, 1, 2 \dots$  do
2:    $\pi' = \mathcal{G}_\varepsilon(\pi_t, \Pi, \mu)$ 
3:   if  $\mathbb{E}_{s \sim d_\mu^{\pi_t}} A^{\pi_t}(s, \pi'(s)) \leq \varepsilon$  then
4:     Return  $\pi_t$ 
5:   end if
6:   Update  $\pi_{t+1} = (1 - \alpha)\pi_t + \alpha\pi'$ 
7: end for
```

The main intuition behind the algorithm is that the stepsize α controls the difference between state distributions of π_t and π_{t+1} . Let us look into the performance difference lemma to get some intuition on this conservative update. From PDL, we have:

$$V_\mu^{\pi_{t+1}} - V_\mu^{\pi_t} = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\mu^{\pi_{t+1}}} A^{\pi_t}(s, \pi_{t+1}(s)) - \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\mu^{\pi_{t+1}}} A^{\pi_t}(s, \pi'(s)),$$

where the last equality we use the fact that $\pi_{t+1} = (1 - \alpha)\pi_t + \alpha\pi'$ and $A^{\pi_t}(s, \pi_t(s)) = 0$ for all s . Thus, if we can search for a policy $\pi' \in \Pi$ that maximizes $\mathbb{E}_{s \sim d_\mu^{\pi_{t+1}}} A^{\pi_t}(s, \pi'(s))$ and makes $\mathbb{E}_{s \sim d_\mu^{\pi_{t+1}}} A^{\pi_t}(s, \pi'(s)) > 0$, then we can guarantee policy improvement. However, at episode t , we do not know the state distribution of π_{t+1} and all we have access to is $d_\mu^{\pi_t}$. Thus, we explicitly make the policy update procedure conservative such that $d_\mu^{\pi_t}$ and the new policy's distribution $d_\mu^{\pi_{t+1}}$ are guaranteed not to be too different. Thus we can hope that $\mathbb{E}_{s \sim d_\mu^{\pi_{t+1}}} A^{\pi_t}(s, \pi'(s))$ is close to $\mathbb{E}_{s \sim d_\mu^{\pi_t}} A^{\pi_t}(s, \pi'(s))$, and the latter is something that we can manipulate using the greedy policy selector.

Below, we formalize the above intuition and show that with a small enough α , we indeed can ensure monotonic policy improvement.

We start from the following lemma which shows that π_{t+1} and π_t are close to each other in terms of total variation distance at any state, and $d_\mu^{\pi_{t+1}}$ and $d_\mu^{\pi_t}$ are close as well.

Lemma 12.1 (Similar Policies imply similar state visitations). *Consider any t , we have that:*

$$\|\pi_{t+1}(\cdot|s) - \pi_t(\cdot|s)\|_1 \leq 2\alpha, \forall s;$$

Further, we have:

$$\|d_\mu^{\pi_{t+1}} - d_\mu^{\pi_t}\|_1 \leq \frac{2\alpha\gamma}{1 - \gamma}.$$

Proof. The first claim in the above lemma comes from the definition of policy update:

$$\|\pi_{t+1}(\cdot|s) - \pi_t(\cdot|s)\|_1 = \alpha \|\pi_t(\cdot|s) - \pi'(\cdot|s)\|_1 \leq 2\alpha.$$

Denote $\mathbb{P}_h^\pi$ as the state distribution resulting from π at time step h with μ as the initial state distribution. We consider

bounding $\|\mathbb{P}_h^{\pi_{t+1}} - \mathbb{P}_h^{\pi_t}\|_1$ with $h \geq 1$.

$$\begin{aligned}
\mathbb{P}_h^{\pi_{t+1}}(s') - \mathbb{P}_h^{\pi_t}(s') &= \sum_{s,a} (\mathbb{P}_{h-1}^{\pi_{t+1}}(s) \pi_{t+1}(a|s) - \mathbb{P}_{h-1}^{\pi_t}(s) \pi_t(a|s)) P(s'|s, a) \\
&= \sum_{s,a} (\mathbb{P}_{h-1}^{\pi_{t+1}}(s) \pi_{t+1}(a|s) - \mathbb{P}_{h-1}^{\pi_{t+1}}(s) \pi_t(a|s) + \mathbb{P}_{h-1}^{\pi_{t+1}}(s) \pi_t(a|s) - \mathbb{P}_{h-1}^{\pi_t}(s) \pi_t(a|s)) P(s'|s, a) \\
&= \sum_s \mathbb{P}_{h-1}^{\pi_{t+1}}(s) \sum_a (\pi_{t+1}(a|s) - \pi_t(a|s)) P(s'|s, a) \\
&\quad + \sum_s (\mathbb{P}_{h-1}^{\pi_{t+1}}(s) - \mathbb{P}_{h-1}^{\pi_t}(s)) \sum_a \pi_t(a|s) P(s'|s, a).
\end{aligned}$$

Take absolute value on both sides, we get:

$$\begin{aligned}
\sum_{s'} |\mathbb{P}_h^{\pi_{t+1}}(s') - \mathbb{P}_h^{\pi_t}(s')| &\leq \sum_s \mathbb{P}_{h-1}^{\pi_{t+1}}(s) \sum_a |\pi_{t+1}(a|s) - \pi_t(a|s)| \sum_{s'} P(s'|s, a) \\
&\quad + \sum_s |\mathbb{P}_{h-1}^{\pi_{t+1}}(s) - \mathbb{P}_{h-1}^{\pi_t}(s)| \sum_{s'} \sum_a \pi_t(a|s) P(s'|s, a) \\
&\leq 2\alpha + \|\mathbb{P}_{h-1}^{\pi_{t+1}} - \mathbb{P}_{h-1}^{\pi_t}\|_1 \leq 4\alpha + \|\mathbb{P}_{h-2}^{\pi_{t+1}} - \mathbb{P}_{h-2}^{\pi_t}\|_1 = 2h\alpha.
\end{aligned}$$

Now use the definition of d_μ^π , we have:

$$d_\mu^{\pi_{t+1}} - d_\mu^{\pi_t} = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h (\mathbb{P}_h^{\pi_{t+1}} - \mathbb{P}_h^{\pi_t}).$$

Add ℓ_1 norm on both sides, we get:

$$\|d_\mu^{\pi_{t+1}} - d_\mu^{\pi_t}\|_1 \leq (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h 2h\alpha$$

It is not hard to verify that $\sum_{h=0}^{\infty} \gamma^h h = \frac{\gamma}{(1-\gamma)^2}$. Thus, we can conclude that:

$$\|d_\mu^{\pi_{t+1}} - d_\mu^{\pi_t}\|_1 \leq \frac{2\alpha\gamma}{1-\gamma}.$$

□

The above lemma states that if π_{t+1} and π_t are close in terms of total variation distance for every state, then the total variation distance between the resulting state visitations from π_{t+1} and π_t will be small up to a effective horizon $1/(1-\gamma)$ amplification.

The above lemma captures the key of the conservative policy update. Via the conservative policy update, we make sure that $d_\mu^{\pi_{t+1}}$ and $d_\mu^{\pi_t}$ are close to each other in terms of total variation distance. Now we use the above lemma to show a monotonic policy improvement.

Theorem 12.2 (Monotonic Improvement in CPI). *Assume $\gamma \in [0.5, 1)$. Consider any episode t . Denote $\mathbb{A} = \mathbb{E}_{s \sim d_\mu^{\pi_t}} \mathcal{A}^{\pi_t}(s, \pi'(s))$. We have:*

$$V_\mu^{\pi_{t+1}} - V_\mu^{\pi_t} \geq \frac{\alpha}{1-\gamma} \left(\mathbb{A} - \frac{2\alpha\gamma}{(1-\gamma)^2} \right)$$

Set $\alpha = \frac{\mathbb{A}(1-\gamma)^2}{4\gamma}$, we get:

$$V_\mu^{\pi_{t+1}} - V_\mu^{\pi_t} \geq \frac{\mathbb{A}^2(1-\gamma)}{8\gamma}.$$

The above lemma shows that as long as we still have positive one-step improvement, i.e., $\mathbb{A} > 0$, then we guarantee that π_{t+1} is strictly better than π_t .

Proof. Via PDL, we have:

$$V_{\mu}^{\pi_{t+1}} - V_{\mu}^{\pi_t} = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\mu}^{\pi_{t+1}}} \alpha A^{\pi_t}(s, \pi'(s)).$$

Recall Lemma 12.1, we have:

$$\begin{aligned} (1-\gamma)(V_{\mu}^{\pi_{t+1}} - V_{\mu}^{\pi_t}) &= \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \alpha A^{\pi_t}(s, \pi'(s)) + \mathbb{E}_{s \sim d_{\mu}^{\pi_{t+1}}} \alpha A^{\pi_t}(s, \pi'(s)) - \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \alpha A^{\pi_t}(s, \pi'(s)) \\ &\geq \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \alpha A^{\pi_t}(s, \pi'(s)) - \alpha \sup_{s,a,\pi} |A^{\pi}(s,a)| \|d_{\mu}^{\pi_t} - d_{\mu}^{\pi_{t+1}}\|_1 \\ &\geq \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \alpha A^{\pi_t}(s, \pi'(s)) - \frac{\alpha}{1-\gamma} \|d_{\mu}^{\pi_t} - d_{\mu}^{\pi_{t+1}}\|_1 \\ &\geq \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \alpha A^{\pi_t}(s, \pi'(s)) - \frac{2\alpha^2\gamma}{(1-\gamma)^2} = \alpha \left(\mathbb{A} - \frac{2\alpha\gamma}{(1-\gamma)^2} \right) \end{aligned}$$

where the first inequality we use the fact that for any two distributions P_1 and P_2 and any function f , $|\mathbb{E}_{x \sim P_1} f(x) - \mathbb{E}_{x \sim P_2} f(x)| \leq \sup_x |f(x)| \|P_1 - P_2\|_1$, for the second inequality, we use the fact that $|A^{\pi}(s,a)| \leq 1/(1-\gamma)$ for any π, s, a , and the last inequality uses Lemma 12.1.

For the second part of the above theorem, note that we want to maximize the policy improvement as much as possible by choosing α . So we can pick α which maximizes $\alpha(\mathbb{A} - 2\alpha\gamma/(1-\gamma)^2)$. This gives us the α we claimed in the lemma. Plug in α back into $\alpha(\mathbb{A} - 2\alpha\gamma/(1-\gamma)^2)$, we conclude the second part of the theorem. $\square$

Note that the above proof relies on two conditions at the same time: (1) $\max_s \|\pi_{t+1}(\cdot|s) - \pi_t(\cdot|s)\|_1 \leq O(\alpha)$, and (2) $\|d_{\mu}^{\pi_t} - d_{\mu}^{\pi_{t+1}}\|_1 \leq O(\alpha)$ which is derived from (1). The two conditions *together* ensure that the *correction term*: $|\mathbb{E}_{s \sim d_{\mu}^{\pi_{t+1}}} \mathbb{E}_{a \sim \pi_{t+1}} A^{\pi_t}(s,a) - \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \mathbb{E}_{a \sim \pi_{t+1}} A^{\pi_t}(s,a)|$, is in order $O(\alpha^2)$:

$$|\mathbb{E}_{s \sim d_{\mu}^{\pi_{t+1}}} \mathbb{E}_{a \sim \pi_{t+1}} A^{\pi_t}(s,a) - \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \mathbb{E}_{a \sim \pi_{t+1}} A^{\pi_t}(s,a)| \leq \|d_{\mu}^{\pi_t} - d_{\mu}^{\pi_{t+1}}\|_1 \max_s |\mathbb{E}_{a \sim \pi_{t+1}} A^{\pi_t}(s,a)| = O(\alpha^2).$$

If one only had the second condition, the monotonic improvement guarantee would not hold since the correction term can only scale linearly in α . We will revisit this point after we introduce TRPO and PPO. This is the key observation on why the practical version of TRPO, which only imposes $\mathbb{E}_{s \sim d_{\mu}^{\pi_t}} \text{KL}(\pi(\cdot|s)|\pi_t(\cdot|s))$ as the trust region constraint, does not guarantee monotonic improvement.

The above theorem indicates that with the right choice of α , we guarantee that the policy is making improvement as long as $\mathbb{A} > 0$. Recall the termination criteria in CPI where we terminate CPI when $\mathbb{A} \leq \varepsilon$. Putting these results together, we obtain the following overall convergence guarantee for the CPI algorithm.

Theorem 12.3 (Local optimality of CPI). *Algorithm 9 terminates in at most $8\gamma/(\varepsilon^2(1-\gamma)^2)$ steps and outputs a policy π_t satisfying $\max_{\pi \in \Pi} \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} A^{\pi_t}(s, \pi(s)) \leq 2\varepsilon$.*

Proof. Note that our reward is bounded in $[0, 1]$ which means that $V_{\mu}^{\pi} \in [0, 1/(1-\gamma)]$. Note that we have shown in Theorem 12.2, every iteration t , we have policy improvement at least $\frac{\mathbb{A}^2(1-\gamma)}{8\gamma}$, where recall $\mathbb{A}$ at episode t is defined as $\mathbb{A} = \mathbb{E}_{s \sim d_{\mu}^{\pi_t}} A^{\pi_t}(s, \pi'(s))$. If the algorithm does not terminate at episode t , then we guarantee that:

$$V_{\mu}^{\pi_{t+1}} \geq V_{\mu}^{\pi_t} + \frac{\varepsilon^2(1-\gamma)}{8\gamma}$$

Since V_{μ}^{π} is upper bounded by $1/(1-\gamma)$, so we can at most make improvement $8\gamma/(\varepsilon^2(1-\gamma)^2)$ many iterations.

Finally, recall that ε -approximate greedy policy selector $\pi' = \mathcal{G}_\varepsilon(\pi_t, \Pi, \mu)$, we have:

$$\max_{\pi \in \Pi} \mathbb{E}_{s \sim d_\mu^{\pi_t}} A^{\pi_t}(s, \pi(s)) \leq \mathbb{E}_{s \sim d_\mu^{\pi_t}} A^{\pi_t}(s, \pi'(s)) + \varepsilon \leq 2\varepsilon$$

This concludes the proof. $\square$

Theorem 12.3 can be viewed as a local optimality guarantee in a sense. It shows that when CPI terminates, we cannot find a policy $\pi \in \Pi$ that achieves local improvement over the returned policy more than ε . However, this does not necessarily imply that the value of π is close to V^* . However, similar to the policy gradient analysis, we can turn this local guarantee into a global one when the restart distribution μ covers d^{π^*} . We formalize this intuition next.

Theorem 12.4 (Global optimality of CPI). *Upon termination, we have a policy π such that:*

$$V^* - V^\pi \leq \frac{2\varepsilon + \epsilon_\Pi}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty,$$

where $\epsilon_\Pi := \mathbb{E}_{s \sim d_\mu^\pi} [\max_{a \in \mathcal{A}} A^\pi(s, a)] - \max_{\pi \in \Pi} \mathbb{E}_{s \sim d_\mu^\pi} [A^\pi(s, \pi(s))]$.

In other words, if our policy class is rich enough to approximate the policy $\max_{a \in \mathcal{A}} A^\pi(s, a)$ under d_μ^π , i.e., ϵ_Π is small, and μ covers d^{π^*} in a sense that $\left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty < \infty$, CPI guarantees to find a near-optimal policy.

Proof. By the performance difference lemma,

$$\begin{aligned} V^* - V^\pi &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi^*}} A^\pi(s, \pi^*(s)) \\ &\leq \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi^*}} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &\leq \frac{1}{1-\gamma} \left\| \frac{d^{\pi^*}}{d_\mu^\pi} \right\|_\infty \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &\leq \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &\leq \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty \left[\max_{\hat{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^{\hat{\pi}}} A^\pi(s, \hat{\pi}(s)) - \max_{\hat{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^{\hat{\pi}}} A^\pi(s, \hat{\pi}(s)) + \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \right] \\ &\leq \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty (2\varepsilon + \epsilon_\Pi), \end{aligned}$$

where the second inequality holds due to the fact that $\max_a A^\pi(s, a) \geq 0$, the third inequality uses the fact that $d_\mu^\pi(s) \geq (1-\gamma)\mu(s)$ for any s and π , and the last inequality uses the definition ϵ_Π and Theorem 12.3. $\square$

It is informative to contrast CPI and policy gradient algorithms due to the similarity of their guarantees. Both provide local optimality guarantees. For CPI, the local optimality always holds, while for policy gradients it requires a smooth value function as a function of the policy parameters. If the distribution mismatch between an optimal policy and the output of the algorithm is not too large, then both algorithms further yield a near-optimal policy. The similarities are not so surprising. Both algorithms operate by making local improvements to the current policy at each iteration, by inspecting its advantage function. The changes made to the policy are controlled using a stepsize parameter in both approaches. It is the actual mechanism of the improvement which differs in the two cases. Policy gradients assume that the policy's reward is a differentiable function of the parameters, and hence make local improvements through gradient ascent. The differentiability is certainly an assumption and does not necessarily hold for all policy classes.

An easy example is when the policy itself is not an easily differentiable function of its parameters. For instance, if the policy is parametrized by regression trees, then performing gradient updates can be challenging.

In CPI, on the other hand, the basic computational primitive required on the policy class is the ability to maximize the advantage function relative to the current policy. Notice that Algorithm 9 does not necessarily restrict to a policy class, such as a set of parametrized policies as in policy gradients. Indeed, due to the reduction to a supervised learning approach (e.g., using the weighted classification oracle CO), we can parameterize the policy class via decision trees, for instance. This property makes CPI extremely attractive. Any policy class over which efficient supervised learning algorithms exist can be adapted to reinforcement learning with performance guarantees.

A second important difference between CPI and policy gradients is in the notion of locality. Policy gradient updates are local in the parameter space, and we hope that this makes small enough changes to the state distribution that the new policy is indeed an improvement on the older one (for instance, when we invoke the performance difference lemma between successive iterates). While this is always true in expectation for correctly chosen stepsizes based on properties of stochastic gradient ascent on smooth functions, the variance of the algorithm and lack of robustness to suboptimal stepsizes can make the algorithm somewhat finicky. Indeed, there are a host of techniques in the literature to both lower the variance (through control variates) and explicitly control the state distribution mismatch between successive iterates of policy gradients (through trust region techniques). On the other hand, CPI explicitly controls the amount of perturbation to the state distribution by carefully mixing policies in a manner which does not drastically alter the trajectories with high probability. Indeed, this insight is central to the proof of CPI, and has been instrumental in several follow-ups, both in the direct policy improvement as well as policy gradient literature.

12.2 Trust Region Policy Optimization

So far, we have seen policy gradient methods and CPI, which all use a small step-size to ensure incremental updates in policies. Another popular approach for incremental policy update is to explicitly force a small change in policies' distribution via a trust region constraint. More specifically, let us go back to the general policy parameterization π_θ . At iteration t with the current policy π_{θ_t} , the ideal version of Trust Region Policy Optimization (TRPO) is interested in the following local trust-region constrained optimization:

$$\begin{aligned} \max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} A^{\pi_{\theta_t}}(s, a) \\ \text{s.t., } \max_s \|\pi_{\theta}(\cdot|s) - \pi_{\theta_t}(\cdot|s)\|_1 \leq \delta. \end{aligned}$$

TRPO shows that the above constrained optimization has the property of monotonic policy improvement, which is similar to Theorem 12.2 from CPI.

Note that in the above constrained optimization problem, the constraint is based on $\max_s$ in the constraints, i.e., it requires the new policy to be close to the old policy at every state. Such a constraint is impossible to optimize for modern MDPs where the number of states is large. Instead of solving the above constrained optimization problem, the practical version of TRPO relaxes the constraint to an average KL constraint:

$$\max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} A^{\pi_{\theta_t}}(s, a) \tag{12.1}$$

$$\text{s.t., } \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \text{KL}(\pi_{\theta}(\cdot|s) || \pi_{\theta_t}(\cdot|s)) \leq \delta. \tag{12.2}$$

Note that after this relaxation from $\max_s$ to $\mathbb{E}_s$, the monotonic improvement guarantee from TRPO does not hold anymore. Monotonic improvement requires stronger conditions $\max_s \|\pi_{\theta}(\cdot|s) - \pi_{\theta_t}(\cdot|s)\|_1 \leq \delta$. Recall that in CPI, we indeed have π_{t+1} being close to π_t under TV distance for all states.

To optimize the above KL constrained objective, TRPO proposes to further linearize the objective and quadratize the

constraint. First-order Taylor expansion on the objective leads to the following linear objective:

$$\max_{\theta} \left\langle \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \nabla_{\theta} \ln \pi_{\theta_t}(a|s) A^{\pi_{\theta_t}}(s, a), \theta \right\rangle. \quad (12.3)$$

Applying the second-order Taylor expansion on the constraint, we get:

$$\mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \text{KL}(\pi_{\theta}(\cdot|s) || \pi_{\theta_t}(\cdot|s)) \approx \frac{1}{2} (\theta - \theta_t)^{\top} F_{\theta_t} (\theta - \theta_t),$$

where the Hessian F_{θ_t} is exactly equal to the Fisher information matrix:

$$F_{\theta_t} = \mathbb{E}_{s, a \sim d_{\mu}^{\pi_{\theta_t}}} \nabla \ln \pi_{\theta_t}(a|s) (\nabla \ln \pi_{\theta_t}(a|s))^{\top}.$$

Note that when we perform Taylor expansion on the KL constraint at θ_t , the zero-th order term and the first order term are exactly zero.

Putting the above together, denote $\Delta = \theta - \theta_t$, now we have:

$$\begin{aligned} & \max_{\Delta} \langle \Delta, \nabla_{\theta} V^{\pi_{\theta_t}} \rangle, \\ & s.t., \Delta^{\top} F_{\theta_t} \Delta \leq \delta, \end{aligned}$$

which gives the following update procedure:

$$\theta_{t+1} = \theta_t + \Delta = \theta_t + \sqrt{\frac{\delta}{(\nabla V^{\pi_{\theta_t}})^{\top} F_{\theta_t}^{-1} \nabla V^{\pi_{\theta_t}}}} \cdot F_{\theta_t}^{-1} \nabla V^{\pi_{\theta_t}},$$

where note that we use the self-normalized learning rate computed using the trust region parameter δ . Note that the above update is essentially NPG.

In practice, one does not explicitly compute F_{θ_t} and perform matrix inverse to compute $F_{\theta_t}^{-1}$ which can be computationally expensive when the matrix is high-dimensional. Instead one can directly compute the vector $F_{\theta_t}^{-1} \nabla V^{\pi_{\theta_t}}$ via the following minimization problem:

$$\arg \min_x \|F_{\theta_t} x - \nabla V^{\pi_{\theta_t}}\|_2^2$$

One can approximately minimize the above objective via the conjugate gradient method. The conjugate gradient method only involves the Hessian-vector product, which can be implemented efficiently via modern automatic differentiation tools without ever explicitly materializing the Hessian matrix.

12.2.1 Proximal Policy Optimization

Scaling TRPO to modern large neural networks is still challenging since conjugate gradient can be slow. The proximal policy optimization (PPO) algorithm further simplifies the TRPO's trust-region constraint by using only first-order gradients, thereby improving scalability.

We first rewrite the objective function in TRPO using importance weighting:

$$\max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \quad (12.4)$$

where we can easily approximate the expectation $\mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)}$ via finite samples $s \sim d^{\pi_{\theta_t}}, a \sim \pi_{\theta_t}(\cdot|s)$.

To make sure $\pi_\theta(a|s)$ and $\pi_{\theta_t}(a|s)$ are not that different, PPO modifies the objective function by clipping the density ratio $\pi_\theta(a|s)$ and $\pi_{\theta_t}(a|s)$:

$$L(\theta) := \mathbb{E}_{s \sim d_\mu^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \min \left\{ \frac{\pi_\theta(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \text{clip} \left(\frac{\pi_\theta(a|s)}{\pi_{\theta_t}(a|s)}; 1 - \epsilon, 1 + \epsilon \right) A^{\pi_{\theta_t}}(s, a) \right\}, \quad (12.5)$$

where $\text{clip}(x; 1 - \epsilon, 1 + \epsilon) = \begin{cases} 1 - \epsilon & x \leq 1 - \epsilon \\ 1 + \epsilon & x \geq 1 + \epsilon \\ x & \text{else} \end{cases}$. The clipping operator ensures that for π_θ , at any state action pair

where $\pi_\theta(a|s)/\pi_{\theta_t}(a|s) \notin [1 - \epsilon, 1 + \epsilon]$, we get zero gradient, i.e., $\nabla_\theta \left[\text{clip} \left(\frac{\pi_\theta(a|s)}{\pi_{\theta_t}(a|s)}; 1 - \epsilon, 1 + \epsilon \right) A^{\pi_{\theta_t}}(s, a) \right] = 0$. The outer min makes sure the objective function $L(\theta)$ is a lower bound of the original objective. PPO then proposes to collect a dataset (s, a) with $s \sim d_\mu^{\pi_{\theta_t}}$ and $a \sim \pi_{\theta_t}(\cdot|s)$, and then perform multiple steps of mini-batch stochastic gradient ascent on $L(\theta)$.

The idea of clipping is also used in LLM RL post-training (e.g., DeepSeek R1 Guo et al. [2025]). The RL algorithm used in DeepSeek R1 is GRPO – Group Relative Policy optimization, is similar to Eq. 12.5 except that it computes the advantage using Monte-Carlo rollouts.

12.2.2 Limitations of Average Trust Region Constraints

In this section, we provide a concrete example to show that the practical version of TRPO (i.e., Eq. 12.1, which uses an average trust-region constraint) does not yield the promised monotonic improvement. This highlights the importance of having the point-wise style trust region constraint $\max_s \|\pi'(\cdot|s) - \pi(\cdot|s)\|_1$. CPI achieves this by mixing the current policy with the greedy policy.

Consider the following simple MDPs. We have three states s_0, s_1 and $s^\dagger$. At state s_0 , we have two actions a_{good} and a_{bad} such that a_{good} leads to s_1 deterministically with reward 0 and a_{bad} leads to the absorbing state $s^\dagger$ with reward 0. At state s_1 , a_{good} leads to $s^\dagger$ with reward 1 and a_{bad} also leads to $s^\dagger$ but with reward -1 . Any action at the absorbing state leads to the absorbing state, and the absorbing state has a reward of 0. The initial state is always s_0 .

Consider the following policy $\pi(a_{bad}|s_0) = 1$ and $\pi(a_{good}|s_1) = 1$. Note that $J(\pi) = 0$ since the moment the agent takes a_{bad} at initial state s_0 , it will be absorbed into $s^\dagger$ forever. Also note that $d^\pi(s_1) = 0$ and $d^\pi(s_0) = 1$, i.e., the agent never visits s_1 .

Now consider the new policy $\pi'(a_{good}|s_0) = \alpha$ and $\pi'(a_{bad}|s_0) = 1 - \alpha$ and $\pi(a_{bad}|s_1) = 1$. The intuition is that π' is close to π under total-variation distance only under s_0 — the state distribution d^π , and yet is very different from π at s_1 — the state that π never visits. Such a policy satisfies the TRPO’s average trust-region constraint as long as α is set properly based on the threshold of the trust-region constraint.

Let us now look at the surrogate loss function of the TRPO, i.e., $\mathbb{E}_{a \sim d^\pi} \mathbb{E}_{a' \sim \pi'(\cdot|s)} A^\pi(s, a)$ which in this case is equal to $\alpha A^\pi(s_0, a_{good}) + (1 - \alpha) A^\pi(s_0, a_{bad}) = \alpha \gamma R$. So π' increases the surrogate objective and is subject to the average trust region constraint. However, it is easy to show that $J(\pi') = -\alpha \gamma R$, indicating that optimizing the surrogate with the average trust region does not guarantee monotonic improvement for any $\alpha > 0$.

The key intuition in the above example is that simply constructing a trust region averaged over the old policy π ’s state distribution is not enough, i.e., the new policy can deviate from π at states that π does not visit. In our case π' is taking the opposite action from π at the state s_1 which is never visited by π . Indeed in our example, if we could keep π' to be the same as π at the state s_1 , π' would be a better policy than π .

12.3 Policy Gradient with Orthogonalization and Row-wise Normalization

While CPI comes with a monotonic policy improvement guarantee, implementing and maintaining an ensemble of policies is computationally intractable for modern neural networks. TRPO aims to eliminate the ensemble but can only achieve the monotonic improvement guarantee under an idealized setting where the trust region is defined over all states. The practical version of TRPO and PPO, in general, cannot guarantee monotonic improvement because they relax the trust region to a constraint averaged over the state distribution induced by the policy. In this section, we introduce new policy gradient methods with orthogonalization and row-wise normalization. We show that the new algorithms induce conservative policy update, i.e., the new policy stays close to the old policy in terms of total variation distance under every state. Compared to standard policy gradient, these new algorithms can make larger updates in the policy's parameters in Euclidean space while making conservative updates in the output space (distributions over actions) of the policies.

12.3.1 Policy Gradient with Orthogonalization

Let us first start with the algorithm. Notation wise, in this section, we use W to represent a 2d weight matrix of the policy, i.e., $\pi_W(a|s)$. We denote ∇_W as the policy gradient, i.e., $\nabla_W V^{\pi_W}$. The algorithm works as follows:

$$\begin{aligned}\Delta &= \text{Newton-Schulz}(\nabla_W), \\ W' &= W + \eta\Delta,\end{aligned}$$

where the Newton-Schulz procedure performs the following procedure iteratively:

$$\begin{aligned}\text{Initialize } X_0 &= \nabla_W / \|\nabla_W\|_F \\ X_{t+1} &= p(X_t)\end{aligned}$$

where $p(X) := aX + bXX^\top X + cXX^\top XX^\top X$ with $a, b, c \in \mathbb{R}$ being scalars. We often design a, b, c such that if we iteratively apply the function $f(x) := ax + bx^3 + cx^5$ to any input $\sigma \in [0, 1]$, the result converges to 1, $(f)^T(\sigma) \rightarrow 1$ as $T \rightarrow \infty$ for any $\sigma \in [0, 1]$. The goal of Newton-Schulz is to approximately orthogonalize ∇_W , i.e., $\Delta = UV^\top$ with $U\Sigma V^\top$ being the SVD of ∇_W . As long as f has the above property, Newton-Schulz can converge to $UV^\top$. To see that, let us say that $X_t = U\Sigma_t V^\top$ (i.e., X_t shares the same singular vectors U, V as the gradient, which can be proved by induction easily). Apply p to X_t , we get:

$$p(X_t) = aU\Sigma_t V^\top + bU\Sigma_t^3 V^\top + cU\Sigma_t^5 V^\top = U(a\Sigma_t + b\Sigma_t^3 + c\Sigma_t^5) V^\top.$$

Since Σ_t is diagonal, this is equivalent to applying the function $f(\sigma)$ to each singular value in Σ_t . Thus as long as $(f)^T(\sigma) \rightarrow 1$ as $T \rightarrow \infty$, the singular values of X_t will converge to 1 as $t \rightarrow \infty$. In practice, with proper choices of a, b, c , 5 iterations are often enough to bring the singular values close to 1.

To gain some intuition, let us compare this to plain policy gradient, which can be understood as steepest ascent with the Frobenius norm, i.e.,

$$\begin{aligned}\Delta' &= \underset{\Delta}{\operatorname{argmax}} \langle \Delta, \nabla_W \rangle, \\ \|\Delta\|_F &\leq 1,\end{aligned}$$

and $W' = W + \eta\Delta'$. Now assume we can perfectly orthogonalize ∇_W , we can interpret the new algorithm as steepest ascent with spectral norm, i.e.,

$$\begin{aligned}\Delta &= \underset{\Delta}{\operatorname{argmax}} \langle \Delta, \nabla_W \rangle, \\ \|\Delta\|_2 &\leq 1,\end{aligned}$$

and $W' = W + \eta\Delta$. It is not hard to show that Δ , i.e., the exact maximizer of the above spectral norm constrained linear optimization problem, is exactly $\Delta = UV^\top$ where U, V are the left and right singular vectors of ∇_W . Note that for any matrix X we have $\|X\|_2 \leq \|X\|_F$. This means that the set of matrices $\|\Delta\|_2 \leq 1$ is larger than the set of matrices satisfying $\|\Delta\|_F \leq 1$. From a gradient-ascent perspective, this means that policy gradient with spectral norm can take larger steps in the policy parameters' Euclidean space than the plain policy gradient. We will show in the next section that this approach ensures a conservative policy update, i.e., the new policy after one step of update is close to the old one in total variation distance across all states. This shares a similar spirit with NPG: this algorithm can take larger steps in the Euclidean space of policy parameters while guaranteeing small changes in the policy's output space.

Analysis

We start by analyzing the ideal version where $\Delta = UV^\top$, i.e., we perform the exact orthogonalization. This $\Delta = UV^\top$ is the solution of the following spectral norm constrained optimization problem:

$$\begin{aligned} \max_{\Delta} \langle \Delta, \nabla_W \rangle, \\ \|\Delta\|_2 \leq 1, \end{aligned}$$

where matrix inner product $\langle A, B \rangle := \text{trace}(AB^\top)$ and $\|X\|_2$ is the spectral norm, i.e., the maximum singular value of X .

To illustrate why the spectral norm constraint induces conservative policy update and monotonic policy improvement, let us first focus on a softmax linear policy, i.e., $\pi_W(\cdot|s) = \text{softmax}(W\phi(s))$ for some state feature $\phi(s)$. Denote η as the learning rate, we have:

$$\pi_{W'} = \text{softmax}((W + \eta\Delta)\phi(s)).$$

Let us assume the feature norm is bounded, i.e., $\max_s \|\phi(s)\|_2 \leq 1$.

Lemma 12.5 (Conservative policy update). *For all s , we have:*

$$\|\pi_{W'}(\cdot|s) - \pi_W(\cdot|s)\|_1 \leq \eta.$$

Note that based on Lemma 12.1 in CPI, this conservative update immediately implies:

$$\|d^{\pi_W} - d^{\pi_{W'}}\|_1 \leq \frac{\eta\gamma}{1-\gamma},$$

where d^{π_W} denotes the state distribution induced by policy π_W . Here we drop the initial state distribution μ in d^{π_W} for notation simplicity. The proof of the above lemma uses the following fact:

Fact 12.6. For any $z, z' \in \mathbb{R}^d$, we have:

$$\|\text{softmax}(z) - \text{softmax}(z')\|_1 \leq \|z - z'\|_2.$$

We provide the proof of the fact at the end of this section. Lemma 12.5 shows that the new policy must be close to the old policy in terms of total variation distance for all states. Note that this is the stronger type of conservative that appeared in the idealized version of TRPO and CPI's policy update. With the above lemma, proving monotonic improvement is quite straightforward.

To show monotonic improvement, let us define the following surrogate objective function:

$$L(W') = \mathbb{E}_{s \sim d^{\pi_W}} \mathbb{E}_{a \sim \pi_{W'}(\cdot|s)} A^{\pi_W}(s, a).$$

Note that by the definition of A^{π_W} , we have $L(W) = 0$. The following lemma provides a lower bound for $L(W')$.

Lemma 12.7. Define $W' = W + \eta\Delta$ with $\|\Delta\|_2 \leq 1$. We have:

$$L(W') \geq \eta \langle \nabla_W L(W), \Delta \rangle - \frac{1.5}{1-\gamma} \eta^2.$$

Proof. Fix a state s . Define $z = W\phi(s)$, $p = \text{softmax}(z)$, $\ell_s(z) = p^\top A$, where we denote the vector A whose dimension is $|\mathcal{A}|$, and $A_a = A^{\pi_W}(s, a)$.

First, we have:

$$\nabla_z \ell_s(z) = (\text{diag}(p) - pp^\top)A.$$

Since $z = W\phi(s)$, we have:

$$\nabla_W \ell_s(W) = ((\text{diag}(p) - pp^\top)A) \phi(s)^\top.$$

For any two different W and W' , we have:

$$\begin{aligned} \|\nabla_W \ell_s(W) - \nabla_{W'} \ell_s(W')\|_* &= \|(\text{diag}(p) - pp^\top)A - (\text{diag}(p') - p'(p')^\top)A\|_2 \|\phi(s)\|_2 \\ &\leq \|(\text{diag}(p) - \text{diag}(p'))A\|_2 + \|(pp^\top - p'(p')^\top)A\|_2 \\ &\leq \frac{3}{1-\gamma} \|p - p'\|_1 \leq \frac{3}{1-\gamma} \|W - W'\|_2, \end{aligned}$$

where $\|X\|_*$ is the nuclear norm (the sum of singular values of X), and the first equality uses the fact that $\nabla_W \ell_s(W) - \nabla_{W'} \ell_s(W')$ is a rank one matrix.

Now using the first part of the results in Lemma 12.13, we have:

$$L(W') \geq L(W) + \eta \langle \nabla_W L(W), \Delta \rangle - \frac{1.5}{1-\gamma} \|W - W'\|_2^2.$$

Using $L(W) = 0$ and set $W' = W + \eta\Delta$ concludes the proof. $\square$

Recall that $\nabla_W L(W) = \nabla_W$, i.e., the policy gradient. Below we show that as long as the algorithm has not converged yet, we have monotonic policy improvement.

Theorem 12.8. Assume there exists $\epsilon > 0$ such that $\langle \nabla_W, \Delta \rangle \geq \epsilon$. Denote $W' = W + \eta\Delta$, we have:

$$V^{\pi_{W'}} - V^{\pi_W} \geq \epsilon^2(1-\gamma)/12,$$

with $\eta = 0.5\epsilon / (\frac{3}{1-\gamma} + \frac{\gamma}{(1-\gamma)^2})$.

Proof. Start with Performance Difference Lemma:

$$\begin{aligned}
(1 - \gamma) (V^{\pi'_W} - V^{\pi_W}) &= \mathbb{E}_{s \sim d^{\pi_W}} \mathbb{E}_{a \sim \pi_{W'}(\cdot|s)} A^{\pi_W}(s, a) \\
&= \mathbb{E}_{s \sim d^{\pi_W}} \sum_a (\pi_{W'}(a|s) - \pi_W(a|s)) A^{\pi_W}(s, a) \\
&= \mathbb{E}_{s \sim d^{\pi_W}} \sum_a (\pi_{W'}(a|s) - \pi_W(a|s)) A^{\pi_W}(s, a) \\
&\quad + \mathbb{E}_{s \sim d^{\pi_W}} \sum_a (\pi_{W'}(a|s) - \pi_W(a|s)) A^{\pi_W}(s, a) \\
&\quad - \mathbb{E}_{s \sim d^{\pi_W}} \sum_a (\pi_{W'}(a|s) - \pi_W(a|s)) A^{\pi_W}(s, a) \\
&\geq \mathbb{E}_{s \sim d^{\pi_W}} \sum_a \pi_{W'}(a|s) A^{\pi_W}(s, a) - \frac{1}{1 - \gamma} \|d^{\pi_W} - d^{\pi_{W'}}\|_1 \max_s \|\pi_W(\cdot|s) - \pi_{W'}(\cdot|s)\|_1 \\
&\geq L(W') - \frac{\eta^2 \gamma}{(1 - \gamma)^2} \\
&\geq \eta \epsilon - \left(\frac{3}{1 - \gamma} + \frac{\gamma}{(1 - \gamma)^2} \right) \eta^2.
\end{aligned}$$

Note that the second inequality above replies the state-wise total variation distance bound.

Setting $\eta = 0.5\epsilon / (\frac{3}{1-\gamma} + \frac{\gamma}{(1-\gamma)^2})$, we get:

$$(1 - \gamma) (V^{\pi'_W} - V^{\pi_W}) \geq \frac{0.25\epsilon^2}{(\frac{3}{1-\gamma} + \frac{\gamma}{(1-\gamma)^2})}$$

This concludes the proof. $\square$

12.3.2 Policy Gradient with Row-wise Normalization

In the previous section, Fact 12.6 implies that the spectral norm is a natural metric to use in steepest ascent. However, we can further tighten the inequality Fact 12.6. Below this new fact shows that softmax is Lipschitz under the ℓ_∞ norm on the input (i.e., logits) space.

Fact 12.9. For any z, z' , we have:

$$\|\text{softmax}(z) - \text{softmax}(z')\|_1 \leq \|z - z'\|_\infty$$

Note that the inequality here is tighter than the one in Fact 12.6. For a linear softmax policy, the logit $z = W\phi(s)$, so the ℓ_∞ norm on the logits naturally implies that we should consider the matrix $\|\cdot\|_{2,\infty}$ norm since

$$\|z - z'\|_\infty = \|W\phi(s) - W'\phi(s)\|_\infty \leq \|W - W'\|_{2,\infty} \|\phi(s)\|_2,$$

where we denote $\|X\|_{2,\infty}$ denotes the maximum row norm $\max_i \|X_{i,:}\|_2$.

More formally, recall that ∇_W denotes the policy gradient, we formulate the following row-wise normalized steepest ascent:

$$\begin{aligned}
&\max_{\Delta} \langle \Delta, \nabla_W \rangle, \\
&\text{s.t., } \|\Delta\|_{2,\infty} \leq 1.
\end{aligned}$$

Note that the above optimization also has a closed-form solution. The optimal Δ can be computed by normalizing every row of Δ_W , i.e., $\Delta_{i,:} = (\nabla_W)_{i,:} / \|(\nabla_W)_{i,:}\|_2$, where we denote $\Delta_{i,:}$ as the i -th row of the matrix Δ . We then update the policy's parameter via $W' = W + \eta\Delta$. Note that this is much simpler than the New-Schultz procedure, which often requires communications between different GPUs when optimizing modern LLMs whose matrices are shared across many GPUs. For row-wise normalization, when matrices are sharded in a row-wise manner, it can be implemented locally on each rank without any communication.

Compare the above update to the steepest ascent with spectral norm, since that for any matrix we have $\|\Delta\|_{2,\infty} \leq \|\Delta\|_2 \leq \|\Delta\|_F$, we see that row-wise normalized steepest ascent allows us to take a larger step in the Euclidean space of the policy's parameters.

Using Fact 12.9, we can easily show that this update ensures the same level of conservativeness as CPI and policy gradient with spectral norm.

Lemma 12.10. *Consider the above update where $W' = W + \eta\Delta$ with Δ being the row-wise normalized version of policy gradient ∇_W . For all state s , we have:*

$$\|\pi_{W'}(\cdot|s) - \pi_W(\cdot|s)\|_1 \leq \eta.$$

The proof of the above lemma uses Fact 12.9, the assumption that $\|\phi(s)\|_2 \leq 1$ which implies $\|\phi(s)\|_\infty \leq 1$, and the inequality $\|\eta\Delta\phi(s)\|_\infty \leq \eta\|\Delta\|_{2,\infty} \leq \eta$.

With the above lemma, we can follow a similar procedure in the above section to prove monotonic improvement again. Starting from the surrogate loss $L(W)$, we have the following lemma.

Lemma 12.11. *Define $W' = W + \eta\Delta$ where $\|\Delta\|_{2,\infty} \leq 1$. We have:*

$$L(W') \geq \eta \langle \nabla_W L(W), \Delta \rangle - \frac{3}{1-\gamma} \eta^2.$$

Proof. Recall the definition of $\ell_s(W)$ and its gradient $\nabla \ell_s(W)$ defined in the proof of Lemma 12.7, we have

$$\begin{aligned} \|\nabla_W \ell_s(W) - \nabla_W \ell_s(W')\|_{2,1} &= \|(\text{diag}(p) - pp^\top - \text{diag}(p') + p'(p')^\top)A\phi(s)^\top\|_{2,1} \\ &= \|(\text{diag}(p) - pp^\top - \text{diag}(p') + p'(p')^\top)A\|_1 \|\phi(s)\|_2 \end{aligned}$$

For $\|(\text{diag}(p) - pp^\top - \text{diag}(p') + p'(p')^\top)A\|_1$, define $\bar{A} = \mathbb{E}_{a \sim p} A_a$ and $\bar{A}' = \mathbb{E}_{a \sim p'} A_a$, we have:

$$\begin{aligned} \|(\text{diag}(p) - pp^\top - \text{diag}(p') + p'(p')^\top)A\|_1 &= \sum_a |p_a A_a - p'_a A_a - (p_a \bar{A} - p'_a \bar{A}')| \\ &\leq \sum_a \frac{1}{1-\gamma} |p_a - p'_a| + \sum_a |p_a \bar{A} - p_a \bar{A}'| + \sum_a |p_a \bar{A}' - p'_a \bar{A}'| \\ &\leq \frac{2}{1-\gamma} \|p - p'\|_1 + \frac{1}{1-\gamma} \sum_a p_a |\bar{A} - \bar{A}'| \\ &\leq \frac{3}{1-\gamma} \|p - p'\|_1, \end{aligned}$$

where the last inequality uses $|\bar{A} - \bar{A}'| \leq \|p - p'\|_1 \|A\|_\infty \leq \|p - p'\|/(1-\gamma)$.

Together with Fact 12.9, the above implies that,

$$\|\nabla_W \ell_s(W) - \nabla_W \ell_s(W')\|_{2,1} \leq \frac{3}{1-\gamma} \|p - p'\|_1 \leq \frac{3}{1-\gamma} \|W\phi(s) - W'\phi(s)\|_\infty \leq \frac{3}{1-\gamma} \|W - W'\|_{2,\infty}.$$

Now using the second result in Lemma 12.13 together with the fact that $L(W) = 0$ by definition, we can conclude the proof. $\square$

Following the exact same proof for Theorem 12.8, we can conclude that policy gradient with ℓ_∞ norm also implies monotonic improvement.

12.3.3 Combining orthogonalization and row-wise normalization for nonlinear policies

Does the conservative policy update condition in Lemma 12.5 hold for nonlinear policies beyond softmax linear? Let us consider a standard multi-layer feedforward neural network, i.e.,

$$h_0 = \phi(s), h_t = g(W_t h_{t-1}) \text{ for } t \in [1, T], z = W_{T+1} h_T, \pi(\cdot|s) = \text{softmax}(z).$$

where g is a point-wise activation function. For this neural network, when we perform policy gradient, we will apply different norms at the different weight matrices. Particularly, for W_{T+1} in the softmax head, we update it via row-wise normalized steepest ascent. For matrices W_t for $t \in [1, T]$, we apply orthogonalized update. The logic behind this design choice is that to control the perturbation in the policy $\pi(\cdot|s)$, we can control the ℓ_∞ norm on the logit perturbation, which is controlled via the ℓ_2 norm on the perturbation on the hidden states, i.e., define $z = W_{T+1} h_T$ and $z' = W_{T+1} h'_T$, we have

$$\|\text{softmax}(z) - \text{softmax}(z')\|_1 \leq \|z - z'\|_\infty \leq \|W_{T+1}\|_{2,\infty} \|h_T - h'_T\|_2,$$

and via induction, $\|h_t - h'_t\|_2$ can be controlled by $\|h_{t-1} - h'_{t-1}\|_2$, i.e.,

$$\|h_t - h'_t\|_2 \leq L_g \|W_t h_{t-1} - W_t h'_{t-1}\|_2 \leq L_g \|W_t\|_2 \|h_{t-1} - h'_{t-1}\|_2.$$

where L_g is the Lipschitz constant of the activation function g . This justifies why we use $\|\cdot\|_{2,\infty}$ norm on the W_{T+1} in the softmax head and use spectral norm on the weight matrices in the hidden layers.

For the analysis below, we consider g being ReLU, which means that $L_g \leq 1$.

Fact 12.12. Consider the above multi-layer MLP policy. We update the parameters as follows: $W'_t = W_t + \eta \Delta_t$ with $\|\Delta_t\|_2 \leq 1$ for $t \in [1, T]$, and $W'_{T+1} = W_{T+1} + \eta \Delta_{T+1}$ with $\|\Delta_{T+1}\|_{2,\infty} \leq 1$. Assume that $\|W_t\|_2 \leq B_t$ for $t \in [1, T]$, i.e., the spectral norm of the hidden layer weight matrices is upper bounded by constants B_i , and $\|W_{T+1}\|_{2,\infty} \leq B_{T+1}$, and the input is bounded $\|s\|_2 \leq c$, for all s . We have:

$$\forall s : \|\pi(\cdot|s) - \pi'(\cdot|s)\|_1 \leq O(\eta),$$

where O hides constants that only dependent on norm of the weight matrices $B_i, i \in [1, T+1]$, the upper bound of the norm of the input s , and the number of layers T .

Proof. We bound $\|h_t - h'_t\|_2$ for $t \in [1, T]$ as follows

$$\begin{aligned} \|h_t - h'_t\|_2 &\leq \|W_t h_{t-1} - W_t h'_{t-1}\|_2 + \|W_t h'_{t-1} - W'_t h'_{t-1}\|_2 \leq B_t \|h_{t-1} - h'_{t-1}\|_2 + \eta \|\Delta_t\|_2 \|h'_{t-1}\|_2 \\ &\leq B_t \|h_{t-1} - h'_{t-1}\|_2 + \eta \|h'_{t-1}\|_2 \\ &\leq (B_t + \eta) \|h_{t-1} - h'_{t-1}\|_2 + \eta \|h_{t-1}\|_2 \end{aligned}$$

Note that:

$$\|h_{t-1}\|_2 \leq \|W_{t-1}\|_2 \|h_{t-2}\|_2 \leq \prod_{j=1}^{t-1} B_j \|s\|_2 \leq c \prod_{j=1}^{t-1} B_j.$$

Now, start from $\|h_T - h'_T\|_2$, and unroll all the way to $t = 0$, we get:

$$\|h_T - h'_T\|_2 \leq \eta c \sum_{k=1}^T \left(\prod_{j=1}^{k-1} B_j \right) \left(\prod_{j=k+1}^T (B_j + \eta) \right).$$

Now for the softmax head, we have:

$$\begin{aligned}
\|\pi(\cdot|s) - \pi'(\cdot|s)\|_1 &\leq \|z - z'\|_\infty \leq \|W_{T+1}h_T - W'_{T+1}h'_T\|_\infty \\
&\leq \|W_{T+1}h_T - W'_{T+1}h_T\|_\infty + \|W'_{T+1}h_T - W'_{T+1}h'_T\|_\infty \\
&\leq \eta c \prod_{j=1}^T B_j + \|W'_{T+1}\|_{2,\infty} \|h_T - h'_T\|_2 \\
&\leq \eta c \prod_{j=1}^T B_j + (B_{T+1} + \eta) \|h_T - h'_T\|_2.
\end{aligned}$$

Plug in the upper bound we have for $\|h_T - h'_T\|_2$, we conclude the proof. $\square$

The above fact immediately implies that for this nonlinear policy, we also have $\|d^\pi - d^{\pi'}\|_1 = O(\eta)$, i.e., the total variation distance between the two resulting state distributions is bounded by $O(\eta)$.

Does the conservative policy update still hold when we use the Newton-Schulz procedure to perform approximate orthogonalization? As long as we pick the function $f(\sigma)$, such that $(f)^5(\sigma) \leq 1$ (assume we perform Newton-Schulz procedure for 5 steps), the stability condition on the policy update and the monotonic improvement guarantee will still hold, simply due to the fact that $\|\Delta\|_2 \leq 1$.

12.3.4 Missing Proofs and Additional Results

We first provide the proof of the fact 12.6 and fact 12.9.

Fact 12.6, 12.9. Consider two vectors z and z' , and define $p = \text{softmax}(z)$, $p' = \text{softmax}(z')$. Define $z_t = z' + tu$ where $u = z - z'$, and $p_t = \text{softmax}(z_t)$. We have:

$$\frac{d}{dt}p_t = (\text{diag}(p_t) - p_t p_t^\top) u.$$

For $\frac{d}{dt}p_t$, we have:

$$\left\| \frac{d}{dt}p_t \right\|_1 = \sum_i |p_{t,i}u_i - p_{t,i}(p_t^\top u)| = \sum_i p_{t,i} |u_i - p_t^\top u|$$

Note that $\mathbb{E}|x - \mathbb{E}x| \leq \sqrt{\mathbb{E}(x - \mathbb{E}x)^2} = \sqrt{\text{Var}(x)} \leq \sqrt{\mathbb{E}x^2}$. Thus,

$$\left\| \frac{d}{dt}p_t \right\|_1 \leq \sqrt{\sum_i p_{t,i}(u_i)^2} \leq \sqrt{\sum_i u_i^2} = \|u\|_2.$$

Now, for $p - p'$, we have:

$$\|p - p'\|_1 = \left\| \int_{t=0}^1 \frac{d}{dt}p_t dt \right\|_1 \leq \int_{t=0}^1 \left\| \frac{d}{dt}p_t \right\|_1 dt \leq \int_{t=0}^1 \|u\|_2 dt = \|z - z'\|_2.$$

For Fact 12.9, we observe:

$$\left\| \frac{d}{dt}p_t \right\|_1 \leq \sqrt{\sum_i p_{t,i}(u_i)^2} \leq \|u\|_\infty.$$

$\square$

Lemma 12.13. Consider a function f . For any W, W' , it has:

$$\|\nabla f(W) - \nabla f(W')\|_* \leq \beta \|W - W'\|_2,$$

then

$$f(W') \geq f(W) + \langle \nabla f(W), W' - W \rangle - \frac{\beta}{2} \|W - W'\|_2^2.$$

If it satisfies the following property:

$$\|\nabla f(W) - \nabla f(W')\|_{2,1} \leq \beta \|W - W'\|_{2,\infty},$$

where $\|X\|_{2,1} := \sum_i \|X_{i,:}\|_2$ (i.e., sum of the ℓ_2 norm of each row in X). Then,

$$f(W') \geq f(W) + \langle \nabla f(W), D \rangle - \beta \|W - W'\|_{2,\infty}^2 / 2.$$

Proof. Denote $D = W' - W$ and define $g(t) = f(W + tD)$. Note that $g'(t) = \langle \nabla f(W + tD), D \rangle$. Note that for $f(W')$, we have:

$$\begin{aligned} f(W') &= f(W) + \int_{t=0}^1 g'(t) dt = f(W) + \int_{t=0}^1 \langle \nabla f(W + tD), D \rangle dt \\ &= f(W) + \langle \nabla f(W), D \rangle + \int_{t=0}^1 \langle \nabla f(W + tD) - \nabla f(W), D \rangle dt \end{aligned}$$

To prove the first result in the lemma, we use the Lipschitzness condition under ℓ_2 norm, i.e.,

$$\begin{aligned} f(W') &\geq f(W) + \langle \nabla f(W), D \rangle - \int_{t=0}^1 \|\nabla f(W + tD) - \nabla f(W)\|_* \|D\|_2 dt \\ &\geq f(W) + \langle \nabla f(W), D \rangle - \int_{t=0}^1 \beta t \|D\|_2^2 dt = f(W) + \langle \nabla f(W), D \rangle - \beta \|D\|_2^2 / 2. \end{aligned}$$

To prove the second part of the lemma, we use the Lipschitzness condition under $\|\cdot\|_{2,\infty}$, i.e.,

$$\begin{aligned} f(W') &\geq f(W) + \langle \nabla f(W), D \rangle - \int_{t=0}^1 \|\nabla f(W + tD) - \nabla f(W)\|_{2,1} \|D\|_{2,\infty} dt \\ &\geq f(W) + \langle \nabla f(W), D \rangle - \int_{t=0}^1 \beta t \|D\|_{2,\infty}^2 dt = f(W) + \langle \nabla f(W), D \rangle - \beta \|D\|_{2,\infty}^2 / 2. \end{aligned}$$

where $\|X\|_{2,1} = \sum_i \|X_{i,:}\|_2$ and is the dual norm of $\|\cdot\|_{2,\infty}$ and the first inequality above uses Holder's inequality. $\square$

12.4 Bibliographic Remarks and Further Readings

The analysis of CPI is adapted from the original one in [Kakade and Langford, 2002]. There have been a few further interpretations of CPI. One interesting perspective is that CPI can be treated as a boosting algorithm [Scherrer and Geist, 2014].

More generally, CPI and NPG are part of family of *incremental* algorithms, including Policy Search by Dynamic Programming (PSDP) [Bagnell et al., 2004] and MD-MPI [Geist et al., 2019]. PSDP operates in a finite horizon setting and optimizes a sequence of time-dependent policies; from the last time step to the first time step, every iteration of PSDP only updates the policy at the current time step while holding the future policies fixed — thus

making an incremental update on the policy. See Scherrer [2014] for more detailed discussion and comparison of some of these approaches. Mirror Descent-Modified Policy Iteration (MD-MPI) algorithm [Geist et al., 2019] is a family of actor-critic style algorithms which is based on regularization and is incremental in nature; with negative entropy as the Bregman divergence (for the tabular case), MD-MPI recovers the NPG in the tabular case (for the softmax parameterization).

The interpretation of NPG as Covariant Policy Search is due to Bagnell and Schneider [2003], as the policy update procedure will be invariant to linear transformations of the parameterization; see Bagnell and Schneider [2003] for a more detailed discussion on this.

The TRPO algorithm is due to Schulman et al. [2015]. The original TRPO analysis provides performance guarantees, largely relying on a reduction to the CPI guarantees. In this Chapter, we make a sharper connection of TRPO to NPG, which was subsequently observed by a number of researchers; this connection provides a sharper analysis for the generalization and approximation behavior of TRPO (e.g. via the results presented in Chapter 11). In practice, a popular variant is the Proximal Policy Optimization (PPO) algorithm [Schulman et al., 2017].

The row-wise normalized and orthogonalized policy gradient methods are motivated by Muon [Jordan et al., 2024]. Muon can be understood as steepest descent with spectral norm constraints and it has been demonstrated that Muon is scalable and can be effective for training modern LLMs [Liu et al., 2025].

Part IV

Further Topics

Chapter 13

Imitation Learning

In this chapter, we study imitation learning. Unlike the Reinforcement Learning setting, in Imitation Learning, we do not have access to the ground truth reward function (or cost function), but instead, we have expert demonstrations. We often assume that the expert is a policy that approximately optimizes the underlying reward (or cost) functions. The goal is to leverage the expert demonstrations to learn a policy that performs as good as the expert.

We consider three settings of imitation learning: (1) pure offline where we only have expert demonstrations and no more real world interaction is allowed, (2) hybrid where we have expert demonstrations, and also is able to interact with the real world (e.g., have access to the ground truth transition dynamics); (3) interactive setting where we have an interactive expert and also have access to the underlying reward (cost) function.

13.1 Setting

We will focus on finite horizon MDPs $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, r, \mu, P, H\}$ where r is the reward function but is unknown to the learner. We represent the expert as a closed-loop policy $\pi^* : \mathcal{S} \mapsto \Delta(\mathcal{A})$. For analysis simplicity, we assume the expert π^* is indeed the optimal policy of the original MDP $\mathcal{M}$ with respect to the ground truth reward r . Again, our goal is to learn a policy $\hat{\pi} : \mathcal{S} \mapsto \Delta(\mathcal{A})$ that performs as well as the expert, i.e., $V^{\hat{\pi}}$ needs to be close to V^* , where V^π denotes the expected total reward of policy π under the MDP $\mathcal{M}$. We denote d^π as the state-action visitation of policy π under $\mathcal{M}$.

We assume we have a pre-collected expert dataset in the format of $\{s_i^*, a_i^*\}_{i=1}^M$ where $s_i^*, a_i^* \sim d^{\pi^*}$.

13.2 Offline IL: Behavior Cloning

We study offline IL here. Specifically, we study the classic Behavior Cloning algorithm.

We consider a policy class $\Pi = \{\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})\}$. We consider the following realizable assumption.

Assumption 13.1. We assume Π is rich enough such that $\pi^* \in \Pi$.

For analysis simplicity, we assume Π is discrete. But our sample complexity will only scale with respect to $\ln(|\Pi|)$.

Behavior cloning is one of the simplest Imitation Learning algorithm which only uses the expert data $\mathcal{D}^*$ and does not require any further interaction with the MDP. It computes a policy via a reduction to supervised learning. Specifically,

we consider a reduction to Maximum Likelihood Estimation (MLE):

$$\text{Behavior Cloning (BC): } \hat{\pi} = \operatorname{argmax}_{\pi \in \Pi} \sum_{i=1}^N \ln \pi(a_i^* | s_i^*). \quad (13.1)$$

Namely we try to find a policy from Π that has the maximum likelihood of fitting the training data. As this is a reduction to MLE, we can leverage the existing classic analysis of MLE (e.g., Chapter 7 in Geer and van de Geer [2000]) to analyze the performance of the learned policy.

Theorem 13.2 (MLE Guarantee). *Consider the MLE procedure (Eq. 13.1). With probability at least $1 - \delta$, we have:*

$$\mathbb{E}_{s \sim d^{\pi^*}} \|\hat{\pi}(\cdot | s) - \pi^*(\cdot | s)\|_{tv}^2 \leq \frac{2 \ln(|\Pi|/\delta)}{M}.$$

We refer readers to Section E, Theorem 21 in Agarwal et al. [2020b] for detailed proof of the above MLE guarantee.

Now we can transfer the average divergence between $\hat{\pi}$ and π^* to their performance difference $V^{\hat{\pi}}$ and V^{π^*} .

One thing to note is that BC only ensures that the learned policy $\hat{\pi}$ is close to π^* under d^{π^*} . Outside of d^{π^*} 's support, we have no guarantee that $\hat{\pi}$ and π^* will be close to each other. The following theorem shows that a compounding error occurs when we study the performance of the learned policy $V^{\hat{\pi}}$.

Theorem 13.3 (Sample Complexity of BC). *With probability at least $1 - \delta$, BC returns a policy $\hat{\pi}$ such that:*

$$V^* - V^{\hat{\pi}} \leq \frac{3}{(1 - \gamma)^2} \sqrt{\frac{\ln(|\Pi|/\delta)}{M}}.$$

Proof. We start with the performance difference lemma and the fact that $\mathbb{E}_{a \sim \pi(\cdot | s)} A^\pi(s, a) = 0$:

$$\begin{aligned} (1 - \gamma) (V^* - V^{\hat{\pi}}) &= \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot | s)} A^{\hat{\pi}}(s, a) \\ &= \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot | s)} A^{\hat{\pi}}(s, a) - \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \hat{\pi}(\cdot | s)} A^{\hat{\pi}}(s, a) \\ &\leq \mathbb{E}_{s \sim d^{\pi^*}} \frac{1}{1 - \gamma} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_1 \\ &\leq \frac{1}{1 - \gamma} \sqrt{\mathbb{E}_{s \sim d^{\pi^*}} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_1^2} \\ &= \frac{1}{1 - \gamma} \sqrt{4 \mathbb{E}_{s \sim d^{\pi^*}} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_{tv}^2}. \end{aligned}$$

where we use the fact that $\sup_{s, a, \pi} |A^\pi(s, a)| \leq \frac{1}{1 - \gamma}$, and the fact that $(\mathbb{E}[x])^2 \leq \mathbb{E}[x^2]$.

Using Theorem 13.2 and rearranging terms conclude the proof. $\square$

For Behavior cloning, from the supervised learning error the quadratic polynomial dependency on the effect horizon $1/(1 - \gamma)$ is not avoidable in worst case Ross and Bagnell [2010]. This is often referred as the distribution shift issue in the literature. Note that $\hat{\pi}$ is trained under d^{π^*} , but during execution, $\hat{\pi}$ makes prediction on states that are generated by itself, i.e., $d^{\hat{\pi}}$, instead of the training distribution d^{π^*} .

13.3 The Hybrid Setting: Statistical Benefit and Algorithm

The question we want to answer here is that if we know the underlying MDP's transition P (but the reward is still unknown), can we improve Behavior Cloning? In other words:

what is the benefit of the known transition in addition to the expert demonstrations?

Instead of a quadratic dependency on the effective horizon, we should expect a linear dependency on the effective horizon. The key benefit of the known transition is that we can test our policy using the known transition to see how far away we are from the expert's demonstrations, and then use the known transition to plan to move closer to the expert demonstrations.

In this section, we consider a statistical efficient, but computationally intractable algorithm, which we use to demonstrate that informationally theoretically, by interacting with the underlying known transition, we can do better than Behavior Cloning. In the next section, we will introduce a popular and computationally efficient algorithm Maximum Entropy Inverse Reinforcement Learning (MaxEnt-IRL) which operates under this setting (i.e., expert demonstrations with a known transition).

We start from the same policy class Π as we have in the BC algorithm, and again we assume realizability (Assumption 13.1) and Π is discrete.

Since we know the transition P , information theoretically, for any policy π , we have d^π available (though it is computationally intractable to compute d^π for large scale MDPs). We have $(s_i^*, a_i^*)_{i=1}^M \sim d^{\pi^*}$.

Below we present an algorithm which we called Distribution Matching with Scheffé Tournament (DM-ST).

For any two policies π and π' , we denote $f_{\pi, \pi'}$ as the following witness function:

$$f_{\pi, \pi'} := \operatorname{argmax}_{f: \|f\|_\infty \leq 1} \left[\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^{\pi'}} f(s, a) \right].$$

We denote the set of witness functions as:

$$\mathcal{F} = \{f_{\pi, \pi'} : \pi, \pi' \in \Pi, \pi \neq \pi'\}.$$

Note that $|\mathcal{F}| \leq |\Pi|^2$.

DM-ST selects $\hat{\pi}$ using the following procedure:

$$\text{DM-ST: } \hat{\pi} \in \operatorname{argmin}_{\pi \in \Pi} \left[\max_{f \in \mathcal{F}} \left[\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*) \right] \right].$$

The following theorem captures the sample complexity of DM-ST.

Theorem 13.4 (Sample Complexity of DM-ST). *With probability at least $1 - \delta$, DM-ST finds a policy $\hat{\pi}$ such that:*

$$V^* - V^{\hat{\pi}} \leq \frac{4}{1 - \gamma} \sqrt{\frac{2 \ln(|\Pi|) + \ln\left(\frac{1}{\delta}\right)}{M}}.$$

Proof. The proof basically relies on a uniform convergence argument over $\mathcal{F}$ of which the size is $|\Pi|^2$. First we note that for all policy $\pi \in \Pi$:

$$\max_{f \in \mathcal{F}} [\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^*} f(s, a)] = \max_{f: \|f\|_\infty \leq 1} [\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^*} f(s, a)] = \|d^\pi - d^{\pi^*}\|_1,$$

where the first equality comes from the fact that $\mathcal{F}$ includes $\arg \max_{f: \|f\|_\infty \leq 1} [\mathbb{E}_{s,a \sim d^\pi} f(s, a) - \mathbb{E}_{s,a \sim d^\star} f(s, a)]$.

Via Hoeffding's inequality and a union bound over $\mathcal{F}$, we get that with probability at least $1 - \delta$, for all $f \in \mathcal{F}$:

$$\left| \frac{1}{M} \sum_{i=1}^M f(s_i^\star, a_i^\star) - \mathbb{E}_{s,a \sim d^\star} f(s, a) \right| \leq 2\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} := \epsilon_{stat}.$$

Denote $\hat{f} := \arg \max_{f \in \mathcal{F}} [\mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s, a) - \mathbb{E}_{s,a \sim d^\star} f(s, a)]$, and $\tilde{f} := \arg \max_{f \in \mathcal{F}} \mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i, a_i)$. Hence, for $\hat{\pi}$, we have:

$$\begin{aligned} \|d^{\hat{\pi}} - d^\star\|_1 &= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \hat{f}(s, a) - \mathbb{E}_{s,a \sim d^\star} \hat{f}(s, a) \leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \hat{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \hat{f}(s_i^\star, a_i^\star) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \tilde{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i, a_i) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\pi^\star}} \tilde{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i, a_i) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\pi^\star}} \tilde{f}(s, a) - \mathbb{E}_{s,a \sim d^\star} \tilde{f}(s, a) + 2\epsilon_{stat} = 2\epsilon_{stat}, \end{aligned}$$

where in the third inequality we use the optimality of $\hat{\pi}$.

Recall that $V^\pi = \mathbb{E}_{s,a \sim d^\pi} r(s, a)/(1 - \gamma)$, we have:

$$V^{\hat{\pi}} - V^\star = \frac{1}{1 - \gamma} (\mathbb{E}_{s,a \sim d^{\hat{\pi}}} r(s, a) - \mathbb{E}_{s,a \sim d^\star} r(s, a)) \leq \frac{\sup_{s,a} |r(s, a)|}{1 - \gamma} \|d^{\hat{\pi}} - d^\star\|_1 \leq \frac{2}{1 - \gamma} \epsilon_{stat}.$$

This concludes the proof. $\square$

Note that above theorem confirms the statistical benefit of having the access to a known transition: comparing to the classic Behavior Cloning algorithm, the approximation error of DM-ST only scales linearly with respect to horizon $1/(1 - \gamma)$ instead of quadratically.

13.3.1 Extension to Agnostic Setting

So far we focused on agnostic learning setting. What would happen if $\pi^\star \notin \Pi$? We can still run our DM-ST algorithm as is. We state the sample complexity of DM-ST in agnostic setting below.

Theorem 13.5 (Agnostic Guarantee of DM-ST). *Assume Π is finite, but $\pi^\star \notin \Pi$. With probability at least $1 - \delta$, DM-ST learns a policy $\hat{\pi}$ such that:*

$$V^\star - V^{\hat{\pi}} \leq \frac{1}{1 - \gamma} \|d^\star - d^{\hat{\pi}}\|_1 \leq \frac{3}{1 - \gamma} \min_{\pi \in \Pi} \|d^\pi - d^\star\|_1 + \tilde{O} \left(\frac{1}{1 - \gamma} \sqrt{\frac{\ln(|\Pi| + \ln(1/\delta))}{M}} \right).$$

Proof. We first define some terms below. Denote $\tilde{\pi} := \arg \min_{\pi \in \Pi} \|d^\pi - d^\star\|_1$. Let us denote:

$$\begin{aligned} \tilde{f} &= \arg \max_{f \in \mathcal{F}} [\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} f(s, a) - \mathbb{E}_{s,a \sim d^\star} f(s, a)], \\ \bar{f} &= \arg \max_{f \in \mathcal{F}} \left[\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i^\star, a_i^\star) \right], \\ f' &= \arg \max_{f \in \mathcal{F}} \left[\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f(s, a)] - \frac{1}{M} \sum_{i=1}^M f(s_i^\star, a_i^\star) \right]. \end{aligned}$$

Starting with a triangle inequality, we have:

$$\begin{aligned}
\|d^{\hat{\pi}} - d^{\pi^*}\|_1 &\leq \|d^{\hat{\pi}} - d^{\tilde{\pi}}\|_1 + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) + \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) + \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\quad + \left[\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] \right] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f'(s,a)] - \frac{1}{M} \sum_{i=1}^M f'(s_i^*, a_i^*) + 2\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} + 2\|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f'(s,a)] - \mathbb{E}_{s,a \sim d^{\pi^*}} [f'(s,a)] + 4\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} + 2\|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq 3\|d^{\pi^*} - d^{\tilde{\pi}}\|_1 + 4\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}},
\end{aligned}$$

where the first inequality uses the definition of $\tilde{f}$, the second inequality uses the fact that $\hat{\pi}$ is the minimizer of $\max_{f \in \mathcal{F}} \mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s,a) - \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*)$, along the way we also use Hoeffding's inequality where $\forall f \in \mathcal{F}$, $|\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} f(s,a) - \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*)| \leq 2\sqrt{\ln(|\mathcal{F}|/\delta)/M}$, with probability at least $1 - \delta$. $\square$

As we can see, comparing to the realizable setting, here we have an extra term that is related to $\min_{\pi \in \Pi} \|d^{\pi} - d^{\pi^*}\|_1$. Note that the dependency on horizon also scales linearly in this case. In general, the constant 3 in front of $\min_{\pi \in \Pi} \|d^{\pi} - d^{\pi^*}\|_1$ is not avoidable in Scheffé estimator Devroye and Lugosi [2012].

13.4 Maximum Entropy Inverse Reinforcement Learning

Similar to Behavior cloning, we assume we have a dataset of state-action pairs from expert $\mathcal{D}^* = \{s_i^*, a_i^*\}_{i=1}^N$ where $s_i^*, a_i^* \sim d^{\pi^*}$. Different from Behavior cloning, here we assume that we have access to the underlying MDP's transition, i.e., we assume transition is known and we can do planning if we were given a cost function.

We assume that we are given a state-action feature mapping $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$ (this can be extended infinite dimensional feature space in RKHS, but we present finite dimensional setting for simplicity). We assume the true ground truth cost function as $c(s,a) := \theta^* \cdot \phi(s,a)$ with $\|\theta^*\|_2 \leq 1$ and θ^* being unknown.

The goal of the learner is to compute a policy $\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})$ such that when measured under the true cost function, it performs as good as the expert, i.e., $\mathbb{E}_{s,a \sim d^{\pi}} \theta^* \cdot \phi(s,a) \approx \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^* \cdot \phi(s,a)$.

We will focus on finite horizon MDP setting in this section. We denote $\rho^{\pi}(\tau)$ as the trajectory distribution induced by π and d^{π} as the average state-action distribution induced by π .

13.4.1 MaxEnt IRL: Formulation and The Principle of Maximum Entropy

MaxEnt IRL uses the principle of Maximum Entropy and poses the following policy optimization program:

$$\begin{aligned} \max_{\pi} & - \sum_{\tau} \rho^{\pi}(\tau) \ln \rho^{\pi}(\tau), \\ \text{s.t.}, & \mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \sum_{i=1}^N \phi(s_i^*, a_i^*)/N. \end{aligned}$$

Note that $\sum_{i=1}^N \phi(s_i^*, a_i^*)/N$ is an unbiased estimate of $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$. MaxEnt-IRL searches for a policy that maximizes the entropy of its trajectory distribution subject to a moment matching constraint.

Note that there could be many policies that satisfy the moment matching constraint, i.e., $\mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$, and any feasible solution is guaranteed to achieve the same performance of the expert under the ground truth cost function $\theta^* \cdot \phi(s, a)$. The maximum entropy objective ensures that the solution is unique.

Using the Markov property, we notice that:

$$\operatorname{argmax}_{\pi} - \sum_{\tau} \rho^{\pi}(\tau) \ln \rho^{\pi}(\tau) = \operatorname{argmax}_{\pi} - \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s).$$

This, we can rewrite the MaxEnt-IRL as follows:

$$\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s), \quad (13.2)$$

$$\text{s.t.}, \mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \sum_{i=1}^N \phi(s_i^*, a_i^*)/N. \quad (13.3)$$

13.4.2 Algorithm

To better interpret the objective function, below we replace $\sum_i \phi(s_i^*, a_i^*)/N$ by its expectation $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$. Note that we can use standard concentration inequalities to bound the difference between $\sum_i \phi(s_i^*, a_i^*)/N$ and $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$.

Using Lagrange multipliers, we can rewrite the constrained optimization program in Eq. 13.2 as follows:

$$\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s) + \max_{\theta} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^{\top} \phi(s, a).$$

The above objective conveys a clear goal of our imitation learning problem: we are searching for π that minimizes the MMD between d^{π} and d^{π^*} with a (negative) entropy regularization on the trajectory distribution of policy π .

To derive an algorithm that optimizes the above objective, we first again swap the min max order via the minimax theorem:

$$\max_{\theta} \left(\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^{\top} \phi(s, a) + \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s) \right).$$

The above objective proposes a natural algorithm where we perform projected gradient ascent on θ , while perform best response update on π , i.e., given θ , we solve the following planning problem:

$$\operatorname{argmin}_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) + \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s). \quad (13.4)$$

Note that the above objective can be understood as planning with cost function $\theta^\top \phi(s, a)$ with an additional negative entropy regularization on the trajectory distribution. On the other hand, given π , we can compute the gradient of θ , which is simply the difference between the expected features:

$$\mathbb{E}_{s,a \sim d^\pi} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a),$$

which gives the following gradient ascent update on θ :

$$\theta := \theta + \eta \left(\mathbb{E}_{s,a \sim d^\pi} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a) \right). \quad (13.5)$$

Algorithm of MaxEnt-IRL MaxEnt-IRL updates π and θ alternatively using Eq. 13.4 and Eq. 13.5, respectively. We summarize the algorithm in Alg. 10. Note that for the stochastic gradient of θ_t , we see that it is the empirical average feature difference between the current policy π_t and the expert policy π^* .

Algorithm 10 MaxEnt-IRL

Input: Expert data $\mathcal{D}^* = \{s_i^*, a_i^*\}_{i=1}^M$, MDP $\mathcal{M}$, parameters β, η, N .

- 1: Initialize θ_0 with $\|\theta_0\|_2 \leq 1$.
 - 2: **for** $t = 1, 2, \dots$, **do**
 - 3: Entropy-regularized Planning with cost $\theta_t^\top \phi(s, a)$: $\pi_t \in \operatorname{argmin}_\pi \mathbb{E}_{s,a \sim d^\pi} [\theta_t^\top \phi(s, a) + \beta \ln \pi(a|s)]$.
 - 4: Draw samples $\{s_i, a_i\}_{i=1}^N \sim d^{\pi_t}$ by executing π_t in $\mathcal{M}$.
 - 5: Stochastic Gradient Update: $\theta' = \theta_t + \eta \left[\frac{1}{N} \sum_{i=1}^N \phi(s_i, a_i) - \frac{1}{M} \sum_{i=1}^M \phi(s_i^*, a_i^*) \right]$.
 - 6: **end for**
-

Note that Alg. 10 uses an entropy-regularized planning oracle. Below we discuss how to implement such entropy-regularized planning oracle via dynamic programming.

13.4.3 Maximum Entropy RL: Implementing the Planning Oracle in Eq. 13.4

The planning oracle in Eq. 13.4 can be implemented in a value iteration fashion using Dynamic Programming. We denote $c(s, a) := \theta \cdot \phi(s, a)$.

We are interested in implementing the following planning objective:

$$\operatorname{argmin}_\pi \mathbb{E}_{s,a \sim d^\pi} [c(s, a) + \ln \pi(a|s)]$$

The subsection has its own independent interests beyond the framework of imitation learning. This maximum entropy regularized planning formulation is widely used in RL as well and it is well connected to the framework of RL as Inference.

As usually, we start from the last time step $H - 1$. For any policy π and any state-action (s, a) , we have the cost-to-go $Q_{H-1}^\pi(s, a)$:

$$Q_{H-1}^\pi(s, a) = c(s, a) + \ln \pi(a|s), \quad V_{H-1}^\pi(s) = \sum_a \pi(a|s) (c(s, a) + \ln \pi(a|s)).$$

We have:

$$V_{H-1}^*(s) = \min_{\rho \in \Delta(\mathcal{A})} \sum_a \rho(a) c(s, a) + \rho(a) \ln \rho(a). \quad (13.6)$$

Take gradient with respect to ρ , set it to zero and solve for ρ , we get:

$$\pi_{H-1}^*(a|s) \propto \exp(-c(s, a)), \forall s, a.$$

Substitute π_{H-1}^* back to the expression in Eq. 13.6, we get:

$$V_{H-1}^*(s) = -\ln \left(\sum_a \exp(-c(s, a)) \right),$$

i.e., we apply a softmin operator rather than the usual min operator (recall here we are minimizing cost).

With V_{h+1}^* , we can continue to h . Denote $Q^*(s, a)$ as follows:

$$Q_h^*(s, a) = r(s, a) + \mathbb{E}_{s' \sim P(\cdot|s, a)} V_{h+1}^*(s').$$

For V_h^* , we have:

$$V_h^*(s) = \min_{\rho \in \Delta(\mathcal{A})} \sum_a \rho(a) (c(s, a) + \ln \rho(a) + \mathbb{E}_{s' \sim P(\cdot|s, a)} V_{h+1}^*(s')).$$

Again we can show that the minimizer of the above program has the following form:

$$\pi_h^*(a|s) \propto \exp(-Q_h^*(s, a)). \quad (13.7)$$

Substitute π_h^* back to Q_h^* , we can show that:

$$V_h^*(s) = -\ln \left(\sum_a \exp(-Q_h^*(s, a)) \right),$$

where we see again that V_h^* is based on a softmin operator on Q^* .

Thus the *soft value iteration* can be summarized below:

Soft Value Iteration:

$$Q_{H-1}^*(s, a) = c(s, a), \quad \pi_{H-1}^*(a|s) \propto \exp(-Q_{H-1}^*(s, a)), \quad V_{H-1}^*(s) = -\ln \left(\sum_a \exp(-Q_{H-1}^*(s, a)) \right), \forall s.$$

We can continue the above procedure to $h = 0$, which gives us the optimal policies, all in the form of Eq. 13.7

13.5 Interactive Imitation Learning: AggreVaTe and Its Statistical Benefit over Offline IL Setting

We study the Interactive Imitation Learning setting where we have an expert policy that can be queried at any time during training, and we also have access to the ground truth reward signal.

We present AggreVaTe (Aggregate Values to Imitate) first and then analyze its sample complexity under the realizable setting.

Again we start with a realizable policy class Π that is discrete. We denote $\Delta(\Pi)$ as the convex hull of Π and each policy $\pi \in \Delta(\Pi)$ is a mixture policy represented by a distribution $\rho \in \mathbb{R}^{|\Pi|}$ with $\rho[i] \geq 0$ and $\|\rho\|_1 = 1$. With

this parameterization, our decision space simply becomes $\Delta(|\Pi|)$, i.e., any point in $\Delta(|\Pi|)$ corresponds to a mixture policy. Notation wise, given $\rho \in \Delta(|\Pi|)$, we denote π_ρ as the corresponding mixture policy. We denote π_i as the i -th policy in Π . We denote $\rho[i]$ as the i -th element of the vector ρ .

AggreVaTe assumes an interactive expert from whom I can query for action feedback. Basically, given a state s , let us assume that expert returns us the advantages of all actions, i.e., *one query of expert oracle at any state s returns $A^*(s, a), \forall a \in \mathcal{A}$.*¹

Algorithm 11 AggreVaTe

Input: The interactive expert, regularization λ

- 1: Initialize ρ_0 to be a uniform distribution.
 - 2: **for** $t = 0, 2, \dots$, **do**
 - 3: Sample $s_t \sim d^{\pi_{\rho_t}}$
 - 4: Query expert to get $A^*(s_t, a)$ for all $a \in \mathcal{A}$
 - 5: Policy update: $\rho_{t+1} = \operatorname{argmax}_{\rho \in \Delta(|\Pi|)} \sum_{j=0}^t \sum_{i=1}^{|\Pi|} \rho[i] (\mathbb{E}_{a \sim \pi_i(\cdot|s_t)} A^*(s_t, a)) - \lambda \sum_{i=1}^{|\Pi|} \rho[i] \ln(\rho[i])$
 - 6: **end for**
-

The algorithm is summarized in Alg. 11.

To analyze the algorithm, let us introduce some additional notations. Let us denote $\ell_t(\rho) = \sum_{i=1}^{|\Pi|} \rho[i] \mathbb{E}_{a \sim \pi_i(\cdot|s_t)} A^*(s_t, a)$, which is dependent on state s_t generated at iteration t , and is a linear function with respect to decision variable ρ . AggreVaTe is essentially running the specific online learning algorithm, Follow-the-Regularized Leader (FTRL) (e.g., see Shalev-Shwartz [2011]) with Entropy regularization:

$$\rho_{t+1} = \operatorname{argmax}_{\rho \in \Delta(|\Pi|)} \sum_{i=0}^t \ell_t(\rho) + \lambda \operatorname{Entropy}(\rho).$$

Denote $c = \max_{s,a} A^*(s, a)$. This implies that $\sup_{\rho,t} \|\ell_t(\rho)\| \leq c$. FTRL with linear loss functions and entropy regularization gives the following deterministic regret guarantees (Shalev-Shwartz [2011]):

$$\max_{\rho} \sum_{t=0}^{T-1} \ell_t(\rho) - \sum_{t=0}^{T-1} \ell_t(\rho_t) \leq c \sqrt{\log(|\Pi|)T}. \quad (13.8)$$

We will analyze AggreVaTe's sample complexity using the above result.

Theorem 13.6 (Sample Complexity of AggreVaTe). *Denote $c = \sup_{s,a} |A^*(s, a)|$. Let us denote ϵ_Π and ϵ_{stat} as follows:*

$$\epsilon_\Pi := \max_{\rho \in \Delta(|\Pi|)} \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*), \quad \epsilon_{stat} := \sqrt{\frac{\ln(|\Pi|)/\delta}{M}} + 2\sqrt{\frac{\ln(1/\delta)}{M}}.$$

With probability at least $1 - \delta$, after M iterations (i.e., M calls of expert oracle), AggreVaTe finds a policy $\hat{\pi}$ such that:

$$V^* - V^{\hat{\pi}} \leq \frac{c}{1 - \gamma} \epsilon_{stat} - \frac{1}{1 - \gamma} \epsilon_\Pi.$$

Proof. At each iteration t , let us define $\tilde{\ell}_t(\rho_t)$ as follows:

$$\tilde{\ell}_t(\rho_t) = \mathbb{E}_{s \sim d^{\pi_{\rho_t}}} \sum_{i=1}^{|\Pi|} \rho_t[i] \mathbb{E}_{a \sim \pi_i(\cdot|s)} A^*(s, a)$$

¹Technically one cannot use one query to get $A^*(s, a)$ for all a . But one can use importance weighting to get an unbiased estimate of $A^*(s, a)$ for all a via just one expert roll-out. For analysis simplicity, we assume one expert query at s returns the whole vector $A^*(s, \cdot) \in \mathbb{R}^{|\mathcal{A}|}$.

Denote $\mathbb{E}_t[\ell_t(\rho_t)]$ as the conditional expectation of $\ell_t(\rho_t)$, conditioned on all history up and including the end of iteration $t - 1$. Thus, we have $\mathbb{E}_t[\ell_t](\rho_t) = \tilde{\ell}_t(\rho_t)$ as ρ_t only depends on the history up to the end of iteration $t - 1$. Also note that $|\ell_t(\rho)| \leq c$. Thus by Azuma-Hoeffding inequality (Theorem A.5), we get:

$$\left| \frac{1}{M} \sum_{t=0}^{M-1} \tilde{\ell}_t(\rho_t) - \frac{1}{T} \sum_{t=0}^{M-1} \ell_t(\rho_t) \right| \leq 2c\sqrt{\frac{\ln(1/\delta)}{M}},$$

with probability at least $1 - \delta$. Now use Eq. 13.8, and denote $\rho^* = \operatorname{argmin}_{\rho \in \Delta(\Pi)} \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho)$, i.e., the best minimizer that minimizes the average loss $\sum_{t=0}^{M-1} \ell_t(\rho)/M$. we get:

$$\frac{1}{M} \sum_{t=0}^{M-1} \tilde{\ell}_t(\rho_t) \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho_t) - 2c\sqrt{\frac{\ln(1/\delta)}{M}} \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{M}} - 2c\sqrt{\frac{\ln(1/\delta)}{M}},$$

which means that there must exist a $\hat{t} \in \{0, \dots, M-1\}$, such that:

$$\tilde{\ell}_{\hat{t}}(\rho_{\hat{t}}) \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{M}} - 2c\sqrt{\frac{\ln(1/\delta)}{M}}.$$

Now use Performance difference lemma, we have:

$$\begin{aligned} (1 - \gamma)(-V^* + V^{\pi_{\rho_{\hat{t}}}}) &= \mathbb{E}_{s \sim d^{\pi_{\rho_{\hat{t}}}}} \sum_{i=1}^{|\Pi|} \rho_{\hat{t}}[i] \mathbb{E}_{a \sim \pi_i(\cdot|s)} A^*(s, a) = \ell_{\hat{t}}(\rho_{\hat{t}}) \\ &\geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{T}} - 2c\sqrt{\frac{\ln(1/\delta)}{T}}. \end{aligned}$$

Rearrange terms, we get:

$$V^* - V^{\pi_{\rho_{\hat{t}}}} \leq -\frac{1}{1 - \gamma} \left[\frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) \right] + \frac{c}{1 - \gamma} \left[\sqrt{\frac{\ln(|\Pi|)/\delta}{M}} + 2\sqrt{\frac{\ln(1/\delta)}{M}} \right],$$

which concludes the proof. $\square$

Remark We analyze the ϵ_{Π} by discussing realizable setting and non-realizable setting separately. When Π is realizable, i.e., $\pi^* \in \Pi$, by the definition of our loss function ℓ_t , we immediately have $\sum_{i=1}^M \ell_t(\rho^*) \geq 0$ since $A^*(s, \pi^*(s)) = 0$ for all $s \in \mathcal{S}$. Moreover, when π^* is not the globally optimal policy, it is possible that $\epsilon_{\Pi} := \sum_{i=1}^M \ell_t(\rho^*)/M > 0$, which implies that when $M \rightarrow \infty$ (i.e., $\epsilon_{stat} \rightarrow 0$), AggreVaTe indeed can learn a policy that outperforms the expert policy π^* . In general when $\pi^* \notin \Pi$, there might not exist a mixture policy ρ that achieves positive advantage against π^* under the M training samples $\{s_0, \dots, s_{M-1}\}$. In this case, $\epsilon_{\Pi} < 0$.

Does AggreVaTe avoids distribution shift? Under realizable setting, note that the bound explicitly depends on $\sup_{s,a} |A^*(s, a)|$ (i.e., it depends on $|\ell_t(\rho_t)|$ for all t). In worst case, it is possible that $\sup_{s,a} |A^*(s, a)| = \Theta(1/(1 - \gamma))$, which implies that AggreVaTe could suffer a quadratic horizon dependency, i.e., $1/(1 - \gamma)^2$. Note that DM-ST provably scales linearly with respect to $1/(1 - \gamma)$, but DM-ST requires a stronger assumption that the transition P is known.

When $\sup_{s,a} |A^*(s, a)| = o(1/(1 - \gamma))$, then AggreVaTe performs strictly better than BC. This is possible when the MDP is mixing quickly under π^* , or Q^* is L -Lipschitz continuous, i.e., $|Q^*(s, a) - Q^*(s, a')| \leq L\|a - a'\|$, with bounded range in action space, e.g., $\sup_{a,a'} \|a - a'\| \leq \beta \in \mathbb{R}^+$. In this case, if L and β are independent of $1/(1 - \gamma)$, then $\sup_{s,a} |A^*(s, a)| \leq L\beta$, which leads to a $\frac{\beta L}{1 - \gamma}$ dependency.

When $\pi^* \notin \Pi$, the agnostic result of AggreVaTe and the agnostic result of DM-ST is not directly comparable. Note that the model class error that DM-ST suffers is algorithmic-independent, i.e., it is $\min_{\pi \in \Pi} \|d^\pi - d^{\pi^*}\|_1$ and it only depends on π^* and Π , while the model class ϵ_Π in AggreVaTe is algorithmic path-dependent, i.e., in addition to π^* and Π , it depends the policies $\pi_1, \pi_2, \dots$ computed during the learning process. Another difference is that $\min_{\pi \in \Pi} \|d^\pi - d^{\pi^*}\|_1 \in [0, 2]$, while ϵ_Π indeed could scale linearly with respect to $1/(1 - \gamma)$, i.e., $\epsilon_\pi \in [-\frac{1}{1-\gamma}, 0]$ (assume π^* is the globally optimal policy for simplicity).

13.6 Bibliographic Remarks and Further Readings

Behavior cloning was used in autonomous driving back in 1989 Pomerleau [1989], and the distribution shift and compounding error issue was studied by Ross and Bagnell [2010]. Ross and Bagnell [2010], Ross et al. [2011] proposed using interactive experts to alleviate the distribution shift issue.

MaxEnt-IRL was proposed by Ziebart et al. [2008]. The original MaxEnt-IRL focuses on deterministic MDPs and derived distribution over trajectories. Later Ziebart et al. [2010] proposed the Principle of Maximum Causal Entropy framework which captures MDPs with stochastic transition. MaxEnt-IRL has been widely used in real applications (e.g., Kitani et al. [2012], Ziebart et al. [2009]).

To the best of our knowledge, the algorithm Distribution Matching with Scheffé Tournament introduced in this chapter is new here and is the first to demonstrate the statistical benefit (in terms of sample complexity of the expert policy) of the hybrid setting over the pure offline setting with general function approximation.

Recently, there are approaches that extend the linear cost functions used in MaxEnt-IRL to deep neural networks which are treated as discriminators to distinguish between experts datasets and the datasets generated by the policies. Different distribution divergences have been proposed, for instance, JS-divergence Ho and Ermon [2016], general f-divergence which generalizes JS-divergence Ke et al. [2019], and Integral Probability Metric (IPM) which includes Wasserstein distance Sun et al. [2019b]. While these adversarial IL approaches are promising as they fall into the hybrid setting, it has been observed that empirically, adversarial IL sometimes cannot outperform simple algorithms such as Behavior cloning which operates in pure offline setting (e.g., see experiments results on common RL benchmarks from Brantley et al. [2019]).

Another important direction in IL is to combine expert demonstrations and reward functions together (i.e., perform Imitation together with RL). There are many works in this direction, including learning with pre-collected expert data Hester et al. [2017], Rajeswaran et al. [2017], Cheng et al. [2018], Le et al. [2018], Sun et al. [2018], learning with an interactive expert Daumé et al. [2009], Ross and Bagnell [2014], Chang et al. [2015], Sun et al. [2017], Cheng et al. [2020], Cheng and Boots [2018]. The algorithm AggreVaTe was originally introduced in Ross and Bagnell [2014]. A policy gradient and natural policy gradient version of AggreVaTe are introduced in Sun et al. [2017]. The precursor of AggreVaTe is the algorithm Data Aggregation (DAgger) Ross et al. [2011], which leverages interactive experts and a reduction to no-regret online learning, but without assuming access to the ground truth reward signals.

The maximum entropy RL formulation has been widely used in RL literature as well. For instance, Guided Policy Search (GPS) (and its variants) use maximum entropy RL formulation as its planning subroutine Levine and Koltun [2013], Levine and Abbeel [2014]. We refer readers to the excellent survey from Levine [2018] for more details of Maximum Entropy RL formulation and its connections to the framework of RL as Probabilistic Inference.

Chapter 14

Linear Quadratic Regulators

This chapter will introduce some of the fundamentals of optimal control for the linear quadratic regulator model. This model is an MDP, with continuous states and actions. While the model itself is often inadequate as a global model, it can be quite effective as a locally linear model (provided our system does not deviate away from the regime where our linear model is reasonable approximation).

The basics of optimal control theory can be found in any number of standards text Anderson and Moore [1990], Evans [2005], Bertsekas [2017]. The treatment of Gauss-Newton and the NPG algorithm are due to Fazel et al. [2018].

14.1 The LQR Model

In the standard optimal control problem, a dynamical system is described as

$$x_{t+1} = f_t(x_t, u_t, w_t),$$

where f_t maps a state $x_t \in \mathbb{R}^d$, a control (the action) $u_t \in \mathbb{R}^k$, and a disturbance w_t , to the next state $x_{t+1} \in \mathbb{R}^d$, starting from an initial state x_0 . The objective is to find the control policy π which minimizes the long term cost,

$$\begin{aligned} \text{minimize} \quad & \mathbb{E}_\pi \left[\sum_{t=0}^H c_t(x_t, u_t) \right] \\ \text{such that} \quad & x_{t+1} = f_t(x_t, u_t, w_t) \quad t = 0, \dots, H. \end{aligned}$$

where H is the time horizon (which can be finite or infinite).

In practice, this is often solved by considering the linearized control (sub-)problem where the dynamics are approximated by

$$x_{t+1} = A_t x_t + B_t u_t + w_t,$$

with the matrices A_t and B_t are derivatives of the dynamics f and where the costs are approximated by a quadratic function in x_t and u_t .

This chapter focuses on an important special case: finite and infinite horizon problem referred to as the linear quadratic regulator (LQR) problem. We can view this model as being an local approximation to non-linear model. However, we will analyze these models under the assumption that they are globally valid.

Finite Horizon LQRs. The finite horizon LQR problem is given by

$$\begin{aligned} \text{minimize} \quad & \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=0}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \right] \\ \text{such that} \quad & x_{t+1} = A_t x_t + B_t u_t + w_t, \quad x_0 \sim \mathcal{D}, \quad w_t \sim N(0, \sigma^2 I), \end{aligned}$$

where initial state $x_0 \sim \mathcal{D}$ is assumed to be randomly distributed according to distribution $\mathcal{D}$; the disturbance $w_t \in \mathbb{R}^d$ follows the law of a multi-variate normal with covariance $\sigma^2 I$; the matrices $A_t \in \mathbb{R}^{d \times d}$ and $B_t \in \mathbb{R}^{d \times k}$ are referred to as system (or transition) matrices; $Q \in \mathbb{R}^{d \times d}$ and $R \in \mathbb{R}^{k \times k}$ are both positive definite matrices that parameterize the quadratic costs. Note that this model is a finite horizon MDP, where the $\mathcal{S} = \mathbb{R}^d$ and $\mathcal{A} = \mathbb{R}^k$.

Infinite Horizon LQRs. We also consider the infinite horizon LQR problem:

$$\begin{aligned} \text{minimize} \quad & \lim_{H \rightarrow \infty} \frac{1}{H} \mathbb{E} \left[\sum_{t=0}^H (x_t^\top Q x_t + u_t^\top R u_t) \right] \\ \text{such that} \quad & x_{t+1} = A x_t + B u_t + w_t, \quad x_0 \sim \mathcal{D}, \quad w_t \sim N(0, \sigma^2 I). \end{aligned}$$

Note that here we are assuming the dynamics are time homogenous. We will assume that the optimal objective function (i.e. the optimal average cost) is finite; this is referred to as the system being *controllable*. This is a special case of an MDP with an average reward objective.

Throughout this chapter, we assume A and B are such that the optimal cost is finite. Due to the geometric nature of the system dynamics (say for a controller which takes controls u_t that are linear in the state x_t), there may exist linear controllers with infinite costs. This instability of LQRs (at least for some A and B and some controllers) leads to that the theoretical analysis often makes various assumptions on A and B in order to guarantee some notion of stability. In practice, the finite horizon setting is more commonly used in practice, particularly due to that the LQR model is only a good local approximation of the system dynamics, where the infinite horizon model tends to be largely of theoretical interest. See Section 14.6.

The infinite horizon discounted case? The infinite horizon discounted case tends not to be studied for LQRs. This is largely due to that, for the undiscounted case (with the average cost objective), we have may infinite costs (due to the aforementioned geometric nature of the system dynamics); in such cases, discounting will not necessarily make the average cost finite.

14.2 Bellman Optimality: Value Iteration & The Algebraic Riccati Equations

A standard result in optimal control theory shows that the optimal control input can be written as a linear function in the state. As we shall see, this is a consequence of the Bellman equations.

14.2.1 Planning and Finite Horizon LQRs

Slightly abusing notation, it is convenient to define the value function and state-action value function with respect to the costs as follows: For a policy π , a state x , and $h \in \{0, \dots, H-1\}$, we define the value function $V_h^\pi : \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$V_h^\pi(x) = \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=h}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \mid \pi, x_h = x \right],$$

where again expectation is with respect to the randomness of the trajectory, that is, the randomness in state transitions. Similarly, the state-action value (or Q-value) function $Q_h^\pi : \mathbb{R}^d \times \mathbb{R}^k \rightarrow \mathbb{R}$ is defined as

$$Q_h^\pi(x, u) = \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=h}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \mid \pi, x_h = x, u_h = u \right].$$

We define V^* and Q^* analogously.

The following theorem provides a characterization of the optimal policy, via the algebraic Riccati equations. These equations are simply the value iteration algorithm for the special case of LQRs.

Theorem 14.1. (*Value Iteration and the Riccati Equations*). *Suppose R is positive definite. The optimal policy is a linear controller specified by:*

$$\pi^*(x_t) = -K_t^* x_t$$

where

$$K_t^* = (B_t^\top P_{t+1} B_t + R)^{-1} B_t^\top P_{t+1} A_t.$$

Here, P_t can be computed iteratively, in a backwards manner, using the following algebraic Riccati equations, where for $t \in [H]$,

$$\begin{aligned} P_t &:= A_t^\top P_{t+1} A_t + Q - A_t^\top P_{t+1} B_t (B_t^\top P_{t+1} B_t + R)^{-1} B_t^\top P_{t+1} A_t \\ &= A_t^\top P_{t+1} A_t + Q - (K_{t+1}^*)^\top (B_t^\top P_{t+1} B_t + R) K_{t+1}^* \end{aligned}$$

and where $P_H = Q$. (The above equation is simply the value iteration algorithm).

Furthermore, for $t \in [H]$, we have that:

$$V_t^*(x) = x^\top P_t x + \sigma^2 \sum_{h=t+1}^H \text{Trace}(P_h).$$

We often refer to K_t^* as the *optimal control gain matrices*. It is straightforward to generalize the above when $\sigma \geq 0$.

We have assumed that R is strictly positive definite, to avoid have to working with the pseudo-inverse; the theorem is still true when $R = 0$, provided we use a pseudo-inverse.

Proof. By the Bellman optimality conditions (Theorem ?? for episodic MDPs), we know the optimal policy (among all possibly history dependent, non-stationary, and randomized policies), is given by a deterministic stationary policy which is only a function of x_t and t . We have that:

$$\begin{aligned} Q_{H-1}(x, u) &= \mathbb{E}[(A_{H-1}x + B_{H-1}u + w_{H-1})^\top Q(A_{H-1}x + B_{H-1}u + w_{H-1})] + x^\top Q x + u^\top R u \\ &= (A_{H-1}x + B_{H-1}u)^\top Q(A_{H-1}x + B_{H-1}u) + \sigma^2 \text{Trace}(Q) + x^\top Q x + u^\top R u \end{aligned}$$

due to that $x_H = A_{H-1}x + B_{H-1}u + w_{H-1}$, and $\mathbb{E}[w_{H-1}^\top Q w_{H-1}] = \text{Trace}(\sigma^2 Q)$. Due to that this is a quadratic function of u , we can immediately derive that the optimal control is given by:

$$\pi_{H-1}^*(x) = -(B_{H-1}^\top Q B_{H-1} + R)^{-1} B_{H-1}^\top Q A_{H-1} x = -K_{H-1}^* x,$$

where the last step uses that $P_H := Q$.

For notational convenience, let $K = K_{H-1}^*$, $A = A_{H-1}$, and $B = B_{H-1}$. Using the optimal control at x , i.e. $u = -K_{H-1}^*x$, we have:

$$\begin{aligned}
V_{H-1}^*(x) &= Q_{H-1}(x, -K_{H-1}^*x) \\
&= x^\top \left((A - BK)^\top Q (A - BK) + Q + K^\top R K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - 2K^\top B^\top QA + K^\top (B^\top QB + R)K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - 2K^\top (B^\top QB + R)K + K^\top (B^\top QB + R)K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - K^\top (B^\top QB + R)K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top P_{H-1}x + \sigma^2 \text{Trace}(Q).
\end{aligned}$$

where the fourth step uses our expression for $K = K_{H-1}^*$. This proves our claim for $t = H - 1$.

This implies that:

$$\begin{aligned}
Q_{H-2}^*(x, u) &= \mathbb{E}[V_{H-1}^*(A_{H-2}x + B_{H-2}u + w_{H-2})] + x^\top Qx + u^\top Ru \\
&= (A_{H-2}x + B_{H-2}u)^\top P_{H-1} (A_{H-2}x + B_{H-2}u) + \sigma^2 \text{Trace}(P_{H-1}) + \sigma^2 \text{Trace}(Q) + x^\top Qx + u^\top Ru.
\end{aligned}$$

The remainder of the proof follows from a recursive argument, which can be verified along identical lines to the $t = H - 1$ case. $\square$

14.2.2 Planning and Infinite Horizon LQRs

Theorem 14.2. *Suppose that the optimal cost is finite and that R is positive definite. Let P be a solution to the following algebraic Riccati equation:*

$$P = A^\top P A + Q - A^\top P B (B^\top P B + R)^{-1} B^\top P A. \quad (14.1)$$

(Note that P is a positive definite matrix). We have that the optimal policy is:

$$\pi^*(x) = -K^*x$$

where the optimal control gain is:

$$K^* = -(B^\top P B + R)^{-1} B^\top P A. \quad (14.2)$$

We have that P is unique and that the optimal average cost is $\sigma^2 \text{Trace}(P)$.

As before, P parameterizes the optimal value function. We do not prove this theorem here though it follows along similar lines to the previous proof, via a limiting argument.

To find P , we can again run the recursion:

$$P \leftarrow Q + A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A$$

starting with $P = Q$, which can be shown to converge to the unique positive semidefinite solution of the Riccati equation (since one can show the fixed-point iteration is contractive). Again, this approach is simply value iteration.

14.3 Convex Programs to find P and K^*

For the infinite horizon LQR problem, the optimization may be formulated as a convex program. In particular, the LQR problem can also be expressed as a semidefinite program (SDP) with variable P , for the infinite horizon case. We now present this primal program along with the dual program.

Note that these programs are the analogues of the linear programs from Section ?? for MDPs. While specifying these linear programs for an LQR (as an LQR is an MDP) would result in infinite dimensional linear programs, the special structure of the LQR implies these primal and dual programs have a more compact formulation when specified as an SDP.

14.3.1 The Primal for Infinite Horizon LQR

The primal optimization problem is given as:

$$\begin{aligned} & \text{maximize} && \sigma^2 \text{Trace}(P) \\ & \text{subject to} && \begin{bmatrix} A^T P A + Q - P & A^T P B \\ B^T P A & B^T P B + R \end{bmatrix} \succeq 0, \quad P \succeq 0, \end{aligned}$$

where the optimization variable is P . This SDP has a unique solution, P^* , which satisfies the algebraic Riccati equations (Equation 14.1); the optimal average cost of the infinite horizon LQR is $\sigma^2 \text{Trace}(P^*)$; and the optimal policy is given by Equation 14.2.

The SDP can be derived by relaxing the equality in the Riccati equation to an inequality, then using the Schur complement lemma to rewrite the resulting Riccati inequality as linear matrix inequality. In particular, we can consider the relaxation where P must satisfy:

$$A^T P A + Q - A^T P B (B^T P B + R)^{-1} B^T P A \succeq P.$$

That the solution to this relaxed optimization problem leads to the optimal P^* is due to the Bellman optimality conditions.

Now the Schur complement lemma for positive semi-definiteness is as follows: define

$$X = \begin{bmatrix} D & E \\ E^T & F \end{bmatrix}.$$

(for matrices D , E , and F of appropriate size, with D and F being square symmetric matrices). We have that X is PSD if and only if

$$D - E^T F^{-1} E \succeq 0.$$

This shows that the constraint set is equivalent to the above relaxation.

14.3.2 The Dual

The dual optimization problem is given as:

$$\begin{aligned} & \text{minimize} && \text{Trace} \left(\Sigma \cdot \begin{bmatrix} Q & 0 \\ 0 & R \end{bmatrix} \right) \\ & \text{subject to} && \Sigma_{xx} = (A \ B) \Sigma (A \ B)^T + \sigma^2 I, \quad \Sigma \succeq 0, \end{aligned}$$

where the optimization variable is a symmetric matrix Σ , which is a $(d+k) \times (d+k)$ matrix with the block structure:

$$\Sigma = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xu} \\ \Sigma_{ux} & \Sigma_{uu} \end{bmatrix}.$$

The interpretation of Σ is that it is the covariance matrix of the stationary distribution. This is analogous to the state-visitation measure for an MDP.

This SDP has a unique solution, say Σ^* . The optimal gain matrix is then given by:

$$K^* = -\Sigma_{ux}^* (\Sigma_{xx}^*)^{-1}.$$

14.4 Policy Iteration, Gauss Newton, and NPG

Note that the noise variance σ^2 does not impact the optimal policy, in either the discounted case or in the infinite horizon case.

Here, when we examine local search methods, it is more convenient to work with case where $\sigma = 0$. In this case, we can work with cumulative cost rather than the average cost. Precisely, when $\sigma = 0$, the infinite horizon LQR problem takes the form:

$$\min_K C(K), \text{ where } C(K) = \mathbb{E}_{x_0 \sim \mathcal{D}} \left[\sum_{t=0}^H (x_t^\top Q x_t + u_t^\top R u_t) \right],$$

where the dynamics evolves as

$$x_{t+1} = (A - BK)x_t.$$

Note that we have directly parameterized our policy as a linear policy in terms of the gain matrix K , due to that we know the optimal policy is linear in the state. Again, we assume that $C(K^*)$ is finite; this assumption is referred to as the system being *controllable*.

We now examine local search based approaches, where we will see a close connection to policy iteration. Again, we have a non-convex optimization problem:

Lemma 14.3. (Non-convexity) *If $d \geq 3$, there exists an LQR optimization problem, $\min_K C(K)$, which is not convex or quasi-convex.*

Regardless, we will see that gradient based approaches are effective. For local search based approaches, the importance of (some) randomization, either in x_0 or noise through having a disturbance, is analogous to our use of having a wide-coverage distribution μ (for MDPs).

14.4.1 Gradient Expressions

Gradient descent on $C(K)$, with a fixed stepsize η , follows the update rule:

$$K \leftarrow K - \eta \nabla C(K).$$

It is helpful to explicitly write out the functional form of the gradient. Define P_K as the solution to:

$$P_K = Q + K^\top R K + (A - BK)^\top P_K (A - BK).$$

and, under this definition, it follows that $C(K)$ can be written as:

$$C(K) = \mathbb{E}_{x_0 \sim \mathcal{D}} x_0^\top P_K x_0.$$

Also, define Σ_K as the (un-normalized) state correlation matrix, i.e.

$$\Sigma_K = \mathbb{E}_{x_0 \sim \mathcal{D}} \sum_{t=0}^{\infty} x_t x_t^\top.$$

Lemma 14.4. (*Policy Gradient Expression*) *The policy gradient is:*

$$\nabla C(K) = 2 \left((R + B^\top P_K B)K - B^\top P_K A \right) \Sigma_K$$

For convenience, define E_K to be

$$E_K = \left((R + B^\top P_K B)K - B^\top P_K A \right),$$

as a result the gradient can be written as $\nabla C(K) = 2E_K \Sigma_K$.

Proof. Observe:

$$\begin{aligned} C_K(x_0) &= x_0^\top P_K x_0 = x_0^\top (Q + K^\top R K) x_0 + x_0^\top (A - BK)^\top P_K (A - BK) x_0 \\ &= x_0^\top (Q + K^\top R K) x_0 + C_K((A - BK)x_0). \end{aligned}$$

Let ∇ denote the gradient with respect to K ; note that $\nabla C_K((A - BK)x_0)$ has two terms as function of K , one with respect to K in the subscript and one with respect to the input $(A - BK)x_0$. This implies

$$\begin{aligned} \nabla C_K(x_0) &= 2RKx_0x_0^\top - 2B^\top P_K(A - BK)x_0x_0^\top + \nabla C_K(x_1)|_{x_1=(A-BK)x_0} \\ &= 2 \left((R + B^\top P_K B)K - B^\top P_K A \right) \sum_{t=0}^{\infty} x_t x_t^\top \end{aligned}$$

where we have used recursion and that $x_1 = (A - BK)x_0$. Taking expectations completes the proof. $\square$

The natural policy gradient. Let us now motivate a version of the natural gradient. The natural policy gradient follows the update:

$$\theta \leftarrow \theta - \eta F(\theta)^{-1} \nabla C(\theta), \text{ where } F(\theta) = \mathbb{E} \left[\sum_{t=0}^{\infty} \nabla \log \pi_\theta(u_t | x_t) \nabla \log \pi_\theta(u_t | x_t)^\top \right],$$

where $F(\theta)$ is the Fisher information matrix. A natural special case is using a linear policy with additive Gaussian noise, i.e.

$$\pi_K(x, u) = \mathcal{N}(Kx, \sigma^2 I) \quad (14.3)$$

where $K \in \mathbb{R}^{k \times d}$ and σ^2 is the noise variance. In this case, the natural policy gradient of K (when σ is considered fixed) takes the form:

$$K \leftarrow K - \eta \nabla C(\pi_K) \Sigma_K^{-1} \quad (14.4)$$

Note a subtlety here is that $C(\pi_K)$ is the randomized policy.

To see this, one can verify that the Fisher matrix of size $kd \times kd$, which is indexed as $[G_K]_{(i,j),(i',j')}$ where $i, i' \in \{1, \dots, k\}$ and $j, j' \in \{1, \dots, d\}$, has a block diagonal form where the only non-zeros blocks are $[G_K]_{(i,\cdot),(i,\cdot)} = \Sigma_K$ (this is the block corresponding to the i -th coordinate of the action, as i ranges from 1 to k). This form holds more generally, for any diagonal noise.

14.4.2 Convergence Rates

We consider three exact rules, where we assume access to having exact gradients. As before, we can also estimate these gradients through simulation. For gradient descent, the update is

$$K_{n+1} = K_n - \eta \nabla C(K_n). \quad (14.5)$$

For natural policy gradient descent, the direction is defined so that it is consistent with the stochastic case, as per Equation 14.4, in the exact case the update is:

$$K_{n+1} = K_n - \eta \nabla C(K_n) \Sigma_{K_n}^{-1} \quad (14.6)$$

One show that the Gauss-Newton update is:

$$K_{n+1} = K_n - \eta (R + B^\top P_{K_n} B)^{-1} \nabla C(K_n) \Sigma_{K_n}^{-1}. \quad (14.7)$$

(Gauss-Newton is non-linear optimization approach which uses a certain Hessian approximation. It can be show that this leads to the above update rule.) Interestingly, for the case when $\eta = 1$, the Gauss-Newton method is equivalent to the policy iteration algorithm, which optimizes a one-step deviation from the current policy.

The Gauss-Newton method requires the most complex oracle to implement: it requires access to $\nabla C(K)$, Σ_K , and $R + B^\top P_K B$; as we shall see, it also enjoys the strongest convergence rate guarantee. At the other extreme, gradient descent requires oracle access to only $\nabla C(K)$ and has the slowest convergence rate. The natural policy gradient sits in between, requiring oracle access to $\nabla C(K)$ and Σ_K , and having a convergence rate between the other two methods.

In this theorem, $\|M\|_2$ denotes the spectral norm of a matrix M .

Theorem 14.5. (*Global Convergence of Gradient Methods*) Suppose $C(K_0)$ is finite and, for μ defined as

$$\mu := \sigma_{\min}(\mathbb{E}_{x_0 \sim \mathcal{D}} x_0 x_0^\top),$$

suppose $\mu > 0$.

- *Gauss-Newton case:* Suppose $\eta = 1$, the Gauss-Newton algorithm (Equation 14.7) enjoys the following performance bound:

$$C(K_N) - C(K^*) \leq \epsilon, \text{ for } N \geq \frac{\|\Sigma_{K^*}\|_2}{\mu} \log \frac{C(K_0) - C(K^*)}{\epsilon}.$$

- *Natural policy gradient case:* For a stepsize $\eta = 1/(\|R\|_2 + \frac{\|B\|_2^2 C(K_0)}{\mu})$, natural policy gradient descent (Equation 14.6) enjoys the following performance bound:

$$C(K_N) - C(K^*) \leq \epsilon, \text{ for } N \geq \frac{\|\Sigma_{K^*}\|_2}{\mu} \left(\frac{\|R\|_2}{\sigma_{\min}(R)} + \frac{\|B\|_2^2 C(K_0)}{\mu \sigma_{\min}(R)} \right) \log \frac{C(K_0) - C(K^*)}{\epsilon}.$$

- *Gradient descent case:* For any starting policy K_0 , there exists a (constant) stepsize η (which could be a function of K_0), such that:

$$C(K_N) \rightarrow C(K^*), \text{ as } N \rightarrow \infty.$$

14.4.3 Gauss-Newton Analysis

We only provide a proof for the Gauss-Newton case (see 14.6 for further readings).

We overload notation and let K denote the policy $\pi(x) = Kx$. For the infinite horizon cost function, define:

$$\begin{aligned} V_K(x) &:= \sum_{t=0}^{\infty} (x_t^\top Q x_t + u_t^\top R u_t) \\ &= x^\top P_K x, \end{aligned}$$

and

$$Q_K(x, u) := x^\top Q x + u^\top R u + V_K(Ax + Bu),$$

and

$$A_K(x, u) = Q_K(x, u) - V_K(x).$$

The next lemma is identical to the performance difference lemma.

Lemma 14.6. (*Cost difference lemma*) Suppose K and K' have finite costs. Let $\{x'_t\}$ and $\{u'_t\}$ be state and action sequences generated by K' , i.e. starting with $x'_0 = x$ and using $u'_t = -K'x'_t$. It holds that:

$$V_{K'}(x) - V_K(x) = \sum_t A_K(x'_t, u'_t).$$

Also, for any x , the advantage is:

$$A_K(x, K'x) = 2x^\top (K' - K)^\top E_K x + x^\top (K' - K)^\top (R + B^\top P_K B)(K' - K)x. \quad (14.8)$$

Proof. Let c'_t be the cost sequence generated by K' . Telescoping the sum appropriately:

$$\begin{aligned} V_{K'}(x) - V_K(x) &= \sum_{t=0}^{\infty} c'_t - V_K(x) = \sum_{t=0}^{\infty} (c'_t + V_K(x'_t) - V_K(x'_t)) - V_K(x) \\ &= \sum_{t=0}^{\infty} (c'_t + V_K(x'_{t+1}) - V_K(x'_t)) = \sum_{t=0}^{\infty} A_K(x'_t, u'_t) \end{aligned}$$

which completes the first claim (the third equality uses the fact that $x = x_0 = x'_0$).

For the second claim, observe that:

$$V_K(x) = x^\top (Q + K^\top R K) x + x^\top (A - BK)^\top P_K (A - BK) x$$

And, for $u = K'x$,

$$\begin{aligned} A_K(x, u) &= Q_K(x, u) - V_K(x) \\ &= x^\top (Q + (K')^\top R K') x + x^\top (A - BK')^\top P_K (A - BK') x - V_K(x) \\ &= x^\top (Q + (K' - K + K)^\top R (K' - K + K)) x + \\ &\quad x^\top (A - BK - B(K' - K))^\top P_K (A - BK - B(K' - K)) x - V_K(x) \\ &= 2x^\top (K' - K)^\top ((R + B^\top P_K B)K - B^\top P_K A) x + \\ &\quad x^\top (K' - K)^\top (R + B^\top P_K B)(K' - K)x, \end{aligned}$$

which completes the proof. □

We have the following corollary, which can be viewed analogously to a smoothness lemma.

Corollary 14.7. (“Almost” smoothness) $C(K)$ satisfies:

$$C(K') - C(K) = -2\text{Trace}(\Sigma_{K'}(K - K')^\top E_K) + \text{Trace}(\Sigma_{K'}(K - K')^\top (R + B^\top P_K B)(K - K'))$$

To see why this is related to smoothness (recall the definition of a smooth function in Equation 9.3), suppose K' is sufficiently close to K so that:

$$\Sigma_{K'} \approx \Sigma_K + O(\|K - K'\|) \quad (14.9)$$

and the leading order term $2\text{Trace}(\Sigma_{K'}(K' - K)^\top E_K)$ would then behave as $\text{Trace}((K' - K)^\top \nabla C(K))$.

Proof. The claim immediately results from Lemma 14.6, by using Equation 14.8 and taking an expectation. $\square$

We now use this cost difference lemma to show that $C(K)$ is gradient dominated.

Lemma 14.8. (*Gradient domination*) *Let K^* be an optimal policy. Suppose K has finite cost and $\mu > 0$. It holds that:*

$$\begin{aligned} C(K) - C(K^*) &\leq \|\Sigma_{K^*}\| \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq \frac{\|\Sigma_{K^*}\|}{\mu^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \end{aligned}$$

Proof. From Equation 14.8 and by completing the square,

$$\begin{aligned} A_K(x, K'x) &= Q_K(x, K'x) - V_K(x) \\ &= 2\text{Trace}(xx^\top (K' - K)^\top E_K) + \text{Trace}(xx^\top (K' - K)^\top (R + B^\top P_K B)(K' - K)) \\ &= \text{Trace}(xx^\top (K' - K + (R + B^\top P_K B)^{-1} E_K)^\top (R + B^\top P_K B)(K' - K + (R + B^\top P_K B)^{-1} E_K)) \\ &\quad - \text{Trace}(xx^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\geq -\text{Trace}(xx^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \end{aligned} \quad (14.10)$$

with equality when $K' = K - (R + B^\top P_K B)^{-1} E_K$.

Let x_t^* and u_t^* be the sequence generated under K^* . Using this and Lemma 14.6,

$$\begin{aligned} C(K) - C(K^*) &= -\mathbb{E} \sum_t A_K(x_t^*, u_t^*) \leq \mathbb{E} \sum_t \text{Trace}(x_t^* (x_t^*)^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &= \text{Trace}(\Sigma_{K^*} E_K^\top (R + B^\top P_K B)^{-1} E_K) \leq \|\Sigma_{K^*}\| \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \end{aligned}$$

which completes the proof of first inequality. For the second inequality, observe:

$$\begin{aligned} \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) &\leq \|(R + B^\top P_K B)^{-1}\| \text{Trace}(E_K^\top E_K) \leq \frac{1}{\sigma_{\min}(R)} \text{Trace}(E_K^\top E_K) \\ &= \frac{1}{\sigma_{\min}(R)} \text{Trace}(\Sigma_K^{-1} \nabla C(K)^\top \nabla C(K) \Sigma_K^{-1}) \leq \frac{1}{\sigma_{\min}(\Sigma_K)^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \\ &\leq \frac{1}{\mu^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \end{aligned}$$

which completes the proof of the upper bound. Here the last step is because $\Sigma_K \succeq \mathbb{E}[x_0 x_0^\top]$. $\square$

The next lemma bounds the one step progress of Gauss-Newton.

Lemma 14.9. (*Gauss-Newton Contraction*) *Suppose that:*

$$K' = K - \eta(R + B^\top P_K B)^{-1} \nabla C(K) \Sigma_K^{-1}, .$$

If $\eta \leq 1$, then

$$C(K') - C(K^*) \leq \left(1 - \frac{\eta \mu}{\|\Sigma_{K^*}\|}\right) (C(K) - C(K^*))$$

Proof. Observe that for PSD matrices A and B , we have that $\text{Trace}(AB) \geq \sigma_{\min}(A)\text{Trace}(B)$. Also, observe $K' = K - \eta(R + B^\top P_K B)^{-1} E_K$. Using Lemma 14.7 and the condition on η ,

$$\begin{aligned} C(K') - C(K) &= -2\eta \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) + \eta^2 \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \sigma_{\min}(\Sigma_{K'}) \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \mu \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \frac{\mu}{\|\Sigma_{K^*}\|} (C(K) - C(K^*)), \end{aligned}$$

where the last step uses Lemma 14.8. □

With this lemma, the proof of the convergence rate of the Gauss Newton algorithm is immediate.

Proof. (of Theorem 14.5, Gauss-Newton case) The theorem is due to that $\eta = 1$ leads to a contraction of $1 - \frac{\mu}{\|\Sigma_{K^*}\|}$ at every step. □

14.5 System Level Synthesis for Linear Dynamical Systems

We demonstrate another parameterization of controllers which admits convexity. Specifically in this section, we consider finite horizon setting, i.e.,

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad x_0 \sim \mathcal{D}, w_t \sim \mathcal{N}(0, \sigma^2 I), \quad t \in [0, 1, \dots, H-1].$$

Instead of focusing on quadratic cost function, we consider general convex cost function $c(x_t, u_t)$, and the goal is to optimize:

$$\mathbb{E}_\pi \left[\sum_{t=0}^{H-1} c_t(x_t, u_t) \right],$$

where π is a time-dependent policy.

While we relax the assumption on the cost function from quadratic cost to general convex cost, we still focus on linearly parameterized policy, i.e., we are still interested in searching for a sequence of time-dependent linear controllers $\{-K_t^*\}_{t=0}^{H-1}$ with $u_t = -K_t^* x_t$, such that it minimizes the above objective function. We again assume the system is controllable, i.e., the expected total cost under $\{-K_t^*\}_{t=0}^{H-1}$ is finite.

Note that due to the fact that now the cost function is not quadratic anymore, the Riccati equation we studied before does not hold, and it is not necessarily true that the value function of any linear controllers will be a quadratic function.

We present another parameterization of linear controllers which admit convexity in the objective function with respect to parameterization (recall that the objective function is non-convex with respect to linear controllers $-K_t$). With the new parameterization, we will see that due to the convexity, we can directly apply gradient descent to find the globally optimal solution.

Consider any fixed time-dependent linear controllers $\{-K_t\}_{t=0}^{H-1}$. We start by rolling out the linear system under the controllers. Note that during the rollout, once we observe x_{t+1} , we can compute w_t as $w_t = x_{t+1} - Ax_t - Bu_t$. With

$x_0 \sim \mu$, we have that at time step t :

$$\begin{aligned}
u_t &= -K_t x_t = -K_t (Ax_{t-1} - BK_{t-1}x_{t-1} + w_{t-1}) \\
&= -K_t w_{t-1} - K_t (A - BK_{t-1})x_{t-1} \\
&= -K_t w_{t-1} - K_t (A - BK_{t-1})(Ax_{t-2} - BK_{t-2}x_{t-2} + w_{t-2}) \\
&= \underbrace{-K_t}_{:=M_{t-1;t}} w_{t-1} - \underbrace{K_t (A - BK_{t-1})}_{:=M_{t-2;t}} w_{t-2} - \underbrace{K_t (A - BK_{t-1})(A - BK_{t-2})}_{:=M_{t-3;t}} x_{t-2} \\
&= -K_t \left(\prod_{\tau=1}^t (A - BK_{t-\tau}) \right) x_0 - \sum_{\tau=0}^{t-1} K_t \prod_{h=1}^{t-1-\tau} (A - BK_{t-h}) w_\tau
\end{aligned}$$

Note that we can equivalently write u_t using x_0 and the noises $w_0, \dots, w_{t-1}$. Hence for time step t , let us do the following re-parameterization. Denote

$$M_t := -K_t \left(\prod_{\tau=1}^t (A - BK_{t-\tau}) \right), \quad M_{\tau;t} := -K_t \prod_{h=1}^{t-1-\tau} (A - BK_{t-h}), \text{ where } \tau = [0, \dots, t-1].$$

Now we can express the control u_t using $M_{\tau;t}$ for $\tau \in [0, \dots, t-1]$ and M_t as follows:

$$u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau,$$

which is equal to $u_t = -K_t x_t$. Note that above we only reasoned time step t . We can repeat the same calculation for all $t = 0 \rightarrow H-1$.

The above calculation basically proves the following claim.

Claim 14.10. *For any linear controllers $\pi := \{-K_0, \dots, -K_{H-1}\}$, there exists a parameterization, $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$, such that when execute π and $\tilde{\pi}$ under any initialization x_0 , and any sequence of noises $\{w_t\}_{t=0}^{H-1}$, they generate exactly the same state-control trajectory.*

We can execute $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$ in the following way. Given any x_0 , we execute $u_0 = M_0 x_0$ and observe x_1 ; at time step t with the observed x_t , we calculate $w_{t-1} = x_t - Ax_{t-1} - Bu_{t-1}$, and execute the control $u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau$. We repeat until we execute the last control u_{H-1} and reach x_H .

What is the benefit of the above parameterization? Note that for any t , this is clearly over-parameterized: the simple linear controllers $-K_t$ has $d \times k$ many parameters, while the new controller $\{M_t; \{M_{\tau;t}\}_{\tau=0}^{t-1}\}$ has $t \times d \times k$ many parameters. The benefit of the above parameterization is that the objective function now is convex! The following claim formally shows the convexity.

Claim 14.11. *Given $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$ and denote its expected total cost as $J(\tilde{\pi}) := \mathbb{E} \left[\sum_{t=0}^{H-1} c(x_t, u_t) \right]$, where the expectation is with respect to the noise $w_t \sim \mathcal{N}(0, \sigma^2 I)$ and the initial state $x_0 \sim \mu$, and c is any convex function with respect to x, u . We have that $J(\tilde{\pi})$ is convex with respect to parameters $\{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$.*

Proof. We consider a fixed x_0 and a fixed sequence of noises $\{w_t\}_{t=0}^{H-1}$. Recall that for u_t , we can write it as:

$$u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau,$$

which is clearly linear with respect to the parameterization $\{M_t; M_{0;t}, \dots, M_{t-1;t}\}$.

For x_t , we can show by induction that it is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{t-1}$. Note that it is clearly true for x_0 which is independent of any parameterizations. Assume this claim holds for x_t with $t \geq 0$, we now check x_{t+1} . We have:

$$x_{t+1} = Ax_t + Bu_t = Ax_t + BM_t x_0 + \sum_{\tau=0}^{t-1} BM_{\tau;t} w_\tau$$

Note that by inductive hypothesis x_t is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{t-1}$, and the part $BM_t x_0 + \sum_{\tau=0}^{t-1} BM_{\tau;t} w_\tau$ is clearly linear with respect to $\{M_t; M_{0;t}, \dots, M_{t-1;t}\}$. Together, this implies that x_{t+1} is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^t$, which concludes that for any x_t with $t = 0, \dots, H-1$, we have x_t is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$.

Note that the trajectory total cost is $\sum_{t=0}^{H-1} c(x_t, u_t)$. Since c_t is convex with respect to x_t and u_t , and x_t and u_t are linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$, we have that $\sum_{t=0}^{H-1} c_t(x_t, u_t)$ is convex with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$.

In last step we can simply take expectation with respect to x_0 and $w_1, \dots, w_{H-1}$. By linearity of expectation, we conclude the proof. $\square$

The above immediately suggests that (sub) gradient-based algorithms such as projected gradient descent on parameters $\{M_t, M_{0;t}, \dots, M_{t-1;t}\}_{t=0}^{H-1}$ can converge to the globally optimal solutions for any convex function c_t . Recall that the best linear controllers $\{-K_t^*\}_{t=0}^{H-1}$ has its own corresponding parameterization $\{M_t^*, M_{0;t}^*, \dots, M_{t-1;t}^*\}_{t=0}^{H-1}$. Thus gradient-based optimization (with some care of the boundness of the parameters $\{M_t^*, M_{0;t}^*, \dots, M_{t-1;t}^*\}_{t=0}^{H-1}$) can find a solution that at least as good as the best linear controllers $\{-K_t^*\}_{t=0}^{H-1}$.

Remark The above claims easily extend to time-dependent transition and cost function, i.e., A_t, B_t, c_t are time-dependent. One can also extend it to episodic online control setting with adversarial noises w_t and adversarial cost functions using no-regret online convex programming algorithms Shalev-Shwartz [2011]. In episodic online control, every episode k , an adversary determines a sequence of bounded noises $w_0^k, \dots, w_{H-1}^k$, and cost function $c^k(x, u)$; the learner proposes a sequence of controllers $\tilde{\pi}^k = \{M_t^k, M_{0;t}^k, \dots, M_{t-1;t}^k\}_{t=0}^{H-1}$ and executes them (learner does not know the cost function c^k until the end of the episode, and the noises are revealed in a way when she observes x_t^k and calculates w_{t-1}^k as $x_t^k - Ax_{t-1}^k - Bu_{t-1}^k$); at the end of the episode, learner observes c^k and suffers total cost $\sum_{t=0}^{H-1} c^k(x_t^k, u_t^k)$. The goal of the episodic online control is to be no-regret with respect to the best linear controllers in hindsight:

$$\sum_{k=0}^{K-1} \sum_{t=0}^{H-1} c^k(x_t^k, u_t^k) - \min_{\{-K_t^*\}_{t=0}^{H-1}} \sum_{k=0}^{K-1} J^k(\{-K_t^*\}_{t=0}^{H-1}) = o(K),$$

where we denote $J^k(\{-K_t^*\}_{t=0}^{H-1})$ as the total cost of executing $\{-K_t^*\}_{t=0}^{H-1}$ in episodic k under c^k and noises $\{w_t^k\}$, i.e., $\sum_{t=0}^{H-1} c(x_t, u_t)$ with $u_t = -K_t^* x_t$ and $x_{t+1} = Ax_t + Bu_t + w_t^k$, for all t .

14.6 Bibliographic Remarks and Further Readings

The basics of optimal control theory can be found in any number of standards text Anderson and Moore [1990], Evans [2005], Bertsekas [2017]. The primal and dual formulations for the continuous time LQR are derived in Balakrishnan and Vandenberghe [2003], while the dual formulation for the discrete time LQR, that we use here, is derived in Cohen et al. [2019].

The treatment of Gauss-Newton, NPG, and PG algorithm are due to Fazel et al. [2018]. We have only provided the proof for the Gauss-Newton case based; the proofs of convergence rates for NPG and PG can be found in Fazel et al. [2018].

For many applications, the finite horizon LQR model is widely used as a model of locally linear dynamics, e.g. Ahn et al. [2007], Todorov and Li [2005], Tedrake [2009], Perez et al. [2012]. The issue of instabilities are largely due to model misspecification and in the accuracy of the Taylor’s expansion; it is less evident how the infinite horizon LQR model captures these issues. In contrast, for MDPs, practice tends to deal with stationary (and discounted) MDPs, due that stationary policies are more convenient to represent and learn; here, the non-stationary, finite horizon MDP model tends to be more of theoretical interest, due to that it is straightforward (and often more effective) to simply incorporate temporal information into the state. Roughly, if our policy is parametric, then practical representational constraints leads us to use stationary policies (incorporating temporal information into the state), while if we tend to use non-parametric policies (e.g. through some rollout based procedure, say with “on the fly” computations like in model predictive control, e.g. Williams et al. [2017]), then it is often more effective to work with finite horizon, non-stationary models.

Sample Complexity and Regret for LQRs. We have not treated the sample complexity of learning in an LQR (see Dean et al. [2017], Simchowitz et al. [2018], Mania et al. [2019] for rates). Here, the basic analysis follows from the online regression approach, which was developed in the study of linear bandits Dani et al. [2008], Abbasi-Yadkori et al. [2011]; in particular, the self-normalized bound for vector-valued martingales [Abbasi-Yadkori et al., 2011] (see Theorem A.9) provides a direct means to obtain sharp confidence intervals for estimating the system matrices A and B from data from a single trajectory (e.g. see Simchowitz and Foster [2020]).

Another family of work provides regret analyses of online LQR problems [Abbasi-Yadkori and Szepesvári, 2011, Dean et al., 2018, Mania et al., 2019, Cohen et al., 2019, Simchowitz and Foster, 2020]. Here, naive random search suffices for sample efficient learning of LQRs [Simchowitz and Foster, 2020]. For the learning and control of more complex nonlinear dynamical systems, one would expect this is insufficient, where strategic exploration is required for sample efficient learning; just as in the case for MDPs (e.g. the UCB-VI algorithm).

Convex Parameterization of Linear Controllers The convex parameterization in Section 14.5 is based on Agarwal et al. [2019] which is equivalent to the System Level Synthesis (SLS) parameterization Wang et al. [2019]. Agarwal et al. [2019] uses SLS parameterization in the infinite horizon online control setting and leverages a reduction to online learning with memory. Note that in episodic online control setting, we can just use classic no-regret online learner such as projected gradient descent Zinkevich [2003]. For partial observable linear systems, the classic Youla parameterization Youla et al. [1976] introduces a convex parameterization. We also refer readers to Simchowitz et al. [2020] for more detailed discussion about Youla parameterization and the generalization of Youla parameterization. Moreover, using the performance difference lemma, the exact optimal control policy for adversarial noise with full observations can be exactly characterized Foster and Simchowitz [2020], Goel and Hassibi [2020] and yields a form reminiscent of the SLS parametrization Wang et al. [2019].

Part V

Appendix

Appendix A

Concentration

This appendix collects a few standard concentration inequalities used throughout the monograph.

A.1 Scalar concentration

Lemma A.1 (Hoeffding's inequality). *Suppose $X_1, \dots, X_n$ are independent, identically distributed random variables with mean μ . Let $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$. If $X_i \in [b_-, b_+]$ almost surely, then for any $\epsilon > 0$,*

$$\Pr(\bar{X}_n - \mu \geq \epsilon) \leq \exp\left(-\frac{2n\epsilon^2}{(b_+ - b_-)^2}\right), \quad \Pr(\mu - \bar{X}_n \geq \epsilon) \leq \exp\left(-\frac{2n\epsilon^2}{(b_+ - b_-)^2}\right).$$

Equivalently,

$$\Pr(|\bar{X}_n - \mu| \geq \epsilon) \leq 2 \exp\left(-\frac{2n\epsilon^2}{(b_+ - b_-)^2}\right).$$

In particular, inverting the two-sided bound yields: with probability at least $1 - \delta$,

$$|\bar{X}_n - \mu| \leq (b_+ - b_-) \sqrt{\frac{\log(2/\delta)}{2n}}.$$

A.2 Sub-Gaussian random variables

Definition A.2 (Sub-Gaussian random variable). A real-valued random variable X is σ -sub-Gaussian if for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp(\lambda(X - \mathbb{E}X))] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right).$$

A centered Gaussian $X \sim \mathcal{N}(0, \sigma^2)$ is σ -sub-Gaussian.

Theorem A.3 (Sub-Gaussian tail bound). *If X is σ -sub-Gaussian, then for all $\epsilon > 0$,*

$$\Pr(X - \mathbb{E}X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right), \quad \Pr(|X - \mathbb{E}X| \geq \epsilon) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Lemma A.4 (Stability under scaling and sums). *If X is σ -sub-Gaussian then for any $c \in \mathbb{R}$, cX is $|c|\sigma$ -sub-Gaussian. If X_1 and X_2 are independent and σ_1 - and σ_2 -sub-Gaussian respectively, then $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ -sub-Gaussian.*

A.3 Martingale concentration

Theorem A.5 (Hoeffding–Azuma inequality (conditional sub-Gaussian form)). *Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration and $(X_t)_{t=1}^N$ a martingale difference sequence, i.e. X_t is $\mathcal{F}_t$ -measurable and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0$. Assume X_t is conditionally σ_t -sub-Gaussian given $\mathcal{F}_{t-1}$, meaning that for all $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[\exp(\lambda X_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma_t^2}{2}\right) \quad \text{a.s.}$$

Then for all $\epsilon > 0$,

$$\Pr\left(\sum_{t=1}^N X_t \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{2 \sum_{t=1}^N \sigma_t^2}\right), \quad \Pr\left(\left|\sum_{t=1}^N X_t\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2 \sum_{t=1}^N \sigma_t^2}\right).$$

Lemma A.6 (Bernstein's inequality (i.i.d.)). *Let $X_1, \dots, X_n$ be independent random variables with $\mu := \mathbb{E}[X_i]$ and $\text{Var}(X_i) \leq \sigma^2$ for all i . Assume $|X_i - \mu| \leq b$ almost surely. Then for all $\epsilon > 0$,*

$$\Pr(\bar{X}_n - \mu \geq \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2 + \frac{2}{3}b\epsilon}\right), \quad \Pr(|\bar{X}_n - \mu| \geq \epsilon) \leq 2 \exp\left(-\frac{n\epsilon^2}{2\sigma^2 + \frac{2}{3}b\epsilon}\right).$$

In particular, with probability at least $1 - \delta$,

$$|\bar{X}_n - \mu| \leq \sqrt{\frac{2\sigma^2 \log(2/\delta)}{n}} + \frac{2b \log(2/\delta)}{3n}.$$

Lemma A.7 (Freedman's inequality (martingale Bernstein)). *Let $(X_t)_{t=1}^n$ be a martingale difference sequence w.r.t. a filtration $(\mathcal{F}_t)_{t \geq 0}$ and assume $|X_t| \leq M$ almost surely. Define the predictable quadratic variation*

$$V_n := \sum_{t=1}^n \mathbb{E}[X_t^2 \mid \mathcal{F}_{t-1}].$$

Then for all $t > 0$,

$$\Pr\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2(V_n + Mt/3)}\right).$$

A.4 A discrete distribution concentration bound

The following bound is a standard application of bounded-differences/McDiarmid (e.g. [McDiarmid, 1989]).

Proposition A.8 (Concentration for discrete distributions). *Let Z be a discrete random variable supported on $\{1, \dots, d\}$ with distribution $q \in \Delta([d])$, and let $\hat{q}$ be the empirical distribution from N i.i.d. samples. Then with probability at least $1 - \delta$,*

$$\|\hat{q} - q\|_2 \leq \frac{1}{\sqrt{N}} + \sqrt{\frac{\log(1/\delta)}{2N}}.$$

Consequently,

$$\|\hat{q} - q\|_1 \leq \sqrt{d} \left(\frac{1}{\sqrt{N}} + \sqrt{\frac{\log(1/\delta)}{2N}} \right).$$

A.5 Self-normalized bounds and least squares

Lemma A.9 (Self-normalized bound for vector-valued martingales; [Abbasi-Yadkori et al., 2011]). *Let $(\varepsilon_t)_{t \geq 1}$ be a real-valued process adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ such that $\mathbb{E}[\varepsilon_t | \mathcal{F}_{t-1}] = 0$ and ε_t is conditionally σ -sub-Gaussian given $\mathcal{F}_{t-1}$. Let $(X_t)_{t \geq 1}$ be an $\mathbb{R}^d$ -valued adapted process, and let $V_0 \in \mathbb{R}^{d \times d}$ be positive definite. Define*

$$V_t := V_0 + \sum_{i=1}^t X_i X_i^\top.$$

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $t \geq 1$,

$$\left\| \sum_{i=1}^t X_i \varepsilon_i \right\|_{V_t^{-1}} \leq \sigma \sqrt{2 \log \left(\frac{\det(V_t)^{1/2} \det(V_0)^{-1/2}}{\delta} \right)}.$$

Equivalently, squaring both sides,

$$\left\| \sum_{i=1}^t X_i \varepsilon_i \right\|_{V_t^{-1}}^2 \leq 2\sigma^2 \log \left(\frac{\det(V_t)^{1/2} \det(V_0)^{-1/2}}{\delta} \right).$$

The following OLS bound above can be derived by writing $\hat{\theta} - \theta^* = \Lambda^{-1} \sum_i x_i \varepsilon_i$ and observing that $\|\hat{\theta} - \theta^*\|_\Lambda = \|\Lambda^{-1/2} \sum_i x_i \varepsilon_i\|_2$ is the Euclidean norm of a d -dimensional sub-Gaussian vector (e.g. see also [Hsu et al., 2012]).

Theorem A.10 (OLS with a fixed design). *Consider a fixed design $\{x_i\}_{i=1}^N \subset \mathbb{R}^d$ and responses $y_i = \langle \theta^*, x_i \rangle + \varepsilon_i$, where $\varepsilon_1, \dots, \varepsilon_N$ are independent, mean-zero, σ -sub-Gaussian random variables. Let*

$$\Lambda := \frac{1}{N} \sum_{i=1}^N x_i x_i^\top$$

and assume Λ is invertible. The ordinary least squares estimator is

$$\hat{\theta} := \Lambda^{-1} \left(\frac{1}{N} \sum_{i=1}^N x_i y_i \right).$$

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|\hat{\theta} - \theta^*\|_\Lambda \leq \sigma \cdot \frac{\sqrt{d} + \sqrt{2 \log(1/\delta)}}{\sqrt{N}},$$

and hence

$$\|\hat{\theta} - \theta^*\|_\Lambda^2 \leq \sigma^2 \cdot \frac{d + 2\sqrt{2d \log(1/\delta)} + 2 \log(1/\delta)}{N}.$$

The following is a standard “fast-rate” bound for squared-loss ERM over a finite function class (e.g. Györfi et al. [2002]), which we include for completeness.

Lemma A.11 (Least-squares generalization bound (finite class)). *Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. with $X_i \sim \nu$ on $\mathcal{X}$ and $Y_i \in [0, B]$ almost surely for some $B > 0$. Let*

$$f^*(x) := \mathbb{E}[Y | X = x]$$

and assume $f^*(x) \in [0, B]$ for all x . Let $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow [0, B]\}$ be finite and define

$$\epsilon_{\text{approx}} := \min_{f \in \mathcal{F}} \|f - f^*\|_{2,\nu}^2.$$

Let $\hat{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n (f(X_i) - Y_i)^2$ be a squared-loss ERM. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\|\hat{f} - f^*\|_{2,\nu}^2 \leq \frac{20 B^2 \ln(2|\mathcal{F}|/\delta)}{n} + 3 \epsilon_{\text{approx}}.$$

Proof. For $f \in \mathcal{F}$ define

$$z_i^f := (f(X_i) - Y_i)^2 - (f^*(X_i) - Y_i)^2.$$

A direct expansion gives

$$z_i^f = (f(X_i) - f^*(X_i))(f(X_i) + f^*(X_i) - 2Y_i).$$

Since $f, f^*, Y_i \in [0, B]$, we have $|z_i^f| \leq B^2$ almost surely. Moreover, using $\mathbb{E}[Y_i | X_i] = f^*(X_i)$,

$$\mathbb{E}[z_i^f | X_i] = (f(X_i) - f^*(X_i))^2, \quad \text{so} \quad \mathbb{E}[z_i^f] = \|f - f^*\|_{2,\nu}^2 =: u_f.$$

Finally,

$$(z_i^f)^2 \leq (f(X_i) - f^*(X_i))^2 \cdot (2B)^2 \Rightarrow \operatorname{Var}(z_i^f) \leq \mathbb{E}[(z_i^f)^2] \leq 4B^2 u_f.$$

Let $L := \ln(2|\mathcal{F}|/\delta)$. Applying Bernstein's inequality to z_i^f and to $-z_i^f$, and taking a union bound over $f \in \mathcal{F}$, yields that with probability at least $1 - \delta$, simultaneously for all $f \in \mathcal{F}$,

$$\left| \frac{1}{n} \sum_{i=1}^n z_i^f - u_f \right| \leq \sqrt{\frac{8B^2 u_f L}{n}} + \frac{2B^2 L}{3n}. \quad (\text{A.1})$$

Let $\tilde{f} \in \operatorname{argmin}_{f \in \mathcal{F}} \|f - f^*\|_{2,\nu}^2$, so $u_{\tilde{f}} = \epsilon_{\text{approx}}$. Since $\hat{f}$ minimizes the empirical squared loss,

$$\frac{1}{n} \sum_{i=1}^n z_i^{\hat{f}} \leq \frac{1}{n} \sum_{i=1}^n z_i^{\tilde{f}}.$$

Using (A.1) twice (once for $\hat{f}$, once for $\tilde{f}$), we get

$$u_{\hat{f}} \leq u_{\tilde{f}} + \sqrt{\frac{8B^2 u_{\tilde{f}} L}{n}} + \frac{4B^2 L}{3n} + \sqrt{\frac{8B^2 u_{\hat{f}} L}{n}}.$$

Let $\alpha := \frac{8B^2 L}{n}$ and $\beta := \frac{2B^2 L}{3n}$, and write $u := u_{\hat{f}}$ and $\epsilon := u_{\tilde{f}} = \epsilon_{\text{approx}}$. Then

$$u \leq \epsilon + \sqrt{\alpha \epsilon} + 2\beta + \sqrt{\alpha u}.$$

Using $\sqrt{\alpha u} \leq \frac{u}{2} + \frac{\alpha}{2}$ gives

$$u \leq 2\epsilon + 2\sqrt{\alpha \epsilon} + 4\beta + \alpha.$$

Using $2\sqrt{\alpha \epsilon} \leq \epsilon + \alpha$ yields

$$u \leq 3\epsilon + 2\alpha + 4\beta.$$

Finally,

$$2\alpha + 4\beta = \left(16 + \frac{8}{3}\right) \frac{B^2 L}{n} = \frac{56}{3} \frac{B^2 L}{n} \leq 20 \frac{B^2 L}{n},$$

which completes the proof. $\square$

A.6 Other Bounds

A.6.1 Covering numbers and uniformization via ϵ -nets

This section collects a few ϵ -net bounds used in Chapter 7. We work with the sup norm $\|f\|_\infty := \sup_{s \in \mathcal{S}} |f(s)|$ for functions $f : \mathcal{S} \rightarrow \mathbb{R}$, the Euclidean norm $\|\cdot\|_2$ for vectors, and the Frobenius norm $\|\cdot\|_F$ for matrices.

Definition A.12 (ϵ -net and covering number). Let $(\mathcal{X}, d)$ be a metric space and let $\mathcal{F} \subseteq \mathcal{X}$. A set $\mathcal{N} \subseteq \mathcal{F}$ is an ϵ -net of $\mathcal{F}$ if for every $f \in \mathcal{F}$ there exists $\tilde{f} \in \mathcal{N}$ such that $d(f, \tilde{f}) \leq \epsilon$. The (internal) covering number is

$$\mathcal{N}(\mathcal{F}, \epsilon, d) := \min \left\{ |\mathcal{N}| : \mathcal{N} \subseteq \mathcal{F} \text{ is an } \epsilon\text{-net of } \mathcal{F} \right\}.$$

Lemma A.13 (Volumetric bound for Euclidean balls). Let $\mathcal{B}_2^m(R) := \{x \in \mathbb{R}^m : \|x\|_2 \leq R\}$. For any $\epsilon \in (0, 2R]$,

$$\mathcal{N}(\mathcal{B}_2^m(R), \epsilon, \|\cdot\|_2) \leq \left(1 + \frac{2R}{\epsilon}\right)^m.$$

Proof. Let $\mathcal{S} \subseteq \mathcal{B}_2^m(R)$ be a maximal ϵ -separated set: $\|x - y\|_2 > \epsilon$ for all distinct $x, y \in \mathcal{S}$. Maximality implies $\mathcal{S}$ is an ϵ -net.

The Euclidean balls $\{x + (\epsilon/2)\mathcal{B}_2^m(1) : x \in \mathcal{S}\}$ are disjoint and lie inside $(R + \epsilon/2)\mathcal{B}_2^m(1)$. Comparing volumes gives $|\mathcal{S}| \leq \left(\frac{R + \epsilon/2}{\epsilon/2}\right)^m = (1 + 2R/\epsilon)^m$. $\square$

Lemma A.14 (Max and clipping are $\|\cdot\|_\infty$ contractions). Let $\{g_a\}_{a \in \mathcal{A}}$ and $\{\tilde{g}_a\}_{a \in \mathcal{A}}$ be real-valued functions on $\mathcal{S}$ and define $G(s) := \max_a g_a(s)$ and $\tilde{G}(s) := \max_a \tilde{g}_a(s)$. Then

$$\|G - \tilde{G}\|_\infty \leq \sup_{a \in \mathcal{A}} \|g_a - \tilde{g}_a\|_\infty.$$

Moreover, for any $M > 0$, the clipping map $\text{clip}_{[0, M]}(x) := \min\{M, \max\{0, x\}\}$ satisfies

$$\|\text{clip}_{[0, M]} \circ u - \text{clip}_{[0, M]} \circ \tilde{u}\|_\infty \leq \|u - \tilde{u}\|_\infty.$$

Proof. For any s ,

$$G(s) - \tilde{G}(s) = \max_a g_a(s) - \max_a \tilde{g}_a(s) \leq \max_a (g_a(s) - \tilde{g}_a(s)) \leq \sup_a \|g_a - \tilde{g}_a\|_\infty,$$

and swapping the roles of g and $\tilde{g}$ yields the two-sided bound. The clipping bound holds because $\text{clip}_{[0, M]}$ is 1-Lipschitz on $\mathbb{R}$. $\square$

Lemma A.15 (Covering number for an optimistic value-function class). Assume $\sup_{s, a} \|\phi(s, a)\|_2 \leq 1$. Fix parameters $L > 0$, $B > 0$, $\lambda > 0$, and $H \geq 1$. For (w, β, Λ) with $\|w\|_2 \leq L$, $\beta \in [0, B]$, and $\Lambda \succeq \lambda I$, define $f_{w, \beta, \Lambda} : \mathcal{S} \rightarrow [0, H]$ by

$$f_{w, \beta, \Lambda}(s) := \text{clip}_{[0, H]} \left(\max_{a \in \mathcal{A}} \left(w^\top \phi(s, a) + \beta \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \right) \right).$$

Let

$$\mathcal{F} := \{f_{w, \beta, \Lambda} : \|w\|_2 \leq L, \beta \in [0, B], \Lambda \succeq \lambda I\}.$$

Then for any $\epsilon \in (0, 1]$,

$$\log \mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_\infty) \leq d \log \left(1 + \frac{6L}{\epsilon}\right) + \log \left(1 + \frac{6B}{\sqrt{\lambda} \epsilon}\right) + \frac{d(d+1)}{2} \log \left(1 + \frac{18B^2 \sqrt{d}}{\lambda \epsilon^2}\right).$$

Remark (adding a known offset). *If one replaces $w^\top \phi(s, a)$ inside the max by $u(s, a) + w^\top \phi(s, a)$ for any fixed bounded function $u : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, the same covering bound holds: u cancels in pairwise differences, and Lemma A.14 still applies.*

Proof. Fix two triples (w, β, Λ) and $(\tilde{w}, \tilde{\beta}, \tilde{\Lambda})$. For (s, a) define

$$g(s, a) := w^\top \phi(s, a) + \beta \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)}, \quad \tilde{g}(s, a) := \tilde{w}^\top \phi(s, a) + \tilde{\beta} \sqrt{\phi(s, a)^\top \tilde{\Lambda}^{-1} \phi(s, a)}.$$

By Lemma A.14,

$$\|f_{w, \beta, \Lambda} - f_{\tilde{w}, \tilde{\beta}, \tilde{\Lambda}}\|_\infty \leq \sup_{s, a} |g(s, a) - \tilde{g}(s, a)|.$$

For fixed (s, a) ,

$$\begin{aligned} |g(s, a) - \tilde{g}(s, a)| &\leq |(w - \tilde{w})^\top \phi(s, a)| + |\beta - \tilde{\beta}| \cdot \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \\ &\quad + \tilde{\beta} \cdot \left| \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} - \sqrt{\phi(s, a)^\top \tilde{\Lambda}^{-1} \phi(s, a)} \right|. \end{aligned}$$

The first term is at most $\|w - \tilde{w}\|_2$ since $\|\phi(s, a)\|_2 \leq 1$. For the second term, $\Lambda \succeq \lambda I$ implies $\phi^\top \Lambda^{-1} \phi \leq 1/\lambda$, so it is at most $|\beta - \tilde{\beta}|/\sqrt{\lambda}$. For the third term, use $|\sqrt{u} - \sqrt{v}| \leq \sqrt{|u - v|}$ for $u, v \geq 0$ and

$$\left| \phi^\top (\Lambda^{-1} - \tilde{\Lambda}^{-1}) \phi \right| \leq \|\Lambda^{-1} - \tilde{\Lambda}^{-1}\|_2 \|\phi\|_2^2 \leq \|\Lambda^{-1} - \tilde{\Lambda}^{-1}\|_F.$$

Thus

$$\sup_{s, a} |g(s, a) - \tilde{g}(s, a)| \leq \|w - \tilde{w}\|_2 + \frac{|\beta - \tilde{\beta}|}{\sqrt{\lambda}} + B \sqrt{\|\Lambda^{-1} - \tilde{\Lambda}^{-1}\|_F}.$$

It suffices to build: (i) an $(\epsilon/3)$ -net for $\{w : \|w\|_2 \leq L\}$ in $\|\cdot\|_2$; (ii) a $(\sqrt{\lambda}\epsilon/3)$ -net for $[0, B]$ in absolute value; and (iii) a $(\epsilon/(3B))^2$ -net for $\{\Lambda^{-1} : \Lambda \succeq \lambda I\}$ in $\|\cdot\|_F$.

For (i), Lemma A.13 with $m = d$ gives size at most $(1 + 6L/\epsilon)^d$. For (ii), an interval cover gives size at most $1 + 6B/(\sqrt{\lambda}\epsilon)$. For (iii), note that Λ^{-1} is symmetric and $\Lambda \succeq \lambda I$ implies $\|\Lambda^{-1}\|_F \leq \sqrt{d}/\lambda$. The set lies in a Frobenius ball of radius $\sqrt{d}/\lambda$ inside the $\frac{d(d+1)}{2}$ -dimensional vector space of symmetric matrices. Applying Lemma A.13 with $m = \frac{d(d+1)}{2}$ and $\epsilon_\Lambda = (\epsilon/(3B))^2$ yields a net of size at most

$$\left(1 + \frac{2(\sqrt{d}/\lambda)}{(\epsilon/(3B))^2} \right)^{\frac{d(d+1)}{2}} = \left(1 + \frac{18B^2\sqrt{d}}{\lambda\epsilon^2} \right)^{\frac{d(d+1)}{2}}.$$

Taking the product net gives an ϵ -net of $\mathcal{F}$. □

Lemma A.16 (Uniform self-normalized bound via an ϵ -net). *Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration and let $(x_t)_{t \geq 0}$ be an adapted sequence in $\mathbb{R}^d$ with $\|x_t\|_2 \leq 1$. Fix $\lambda > 0$ and define*

$$\Lambda_n := \lambda I + \sum_{t=0}^{n-1} x_t x_t^\top.$$

Let $\mathcal{F}$ be a class of functions $f : \mathcal{S} \rightarrow [0, H]$, and for each $f \in \mathcal{F}$ let $(\varepsilon_t(f))_{t \geq 0}$ be a sequence such that $\varepsilon_t(f)$ is $\mathcal{F}_{t+1}$ -measurable, $\mathbb{E}[\varepsilon_t(f) \mid \mathcal{F}_t] = 0$, and $|\varepsilon_t(f)| \leq H$ almost surely. Assume moreover that for all $f, g \in \mathcal{F}$,

$$|\varepsilon_t(f) - \varepsilon_t(g)| \leq 2\|f - g\|_\infty \quad \text{almost surely for all } t. \quad (\text{A.2})$$

Fix $\epsilon \in (0, 1]$ and let $\mathcal{N}_\epsilon \subseteq \mathcal{F}$ be an ϵ -net in $\|\cdot\|_\infty$. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $n \geq 1$ and all $f \in \mathcal{F}$,

$$\left\| \sum_{t=0}^{n-1} x_t \varepsilon_t(f) \right\|_{\Lambda_n^{-1}} \leq H \sqrt{2 \log \left(\frac{\det(\Lambda_n)^{1/2} \det(\lambda I)^{-1/2} |\mathcal{N}_\epsilon|}{\delta} \right)} + 2\epsilon\sqrt{n}.$$

Proof. Apply Lemma A.9 (Appendix A) to each fixed $\tilde{f} \in \mathcal{N}_\epsilon$ with $V_0 = \lambda I$ and failure probability $\delta/|\mathcal{N}_\epsilon|$. Since $|\varepsilon_t(\tilde{f})| \leq H$, each sequence is conditionally H -sub-Gaussian. A union bound implies that with probability at least $1 - \delta$, for all $n \geq 1$ and all $\tilde{f} \in \mathcal{N}_\epsilon$,

$$\left\| \sum_{t=0}^{n-1} x_t \varepsilon_t(\tilde{f}) \right\|_{\Lambda_n^{-1}} \leq H \sqrt{2 \log \left(\frac{\det(\Lambda_n)^{1/2} \det(\lambda I)^{-1/2} |\mathcal{N}_\epsilon|}{\delta} \right)}.$$

Fix any $f \in \mathcal{F}$ and choose $\tilde{f} \in \mathcal{N}_\epsilon$ with $\|f - \tilde{f}\|_\infty \leq \epsilon$. Write

$$\sum_{t=0}^{n-1} x_t \varepsilon_t(f) = \sum_{t=0}^{n-1} x_t \varepsilon_t(\tilde{f}) + \sum_{t=0}^{n-1} x_t (\varepsilon_t(f) - \varepsilon_t(\tilde{f})).$$

For the second term, define $u_t := \varepsilon_t(f) - \varepsilon_t(\tilde{f})$. By (A.2), $|u_t| \leq 2\epsilon$, hence $\|u\|_2 \leq 2\epsilon\sqrt{n}$ for $u = (u_0, \dots, u_{n-1}) \in \mathbb{R}^n$. Let $X = [x_0, \dots, x_{n-1}] \in \mathbb{R}^{d \times n}$ so that $\Lambda_n = \lambda I + X X^\top$ and $\sum_{t=0}^{n-1} x_t u_t = X u$. Then

$$\|X u\|_{\Lambda_n^{-1}}^2 = u^\top X^\top (\lambda I + X X^\top)^{-1} X u.$$

By the Woodbury identity, $X^\top (\lambda I + X X^\top)^{-1} X = M(I + M)^{-1}$ with $M := \lambda^{-1} X^\top X \succeq 0$, so $M(I + M)^{-1} \preceq I$. Therefore $\|X u\|_{\Lambda_n^{-1}}^2 \leq \|u\|_2^2$ and $\|X u\|_{\Lambda_n^{-1}} \leq 2\epsilon\sqrt{n}$. Combine with the triangle inequality. $\square$

We now package the ϵ -net machinery into a single stage-wise uniform confidence statement tailored to Chapter 7.

Corollary A.17 (Uniform confidence over an optimistic value-function class). *Fix a stage h and an episode budget $K \geq 2$. Let $(\mathcal{F}_i)_{i \geq 0}$ be the filtration generated by the interaction up to (and including) (s_h^i, a_h^i) at stage h of episode i . Define*

$$x_i := \phi(s_h^i, a_h^i) \in \mathbb{R}^d, \quad s_{i+1} := s_{h+1}^i \in \mathcal{S}, \quad \Lambda_k := \lambda I + \sum_{i=0}^{k-1} x_i x_i^\top,$$

with $\lambda \geq 1$ and $\|x_i\|_2 \leq 1$.

Let $\mathcal{F}$ be the optimistic value-function class from Lemma A.15, so $\mathcal{F} \subseteq \{f : \mathcal{S} \rightarrow [0, H]\}$. Assume a linear conditional expectation model at stage h : for every $f \in \mathcal{F}$ there exists $w_f^* \in \mathbb{R}^d$ such that

$$\mathbb{E}[f(s_{i+1}) \mid \mathcal{F}_i] = x_i^\top w_f^* \quad \text{for all } i \geq 0, \quad (\text{A.3})$$

and $\|w_f^*\|_2 \leq H\sqrt{d}$ for all $f \in \mathcal{F}$.

For each $k \geq 1$ and $f \in \mathcal{F}$, define the ridge estimator

$$\hat{w}_f^k := (\Lambda_k)^{-1} \sum_{i=0}^{k-1} x_i f(s_{i+1}), \quad (\hat{P}^k f)(s, a) := \phi(s, a)^\top \hat{w}_f^k, \quad (P^* f)(s, a) := \phi(s, a)^\top w_f^*.$$

Fix $\delta \in (0, 1)$ and set $\epsilon := 1/\sqrt{K}$. Then with probability at least $1 - \delta$, simultaneously for all $k \in \{1, \dots, K\}$, all $f \in \mathcal{F}$, and all $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$|(\widehat{P}^k f)(s, a) - (P^* f)(s, a)| \leq \beta_{K, \delta} \|\phi(s, a)\|_{(\Lambda_k)^{-1}},$$

where one may take

$$\beta_{K, \delta} := H \sqrt{d \log \left(1 + \frac{K}{\lambda}\right) + 2 \log \left(\frac{\mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_\infty)}{\delta}\right)} + 2 + \sqrt{\lambda} H \sqrt{d}.$$

Moreover, by Lemma A.15 and $\epsilon = 1/\sqrt{K}$,

$$\log \mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_\infty) \leq d \log(1 + 6L\sqrt{K}) + \log \left(1 + \frac{6B\sqrt{K}}{\sqrt{\lambda}}\right) + \frac{d(d+1)}{2} \log \left(1 + \frac{18B^2\sqrt{d}K}{\lambda}\right),$$

so for polynomially bounded (L, B) one has $\beta_{K, \delta} = \widetilde{O}(Hd)$ as a function of $(K, 1/\delta)$.

Proof. For $f \in \mathcal{F}$, define the noise sequence

$$\varepsilon_i(f) := f(s_{i+1}) - \mathbb{E}[f(s_{i+1}) \mid \mathcal{F}_i].$$

Then $(\varepsilon_i(f))_{i \geq 0}$ is a martingale difference sequence and $|\varepsilon_i(f)| \leq H$ almost surely. For any $f, g \in \mathcal{F}$,

$$|\varepsilon_i(f) - \varepsilon_i(g)| \leq |f(s_{i+1}) - g(s_{i+1})| + \left| \mathbb{E}[f(s_{i+1}) - g(s_{i+1}) \mid \mathcal{F}_i] \right| \leq 2\|f - g\|_\infty,$$

so (A.2) holds.

Apply Lemma A.16 to the class $\mathcal{F}$ with this choice of $\varepsilon_i(\cdot)$ and with $\epsilon = 1/\sqrt{K}$. On its high-probability event (probability at least $1 - \delta$), for all $k \leq K$ and all $f \in \mathcal{F}$,

$$\left\| \sum_{i=0}^{k-1} x_i \varepsilon_i(f) \right\|_{\Lambda_k^{-1}} \leq H \sqrt{2 \log \left(\frac{\det(\Lambda_k)^{1/2} \det(\lambda I)^{-1/2} \mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_\infty)}{\delta} \right)} + 2\epsilon\sqrt{k}.$$

Since $\Lambda_k \preceq (\lambda + k)I$, we have $\det(\Lambda_k) \det(\lambda I)^{-1} \leq (1 + k/\lambda)^d \leq (1 + K/\lambda)^d$ and $2\epsilon\sqrt{k} \leq 2$. Hence the above is at most

$$H \sqrt{d \log \left(1 + \frac{K}{\lambda}\right) + 2 \log \left(\frac{\mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_\infty)}{\delta}\right)} + 2.$$

Next, using (A.3), write $f(s_{i+1}) = x_i^\top w_f^* + \varepsilon_i(f)$ and compute

$$\widehat{w}_f^k - w_f^* = \Lambda_k^{-1} \sum_{i=0}^{k-1} x_i \varepsilon_i(f) - \lambda \Lambda_k^{-1} w_f^*.$$

Therefore, for any (s, a) ,

$$\begin{aligned} |(\widehat{P}^k f - P^* f)(s, a)| &= |\phi(s, a)^\top (\widehat{w}_f^k - w_f^*)| \\ &\leq \|\phi(s, a)\|_{\Lambda_k^{-1}} \cdot \left\| \sum_{i=0}^{k-1} x_i \varepsilon_i(f) \right\|_{\Lambda_k^{-1}} + \lambda |\phi(s, a)^\top \Lambda_k^{-1} w_f^*|. \end{aligned}$$

Since $\Lambda_k \succeq \lambda I$,

$$\lambda |\phi^\top \Lambda_k^{-1} w_f^*| \leq \sqrt{\lambda} \|\phi\|_{\Lambda_k^{-1}} \|w_f^*\|_2.$$

Combine the bounds and use $\|w_f^*\|_2 \leq H\sqrt{d}$. □

A.6.2 Uniform convergence primitives for Bellman-rank losses

This appendix collects standard high-probability uniform convergence bounds that can be used as $\varepsilon_{\text{gen}}(m, \mathcal{H}, \delta)$ in Assumption 8.5. All statements are for a fixed stage h (the chapter union-bounds over $h \in \{0, \dots, H-1\}$).

Notation. Fix a loss family $\{\tilde{\ell}(s, a, s'; g) : g \in \mathcal{H}\}$ and a distribution ν over (s, a, s') . Given i.i.d. data $\mathcal{D} = \{(s_i, a_i, s'_i)\}_{i=1}^m \sim \nu^m$, write

$$\mathbb{E}_{\mathcal{D}}[\tilde{\ell}(\cdot; g)] := \frac{1}{m} \sum_{i=1}^m \tilde{\ell}(s_i, a_i, s'_i; g), \quad \mathbb{E}_{\nu}[\tilde{\ell}(\cdot; g)] := \mathbb{E}_{(s, a, s') \sim \nu}[\tilde{\ell}(s, a, s'; g)].$$

We will apply the lemmas below with either $\tilde{\ell} = \ell_h(\cdot; g)$ (the Q -version) or $\tilde{\ell} = \tilde{\ell}_h^V(\cdot; g)$ (the importance-weighted V -version from (8.3) in the chapter).

Finite hypothesis classes

Lemma A.18 (Finite class: Hoeffding + union bound). *Assume $|\mathcal{H}| < \infty$ and $\|\tilde{\ell}(\cdot; g)\|_{\infty} \leq B$ for all $g \in \mathcal{H}$. Then with probability at least $1 - \delta$,*

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\tilde{\ell}(\cdot; g)] - \mathbb{E}_{\nu}[\tilde{\ell}(\cdot; g)]| \leq B \sqrt{\frac{2 \ln(2|\mathcal{H}|/\delta)}{m}}.$$

Proof. Fix $g \in \mathcal{H}$ and apply Hoeffding's inequality to the bounded random variables $\tilde{\ell}(s_i, a_i, s'_i; g) \in [-B, B]$. Union bound over $g \in \mathcal{H}$. $\square$

Linear function classes

We record convenient bounds for Bellman-rank losses induced by linear Q -classes. Fix a feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ with $\|\phi(s, a)\|_2 \leq 1$ for all (s, a) . For parameters $W > 0$ and $H \geq 1$, define the stage-wise linear class

$$\mathcal{G}_h(\phi, W) := \left\{ Q_{h,w}(s, a) = \langle w, \phi(s, a) \rangle : \|w\|_2 \leq W, \|Q_{h,w}\|_{\infty} \leq H \right\}.$$

At a fixed stage h , we will view a hypothesis g as specifying a pair of weights (w_h, w_{h+1}) , each lying in the above ball, and use the one-step Bellman residual

$$\ell_h(s, a, s'; g) := \langle w_h, \phi(s, a) \rangle - r_h(s, a) - \max_{a' \in \mathcal{A}} \langle w_{h+1}, \phi(s', a') \rangle.$$

For the V -version with uniform exploration at time h , define the importance-weighted loss

$$\tilde{\ell}_h^V(s, a, s'; g) := \frac{\mathbf{1}\{a = \pi_{h,g}(s)\}}{1/A} \ell_h(s, a, s'; g), \quad \pi_{h,g}(s) \in \arg \max_{a \in \mathcal{A}} \langle w_h, \phi(s, a) \rangle.$$

Lemma A.19 (Linear classes: uniform convergence for ℓ_h and $\tilde{\ell}_h^V$). *Fix $\delta \in (0, 1)$ and let $\mathcal{D}$ be m i.i.d. samples from ν . Assume $r_h(s, a) \in [0, 1]$ and $\|Q_{h,w}\|_{\infty} \leq H$ for all w in the class. Then:*

1. (*Q -loss*) With probability at least $1 - \delta$,

$$\sup_g |\mathbb{E}_{\mathcal{D}}[\ell_h(\cdot; g)] - \mathbb{E}_{\nu}[\ell_h(\cdot; g)]| \leq 3H \sqrt{\frac{2d \ln(1 + 4Wm) + \ln(1/\delta)}{m}}.$$

2. (*V-loss*) With probability at least $1 - \delta$,

$$\sup_g |\mathbb{E}_{\mathcal{D}}[\tilde{\ell}_h^V(\cdot; g)] - \mathbb{E}_{\nu}[\tilde{\ell}_h^V(\cdot; g)]| \leq 3AH \sqrt{\frac{2d \ln(1 + 4AWm) + d \ln(emA) + \ln(1/\delta)}{m}}.$$

The suprema are over all stage-local pairs (w_h, w_{h+1}) with $\|w_h\|_2, \|w_{h+1}\|_2 \leq W$ and $\|Q_{h,w_h}\|_{\infty}, \|Q_{h+1,w_{h+1}}\|_{\infty} \leq H$.

Proof. We use covering-number arguments.

For the Q -loss, note that $|\ell_h| \leq |Q_{h,w_h}| + |r_h| + |V_{h+1,w_{h+1}}| \leq 2H + 1 \leq 3H$. Let $\mathcal{N}_{\epsilon}$ be an ϵ -net of the Euclidean ball $\{w : \|w\|_2 \leq W\}$; a volumetric bound gives $|\mathcal{N}_{\epsilon}| \leq (1 + W/\epsilon)^d$. Consider the finite set $\mathcal{N}_{\epsilon} \times \mathcal{N}_{\epsilon}$ for the pair (w_h, w_{h+1}) . For any fixed net pair $(\bar{w}_h, \bar{w}_{h+1})$, Hoeffding's inequality gives

$$|\mathbb{E}_{\mathcal{D}}[\ell_h(\cdot; (\bar{w}_h, \bar{w}_{h+1}))] - \mathbb{E}_{\nu}[\ell_h(\cdot; (\bar{w}_h, \bar{w}_{h+1}))]| \leq 3H \sqrt{\frac{\ln(2/\delta')}{2m}}$$

with probability at least $1 - \delta'$. Union bounding over net pairs and taking δ' so that the overall failure probability is δ yields a bound with $\ln |\mathcal{N}_{\epsilon} \times \mathcal{N}_{\epsilon}| \leq 2d \ln(1 + W/\epsilon)$. Finally, if (w_h, w_{h+1}) is within ϵ (in $\|\cdot\|_2$) of $(\bar{w}_h, \bar{w}_{h+1})$, then $\sup_{s,a} |\langle w_h - \bar{w}_h, \phi(s, a) \rangle| \leq \epsilon$ and $\sup_{s',a'} |\max_{a'} \langle w_{h+1} - \bar{w}_{h+1}, \phi(s', a') \rangle| \leq \epsilon$, so $|\ell_h(\cdot; g) - \ell_h(\cdot; \bar{g})| \leq 2\epsilon$ pointwise. Taking $\epsilon = 1/(4m)$ and simplifying constants yields the stated bound.

For the V -loss, we again have $|\tilde{\ell}_h^V| \leq A|\ell_h| \leq 3AH$. The indicator $\mathbf{1}\{a = \pi_{h,g}(s)\}$ depends on w_h , so we first control the number of distinct activation patterns it can induce on the sample. Fix the sample $\{(s_i, a_i)\}_{i=1}^m$. For each i , the event that a_i is greedy under w_h is

$$\langle w_h, \phi(s_i, a_i) \rangle \geq \langle w_h, \phi(s_i, b) \rangle \quad \text{for all } b \in \mathcal{A},$$

which is described by at most $A-1$ halfspace constraints in w_h . Across $i = 1, \dots, m$ there are at most $m(A-1) \leq mA$ corresponding hyperplanes, and a standard arrangement bound implies that $\mathbb{R}^d$ is partitioned into at most $(emA)^d$ regions on which all indicators $\mathbf{1}\{a_i = \pi_{h,g}(s_i)\}$ are constant. This contributes the additive term $d \ln(emA)$.

Condition on any fixed activation pattern of the sample. On this event, the indicator is fixed on each sample point, and the remaining dependence of $\tilde{\ell}_h^V$ on (w_h, w_{h+1}) is $O(A)$ -Lipschitz on the sample. We can therefore repeat the same ϵ -net argument as above, now with an effective net radius $\epsilon \simeq 1/(Am)$, and union bound over both the net pairs and the (at most) $(emA)^d$ activation patterns. This yields the stated bound. $\square$

Unions of linear classes (feature selection)

Many representation-selection settings take $\mathcal{H}$ to be a union over candidate feature maps. Let Φ be a finite set of feature maps $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^s$ with $\|\phi(s, a)\|_2 \leq 1$. Define a union hypothesis class $\mathcal{H} = \bigcup_{\phi \in \Phi} \mathcal{H}(\phi)$, where $\mathcal{H}(\phi)$ is the linear class induced by ϕ with $\|w\|_2 \leq W$ and $\|w^\top \phi\|_{\infty} \leq H$.

Lemma A.20 (Union of linear classes: extra $\log |\Phi|$). *Under the assumptions of Lemma A.19 with dimension $d = s$, the same bounds hold with $\ln(1/\delta)$ replaced by $\ln(|\Phi|/\delta)$, i.e. with probability at least $1 - \delta$,*

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\tilde{\ell}(\cdot; g)] - \mathbb{E}_{\nu}[\tilde{\ell}(\cdot; g)]| \leq \varepsilon_{\text{gen}}(m, \mathcal{H}, \delta),$$

where one may take

$$\varepsilon_{\text{gen}}(m, \mathcal{H}, \delta) \lesssim \begin{cases} H \sqrt{\frac{s \ln(1 + Wm) + \ln(|\Phi|/\delta)}{m}} & \text{for the } Q\text{-loss,} \\ AH \sqrt{\frac{s \ln(1 + AWm) + s \ln(emA) + \ln(|\Phi|/\delta)}{m}} & \text{for the } V\text{-loss.} \end{cases}$$

Proof. Apply Lemma A.19 to each fixed $\phi \in \Phi$ with failure probability $\delta/|\Phi|$, then union bound over Φ . $\square$

Specialization to sparse feature selection. If Φ indexes all s -subsets of $[D]$, then $|\Phi| = \binom{D}{s}$ and $\ln |\Phi| \leq s \ln(eD/s)$. Plugging this into Lemma A.20 yields dependence polynomial in s and polylogarithmic in D once substituted into Theorem 8.6.

Appendix B

Linearly realizable Q^* is exponentially hard

This appendix records a (trajectory-tree) hard instance in which Q^* is *exactly* linear in a given feature map, yet identifying a near-optimal policy requires exponentially many queries in $\min\{d, H\}$ (once the action set is taken to be exponentially large in d).

Theorem B.1 (Linear Q^* realizability can be exponentially hard (generative model)). *Fix integers $d \geq 1$ and $H \geq 1$. Consider any algorithm $\mathcal{A}$ that has access to a generative model and is given as input a feature map $\phi : \mathcal{S} \times \mathcal{A} \times [H] \rightarrow \mathbb{R}^d$. There exist absolute constants $c, C > 0$ and a finite-horizon MDP M with horizon H and action set size $|\mathcal{A}| = \exp(cd)$, together with a feature map ϕ , such that Assumption 3.9 holds (i.e. for each $h \in [H]$, $Q_h^*(\cdot, \cdot)$ is linear in $\phi(\cdot, \cdot, h)$), and the following is true.*

If $\mathcal{A}$ outputs a policy π satisfying

$$V_0^\pi(s_0) \geq V_0^*(s_0) - 0.05$$

with probability at least 0.1, then $\mathcal{A}$ must make at least

$$\exp(\Omega(\min\{d, H\}))$$

calls to the generative model.

The construction presented here follows from that in Weisz et al. [2021a].

B.1 Construction

Fix a horizon $H \geq 1$ and a feature dimension d . Let $\rho \in (0, 1/6]$ be a constant and set $A := |\mathcal{A}|$.

Nearly orthogonal vectors. Assume we are given unit vectors $\{v(a)\}_{a \in [A]} \subset \mathbb{R}^d$ such that

$$\forall a \neq a', \quad |\langle v(a), v(a') \rangle| \leq \rho.$$

Such a set exists for $A \leq \exp(c_0 \rho^2 d)$ for an absolute constant c_0 (e.g. a standard spherical packing / JL-type argument). We will take $A = \lfloor \exp(cd) \rfloor$ with c small enough.

State space. For each layer $h \in \{0, 1, \dots, H\}$, define the set of nonterminal states

$$\mathcal{S}_h^{\text{nt}} := \{\overline{a_{1:h}} : a_1, \dots, a_h \in [A] \text{ are all distinct}\},$$

with $\mathcal{S}_0^{\text{nt}} = \{\bar{\emptyset}\}$. Let f be an absorbing terminal state and set $\mathcal{S}_h := \mathcal{S}_h^{\text{nt}} \cup \{f\}$. The action space is $\mathcal{A} = [A]$, and the initial state is $s_0 = \bar{\emptyset}$.

MDP family. The family is indexed by $a^* \in [A]$. We write $\mathcal{M}_{a^*}$ for the MDP with distinguished index a^* .

Transitions. For each $h \leq H - 1$: if $s_h = f$ then $s_{h+1} = f$ surely. If $s_h = \overline{a_{1:h}}$ is nonterminal and the agent chooses $a \in [A]$, then

$$s_{h+1} := \begin{cases} f, & a \in \{a_1, \dots, a_h\} \quad (\text{repeat an action already on the path}) \\ f, & a = a^* \quad (\text{the distinguished action}) \\ \overline{a_{1:h}a}, & \text{otherwise.} \end{cases}$$

Thus along any *nonterminal* trajectory, actions are distinct and never equal to a^* .

Bias sequence. Set $b_0 := 1$ and define recursively for $h = 0, 1, \dots, H - 1$,

$$b_{h+1} := 3\rho b_h, \quad \text{so that} \quad b_h = (3\rho)^h,$$

and note $3\rho \leq 1/2$ when $\rho \leq 1/6$.

Feature map. Define $\phi_h : \mathcal{S}_h \times \mathcal{A} \rightarrow \mathbb{R}^d$ recursively as follows. For the terminal state, set $\phi_h(f, a) = 0$ for all h and a . For the root,

$$\phi_0(\bar{\emptyset}, a_1) := b_0 v(a_1) = v(a_1).$$

For $h \geq 1$ and a nonterminal state $s_h = \overline{a_{1:h}}$, define for each $a \in [A]$,

$$\phi_h(\overline{a_{1:h}}, a) := \begin{cases} 0, & a \in \{a_1, \dots, a_h\} \\ \left(\langle \phi_{h-1}(\overline{a_{1:h-1}}, a_h), v(a) \rangle + b_h \right) v(a), & a \notin \{a_1, \dots, a_h\}. \end{cases}$$

We view the global feature map as $\phi(s, a, h) := \phi_h(s, a)$.

Candidate parameter and Q . Fix $a^* \in [A]$ and set $\theta^* := v(a^*)$. Define

$$\tilde{Q}_h(s, a) := \langle \phi_h(s, a), \theta^* \rangle, \quad h = 0, 1, \dots, H - 1,$$

and $\tilde{V}_h(s) := \max_{a \in \mathcal{A}} \tilde{Q}_h(s, a)$, with $\tilde{V}_H \equiv 0$.

Mean rewards. Define deterministic mean rewards $r_h : \mathcal{S}_h \times \mathcal{A} \rightarrow \mathbb{R}$ as follows. For each $h = 0, 1, \dots, H - 2$,

$$r_h(s, a) := \begin{cases} \tilde{Q}_h(s, a), & \text{if the transition sends } s_{h+1} = f, \\ -b_{h+1}, & \text{if the transition sends } s_{h+1} \in \mathcal{S}_{h+1}^{\text{nt}}. \end{cases}$$

For the last layer $h = H - 1$, set

$$r_{H-1}(s, a) := \tilde{Q}_{H-1}(s, a).$$

Observed rewards (Gaussian noise). When the learner queries (s, a, h) , the generative model returns the next state according to the deterministic transition above and a noisy reward

$$R_h = r_h(s, a) + \xi_h, \quad \xi_h \sim \mathcal{N}(0, 1) \text{ i.i.d.},$$

independently across all queries.

B.2 Key inequalities: a^* is greedy everywhere

Lemma B.2 (Fresh-direction correlation bound). *For any $t \in \{0, 1, \dots, H-1\}$, any nonterminal $\overline{a_{1:t}} \in \mathcal{S}_t^{\text{nt}}$, any action $a_{t+1} \notin \{a_1, \dots, a_t\}$, and any $a' \notin \{a_1, \dots, a_t, a_{t+1}\}$, we have*

$$|\langle \phi_t(\overline{a_{1:t}}, a_{t+1}), v(a') \rangle| \leq \frac{1}{2} b_{t+1}.$$

Proof. We proceed by induction on t .

Base case $t = 0$. $\phi_0(\overline{\emptyset}, a) = v(a)$, hence for $a' \neq a$,

$$|\langle \phi_0(\overline{\emptyset}, a), v(a') \rangle| = |\langle v(a), v(a') \rangle| \leq \rho = \frac{1}{3} b_1 \leq \frac{1}{2} b_1,$$

since $b_1 = 3\rho$.

Inductive step. Assume the claim holds for $t-1 \geq 0$. Let $a' \notin \{a_1, \dots, a_{t+1}\}$. By definition (and since a_{t+1} is not a repeat),

$$\phi_t(\overline{a_{1:t}}, a_{t+1}) = \left(\langle \phi_{t-1}(\overline{a_{1:t-1}}, a_t), v(a_{t+1}) \rangle + b_t \right) v(a_{t+1}).$$

Thus

$$\begin{aligned} |\langle \phi_t(\overline{a_{1:t}}, a_{t+1}), v(a') \rangle| &= |\langle v(a_{t+1}), v(a') \rangle| \left| \langle \phi_{t-1}(\overline{a_{1:t-1}}, a_t), v(a_{t+1}) \rangle + b_t \right| \\ &\leq \rho \left(\frac{1}{2} b_t + b_t \right) = \frac{3}{2} \rho b_t = \frac{1}{2} b_{t+1}, \end{aligned}$$

using the induction hypothesis (with $a' = a_{t+1}$ at level $t-1$) and the recursion $b_{t+1} = 3\rho b_t$. $\square$

Lemma B.3 (Greedy action is a^* everywhere). *For every $h \in \{0, 1, \dots, H-1\}$ and every nonterminal state $s_h \in \mathcal{S}_h^{\text{nt}}$,*

$$\tilde{Q}_h(s_h, a^*) \geq \frac{1}{2} b_h \quad \text{and} \quad \max_{a \neq a^*} \tilde{Q}_h(s_h, a) \leq \frac{1}{4} b_h,$$

hence $a^* \in \arg\max_a \tilde{Q}_h(s_h, a)$.

Proof. Fix h and $s_h = \overline{a_{1:h}} \in \mathcal{S}_h^{\text{nt}}$.

Lower bound for a^ .* If $h = 0$, then $\tilde{Q}_0(\overline{\emptyset}, a^*) = \langle v(a^*), v(a^*) \rangle = 1 = b_0$. If $h \geq 1$, then along any nonterminal path we have $a^* \notin \{a_1, \dots, a_h\}$, so

$$\tilde{Q}_h(s_h, a^*) = \langle \phi_h(s_h, a^*), v(a^*) \rangle = \langle \phi_{h-1}(\overline{a_{1:h-1}}, a_h), v(a^*) \rangle + b_h.$$

By Lemma B.2 applied at time $t = h-1$ with $a_{t+1} = a_h$ and $a' = a^*$, $|\langle \phi_{h-1}(\overline{a_{1:h-1}}, a_h), v(a^*) \rangle| \leq \frac{1}{2} b_h$. Hence $\tilde{Q}_h(s_h, a^*) \geq \frac{1}{2} b_h$.

Upper bound for $a \neq a^*$. Let $a \neq a^*$. If $a \in \{a_1, \dots, a_h\}$, then $\phi_h(s_h, a) = 0$ by construction, so $\tilde{Q}_h(s_h, a) = 0 \leq \frac{1}{4}b_h$.

Otherwise $a \notin \{a_1, \dots, a_h\}$, and

$$\tilde{Q}_h(s_h, a) = \langle \phi_h(s_h, a), v(a^*) \rangle = \left(\langle \phi_{h-1}(\overline{a_{1:h-1}}, a_h), v(a) \rangle + b_h \right) \langle v(a), v(a^*) \rangle.$$

By Lemma B.2 (with $t = h - 1$, $a_{t+1} = a_h$, $a' = a$), $|\langle \phi_{h-1}(\overline{a_{1:h-1}}, a_h), v(a) \rangle| \leq \frac{1}{2}b_h$. Also $|\langle v(a), v(a^*) \rangle| \leq \rho$. Therefore,

$$\tilde{Q}_h(s_h, a) \leq \left(\frac{1}{2}b_h + b_h \right) \rho = \frac{3}{2}\rho b_h \leq \frac{1}{4}b_h,$$

using $\rho \leq 1/6$. □

B.3 Realizability and Bellman optimality

Lemma B.4 (Bellman optimality of $\tilde{Q}$). *In $\mathcal{M}_{a^*}$, the function $\tilde{Q}$ satisfies the Bellman optimality equations: for all $h \in \{0, \dots, H - 1\}$ and $(s, a) \in \mathcal{S}_h \times \mathcal{A}$,*

$$\tilde{Q}_h(s, a) = r_h(s, a) + \mathbb{E} \left[\tilde{V}_{h+1}(s_{h+1}) \mid s_h = s, a_h = a \right].$$

Consequently, $\tilde{Q} = Q^*$ and the greedy policy is optimal.

Proof. We verify the Bellman equation case-by-case.

If $h = H - 1$, then $\tilde{V}_H \equiv 0$ and $r_{H-1}(s, a) = \tilde{Q}_{H-1}(s, a)$ by definition.

Now fix $h \leq H - 2$. If the transition sends $s_{h+1} = f$, then $\tilde{V}_{h+1}(f) = 0$ and $r_h(s, a) = \tilde{Q}_h(s, a)$ by definition.

Otherwise $s_{h+1} \in \mathcal{S}_{h+1}^{\text{nt}}$. By Lemma B.3, the greedy action at s_{h+1} is a^* , hence

$$\tilde{V}_{h+1}(s_{h+1}) = \tilde{Q}_{h+1}(s_{h+1}, a^*) = \langle \phi_{h+1}(s_{h+1}, a^*), \theta^* \rangle.$$

By the feature recursion and $\theta^* = v(a^*)$,

$$\langle \phi_{h+1}(s_{h+1}, a^*), \theta^* \rangle = \langle \phi_h(s, a), \theta^* \rangle + b_{h+1} = \tilde{Q}_h(s, a) + b_{h+1}.$$

Therefore, using $r_h(s, a) = -b_{h+1}$ on this branch,

$$r_h(s, a) + \tilde{V}_{h+1}(s_{h+1}) = (-b_{h+1}) + (\tilde{Q}_h(s, a) + b_{h+1}) = \tilde{Q}_h(s, a).$$

This establishes Bellman optimality, hence $\tilde{Q} = Q^*$. □

B.4 Proof sketch of Theorem B.1

Proof sketch of Theorem B.1. We instantiate M as $\mathcal{M}_{a^*}$ with a^* drawn uniformly from $[A]$. The feature map ϕ is shared across all instances; the instance index a^* only appears through the parameter $\theta^* = v(a^*)$ (and hence through the mean rewards $r_h(s, a) = \langle \phi_h(s, a), \theta^* \rangle$ at the last layer and on terminal transitions).

Step 1: Near-optimality forces identifying a^* at the root. By Lemma B.3, at the start state $s_0 = \bar{\emptyset}$ we have

$$Q_0^*(s_0, a^*) = \tilde{Q}_0(s_0, a^*) = 1, \quad \max_{a \neq a^*} Q_0^*(s_0, a) = \max_{a \neq a^*} \tilde{Q}_0(s_0, a) \leq \rho \leq \frac{1}{6}.$$

Thus any policy that fails to choose a^* at s_0 suffers a constant value gap (at least $1 - \rho \geq 5/6$ at the root). In particular, producing a 0.05-optimal policy with constant probability requires outputting a^* at s_0 with constant probability, i.e. identifying a^* with constant probability.

Step 2: Unless the algorithm ever queries a^* before the last layer, the transcript depends on a^* only through last-layer rewards. Let Z denote the full transcript of T generative-model queries and observations. Define the event

$$E := \{\text{the algorithm makes at least one query with action } a = a^* \text{ at some } h \leq H - 2\}.$$

We claim that on the complementary event E^c , all observations from layers $h \leq H - 2$ are independent of a^* . Indeed, fix any query (s, a, h) with $h \leq H - 2$ and $a \neq a^*$:

- If a repeats an action already on the path encoded by s , then by construction $\phi_h(s, a) = 0$, the next state is f , and the (mean) reward is $r_h(s, a) = \tilde{Q}_h(s, a) = 0$, independent of a^* .
- Otherwise (a fresh action $a \neq a^*$), the transition is to a nonterminal state and the mean reward is the constant $r_h(s, a) = -b_{h+1}$, again independent of a^* .

Thus, conditioned on E^c , the only dependence of Z on a^* comes from queries at the last layer $h = H - 1$ (where the mean reward is $\tilde{Q}_{H-1}(s, a) = \langle \phi_{H-1}(s, a), \theta^* \rangle$).

Step 3: Hitting a^* early costs $\Omega(A)$ queries. Let N_{a^*} be the number of times the algorithm queries action a^* at layers $h \leq H - 2$. Then $\mathbf{1}_E \leq N_{a^*}$, so

$$\Pr(E) \leq \mathbb{E}[N_{a^*}].$$

Averaging over the uniform draw of a^* and using $\sum_{a=1}^A N_a \leq T$,

$$\mathbb{E}[N_{a^*}] = \frac{1}{A} \sum_{a=1}^A \mathbb{E}[N_a] \leq \frac{T}{A}.$$

Hence if $T \ll A$, then $\Pr(E)$ is small: with high probability the algorithm never queries a^* at any layer $h \leq H - 2$.

Step 4: Conditioned on E^c , learning a^* from last-layer “leakage” costs $\Omega((\log A)/b_{H-1}^2)$ queries. On E^c , the entire a^* -dependence comes from last-layer reward observations. In our construction rewards are observed with i.i.d. $\mathcal{N}(0, 1)$ noise, so each last-layer query yields an observation distributed as $\mathcal{N}(\mu_{a^*}, 1)$ where $\mu_{a^*} = \langle \phi_{H-1}(s, a), v(a^*) \rangle$.

Moreover, by the feature recursion and Lemma B.2, whenever $\phi_{H-1}(s, a) \neq 0$ its norm is on the order of b_{H-1} (in fact $\|\phi_{H-1}(s, a)\|_2 \leq \frac{3}{2}b_{H-1}$), so for any two instances $a^* \neq a^{*'}$ and any fixed last-layer query we have

$$|\mu_{a^*} - \mu_{a^{*'}}| = |\langle \phi_{H-1}(s, a), v(a^*) - v(a^{*'}) \rangle| \leq 3b_{H-1}.$$

Thus the per-query KL divergence between the two corresponding Gaussians is

$$\text{KL}(\mathcal{N}(\mu_{a^*}, 1) \parallel \mathcal{N}(\mu_{a^{*'}}, 1)) = \frac{(\mu_{a^*} - \mu_{a^{*'}})^2}{2} \leq O(b_{H-1}^2).$$

By the chain rule for KL (or mutual information) and adaptivity, the total information in T queries satisfies $\text{KL}(P_{a^*}^Z \parallel P_{a^{*'}}^Z) \leq O(Tb_{H-1}^2)$ for all $a^* \neq a^{*'}$. A standard Fano-type bound then implies that to identify a^* among A possibilities with constant probability, it is necessary that

$$T \gtrsim \frac{\log A}{b_{H-1}^2}.$$

Since $b_{H-1} = (3\rho)^{H-1} \leq 2^{-(H-1)}$ (for $\rho \leq 1/6$), this term is $\exp(\Omega(H))$ up to polynomial factors.

Step 5: Combine and conclude. Putting Steps 3–4 together yields a lower bound of the form

$$T \geq \Omega\left(\min\left\{A, \frac{\log A}{b_{H-1}^2}\right\}\right).$$

With $A = \exp(\Theta(d))$ (from the packing argument) and $b_{H-1} = (3\rho)^{H-1}$, this implies

$$T \geq \exp(\Omega(\min\{d, H\})) \quad (\text{up to polynomial factors}).$$

In particular, taking $H = \Theta(d)$ recovers $T \geq \exp(\Omega(d))$ as stated. By Step 1, such a sample size is necessary for producing a 0.05-optimal policy with constant probability. $\square$

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